BOW_ECON_PRODUCT_BIBLE.md

BOW ECONOMICS PRODUCT BIBLE

Canonical as of: September 4, 2026 **Product deadline:** October 1, 2026 **Bible version: 1.2** **Repo inspected:** `bow-economics-live` @ `eadc041` (2026-09-03) **Owner:** Chief Product Architect (this document is the authority; build prompts are not)

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0. HOW TO USE THIS BIBLE

0.1 What this document is

This is BOW Economics' operating memory. It is not a strategy memo, a pitch, or a curriculum binder. It is the file a future session reads in order to continue the product intelligently without reconstructing hundreds of messages of history.

It answers four questions permanently:

1. What have we decided, and why?
2. What is still empirically open, and what do we do in the meantime?
3. What have we killed, so we stop reinventing it?
4. What should we attack next?

0.2 Status vocabulary — every material issue carries exactly one

Status	Meaning
FOUNDER LOCKED	The founder explicitly decided it. Constitutional.
OPERATING CANON	A strong architect/product ruling BOW currently builds against — real authority, but not founder law. Replaceable by a demonstrably better answer without a founder call.
LOCKED *(legacy label from v1.0)*	Read as OPERATING CANON unless §32 re-labels it FOUNDER LOCKED. v1.0 marked many architect rulings LOCKED; that overstated their authority and v1.1 corrects it.

Status	Meaning
OPERATING DEFAULT	Not proven permanently, but this is what BOW does now. Build against it.
EXPERIMENT	Genuinely empirical. Carries a current default, the evidence needed, the test, and the trigger to change.
DEFERRED BY DESIGN	Explicitly out of current scope. Carries why, when to reconsider, and what behavior replaces it now.
KILLED	Considered and rejected. Carries the reason so it is not reinvented.

There are **no naked TBDs in this document**. Where evidence is absent, the absence is stated and a current behavior is still specified. Where a number is a guess, it says so.

0.3 Source-of-truth hierarchy

When sources conflict, resolve in this order:

1. **Founder rulings issued this session** — they outrank everything below.
2. **Prior explicit founder decisions**, whether recorded here as FOUNDER LOCKED or in the repo's `docs/PRODUCT_DECISIONS.md` with a Founder owner.
3. **Verified real-world fact** (NBA/CBA, research, market data) — authority over what is true. A product decision that requires a false fact is invalid; change the mechanic, not the fact.
4. **This bible** — the synthesised, prosecuted product constitution.
5. **The repository** — authority over what exists today, never over what should exist.
6. **Architect judgments and brainstorms** — the lowest rung, however confidently written.

0.3.1 This bible and the repo's decision log are parallel, not nested

The repo carries its own standing log, `docs/PRODUCT_DECISIONS.md` (D1–D47), plus `CLAUDE.md`, `TRACK_101_MAP.md` and `RAMAZ_READINESS.md`. Those are **equally binding** and closer to the code. **This bible is the product constitution; that log is the build law.** Neither silently overrides the other. Where they conflict, resolve by the hierarchy above and record the conflict in §32 — never by quietly rewriting one to match the other.

Implementation never silently overwrites product truth, and product truth never silently overwrites implementation reality. Both are recorded; the gap is the work.

0.4 Conflict-resolution priority

When principles collide, prefer in this order: **economic truth → student comprehension → meaningful student agency → student pull → classroom pacing → teacher leverage → NBA authenticity → persistence/social energy → visual quality → implementation elegance.**

This is a tiebreaker, not a formula. Full NBA complexity that destroys comprehension loses to honest simplification. Fun built on false economics is killed. A prettier architecture never justifies destroying a working runtime fallback.

0.5 Update protocol

A build result updates §36 (Current Implementation Truth) and, if it produced evidence, §33 (Experiment Register). It updates §32 (Decision Ledger) **only** when the recorded reopening trigger for that decision has actually fired. Progress is not evidence. A passing test is not a proven hypothesis. Real students are the only proof of pull.

Every material change appends a row to §39 (Change Log) and increments the version in the header. **There is one bible.** No `_V2_FINAL`. History lives in the Decision Ledger and the Change Log.

0.6 Epistemic labels used throughout

`[FACT]` sourced and dated · `[JUDGMENT]` product architect's reasoning · `[FOUNDER]` founder decision · `[HYPOTHESIS]` operating belief not yet tested.

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1. EXECUTIVE NORTH STAR

What BOW Economics is. A six-week live social economics experience in which 8–16 students each run one real NBA franchise inside one shared league, make consequential decisions that affect each other, experience the consequences, argue about them, and are then explicitly taught the economics they just lived.

Who it is for. Grades 5–6 and grades 7–8, taught as separate cohorts on a shared simulation. Sports knowledge is never required; sports fluency earns recognition and a small speed advantage, never an information advantage.

What a student should feel. *I ran an NBA franchise for six weeks, and what I did mattered to other people.* Not six lessons. One season.

What a parent should understand. Their child made real decisions under real constraints, defended them out loud, and can now explain scarcity, opportunity cost, and trade to an adult using their own example. The evidence is the child's own words, not a score.

Sessions are 60 minutes, online-first, 12–16 desks. Pairs on one device is the default play mode; solo is supported. Both grade bands (5–6 and 7–8) are in scope, run as separate classes on a shared simulation with a grade-depth layer.

What an instructor should be able to do. A trained instructor (18+, typically 20–24) who has never run a class like this should be able to run an extraordinary one, because the software carries complexity — 16 franchise histories, market resolution, legality, evidence detection — and the human carries meaning: tension, argument, timing, the reveal. The founder is not a runtime dependency.

What must be true by October 1, 2026.

1. All six weeks are runnable end-to-end for both grade bands with persistent franchise state.
2. Every week names one economic concept explicitly and produces class-visible evidence for it.
3. A trained non-founder instructor completes a full dry run of Weeks 1 and 3.
4. The free public playable is live and collects nothing.
5. The Ramaz pilot can start with the safety, consent, and evidence instrumentation already in place.

The two-year strategy, stated here because it should govern build decisions now. The only asset in this product a well-funded competitor cannot copy in a quarter is **a league with a history.** The code, the curriculum and the pricing are all cloneable. A franchise a student inherits next year, and a set of rules written by last year's cohort that this year's cohort must live under (C-014, C-015), is not. That is Tier 3 for October, and **the data model is built for it now**, because retrofitting franchise lineage into a schema is the kind of thing that never happens.

One sentence. *BOW puts kids in charge of a system adults normally control, lets another kid take what they wanted, and then teaches them the word for what just happened.*

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2. CANONICAL PRODUCT THESIS

2.1 What it is

BOW Economics is one continuous six-week NBA franchise experience — Modules 1 and 2 of Track 101. `[FOUNDER LOCKED]` Modules (M1/M2) remain useful for code and curriculum organization. They are invisible to the student. There is no reset between them.

The core loop, canonical:

> **ENTER THE WORLD → SEE THE PROBLEM → MAKE THE BET → COMMIT → THE WORLD MOVES → ANOTHER STUDENT CHANGES MY OPTIONS → CLASS EVIDENCE → ARGUE → NAME THE ECONOMICS → TRANSFER → THE STORY CONTINUES.**

2.2 The social principle, and the grammar inside it `[FOUNDER LOCKED — revised in v1.1]`

The principle is: OTHER HUMANS MATERIALLY CHANGE MY DECISION ENVIRONMENT.

The contested claim — one shared object, several desks that want it, a clock, a public resolution, a visible loser — is **one especially strong recurring grammar inside that principle**, not the universal metaphysics of BOW. v1.0 overreached by forcing every week into one object / one winner / one loser. Weeks 5 and 6 are not shaped like that and should not be.

The permitted repertoire is wider and all of it is already in the built course: **competition · exchange · negotiation · externalities · coalition formation · voting · shared pools · information · institutional rules.**

The old framing, kept because the escalation is still true:

2.2.1 The escalation (retained)

`[JUDGMENT, LOCKED — this is the sharpened thesis]`

Every education product borrows an interaction grammar. Problem sets. Levels. Conversations. Dashboards. All four are single-player grammars, played against a system with infinite patience and infinite supply.

BOW's grammar is **the contested claim**: *one shared object, several children who want it, a clock, a public resolution, a visible loser.* It is the only grammar in this document that cannot be faked by self-serve software, because the instant the rival is simulated the gasp disappears.

The entire six weeks is one primitive escalating in abstraction:

Week	The contested object	The economics it produces
1	**A roster slot, and the players you cannot also have**	Scarcity, opportunity cost

Week	The contested object	The economics it produces
2	**A scarce target, a counterparty's attention, and the price of undoing**	Cost of revision, hidden information, scarce attention
3	**A free agent whose ask moves because of what the room did**	Market price under strategic interaction
4	**Your own payroll, and the night's crowd**	Revenue, and what it takes to clear a bill you created
5	**One shared pool that everyone filled unequally**	Externality, free riding, joint product
6	**The rule itself**	Institutional design, incentives

A ten-year-old who understands *"someone took the thing I wanted, in front of me, and I had to decide what to do next"* in Week 1 has been handed the cognitive template for constitutional design in Week 6. That is a spine. Spines are what nobody else has.

The commitment this implies, and it is uncomfortable: children will lose things in public. Some sessions will be loud. The product is structurally worse for a desk that joins late.

The rule, corrected in v1.1 `[FOUNDER LOCKED — D-002]`: the paid six-week course is fundamentally **live and social**, and **must not be diluted to make solo parity possible.** That is not the same as banning every non-social surface. Solo, public-demo, playable and rehearsal experiences **may** exist — the repo already requires a full teacher rehearsal walkable with zero desks — and they are allowed to be smaller and less powerful. What is forbidden is redesigning the flagship so a single player could get the same thing.

2.3 What it is not

Not standards-alignment software. Not an LMS. Not a quiz platform. Not a worksheet wrapped in animation. Not a fantasy league. Not analytics. Not a career-exploration course. **And not MyGM with economics sprinkled on top** — see §2.4.

2.4 The Anti-MyGM Gate `[LOCKED]`

Every proposed mechanic must answer:

> **What economic reasoning does this create or strengthen?**

and at least one of: what consequence does it create · what social interdependence · what later persistence · what evidence of understanding · what instructor leverage.

If the only answer is *"it feels more like NBA 2K,"* the mechanic is killed or demoted. The NBA fantasy is the vehicle. The economic reasoning is the intellectual product.

2.5 The Anti-School Gate `[LOCKED]`

Applied honestly to BOW's own rituals, because three of them are school moves in costume and pretending otherwise would be the exact failure this gate exists to catch:

- **The commitment capture is an exit ticket**, and it survives — because its four fields (*what we chose / gave up / expect / how sure*) are a front-office record a real GM would keep, and because it is the only instrument that makes a four-week delayed consequence *felt* at age ten. It is signed, kept, and **handed back**. An exit ticket is collected and never seen again; that difference is the whole justification, and if BOW ever stops handing them back, the ritual should be killed.
- **The transfer block is think-pair-share.** It survives as **THE OWNER'S QUESTION** — asked in role, about somebody else's problem, never as *"what did you learn today."* The register is the entire defense and it is thin. Watch it.
- **The rationale gate is a distributed worksheet.** Its defense is that it **blocks a transaction the student wants to make**, rather than interrupting one they don't. Two inputs, never four. If it grows a third field, it has become homework.

Test the opposite too. If an interaction exists only because *schools normally ask students to answer this question*, challenge it. Can the world reveal the concept? Can it be a press conference? Can the instructor provoke it live? Can the evidence come from an authentic decision?

Concrete rule that operationalizes both gates `[FOUNDER LOCKED — D-003, revised]`: do not name the economic concept before the experience earns it. During operation the surface speaks front-office language — cap room, dead cap, the ask, the bill, the vote. At NAMING the instructor says the word, on the board, built from the room's own sentences. **Afterwards the term returns deliberately** — in Previously On, the retrieval question, Decision Replay, franchise history, the season recap and the Week-6 finale deck. Retrieval is the point; hiding the vocabulary forever would have been a different mistake.

2.6 Current product unit and future direction

The product unit is **the six-week course**, sold as a season. `[LOCKED]`

BOW Economics, BOW Financial Literacy, and BOW Decision Challenges remain distinct products and are not merged. `[FOUNDER, LOCKED]` Shared infrastructure (identity, classes, runtime, projector, persistence, reasoning capture, design system) may emerge; shared infrastructure is not a merged product.

The long-term brand promise — *BOW = I get to run something real* — is preserved as a design principle, not as a build target. **Make the NBA experience extraordinary first.** `[LOCKED]`

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3. SUCCESS TESTS

Separated into what is proven and what is not. Nothing in the "proven" column is proven by a test suite.

3.1 PROVEN

Nothing about student experience is proven. No lesson has been run with real students; none has been timed against a real 50–60 minute period. The repo reserves *classroom-proven* for what has survived a real class, and its readiness ladder tops out at **classroom-ready candidate** for all six lessons (repo D10).

What **is** established, repo-verified this session: six registered lesson modules covering all six weeks, each a classroom-ready candidate with independent fresh-context gate certification; 595+ passing tests; Playwright proofs including the full Week 1 → 2 → 3 chain played for real through the API; concurrency and latency harnesses at 16 and 32 desks; teacher-transfer gates returning READY on all three Module 2 lessons. **That is implementation progress and it is substantial. It is not evidence of pull, learning or transfer.**

3.2 NOT YET PROVEN — the seven tests

#	Test	Passes when	Measured by
T1 Learning	Students own the course's economic concepts	**≥60% of students give *strong* evidence (§14.3) on ≥2 concepts, and ≥85% on ≥1**	Four capture streams, deliberately sized so the test is computable: (a) six commitment captures per student — opportunity cost, every student, every week; (b) the **rotating weekly 90-second Front Office Review**, which at 2 franchises per

#	Test	Passes when	Measured by
			week from W2 gives ~10 observations per cohort; (c) the **transfer block's three shares-out**, logged with a one-tap concept code — 15 logged utterances per cohort; (d) the 4-student rotating sample. Together ≈ **4–5 observations per student.** The earlier ≥3-of-6 threshold was not computable from 1.5 observations per student and has been corrected rather than defended.
T2 Pull	Kids want the next session	Session-6 attendance ≥80% (c1), ≥85% (c2), ≥88% (c3). **Note the base rate honestly: comparable programs lose 20–35% by week four, so an 80% target is already an outperformance claim and the <75% stop condition sits inside the industry's normal band.** A first cohort at 76% is not a failure signal; it is the industry mean. The stop condition applies from cohort 2 onward.	Attendance

#	Test	Passes when	Measured by
T3 Continuation	Kids ask for more BOW	Kid-Ask Rate ≥50% (c1), ≥60% (c2), ≥65% (c3)	One SMS to parent after Session 5
T4 Teacher transfer	A non-founder runs it well	A trained instructor's cohort lands within 1 SD of the founder's on attendance and airtime distribution	Instructor telemetry (§17.5)
T5 Classroom reliability	The runtime does not break the class	Zero sessions lost to state, sync, or transaction faults across a full cohort; 16 desks nominal, 32 stress	Event history + incident log
T6 Parent value	Parents can say what they bought	≥1.0 referral actions per family by cohort 3	Share/waitlist attribution
T7 Business evidence	The model can exist without paid acquisition	Repeat-course purchase ≥50% at n≥100 families	Enrollment

T2 and T3 are the load-bearing tests. A product that teaches economics but that kids do not want to return to is insufficient. A product kids love that teaches no credible economics is insufficient. The objective is **LEARNING × PULL × CONTINUATION**. `[FOUNDER, LOCKED]`

T7 is deliberately deferred as an operating metric. At 42 families across three cohorts, a repeat-rate estimate carries roughly ±15 points of noise; steering on it means steering on a coin flip. Report it quarterly; run the business on T2, T3, T6. `[OPERATING DEFAULT — D-041]`

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4. SIX-WEEK EXPERIENCE MAP

v1.1 REBASE. v1.0 wrote this map from founder prose before the repository was inspected. The repository (`bow-economics-live`) contains **six built, independently gate-certified lessons**, and Module 1's arc is **not** the one v1.0 described. The built arc is canonical. Where v1.0's map and the repo disagree, the repo wins (§0.3, and the founder rulings D17/D18 that produced it). The pedagogy, the reveal discipline and the grade-band contracts from v1.0 survive intact; the week identities do not.

4.0 The product unit, restated against reality `[OPERATING CANON]`

The six-week course is **Module 1 (weeks 1–3) + Module 2 (weeks 4–6)** of Track 101. Modules 3 and 4 exist in the repo's founder-fixed four-module spine (repo D1/D2) and are **not** part of the October 1 product — they are the second course, and that is what makes repeat purchase (§27.4) structurally possible.

Week	Module / lesson id	Built title	Repo status
1	`m1l1-draft-day`	**Draft Day**	classroom-ready candidate — STRONG, round-3 certified
2	`m1l2-trade-deadline`	**The Trade Deadline: Undo Isn't Free**	classroom-ready candidate — STRONG, 2 e2e proofs
3	`m1l3-free-agency`	**Free Agency: The Signing Window** *(module finale)*	classroom-ready candidate — STRONG, 4 e2e proofs incl. full L1→L2→L3 chain
4	`m2l1-full-house`	**Full House** → reframed **THE BILL COMES DUE**	classroom-ready candidate — STRONG ×3, TRANSFER READY
5	`m2l2-host-league`	**You Don't Play Alone**	classroom-ready candidate — STRONG, TRANSFER READY
6	`m2l3-write-rule`	**Writing the Rule**	classroom-ready candidate — STRONG, TRANSFER READY

Nothing above is classroom-proven (repo D10). Six certified candidates is a much stronger starting position than v1.0 assumed, and it changes the October 1 job from *build six weeks* to

make six built weeks feel like one premium product, and close the seams and primitives that do not yet exist.

4.0.1 Architecture decisions that govern all six weeks

D-010 — MULTIPLE TRUTHFUL VIEWS, NO MACHINE VERDICT. **`[FOUNDER LOCKED – principle]`** There is no BOW-created composite decision-quality score and no universal machine-declared winner. The franchise's position is shown through several truthful, separately-legible readings. **The specific four-ladder set (WINS · CAP HEALTH · ASSET VALUE · FAN SUPPORT) is `[OPERATING CANON]`, not constitutional** — a substantially better truthful representation may replace it. What may never return is a single composite number BOW invented and ranked children on. Consistent with repo D4 (no XP, levels, badges, leaderboards).

Each ladder is a real, publicly checkable quantity, displayed as components and never summed:

Ladder	What it is
WINS / RESULT	The lesson's own outcome (standings, gate, net)
CAP HEALTH	Room against the line, dead cap carried, tools still in hand — components, never one score
ASSET VALUE	Unspent picks · years of control · room next season — three columns, sorted, never summed
FAN SUPPORT	Fill and renewals — two numbers

D-011 — STATED INTENT THAT IS ALLOWED TO CHANGE. **`[FOUNDER LOCKED, revised in v1.1]`** Students state a strategic intent early and the product holds them to it *as evidence, not as a class*. **CONTEND / RETOOL / REBUILD** remains useful language. It is **not** an RPG character class, it does not gate legal moves, and it may be changed. The deeper truth the product teaches is: **opening plan → the world changes → change course if justified.** A strategy change is first-class reasoning evidence and a natural Press Conference trigger — *"You said you were rebuilding in Week 1. You just traded two picks. What changed?"*

D-012 — ROTATING LEVERAGE AND VIABLE PATHS. **`[FOUNDER LOCKED, replaces the Peak Week Guarantee]`** Every franchise stays strategically alive; different constraints create different leverage at different moments; no desk is doomed for six weeks; legitimate league mechanisms may rebalance opportunity. **There is no scripted week in which each**

franchise is guaranteed to "win." No obvious charity, no hidden trophy week. The universe stays economically truthful, and a desk that played badly is allowed to be behind.

D-013 — DIEGETIC RECOVERY ONLY. `[OPERATING CANON]` Recovery mechanisms must be things the real league contains — expiring contracts, the buyout market, the cap rise, seller leverage at a deadline, revenue sharing. The teacher never gives help. The repo already honours this: L1's ADAPT budget equals remaining cap room (repo D15), L2's rescue window is restricted to open-slot teams with a brute-forced ≥2-affordable guarantee (repo D17d), L3's cap rise is an institution adjusting (repo D18).

D-014 — DIFFERENT ENERGY PROFILES. **`[OPERATING CANON — vocabulary, not taxonomy]`** Six weeks should not have identical spectacle. Weeks 1, 3 and 6 carry the loudest beats; weeks 2, 4 and 5 are denser and more argumentative. **EVENT / OPERATE is architecture vocabulary, not law** — never preserve a weak session because its label says OPERATE, and never inflate a good quiet week for symmetry.

D-015 — ANCHOR + SUPPORTING + RECOGNITION. **`[FOUNDER LOCKED, replaces "six concepts"]`** v1.0's "six owned concepts, full stop" under-described an economics course. Three tiers (§5):

- **MUST OWN — 6–8 anchors across the course.** One anchor per week, retrieved every subsequent week, assessed.
- **MUST UNDERSTAND / USE — 12–18 supporting concepts.** Explicitly taught and used; recognised and applied, not drilled for recall.
- **RECOGNITION — additional sports-business and economic vocabulary** met in context. Do not create thirty equal objectives. Do not claim a term was learned because it appeared.

D-016 — PUBLIC STATE, PRIVATE INTENT. `[OPERATING CANON]` Public to the room: rosters, cap/dead-cap position, ticket prices, gate, published asks, league feed. Private: targets, offers in flight, rationales, predictions, **and roster needs until a desk publishes one.** Publishing a need is itself a decision — it attracts the right calls and tells fifteen rivals where you are weak. Structurally enforced by the repo: `boardView` is never handed a seat identity; `studentView` never leaks another seat.

D-017 — THE SIM'S ONLY JOB IS CONSEQUENCE. `[OPERATING CANON]` No box scores, no per-player statlines. Weekly output is a franchise's own result line, its books, and one causal sentence. Every minute reading simulated basketball is a minute not deciding, and BOW cannot win a fight with NBA 2K.

D-018 — CONCRETENESS FADING ACROSS THE ARC. `[OPERATING CANON]` Week 1 renders franchises with names, dollars, contract cards. Week 6 renders them as market-size groups and dots on an incentive plane. Concrete-to-abstract on one underlying structure supports transfer better than either alone.

D-019 — THE SESSION SHAPE. `[FOUNDER LOCKED — 60 minutes]` Standard session is **60 minutes**. Phase budgets are structural, not robotic timestamps: the software protects the economics from gameplay expansion rather than enforcing a stopwatch.

Share of session	Block	Non-negotiable
~0–8%	**PREVIOUSLY ON + RETRIEVAL** — this desk's three facts, plus one retrieval question on a prior week's anchor	The retrieval question. It is the only spacing mechanism the course has.
~8–17%	**SETUP** — the decision and the constraint. **Do not name the concept.**	Under five minutes. Longer means the mechanics are too complicated.
~17–42%	**PLAY / DECIDE**	A visible close. See the protection rule below.
~42–55%	**REVEAL** — curated contrast, 4–6 desks, one variable, sorted	Written prediction *before* the screen changes.
~55–63%	**STUDENT DESCRIPTION** — instructor types student words	The naming screen will not advance without it.
~63–80%	**NAMING** — the anchor, attached to the words already on the board, plus one non-example	Dialogue, not monologue.
~80–92%	**TRANSFER — the Owner's Question**, asked in role	Non-negotiable.
~92–100%	**DECISION CAPTURE + PRESS CONFERENCE**	The forgone-alternative field.

The economics protection rule `[FOUNDER LOCKED]`. The teacher has bounded flexibility over pacing; the naming and interpretation of the economics cannot be skipped because the market was fun. `/teach` therefore surfaces, from the server clock (repo D42 already carries a class clock):

ECONOMICS AT RISK — 11 MINUTES REMAIN. FINAL CALL.

This replaces v1.0's un-extendable timer, which was too rigid for a live room, with the thing that timer was actually protecting.

Week 6 length. Week 6 runs **60 minutes like every other week by default.** The 75–90 minute finale is an explicit `[OPERATING DEFAULT]` exception available only where the school gives a double period, and Week 6 must be fully walkable in 60. The durable shape is proportional, not absolute: **the vote lands at roughly 55% of the session, and consequence occupies the remainder.** The repo already flags Week 6's reveal half as over-budget against a 7–15 minute debrief standard — that gap is real and is Wave 7 work, not a reason to normalise 90 minutes.

4.1 WEEK 1 — **DRAFT DAY** *(`m1l1-draft-day`)*

Student fantasy. *I have five slots, one hundred million dollars, and more players I want than I can fit.*

Built mechanic (preserve). A reversible **Roster Wall** with a live **Foregone Panel** — every placement stays editable pre-lock, so opportunity cost is *ambient*, not an end-of-round report (repo D13). $100M cap, $10M steps, five slots (repo D11e), cap inviolable through every reachable sequence. A lock ritual, then a **shock**, then **ADAPT** with a budget that equals remaining cap room and never breaks the cap (repo D15), then a **RISK BUFFER** whose claim is mechanically true rather than asserted (repo D16).

Central tension. *I want all of them → why can't I just do that? → taking him means not taking her → that is the tradeoff.*

Do not brief the constraint — let them collide with it. The Foregone Panel already does this: the cost of a placement is visible *while* the student is deciding, not afterward. Protect that. Twenty minutes of a ten-year-old studying a cap sheet before touching anything is a study hall with a basketball skin.

Social interaction. Currently the weakest week on the founder's principle that *other humans materially change my decision environment*. Draft Day is a constrained-allocation problem against the cap, not against the room. **Wave 2 requirement:** add one truthful cross-desk consequence — the strongest candidate is a **shared scarce pool at the top of the board**, where the very best players are genuinely finite across the room, resolved simultaneously and sealed so that speed never decides (founder ruling 10). This must be prosecuted against L1's certified cap-inviolability and its ADAPT guarantees before it ships.

Economics. ANCHOR: OPPORTUNITY COST. Supporting: scarcity · constrained allocation · substitutes · risk buffer · the cap as an institution. Misconceptions attacked: opportunity cost = money spent (§14); more options is always better.

Class evidence. Simultaneous display of 4–6 desks sorted by *what each gave up to get what they got.* The contrast that lands: a tightly constrained desk that had to be decisive often ends better than one that dithered.

Teacher moves. Name the market. Name opportunity cost from student words. Seed decision-quality-vs-outcome in the closing minute — this is classroom-management infrastructure for Week 2, not just sequencing.

Projector beats. Lobby and desks filling · lock ritual · the shock landing across the room · the reveal sorted by forgone value.

Grade bands. 5–6: options given, constraint stated, concept withheld; two variables traded off; cap slices alongside dollars; no negatives. **7–8:** options partly self-generated, the constraint inferred from the books, three variables, the counterfactual generated in writing before the reveal, and a heavier obligation to justify. (Repo D22 program 2 builds this from birth; repo D38 names the two attachment points.)

State out. Roster · payroll · cap room · dead cap exposure · risk buffer · stated intent · shock response · the forgone set (which is the whole opportunity-cost record).

Transfer — the Owner's Question. One Saturday, three things you want, one you can do. Name the *one* you gave up that hurts most — not "the other two."

Failure / recovery. Absent desk: honest stock franchise, never a broken one (repo CLAUDE.md §9). Pairs default, solo supported (repo D11b).

Return hinge to W2 `[build]`. Every desk seals a falsifiable claim about its roster plus a confidence, opened in Week 5. Then, flat: *"One of the players in this room will not be worth what you paid. We find out Thursday."*

October 1 status. BUILT. Work remaining: shared scarce pool (social), sealed prediction capture, Decision Replay hooks, four-ladder display, and the two recorded M1 defects (see §36).

4.2 WEEK 2 — **THE TRADE DEADLINE: UNDO ISN'T FREE** *(`m1l2-trade-deadline`)*

Student fantasy. *My team from last week has a problem, and fixing it costs something.*

Built mechanic (preserve). L1's franchise state carries in through an opaque seed; desks **claim their own franchise**; a deterministic, path-dependent **midseason report** surfaces L1's busts and gems (no random injury — L1 already owns involuntary loss). Three real paths: **STAND PAT** (an affirmative, reasoned lock), **CUT + VETERAN** (safe, known value), **CUT + SEALED BID** on four scarce named targets against the room *and* a hidden per-seller reserve. **The cut commits at submission** — a losing bidder keeps the hole and the dead cap — with a restricted, brute-force-verified rescue window that closes the wait-and-see exploit. Dead cap ≈10% of the cut player's value. Teacher-paced per-target reveal theatre. (Repo D17.)

Central tension. Revising a commitment is not free, and you are bidding blind against people you can see.

THE ONLINE TRADE DEADLINE WAR ROOM `[OPERATING CANON — highest-priority new surface]`. Week 2 is where *other humans materially change my decision environment* becomes visceral, and the repo currently has **no desk-to-desk interaction anywhere**. The War Room (Mockup 3) lands here, as an addition that must not break L2's certified sealed-bid economics:

- a deadline clock the whole room shares;
- **a small number of active negotiation lines per desk** — this is the mechanic that makes **attention itself scarce**, which is the genuinely new economics online can teach better than a physical trade floor;
- **incoming calls** from other desks that must be taken or declined, each one costing a line;
- an offer builder, one counter, a decline that carries a reason;
- thread timeouts that return a line;
- public NEED / AVAILABLE signals a desk chooses to publish (D-016);
- completed trades announced to the room as league events.

Fable must froth this before building (§ build plan Wave 4): whether desk-to-desk trading is a fourth deadline path inside L2, a distinct pre-deadline window, or its own surface. **Default: a fourth path inside L2.** The target sentence is the acceptance condition: *"I only have two lines open, Miami is calling, Boston hasn't answered, the clock is running, and another trade just changed what I can do."*

Economics. ANCHOR: THE COST OF UNDOING A COMMITMENT (dead cap / sunk-but-not-free). Supporting: path dependence · bidding under hidden information · winner's curse · scarce attention as a real constraint · decision quality vs. outcome. Misconceptions attacked: sunk cost · outcome bias · "revision is free."

The anti-outcome-bias drill. Two desks that made the *same* call and got *different* results, side by side, with the room rating the decision **before** the outcome is revealed and again after.

Outcome bias survives being told the rule; it only moves under repeated structured practice on uncomfortable cases. This drill returns in W3, W5 and W6, and is the seed of **THE TAPE** (§12).

Teacher moves. Run the near-miss pair. Refuse to let the room evaluate by result. Read one desk its own Week-1 forgone list before it decides.

Projector beats. The midseason report landing across the room · per-target reveal theatre · the dead-cap bite · completed trades as league events.

Grade bands. 5–6: trade from proposed structures; one counter; guided builder. **7–8:** originate offers; more lines but a harder justification burden; must concede the strongest cost of their own decision.

State out. Post-deadline roster · dead cap carried · cap room · who dealt with whom · who refused · the reasons given.

Transfer. Your friend didn't study, guessed, got a B. Good decision? You studied three hours and got a C. Was *that*?

Return hinge to W3 `[build]`. Actual pending interest: *"Three teams asked about your best player."* Show three crests, not who. An inbox with unread mail is the strongest return hinge any product has, and it is nearly free.

October 1 status. BUILT + the War Room is the largest net-new surface in the course.

4.3 WEEK 3 — **FREE AGENCY: THE SIGNING WINDOW** *(`m1l3-free-agency` — module finale)*

Student fantasy. *Everyone in this room is constrained too, and their choices are changing my options.*

Built mechanic (preserve). A **four-day market inside PLAY**: one binding sealed offer per desk per day or an explicit hold; teacher-paced day closes resolving every agent simultaneously; top offer at or above the public ask signs **at the offer price**; unsigned agents' asks move with demand (0 offers −$10M · 1 offer −$5M · **2+ offers +$5M**, floor $10M); day 4 desperation lets a top offer sign below ask. The cap rises **$100M → $130M** — an institution adjusting — which guarantees every carried franchise $30–80M of room while fully preserving L1/L2 path dependence including dead cap. Hidden playoff factors on eight fixed agents feed a staged module finale: window recap → factor reveals → standings → semis/final → computed **GM Awards → COUNTERFACTUAL** (personal what-ifs plus class-level patience-dividend and dead-cap-drag cards) → SYNTHESIS closing the three-lesson arc. Durable rule: **withdrawing**

an offer ends that desk's market day — editing is free, but talk is only cheap until you want your slot back. (Repo D18.)

This is where "same player, different cost" actually lives. v1.0 named an engine called *The Same Line* and placed it in Week 1. The built product delivers the same economic truth in a better place: by Week 3 every desk arrives with a different roster, different dead cap and different room, so **the identical free agent is a different decision for every desk in the room** — and the ask moves because of what the room does. Keep the truth; retire the invented name.

Sealed resolution stays. Founder ruling 10: major free-agency resolution is sealed and simultaneous so reaction speed never decides an outcome. The live, urgent, human-versus-human energy in Module 1 belongs to Week 2's War Room; Week 3's drama is the **day close** and the **ask moving**.

Economics. ANCHOR: MARKET-DETERMINED PRICES UNDER STRATEGIC INTERACTION. Supporting: opportunity cost of timing · patience dividend · path dependence across the whole module · decision quality vs. outcome luck · the cap rise as institutional design (which is also the Week-6 seed). Misconceptions attacked: zero-sum bias · "later is worse" · outcome bias.

The zero-sum correction, stated explicitly. For three weeks "competition" has meant *athletic* competition, where somebody loses. In this market, competition among buyers changed prices and made agents better off. Those are two different words that sound the same. Zero-sum bias is only reduced by explicit statement of the structure — so state it, in Week 3, on the board.

Known defect to repair (repo). `freeAgency.ts` breaks a tie between two identical sealed bids on `submittedAt` — the server's record of which Chromebook was faster. That is the one thing a sealed bid exists to rule out. **Replace with an economically honest tie-break** (cap room, or the agent choosing on a published preference), per §36.

Teacher moves. Post the ask movement and say nothing for thirty seconds. Run the zero-sum contrast explicitly. Run the counterfactual cards without commentary and let the room react first.

Projector beats. The ticker at day close · asks moving · signings landing · the staged finale · GM Awards · the counterfactual.

Grade bands. 5–6: a shortlist and a guided offer. **7–8:** set the price freely, read the ask movement as a signal, and price against what the room will do next.

State out. Final roster · payroll at $130M cap · dead cap · room · who signed whom · who walked · the counterfactual record · **the payroll that becomes Week 4's bill.**

Transfer. You and a friend both want the last ticket. Who ends up paying more, and why is that not the seller's doing?

Return hinge to W4 `[FOUNDER LOCKED – build]`. Not a greeting. The invoice:

> **DETROIT OWES $184.2M. DETROIT HAS $161.9M.**

Every desk short. Closing line: *"Every one of you owes more than you have. Next Thursday you find out how to get it."*

October 1 status. BUILT. Work remaining: the honest tie-break, the M1→M2 seed (below), Press Conference and Tape hooks, and the e2e client race in `E2E_L3_FLAKE_NOTE.md`.

4.4 WEEK 4 — **THE BILL COMES DUE** *(`m2l1-full-house`)* `[FOUNDER LOCKED]`

Student fantasy. *The team I built is expensive. Now I have to pay for it.*

Built mechanic (preserve). Price the night blind against **real NBA night cards**; two non-collapsible books — **CASH** and **RENEWALS** — that cannot be reduced to one number; a hidden demand curve; the **Two Peaks** reveal; the **GATE CALL** free commitment that removes dead air (repo D28); path dependence across nights via the repeat card; a measured, honest arena fill encoding (repo D25, D21 FD-3); the computed frontier drawn in both books' own units and **never captioned as "you gave up almost nothing"** (repo D21 FD-1).

THE SEAM — the single most important build in the course. `[FOUNDER LOCKED]` The repo states plainly that **M1→M2 deliberately does not seed.** Week 4 today would work just as well with a generic team. That is exactly the failure the founder's Week-4 lock forbids.

Requirement: the Week-3 franchise seeds Week 4 through the runtime's existing opaque-seed mechanism, and **the payroll the desk built in Weeks 1–3 becomes the bill it must cover in Week 4** — by name, in dollars, with the players visible. The cost line is not invented; it already exists. This is the carry graph made visceral, and it is the difference between a course and six lessons.

Say the reason precisely, because the obvious version is wrong. With a fixed payroll and near-zero marginal cost per seat — which is what gate revenue is — revenue maximisation *is* profit maximisation. The payroll line does **not** move the optimal price. What it adds is a **solvency constraint**: a number you must clear or you lose the player you fought for in Week 1. That is dramatic, it is true, and claiming instead that costs move the optimal price would teach the sunk-cost fallacy this course exists to attack.

Economics. ANCHOR: REVENUE = PRICE × QUANTITY, AND REVENUE IS NOT THE SAME AS CLEARING YOUR BILL. Supporting: demand shifters · rising opportunity cost between two books · the total-revenue test (elasticity by another name) · path dependence · the difference between a full building and a good night. Misconceptions attacked: revenue = profit · price is set by cost · a sellout means you priced right.

On elasticity. *"When I raised the price, did total money go up or down?"* is the total-revenue test — a standard operationalisation, not a softening. Three ways it becomes a lie, all forbidden: making elasticity a property of the good ("luxuries are elastic"); over-generalising one trial into a rule; and implying revenue maximisation is the goal. **5–6 teach the revenue test and do not say the word. 7–8 name it**, define it as *how strongly quantity reacts when price changes*, and do **no percent-change arithmetic** — percent-change is the documented wall for this band, and the formula would consume the session teaching arithmetic.

Class evidence — THE MIRROR. The room's own pricing decisions from the first half become the dataset it analyses in the second. Desks as dots, price against gate, sorted. The curve is theirs. Controlled for market size, or the class cannot tell "different starting price" from "different city."

Teacher moves. Prediction before reveal. Student description before naming. The identical-cost contrast: two desks, same building costs, different right prices — *what differs is the buyers, not the costs.*

Projector beats. The invoices landing across the room · the fill drawing at true magnitude · the Two Peaks · the Mirror filling in live.

Grade bands. 5–6: price from a small menu; cost shown in slices; the shortfall stated as *"you are $22M short,"* never as a negative. **7–8:** price freely across night classes, carry the payroll, and get a second reprice round so the price trap can be tested in both directions on the same franchise.

State out. Gate · renewals · whether the bill cleared · what was sold under duress · fan support delta · price history (the input to Week 5's pool).

Transfer. The fundraiser cost $400 to run. Does that change which price you pick? *(No — it changes whether you clear your costs. Students who say it changes the price have found the sunk-cost trap, which is the point of the question.)*

Return hinge to W5 `[build]`. The room's gate spread revealed at the buzzer, top to bottom, with one line: *"Next Thursday, some of you are going to find out you've been paying for somebody else's building."*

October 1 status. BUILT — plus the M1→M2 seam, which is founder-locked and does not exist.

4.5 WEEK 5 — **YOU DON'T PLAY ALONE** *(`m2l2-host-league`)*

Student fantasy. *My building is not an island, and I am not sure I like that.*

Built mechanic (preserve). Host the league: **the visiting club's Draw is the dominant term in your own gate.** Reinvest-in-Draw as an intertemporal trade (fast gate money now against slow Draw money later). A decomposition reveal that shows where the money in your building actually came from, in two instruments: **DEALT** totals (the Draw you were handed) and **BY-CHOICE** figures recomputed against a never-reinvested counterfactual (repo gate repair g). Real NBA clubs as desks. An equal national check that clears every club's bill from every reachable state.

THE POOL — the online-native ritual `[FOUNDER LOCKED in principle]`. The underlying mechanic is: **unequal contributions visibly enter one common pool, and the room decides how it comes out.** Do not anchor the design to a literal classroom bowl — BOW Economics is online-first. Build the ritual for the screen:

> Sixteen contributions animate into one pool, each labelled with its desk and its size, so the inequality is *seen* rather than described. The total builds. The screen stops. `/teach` says: **WHO GETS IT?**

A cheesy bowl animation fails this. The requirement is that the shared resource feel *real* and the inequality feel *unfair to somebody in the room*. **The split vote is binding and executes in-session** — the money moves on screen before anyone leaves. Week 5 decides *this year's* split; Week 6 writes the rule that governs every year. Those are different acts and the escalation between them is the point.

Economics. ANCHOR: EXTERNALITY. Supporting: joint product · free riding · market size · revenue sharing incentives · the equity–efficiency tension. Misconceptions attacked: "an externality is a side effect" · "fair means equal" · "sharing is fair, therefore good."

The definition BOW uses, and the error it must not make. `[FOUNDER LOCKED — the one economics correction from v1.1's prosecution]` An externality is not "a side effect." The defining features are that the affected party **had no say** and **no price mediated the effect.** The diagnostic taught instead of a definition: ***Who was at the table, and who got the bill?***

A rival outbidding you in Week 3 is NOT an externality. You were at the table; it ran through a market you were both in. That is a *pecuniary* externality — the textbook name for ordinary competition, and the most common error in undergraduate economics. Seeding Week 5's anchor with its own counterexample would teach the misconception the course exists to attack.

Qualifying cases are effects that run **outside any market the harmed desk was in**: a club that rests its stars degrading a night other clubs' fans paid for; a weak visiting Draw; a decision whose cost lands on people who never chose. Students *will* raise the outbidding case, and the instructor naming why it is competition rather than externality is one of the best teaching moments in the course.

Both halves, always. Sharing makes small markets viable **and** weakens every club's incentive to build its own market. If Week 5 leaves students believing sharing is unambiguously good, BOW has taught a political position, not an economic concept.

Teacher moves. Show the pool. Ask who gets it. Say nothing for as long as it takes. Split the board: ***does it work?*** | ***is it fair?*** — answered separately, and one answer may never serve for both. The best moment of the week is a student saying *"it works but I don't like it."*

Projector beats. Contributions filling the pool · the decomposition reveal · the by-choice counterfactual · the binding split executing · the sealed Week-1 predictions opening.

Grade bands. 5–6: vote between two clearly stated splits; the projector performs the aggregation their cognition cannot yet perform internally — the single best use of the board in the course and the reason Week 6 works for this band at all. **7–8:** propose the formula, then defend it against the desk it hurts most.

State out. Contribution and draw per desk · the executed split · reinvestment choices · declared bloc · calibration of the Week-1 predictions.

Transfer. Group project, five people, one grade. Predict what happens. Now design a rule that fixes it — then predict how someone beats your rule.

Return hinge to W6 `[build]`. *"You just spent an hour arguing about a rule nobody has to follow. Next Thursday you write one that everybody does — and then you live in it."*

October 1 status. BUILT. Work remaining: the online pool ritual, the binding split, the prediction replay, and the open sports-reality and projector findings in §36.

4.6 WEEK 6 — **WRITING THE RULE** *(`m2l3-write-rule`)*

Student fantasy. *I don't play the game any more. I write it.*

Built mechanic (preserve). Rule dials → a **sealed two-thirds vote** (sealed at the close of round 3; a post-close proposal is refused by the reducer; a desk with no number **abstains** rather than being recorded at a number nobody proposed) → a **lived-under season** in which the room's own rule actually governs → the **Kings 22-8 commit-then-reveal capstone**

(outcome vs. decision quality) → the revisit-able **"Economics You Learned"** finale. L2→L3 seeds opaquely. A live in-band gauge on the board: *"right now N of M would pass; K are needed."* (Repo D20, and the wave-4 repairs.)

The shape is already right. v1.0 argued that ending on a vote is weak because deliberation is not a climax for twelve-year-olds — consequence is. **The built lesson already votes and then lives under the rule.** Preserve that ordering and strengthen it:

1. **Private, dollar-denominated stakes** delivered before debate opens, so a desk knows what the proposal is worth *to it* and must persuade someone to vote against their own interest. That is the actual lesson of institutional design.
2. **A threshold that requires defectors.** Two-thirds already does this. Engineer the natural split so unanimity is not the likely outcome — a vote that passes 16–0 is a mock UN.
3. **The vote must change something they then see.** Already true. Protect it.
4. **The counterfactual — what the other rule would have produced.** Thirty seconds, and the entire concept of institutional design delivered as a gotcha, which is the only register in which twelve-year-olds accept intellectual content.

THE FRANCHISE AUCTION `[EARNED / EXPERIMENTAL — not architecture-critical]`. A closing valuation in which the room prices each other's franchises is promising and stays *inside* Week 6's design as an option, **not as a dependency.** If it ships it must be guarded against: popularity contest (sealed bids, equal budgets stated in Week 2, anonymous), sports-knowledge bias, current-wins bias, and any reading of the clearing price as a secret universal winner. **The framing is fixed: the price is the market's opinion, and the market can be wrong.** The closing question is *"which franchise did the market get wrong?"* — which converts a ranking into an argument the lowest-priced desk can win. If the auction cannot be built without becoming a scoreboard, it does not ship, and Week 6 is not weakened by its absence.

Economics. ANCHOR: INCENTIVES AND RULES (institutional design). Supporting: differential incentive response · capacity-bound immobility · self-interest vs. collective outcome · unintended consequences · outcome vs. decision quality (the Kings capstone). Misconceptions attacked: "good rules make things fair" — institutional design is about aligning private incentive with collective outcome; fairness is a separate question reasonable people answer differently.

5–6: one step — *"if we pass this, what will the other fifteen desks do next week?"* **7–8:** the real thing — *"design a rule that still works even if every GM is completely selfish,"* predict how it gets gamed, and revise once before the vote.

Teacher moves. Refuse to say which rule is better. Split the board — *does it work? / is it fair?* Name every economic term a student uses correctly, out loud, as it happens. Run the counterfactual with no commentary and let the room react first.

Projector beats. The dials and the in-band gauge · the sealed vote and the roll call · the season running under the room's own rule · the counterfactual · the Kings capstone in two presses · the finale deck.

State out. The signed rule · the lived-under result · the counterfactual · the finale deck (a high-water mark, not a live page — repo D47) · everything the parent artifact and the student's own recap read from.

Transfer. Your class writes its own phone rule. Write it, then write how a clever classmate gets around it, then fix it.

Return hinge — after the course. Six weeks ending in "good job everyone" is a waste. Every desk leaves with a **season card**: their franchise, their stated intent and whether it changed, three defining decisions, where they finished on each truthful reading, and one line of specific praise generated from their actual choices. **And the Course 2 seat is on the same page** (§27.4).

October 1 status. BUILT. Work remaining: private stakes cards, the auction if it earns it, the timing gap on the reveal half, and the open economics ruling in §36.5.

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5. ECONOMICS SPINE

5.1 The three tiers `[FOUNDER LOCKED — D-015]`

MUST OWN — the anchors. One per week, retrieved in every subsequent week, assessed against §14's evidence standard.

Wk	Lesson	ANCHOR
1	Draft Day	**OPPORTUNITY COST**
2	The Trade Deadline	**THE COST OF UNDOING A COMMITMENT** *(dead cap; revision is not free)*
3	Free Agency	**MARKET-DETERMINED PRICE UNDER STRATEGIC INTERACTION**
4	The Bill Comes Due	**REVENUE = PRICE × QUANTITY** *(and revenue is*

Wk	Lesson	ANCHOR
		not the same as clearing your bill)
5	You Don't Play Alone	**EXTERNALITY**
6	Writing the Rule	**INCENTIVES AND RULES** *(institutional design)*

Plus two anchors that are **taught across weeks rather than owned by one**, which is what makes 6–8 rather than 6:

Anchor	Lived	Named	Retrieved
DECISION QUALITY ≠ OUTCOME	W1 shock → W2 near-miss pairs	W2	W3 counterfactual · W6 Kings capstone
PATH DEPENDENCE	W1→W2 carry	W3 module finale	W4 the bill · W5 reinvestment

5.2 MUST UNDERSTAND / USE — the supporting set (14)

Explicitly taught, used in play, recognised and applied — **not** drilled for recall and not counted as anchors.

scarcity · constrained allocation · substitutes · risk buffer · sunk cost · hidden information and the winner's curse · scarce attention · patience / timing cost · demand shifters · the total-revenue test (elasticity) · joint product · free riding · market size · the equity–efficiency tradeoff.

5.3 RECOGNITION

Met in context, never assessed: the cap as an institution · dead cap · the ask · renewals vs. cash · the national check · draw · reserve price · two-thirds majority · counterfactual · veil of ignorance.

5.4 Concept map — lived, named, reused, assessed

Anchor	First lived	Named	Used again	Assessed by	Likely misconception	5–6	7–8
Opportunity cost	W1 Foregone Panel	W1	W2 (dead cap) · W3 (timing) · W4 (the bill) · W6 (the vote's cost)	The forgone alternative named unprompted; "money" is not an acceptable answer	Opportunity cost = money spent. **Never assess it with multiple choice** — even professional economists fail that format.	Names a specific alternative when asked	Names it unprompted; identifies non-money costs (a slot, flexibility, a year of a prime)
Cost of undoing	W2 cut fee	W2	W3 (dead-cap drag) · W4 (payroll)	Explains why the cut cost more than the player	"Revision is free"; sunk-cost reasoning	Says fixing it cost something and names what	Distinguishes the sunk part from the forward-looking part
Market price under strategic interaction	W3 asks moving	W3	W4 (pricing) · W5 (pool)	Predicts ask movement before a day close and explains it by what the room did	Zero-sum bias; price set by cost	Reads the movement and reacts	Prices against what rivals will do next
Revenue = P × Q	W4 the Mirror	W4	W5 (contributions) · W6	Predicts direction before the	Revenue = profit; a sellout means	Predicts quantity falls and checks	Adds the total-revenue test and the

Anchor	First lived	Named	Used again	Assessed by	Likely misconception	5–6	7–8
			(valuation)	reveal; separates gate from clearing the bill	you priced right	the product	point-on-the-curve caveat
Externality	W2 ripple *(qualifying kind only) →* W5 pool	W5	W6 (the rule)	Identifies someone affected who had no say **and correctly rejects a non-example — specifically, says that a rival outbidding you is competition, not an externality**	"It's a side effect"; treating ordinary competition as an externality	Who was at the table? Who got the bill?	Adds: therefore the decision-maker does **too much** of it
Incentives and rules	W6 the vote and what follows	W6	Course climax	Predicts an unintended response; designs a rule that survives selfish	"Good rules make things fair"	One step ahead	A rule that works when everyone is selfish

Anchor	First lived	Named	Used again	Assessed by	Likely misconception	5–6	7–8
				play and names its cost			
Decision quality ≠ outcome	W1 shock; W2 near-miss	W2	W3 · W5 · W6	**Rates a failed decision as good, or a successful one as bad, and defends it**	Outcome bias — which survives being told the rule	Changes judgment when told what was known then	Volunteers it about their *own* decision, in front of peers
Path dependence	W1→W2 carry	W3 finale	W4 · W5	Explains a current constraint by naming an earlier decision	"Every week starts fresh"	Points at the earlier choice	Traces two steps and prices the future against the present

5.5 Deliberate scope statement [OPERATING CANON]

BOW Economics Track 101 is a microeconomics and institutions course. It does not teach money, inflation, interest rates, income, growth, or macroeconomics. Say this proactively in parent materials rather than being caught by it.

The back half teaches **above grade level**: externalities, public goods, free riding and institutional design are conventionally benchmarked at grade 12, not grade 8. That is a real marketing fact and a real caution — those concepts are benchmarked late because they require holding an individual decision and an aggregate consequence in mind at once, which is why Week 5 hands the aggregation to the board for the 5–6 band.

Comparative advantage is deliberately excluded [KILLED — K-001]. The most-failed concept in introductory economics for adults; it would consume a session and produce a memorised ratio trick. **Differing valuations** is what actually drives a trade and is the genuine foundation of gains from trade.

5.6 Transfer discipline — the Owner's Question `[OPERATING CANON]`

A transfer prompt must pass all three tests or it is fake:

1. **Mapping** — the correspondence can be written element by element (constraint, alternatives, decision-maker, affected party). "It's kind of like that" fails.
2. **Necessity** — the concept is *required* to answer. If common sense suffices, you are testing common sense.
3. **Discrimination** — the answer would change if the concept did not apply.

The most common fake: any prompt with money and a choice gets labelled opportunity cost. *"You have $20, buy a game or a hoodie?"* fails necessity — the student answers by preference and never names an alternative. The repair is a binding constraint that makes the forgone thing painful and specific.

Register matters as much as content. The prompt is asked **in role** — *"your owner's brother-in-law runs a bakery and asked how you decide anything"* — never as *"what did you learn today."* That single reframe is the difference between a transfer block and a worksheet, and it is thin enough to need watching.

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6. AUTHORITY PROGRESSION

6.1 The principle `[FOUNDER LOCKED]`

Other humans materially change my decision environment, and the scale of the system a desk is trusted with grows across six weeks. Students do not experience authority as *scope* — they experience it as **what and who they are allowed to act against**. The progression must therefore be legible in the mechanic, not asserted in the framing.

6.2 The progression as built + as it must become

Week	Lesson	What the desk is trusted with	Human-vs-human surface	Status
1	Draft Day	A roster under a hard cap	**Weak today** — the cap is the only adversary	**Wave 2:** a genuinely finite top of the board, sealed

Week	Lesson	What the desk is trusted with	Human-vs-human surface	Status
2	Trade Deadline	Revising a commitment, at a price	Sealed bids against the room's real demand plus a hidden reserve	**Wave 4:** the War Room adds direct desk-to-desk negotiation and scarce attention
3	Free Agency	Pricing into a market that moves because of the room	The ask moves with the room's own demand	Built
4	The Bill Comes Due	The business of the team you built	Weakest of M2 — pricing is against a curve	**Wave 5:** the Mirror makes the room the dataset; the invoice makes the stakes personal
5	You Don't Play Alone	Money that is partly other desks' doing	The pool, the binding split, the reinvestment externality	**Wave 6:** the online pool ritual
6	Writing the Rule	The rule every desk lives under	The sealed two-thirds vote and the season under it	Built

The rule that keeps the arc from sagging `[OPERATING CANON]`: every week must contain at least one place where **another desk's action changes what this desk can do**, and the desk can feel it happen. Weeks 1 and 4 are the two currently thin on this and they are where the social work goes.

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7. FRANCHISE PERSISTENCE MODEL

Rule: maximum meaningful continuity, minimum useless simulation baggage. `[FOUNDER LOCKED]`

Carry something **mechanically** only when it changes later options, incentives, economics or consequences. Carry it **narratively** when it strengthens ownership or memory but has no mechanical role.

7.1 How the repo actually does it — preserve this

Cross-lesson continuity travels through **one opaque seed**: `createSession` may name a source session, and the runtime hands the receiving module a `{ lessonModuleId, state }` blob it never inspects. Only the receiving module knows the shape, and it must treat everything about the seed — including its presence — as untrusted input to validate. Desks then **claim** their own franchise; a seat without valid prior state gets an honestly-labelled stock franchise, never a broken one.

This is a good design and it is not to be replaced by a global franchise table. **Extend it; do not generalise it.** (Repo CLAUDE.md §12: no premature generic engine.)

7.2 The chain, current and required

Seam	Today	Required
W1 → W2	**Built** (roster, cap, dead-cap exposure)	Keep
W2 → W3	**Built** (books, dead cap, room; falls back to W1, then stock)	Keep
W3 → W4	**DOES NOT EXIST —** M1→M2 deliberately does not seed	**FOUNDER LOCKED: build it.** The payroll and roster built in weeks 1–3 become the bill in Week 4. This is §4.4 and it is the single most important seam in the course.
W4 → W5	Deliberately does not seed	**Build it**: Week 4's gate, price history and fan support are Week 5's contribution base
W5 → W6	**Built** (opaque seed)	Keep

7.3 Persists mechanically

Roster and contracts · payroll and cap position · dead cap carried · room · stated intent and whether it changed · assets · fan support · gate and renewals · reinvestment · the adopted rule.

7.4 Persists narratively (recognition only)

Missed targets and who took them · bidding wars · refused trades · direction changes · the sealed prediction · Press Conference lines · the Tape entries a desk appeared in.

Derived factual descriptions are allowed — *"this desk has been the most active buyer"*, *"this desk has not spent a pick."* **Invented RPG statistics are killed** — no AGGRESSION = 91. Only claims grounded in recorded behaviour. Consistent with repo D4.

7.5 Deliberately does NOT persist

Per-game simulated results · per-player statlines · clickstream · UI state · anything whose only purpose is to make persistence look impressive.

7.6 The memory problem, and the countermeasures `[OPERATING CANON]`

Seven days is a long time for a ten-year-old. By Week 4 a desk will not reliably remember what it did in Week 1 — which means the payoff the whole design banks on lands on an audience that has forgotten the decision.

1. **PREVIOUSLY ON runs every week without exception** — this desk, this move, this standing. **Three facts.** The runtime already has the raw material: `classEvents` and the WHILE YOU WERE AWAY recap.
2. **THE TAPE (§12.3) is the real countermeasure.** Reconstructing the information state at the moment of the decision is what makes a four-week-old choice *felt* rather than merely recorded.

7.7 State-heavy, outcome-light `[OPERATING CANON]`

Carry everything that makes it *your* franchise. Do not let the **outcome** compound without limit. Every recovery mechanism the real league contains exists because the real league had this problem; use theirs (D-013), and note the repo already does — the Week-3 cap rise is exactly this.

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8. SIX-WEEK CARRY GRAPH

Read as: **decision → immediate effect → next week → later → where it reappears → mechanical (M) or recognition (R).** Seams marked **[NEW]** do not exist yet.

#	Decision (week)	Immediate	Next week	Later	Reappears as	M/R
1	**Roster built under the cap** (W1)	Five slots filled; forgone set recorded	Midseason report is about *these* players	Payroll	W2 report · **W4 invoice [NEW]**	**M**
2	**What was forgone** (W1)	Foregone Panel	—	—	W2 Tape · W3 counterfactual · parent artifact	**M + R**
3	**Risk buffer kept or spent** (W1)	Room left after lock	Rescue affordability	Room at the cap rise	W2 aftermath · W3 day-1 viability	**M**
4	**Stated intent** (W1)	Public; changeable	Read back at the deadline	—	Press Conference: *"you said you were rebuilding"*	**R**
5	**Sealed prediction** (W1) `[NEW]`	—	—	—	**Opened W5**; calibration for 7–8	**R**
6	**Cut / stand pat** (W2)	Dead cap ≈10%; hole or not	Books into free agency	Payroll	W3 dead-cap-drag card · **W4 bill [NEW]**	**M**
7	**Sealed bid won or lost** (W2)	Target or hole + dead cap	Roster need into W3	—	W3 market behaviour · Tape near-miss pairs	**M**

#	Decision (week)	Immediate	Next week	Later	Reappears as	M/R
8	**Trade with another desk** (W2) `[NEW]`	Both rosters change	Both desks' W3 needs change	—	League feed · Press Conference · W5 grievance	**M**
9	**Offer / hold each day** (W3)	Signs at your price, or the ask moves	—	Final payroll	W3 patience-dividend card · **W4 invoice [NEW]**	**M**
10	**Withdrawing an offer** (W3)	Ends that desk's market day	—	—	W3 counterfactual	**M**
11	**Final payroll at the $130M cap** (W3)	The books	**The bill**	Solvency	**W4 minute 0 [NEW]**	**M**
12	**Night price** (W4)	Gate and renewals	Contribution base	Pool size	**W5 pool [NEW]** · W6 valuation	**M**
13	**Whether the bill cleared** (W4) `[NEW]`	Kept or sold the player	Fan support	—	W5 stakes · W6 argument	**M**
14	**Reinvest in Draw** (W5)	Slow money later	Everyone else's gate	—	W5 by-choice decomposition · W6 incentive argument	**M**

#	Decision (week)	Immediate	Next week	Later	Reappears as	M/R
15	**The binding split** (W5) `[NEW]`	Money moves in-session	Bloc identity	—	W6: shown beside their W6 vote	**M**
16	**The vote** (W6)	The adopted rule	—	—	W6 season under it · counterfactual · season card	**M**

The three load-bearing carries. #11 (Week-3 payroll becomes Week-4's bill) · #2 (the forgone set is the opportunity-cost record the whole course reads from) · #12→#15 (Week-4 gate becomes Week-5's contribution and Week-6's argument). If only three things carry, these three carry.

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9. LIVING LEAGUE / SOCIAL SYSTEM

9.1 The founding commitment `[FOUNDER LOCKED — restated in v1.1]`

Other humans materially change my decision environment. This is the principle. **The contested claim** — one shared object, several desks that want it, a clock, a public resolution, a visible loser — is *one especially strong recurring grammar inside it*, not the universal metaphysics of the product. Do not force every week into one object / one winner / one loser.

The full permitted repertoire, all of which appear in the built course: **competition · exchange · negotiation · externalities · coalition formation · voting · shared pools · information · institutional rules.**

The repo already encodes the discipline: *real student-to-student interaction only when another participant materially changes the economics — never multiplayer for spectacle.*

9.2 ONLINE IS THE FLAGSHIP ENVIRONMENT `[FOUNDER LOCKED — v1.1 override]`

v1.0 repeatedly treated a physical trade floor as the superior version and online as a compromised fallback. **That product hierarchy is deleted.** BOW Economics is an online-first live social product. The design question is never *how do we imitate children walking around a classroom*; it is:

> **What can twelve to sixteen connected computers do better than a physical trade floor?**

Answers the built product should exploit: simultaneity without shouting · sealed resolution that removes reflex advantage · a shared clock everyone actually sees · scarce, countable, visible negotiation lines · incoming calls that cost something to take · a league feed that is genuinely live · information asymmetry that is *enforced* rather than politely respected · reveal choreography paced to the room instead of to the loudest voice · and a projector that is a broadcast surface rather than a whiteboard.

In-person sessions run the same product. Where the room adds something — argument, volume, a teacher reading faces — it is the *room* adding it, not a different build.

9.3 The shared market — how the repo resolves it, and why

Sealed and simultaneous, never click-order, never turn-taking. **`[FOUNDER LOCKED]`** Reaction speed must never decide an outcome. Turn-taking is one desk acting and fifteen watching, and the last mover has perfect information. The built M1 L2 sealed-bid market and M1 L3 four-day window both honour this.

But the whole course is not submit → wait → reveal. The contrast is deliberate: **Week 2 is urgent and live** (the War Room, calls, lines, a running clock) **around a sealed settlement**; **Week 3 is a paced market** whose drama is the day close and the ask moving. Live reactions, agent responses and social beats belong wherever they add real economics.

One economically honest tie-break is required and currently missing. `freeAgency.ts` breaks identical sealed bids on request arrival time. Fix it (§36).

9.4 The War Room — the flagship social surface

Specified in §4.2 and Mockup 3. The economics it adds that nothing else in the course teaches: **attention is a scarce resource, and spending it is a real cost.** Two open lines, four desks that want you, and a clock — choosing whom *not* to talk to is an opportunity-cost decision with a face on it.

9.5 Communication design `[FOUNDER LOCKED — safety]`

No free-form private chat between students, ever, in any cohort. All negotiation is routed by the product itself: structured offers, a bounded set of intent tags, a decline that carries a reason, a counter. This is not merely the safe option — it is the pedagogically better one, because a rationale expressed as a *selected structured claim* is legible to the teacher, to the board, and to the parent artifact, and free prose is not. Generic video breakout rooms are not the answer and are not to be used as the central mechanism.

9.6 Social uncertainty

Seeded and reproducible; the same seed replays. Teacher-triggered predefined shocks are available and never required (repo D22). **No random chaos.** A good decision can get a bad outcome and a bad decision can get lucky — that is the Week-2 anchor, not a defect.

9.7 Signals and reputation

Bounded public signals only: published NEED / AVAILABLE, stated intent, the league feed, the ask. Free-form status broadcasting stays **DEFERRED**. **Priced secrecy** — a desk pays to hide one thing for one week, making asymmetric information a mechanic rather than an accident — is a **frontier idea (§30)**, 7–8 first.

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10. M1 PRODUCT SPEC — WEEKS 1–3

M1 is **"The Cap"**: scarcity, opportunity cost, constrained allocation, the cap as an institution. To the student it is the first half of the season, never "Module 1."

The arc, in the student's words: *I can't have everything* (L1) → *my past constrains me* (L2) → *everyone else is constrained too, and their choices change my opportunities* (L3).

10.1 What is built and certified

Full mechanics in §4.1–4.3. All three are **classroom-ready candidates** with independent fresh-context verification, repaired findings, 313/313 module tests at L3 including cap-inviolability at $130M and day-1 viability property tests, and four rerunnable Playwright proofs including the full L1→L2→L3 chain played for real through the API.

10.2 The numeric scale `[OPERATING CANON + experiment]`

The product runs a **scaled cap: $100M, $10M steps, five slots**, rising to $130M in L3. The real 2026-27 NBA figures (§22) are the **truth ledger** for facts, cases, quotes and copy — not the play scale.

This follows the repo's own rule: **simplify the interface before simplifying the economics.** Five slots and round numbers let a ten-year-old hold the whole board; $164,961,000 and fifteen roster spots do not. **The 7–8 band differentiates by authority, ambiguity and justification burden — not by restoring real magnitudes** (§21). Whether real magnitudes improve or damage the 7–8 experience is **X-014**.

10.3 What M1 still needs

A truthful cross-desk scarcity in L1 · the War Room in L2 · an honest sealed-bid tie-break in L3 · the claim-release path in `tradeDeadline.ts` · the L3 client race · Press Conference and Tape hooks · the sealed prediction · **the seed into Week 4.**

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11. M2 PRODUCT SPEC — WEEKS 4–6

M2 is **"Money in Motion"**: revenue, incentives, path dependence, institutions. It was won on evidence in a three-candidate architecture war; the earlier "Box Office" prototype was **killed on evidence** and its operate-don't-choose patterns absorbed. `boxOffice.ts` remains in the tree as history and is deregistered from the picker — **do not resurrect it.**

11.1 What is built and certified

Full mechanics in §4.4–4.6. All three lessons passed independent gates for gameplay (STRONG ×3), economic truth, **teacher transfer (READY ×3)**, classroom/projector and sports reality. Module discipline worth preserving verbatim: **every registered claim is a computed, audited atom** — value, quantifier, bound, noun, level, plus rendered-string drift, mutation-proven. A claim-carrying surface added without registering it is invisible to the audit; that is the one way to reintroduce the defect class.

11.2 What changes in v1.1

Lesson	Preserve	Change
L1 Full House	Two books, hidden curve, Two Peaks, gate call, measured fill, the computed frontier and its forbidden caption	**Opens on the desk's own invoice, carried from Week 3.** Fixed cost is the payroll it built. Add the Mirror. Add the shortfall decision and its public consequence.
L2 You Don't Play Alone	Visitor-Draw dominance, reinvest trade, DEALT vs	**The online pool ritual and the binding in-session split.** Prediction replay. Close the

Lesson	Preserve	Change
	BY-CHOICE decomposition, real clubs	open sports-reality and projector findings.
L3 Writing the Rule	Dials, sealed two-thirds vote, lived-under season, Kings capstone, finale deck	**Private stakes cards.** The counterfactual made a headline beat. Fix the reveal-half timing. The franchise auction only if it earns it.

11.3 The premium visual clause

M2's visual quality was graded **SERVICEABLE-NOT-PREMIUM** three times and left open for the founder as explicitly non-blocking for classroom use. **v1.1 rules it in scope**: the five north-star mockups (§23, `docs/design/BOW_VISUAL_NORTH_STARS.md`) are the standard, and Weeks 4–6 are not permitted to look like a different product from Weeks 1–3.

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12. THE BOW PRESS CONFERENCE SYSTEM `[FOUNDER LOCKED — elevated in v1.1]`

Reasoning must feel native to the world. Never *"Define opportunity cost."* Always *"Why did you walk away?"*

This is **not a cute podium feature.** It is one of BOW's signature interaction primitives, and it is the surface that simultaneously produces drama, authentic reasoning, economics, assessment, student voice, teacher leverage and parent-visible evidence. It does not exist in the repo today.

12.1 The three instruments, distinguished

A. THE COMMITMENT CAPTURE — private, cheap, constant. Before a consequential commit lands, the desk records a small, sparse combination of: **the choice · the forgone alternative · intent · what they expect · how sure they are · what they are risking.** Two inputs at most on any one commit — a chip and a short line. Not four fields; not a form. **The manifestation varies by week** (§13); the principle does not.

B. THE PRESS CONFERENCE — public, instructor-called, one or two per session.

> The board goes dark. One line: **LEAGUE PAUSED — PRESS CONFERENCE.**
> Under it, a desk. Every play surface locks. The desk's real decision comes up on

the board, with what they knew at the time. The instructor, deadpan: *"You had the room's most cap space and you walked away at $31.4M. Why?"* The room watches. Three hands are already up.

C. THE FRONT OFFICE REVIEW — the extended version at a module finale, plus a 90-second rotating version every week from Week 2 onward. Running the best assessment instrument in the product exactly once wastes it.

12.2 Selection — the rule that makes it safe and interesting `[FOUNDER LOCKED]`

The console proposes candidates from **the state the lesson already holds**, ranked by:

interesting reasoning · contrast with another desk · risk taken · a reversal · genuine ambiguity.

Never by: the best desk · the highest number · correctness. A desk that is behind is *more* likely to be interesting, not less.

Protection against humiliation is structural, not advisory. Questions attack the decision, never the person. The instructor takes the first question to model tone. A desk's first Press Conference of the course is invited, never cold-called. A desk may decline once per course, silently, at no cost. Student names on shared surfaces stay fictional (repo CLAUDE.md §11).

12.3 THE TAPE — DECISION REPLAY `[FOUNDER LOCKED — think bigger than the near-miss pair]`

Core principle: reconstruct the information state at the moment the decision was made.

The Tape is what turns a Press Conference from an opinion into an argument, and it is potentially BOW's strongest cross-product primitive. Replay types:

Type	What it shows	Where it lands
NEAR MISS	Two desks, the same call, different outcomes	W2, W3, W5, W6
LUCKY BAD DECISION	Weak reasoning, good result	W2, W6 (Kings capstone)
UNLUCKY GOOD DECISION	Strong reasoning, bad result	W2, W3
STRATEGY DIVERGENCE	Similar situation, opposite strategies	W1, W3, W5

Type	What it shows	Where it lands
PREDICTION REPLAY	What the desk expected, against what happened	W5 (the sealed Week-1 prediction), W6
COUNTERFACTUAL	The realistic alternative that existed *at the time*	Already built in every module — this is the foothold

THE TAPE IS NOT UNDO. History stays history. It reconstructs: **what was known · what was unknown · what options existed · what was chosen · what was predicted · what happened.** It never re-opens a committed decision, and it never re-renders the past with knowledge the desk did not have.

Build discipline. `COUNTERFACTUAL` is already a canonical phase and every module implements something in it. **Build the Tape inside one lesson first** — Week 2, where near-miss pairs and the dead-cap bite make it richest — prove it, then extend. Do not extract a generic replay engine from one data point (repo CLAUDE.md §12).

12.4 Economics vocabulary in the student UI `[FOUNDER LOCKED — v1.1 replaces the absolute ban]`

Old rule (retired): economic terms never appear in student UI. **New rule: do not name the economic concept before the experience earns it.**

- **During operation:** diegetic front-office language only. Cap room, dead cap, the ask, the bill, the vote.
- **After NAMING:** the term may and should appear — in **Previously On**, the retrieval question, **Decision Replay**, franchise history, the season recap, the review, support copy, and the Week-6 finale deck.

Retrieval is the point. The goal was never to hide economics forever; it was to stop the product from handing over the word before the room has earned it.

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13. DECISION MEMORY MODEL `[FOUNDER LOCKED — principle, not a fixed form]`

Lock the principle: important decisions are remembered richly enough that an old decision can come back later and mean something. Do **not** ship six identical exit tickets — that is the fastest way to make a live product feel like school.

13.1 The episode

Field	Source	Student-entered?
CONTEXT — what was true at the moment	Canonical state snapshot	No
INTENT — what they were trying to do	Stated intent + a chip	Chip only
BELIEF — what they expected, how sure	Confidence tier / sealed prediction	Sparingly
OPTIONS — what was realistically available	Derived: legal, affordable, on the board	No
DECISION — what they committed	Transaction record	No
FORGONE — what they knowingly gave up	**The one required student input**	**Yes**
RESULT — what happened	Module reduce + reveal	No
LATER RELEVANCE — how it came back	Carry graph (§8)	No

Seven of eight fields derive from state the modules already hold. The desk types a short line and picks a chip. That is the whole burden.

13.2 Manifestation varies by week `[OPERATING CANON]`

W1 the Foregone Panel is already a live, ambient record — capture the *why* at lock. W2 a reason chip on the cut and on every declined offer. W3 the hold-or-offer call and what the desk expects the ask to do. W4 the gate call is already a free commitment — attach the belief to it. W5 the split position and the reinvestment reason. W6 the proposed number and the predicted response. **Six different shapes, one principle.**

13.3 What it powers

Press Conference selection · The Tape · the Front Office Review · Week-6 private stakes · the parent artifact · the student's season card · assessment evidence (§14).

13.4 Never stored

Clickstream, dwell time, keystroke telemetry, anything that reads as surveillance of a child. Meaningful events only (repo D22).

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14. ASSESSMENT / EVIDENCE MODEL

14.1 The three principles `[FOUNDER, LOCKED]`

Outcome ≠ decision quality. Preference ≠ mastery. Calculation ≠ decision.

BOW never machine-declares DECISION QUALITY = 87. `[LOCKED — D-035]` It truthfully describes roster position, flexibility, assets, revenue, decisions, adaptations, and reasoning. Final decision quality is **defended and argued**, never asserted as objective machine truth.

14.2 Why this is empirically right, not merely a philosophical preference

`[SOURCED: Schwartz & Bransford]` In their studies the **verification measure — true/false statements about the target concepts — was at ceiling across all conditions** and could not distinguish deep understanding from surface familiarity. The **prediction task** separated conditions cleanly (0.44 vs. 0.11 probability of relevant predictions).

Recognition-format assessment is not merely a weak measure of conceptual understanding; in a well-run experiment it was incapable of detecting a large real difference. A multiple-choice item would have reported that both groups learned equally. They had not.

Therefore BOW assesses by prediction and by application to novel cases. That sentence belongs in the parent-facing material, and it is a stronger argument than "we object to scores."

14.3 Weak vs. strong evidence

Weak — does not count	**Strong — counts**
Names the term	Applies it to a case with no NBA content in it
Picks it from four options	**Identifies the forgone alternative unprompted**
Defines it back in your words	Defines it in their own words *and* rules out a non-example

Weak — does not count	Strong — counts
Says "judge the decision, not the outcome"	**Rates a failed decision as good, or a successful one as bad, and defends it**
Agrees when you raise the counterfactual	**Revises their own earlier judgment** when the counterfactual is raised
Says trade helps both sides	Explains *why* both gain — differing valuations — without prompting

The strongest single signal, and the cheapest to observe: unprompted use of a concept in a week when it is not the topic. A student who says *"wait, that's an opportunity cost"* during Week 4 has transferred. That is worth more than any Week-1 exit ticket. Instructors are trained to log exactly this.

Per-concept three-level performance descriptions (emerging / solid / strong), written as observable behavior, are in §5.2 and expanded in the instructor pack.

14.4 The three instruments — all lightweight

1. **THE COMMITMENT CAPTURE** (~20 seconds, on consequential commits). *What we chose · what we gave up · what we expect · how sure we are* — **at most two inputs on any one commit**, and its shape changes by week (§13.2). **Digital-first**, because the product is. Returned to the desk through **The Tape**, not as a filed worksheet.
2. **THE 4-STUDENT SAMPLE** (instructor, during work time). Do not observe sixteen. Observe **four**, rotating, one note each on this week's primary concept. Over six sessions every student is sampled ~1.5 times and every student has six receipts. That is enough for an honest narrative and costs the instructor nothing they were not already doing while circulating.
3. **THE FRONT OFFICE REVIEW**, full in W3 and 90 seconds weekly from W2.

14.5 What goes to parents

Two things, and they are deliberately different objects.

The forwarding artifact (§26.2) — one decision, one season visual, mobile, shareable, shipped at Session 5.

The learning note — a short narrative per concept, grounded in that child's decisions: *"Maya reliably identifies what her franchise gave up, and in Week 5 she was the first student to point out that a successful trade had been a poor decision"* — plus the child's own words and two photographed receipts. It is delivered as **plain text in an email at the end of the course**, not as a designed artifact, because it answers *"what economics did my child learn"* and that question deserves an honest paragraph rather than a layout. **No number.** And say why, in one sentence, because the reason is a differentiator.

Designing the learning note as an artifact is what §26.2 rejects; **not sending one at all** would leave the only question a paying parent actually asks unanswered. The email is Tier 1; the design is not.

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15. FRONT OFFICE REVIEW

Where it lives now. The extended version belongs at a **module finale** — Week 3 closes Module 1 and already ships a staged finale with GM Awards and counterfactuals, and Week 6 closes the course. The 90-second rotating version runs every week from Week 2. v1.0 placed the full event in Week 3 for a different arc; in the built arc Week 3 *is* the Module 1 finale, so the placement survives the rebase intact.

Purpose. The module climax and the course's primary assessment instrument. The student reconstructs and defends the strategy they actually followed — not the one they now wish they had.

What is reconstructed for them (from canonical state, not memory): opening strategy · major moves · players missed and who took them · signings · changed conditions · trades · assets spent · flexibility preserved · adaptations.

The five questions.

- **STRATEGY** — What was your plan?
- **ALTERNATIVE** — What realistic path did you reject?
- **RISK** — What did you knowingly risk?
- **ADAPTATION** — What changed, and how did you respond?
- **DECISION QUALITY** — Was the strategy good *given what you knew then*?

Structure: the distributed four-part format in §4.3 — private one-card defense, pair interrogations, four console-selected hot seats, aggregate reveal. Sixteen minutes, every student speaks, four perform.

Grade bands. 5–6: structured but real; two scripted questions prepared in advance; a defense is three chips plus twelve words. 7–8: deeper evidence, one adversarial and one unseen question, and a **counterfactual burden** — they must name the alternative they rejected and concede the strongest cost of their own decision.

This is meaningful but it is not the six-week climax. Week 6 is.

16. THE FINALE — WRITING THE RULE

Full spec in §4.6. The essentials, restated as contract:

The intellectual opportunity. *What rule is best for MY franchise?* versus *What rule creates the league we actually want?* — with the second question only becoming real because the student spent five weeks answering the first.

Earned authority. Week 1: make a roster decision. Week 6: help write the rules everyone lives under. That progression must land, and it lands only because the incentives are **derived from the student's own franchise history**, never assigned as debate roles. `[LOCKED]`

The four non-negotiables.

1. Private, dollar-denominated stakes cards before debate opens.
2. A threshold (11 of 16) that requires defectors, with the natural split engineered near 9–7.
3. The vote at minute 40, with fifty minutes of consequence after it.
4. **The counterfactual reveal** — what the other proposal would have produced. Thirty seconds, and it is the entire concept of institutional design delivered as a gotcha, which is the only register in which twelve-year-olds accept intellectual content.

Explicit economics. This is the one week where terminology is used freely and openly by students, because they have five weeks of experience underneath it. Incentives, externality, free riding, opportunity cost, and decision quality vs. outcome should all appear in student speech, and the instructor should mark each time one does.

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17. `/teach` — DIRECTOR + COMMISSIONER

Target scale: ADMIN → GUIDED → **DIRECTOR** *(hard minimum)* → **AMPLIFIED** *(ambition)*. `[FOUNDER, LOCKED]`

17.1 The operating principle

> **SOFTWARE RUNS COMPLEXITY. HUMAN RUNS MEANING.**

The software does what humans are bad at: maintain canonical state, remember sixteen histories, detect contrasts, resolve markets, enforce legality, surface evidence. The instructor does what humans are good at: create tension, ask why, provoke argument, notice emotion, let discussion breathe, teach the economics, make reveals land.

Buttons are not power. The bigger opportunity is that the software *finds the interesting classroom evidence* and the human turns it into meaning. This is potentially BOW's largest moat, and it is what makes a trained non-founder able to run an extraordinary class.

17.1.1 What the repo already gives the console — preserve it

Per-session bearer-key auth · phase control (advance / reveal / restore / end) with a manual fallback behind every synchronised reveal · **TIME CUT with FINAL CALL and CLOSE NOW on server time, and a per-desk named fallback the module itself declares** · a room view carrying committed decisions and teacher-private diagnostics · a class clock on the server's clock · a projector-truth indicator that says whether a board is really watching · device-health and undo-depth reporting · a full **cold rehearsal** of any lesson with zero desks, every stand-in marked `REHEARSAL —` · and a director panel vocabulary already moving toward NOW / WATCH FOR / DON'T EXPLAIN YET / ASK / TRIGGER / SYNTHESIS.

17.1.2 The economics protection rule `[FOUNDER LOCKED]`

The console watches the clock against the session's remaining economics and says so:

> **ECONOMICS AT RISK — 11 MINUTES REMAIN. FINAL CALL.**

The teacher keeps bounded flexibility. **The naming and interpretation of the economics is what the product refuses to let the market eat.** This replaces v1.0's un-extendable timer.

17.2 What the instructor controls (Commissioner powers)

FINAL CALL · CLOSE NOW · CLOSE MARKET · OPEN TRADE WINDOW · SPOTLIGHT DEAL · **CALL PRESS CONFERENCE (lock all screens)** · TRIGGER AUTHORED CONDITION · COMPARE STRATEGIES · CALL FRONT OFFICE REVIEW · CALL BOARDROOM · REOPEN (pre-reveal only).

17.3 What the software detects automatically (Director intelligence)

- **DISCUSSION OPPORTUNITY** — *"Boston protected its first-round pick while New York spent one for similar short-term improvement."*
- **MARKET COLLISION** — *"Four franchises are chasing the same position."*
- **CONTRAST** — *"Chicago and Miami faced similar roster problems and chose opposite strategies."*
- **THE NEAR-MISS PAIR** — same decision, different outcome. Surfaced automatically every week from W2, and the entry point to **The Tape** (§12.3).
- **PRESS CONFERENCE CANDIDATES** — ranked by interesting reasoning, contrast, risk, reversal and ambiguity; **never** by who is winning (§12.2).
- **THE SILENT DESK** — in the War Room, the desks nobody has called, named before the window closes.
- **THE SILENT FRANCHISE** — no action for 8 minutes (§17.5).

- **THE UNPROMPTED TRANSFER** — a student used a prior week's concept this week. One-tap log.

17.4 The seven things the software must encode `[LOCKED — this is the instructor-independence spec]`

1. **A per-minute runsheet** on the instructor's screen: what is on students' screens now, what to say, what is next, elapsed vs. planned. A teleprompter, not a lesson plan.
2. **The rationale gate**, unskippable, even when behind schedule.
3. **Guaranteed constraint binding** — the scenario generator ensures every student hits a cap wall by Week 3. Enforced, not emergent.
4. **Airtime telemetry** — a live panel showing who has spoken or acted and who has not, nudging after 8 minutes of silence. This mechanically fixes the single most common failure of weak facilitators.
5. **Contextual Socratic prompts** — when a student commits a trade, the instructor's screen shows two questions specific to *that* trade. This is how a mediocre instructor sounds insightful.
6. **The naming screen will not advance until the instructor has typed student language into it.** `[The most important pedagogical enforcement in the product. Instruction that builds on student-generated solutions is the strongest moderator in the problem-solving-before-instruction literature: g = 0.56 vs. 0.20. A pre-baked slide reading "OPPORTUNITY COST: the value of the next best alternative" delivers the 0.20 version and engineers away most of the effect.]`
7. **Scripted wait time** — a literal instruction reading "wait 7 seconds." Under pressure with sixteen kids, silence feels like failure and adults fill it. Make the product enforce the pedagogy.

17.5 Quality monitoring without surveillance `[LOCKED]`

Recordings exist for safety and are **not routinely reviewed** — routine review is surveillance and corrodes trust with a young workforce. Monitor four proxy signals the software already produces: **session attendance curve** (the week-2-to-4 shape is the strongest quality proxy available) · **rationale completion rate and median length** · **airtime distribution** (Gini across students) · **the end-of-course "would you take another BOW course?" question, asked of students, not parents.**

More than ~1 SD deviation on attrition or airtime triggers a **coaching conversation, not a recording review.** Recordings are reviewed only on a safety flag or a parent complaint, and the instructor handbook says so.

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18. `/board` — THE PROJECTOR

`/play` = MY FRANCHISE (private). `/board` = THE LEAGUE (public). `[FOUNDER LOCKED]` `boardView` is structurally never handed a seat identity.

Four rules the repo already enforces and this bible adopts verbatim. (1) **A computed finding may be split by surface, never deleted to fit** — the board gets the short rendering, the teacher's script gets the whole argument, and a test fails if a finding reaches neither. (2) **Drawn magnitude must equal modelled magnitude, and it is measured in a browser, not eyeballed** — by an instrument that poisons itself before its result is believed. (3) **The student device paces with the board, never ahead of it** — numbers belonging to an unpressed beat are *unsent*, not merely unrendered. (4) **A surface never moves backwards.**

18.1 What the projector is for

Making students physically look up. It is the league's breaking-news layer and the course's contrasting-cases instrument — and the second of those is the more valuable and the more neglected.

18.2 The contrasting-cases contract `[LOCKED — D-036]`

BOW has accidentally built a textbook contrasting-cases affordance and must treat it as core, not as a visualization feature. Seven rules:

1. **Simultaneous, never sequential.** The mechanism is noticing what varies. Sequential reveal is storytelling; simultaneous display is contrast.
2. **Curate to 4–6 franchises, not 16.** Sixteen franchises differing on everything is noise. The software must support *selecting and sorting* far more than it supports *showing everything.*
3. **Vary one dimension at a time**, sorted by this week's concept. If the axis is not obvious in two seconds from the back of the room, it is not working.
4. **Prediction before reveal**, in writing. This is the discriminating measure, and it creates the readiness for telling.
5. **Student description before naming.** Ask "what's different between these two?" and put their words up first.
6. **Build the near-miss pair.** Same decision, different outcome, side by side. If BOW builds exactly one custom display, build this one — it is the only instructional move with a real chance against outcome bias, and it is reusable in W2, W3, W5 and W6.
7. **Fade the concreteness across six weeks.** W1: team names, player names, dollar signs. W6: an abstract 2×2 of incentive structures with franchises as labeled dots.

18.3 Spectacle beats — four per session, no more

(1) The signature decision. (2) The consequence. (3) The class reveal. (4) The intellectual reveal. Roughly **75% restraint, 25% spectacle.** An important moment produces *"oh."* Not *"next slide."*

18.4 What the projector must never do

Decorative dashboards · continuous low-value tickers · leaking private reasoning (nothing auto-publishes; the instructor selects what goes up) · generic scoreboards · a spreadsheet at any size.

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19. THE LEAGUE FEED AND THE DESK'S INBOX

Hard rule: if the phone lights up, the kid should care. `[FOUNDER, LOCKED]`

Permitted events, and this is the complete list:

Event	Week	Why it earns a light
Agent response to your offer	W1	Directly changes your decision
Your personal headline	W2	It is about your franchise, by name
A ripple with a named cause	W2	Another human changed your world
Buyout-market opportunity	W2	A live, contested chance
Incoming trade offer	W3	The strongest single notification in the product
Counter received / offer declined with reason	W3	Actionable
Inbox-full bounce	W3	Funny, informative, teaches attention scarcity
The invoice	W4	The week's entire premise
Shortfall warning	W4	Consequence is imminent

Event	Week	Why it earns a light
Your private stakes card	W6	Changes how you vote

Killed: ambient market chatter · "your team practiced today" · streaks · badges · reminders · any notification that does not change a decision. `[K-006]`

Cap: no more than five inbox events per student per session, excluding trade-thread replies during the W3 window.

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20. GRADE 5–6 PRODUCT CONTRACT

Do not baby them. They make authentic consequential decisions. What changes is representation and structure, not stakes.

20.1 The correction that shapes this entire section `[SOURCED — highest-priority pedagogical finding in this pass]`

The meta-analytic evidence for experience-before-instruction is **positive for grades 6–10 (g = 0.50) and negative for grades 2–5 (g = −0.09).** Fifth graders sit inside the negative band and sixth graders sit at its edge. The most plausible mechanism is that younger students lack the domain knowledge and self-regulation to structure an open problem, so exploration time is spent working out *what the task is* rather than generating conceptual variation.

The fix preserves BOW's product rule. For 5–6, hand them the **structure** of the decision — here are your two options, here is your constraint, here is your deadline — and withhold only the **concept.** They still experience before they can name. What is removed is the unstructured search space, which is the part that hurts younger learners. **This is also BOW's best differentiation axis, and it is free.** `[LOCKED — D-037]`

20.2 Hard constraints

Constraint	Rule
Variables traded off	**2** (3 by W5). Variables *on the card* are free; only variables that must be weighed against each other count.
Percentages	Benchmark fractions only (½, ¼, □, "about a third"). Never compute a percentage of a large number.

Constraint	Rule
Ratios	Never compare two ratios with different denominators. Pre-normalize all rates.
Negative numbers	**Never shown, anywhere, including derived figures.** "You are $4M over the line," never "−$4M." A Week-4 shortfall reads *"you are $22M short"*, never a negative profit. CAP HEALTH for an over-apron franchise reads *"$6M past the first line."* Show every line physically.
Probability	Three named tiers in words: **safe / risky / long-shot.** No expected-value multiplication.
Large dollar amounts	$30M is a word, not a quantity, and sits in the same mental bin as $300M. Use one constant unit all six weeks: **CAP SLICES**, where one slice ≈ 10% of the cap. Money becomes counting. Real NBA dollar figures remain visible alongside.
Counterfactuals	The instructor or the interface supplies the alternative; the student evaluates it. Do not expect them to generate one about a choice they are emotionally invested in.
Delayed consequence	Four weeks is not felt unless the moment is reconstructed. **The Tape** is that bridge, and PREVIOUSLY ON is its weekly minimum.
Reading	No screen requires more than ~40 words to make a decision, **and no screen shows more than six player cards at once.** The 18-player board is paged in tiers of six for this band, with the agent's priority as a single icon rather than a four-line card. Sixteen three-field cards on one screen is the "wall of cards" §23 forbids.
New terms	One primary + two secondary per session. Hard cap.

Constraint	Rule
Franchise unit	**The desk is the economic agent. Pairs on one device are the default; solo is supported** (repo D11b). 12–16 desks per class, which is what every number in this bible is calibrated to. Within a pair, split the information — one partner holds the books, the other holds the scouting — so the decision is impossible without both talking.
Reversibility	Most decisions adjustable next week.
Aggregation	The projector performs the aggregation their cognition cannot yet perform internally. This is the single best use of the projector for this band and the reason Week 6 works for them.

20.3 What 5–6 still gets, undiluted

Real franchises, real players, real dollar amounts, real consequences, a real rival taking their target, a real trade with another human, a real vote, and a real public defense of their own decision.

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21. GRADE 7–8 PRODUCT CONTRACT

The difference is not harder arithmetic. It is more GM power. `[FOUNDER, LOCKED — and it is pedagogically correct]`

21.1 Why "more power, not harder math" is right

1. What actually changes between 10 and 14 is capacity to hold more variables, tolerate ambiguity, reason about others' reasoning, and generate counterfactuals unprompted. Harder arithmetic exercises none of these.
2. Computation consumes the same working memory the economic reasoning needs. A student computing 14% of $118M is not thinking about the tradeoff.
3. `[SOURCED: NAEP 2024 — 41% of U.S. eighth-graders scored below NAEP Basic in mathematics; 27% at or above Proficient.]` Even allowing for enrichment self-selection, **design assuming a meaningful minority of eighth-graders cannot fluently compute a percentage of a large number under time pressure in front of peers.** Differentiating upward via arithmetic sorts rather than teaches.

The caveat that matters: "more power" means more *authority*, not more *buttons*. Twelve options where 5–6 gets four is increased search cost, not depth, and it blows the variable budget. The right form is **fewer, deeper, more consequential choices with a heavier obligation to justify them.** A seventh grader making one irreversible decision they must defend under questioning is doing far more cognitive work than one making six reversible ones.

21.1.1 Where the band attaches, technically `[repo D38]`

If and only if a second band exists, it attaches at exactly two places: the `createSession` input carried onto the session row, and the `initialState` context every module already receives. Nothing else in the runtime needs to know. **Do not ship an unused switch**, and do not build a fork nobody has designed.

21.2 The differentiation matrix

Axis	**5–6**	**7–8**
Problem structure	Options given; constraint stated	Options partly self-generated; constraint must be inferred from the data
Information load	3–4 fields per player; pre-normalized	6–8 fields; some raw, requiring interpretation
Variables traded off	2 (3 by W5)	3 (4 by W5)
Numeric form	Cap slices; no negatives; no percent computation	Whole percentages and negatives permitted. **No percent-change.**
Ambiguity	Complete information; a defensible answer exists	Deliberately incomplete; some information must be requested, some does not exist
Counterfactual burden	Instructor supplies; student evaluates	**Student generates**, in writing, before the reveal
Order of reasoning	First-order: what happens to us?	**Second-order: what will the other fifteen GMs do in response?**
Argument standard	State the choice; name one thing given up	State it, name what was given up, **and concede the**

Axis	5–6	7–8
		strongest cost of their own decision
Time horizon	2 weeks, bridged by the receipt	4 weeks; must explicitly price future against present
Reversibility	Mostly adjustable	**At least one irreversible decision per franchise per course**
Vocabulary	Recognize and use in context	Define, apply to a novel non-NBA case, **and identify a non-example**
Franchise unit	Pairs	**Individual**
Front Office Review	2 friendly scripted questions	4 questions, one adversarial, one unseen

21.3 Week by week: what "more power" actually means

Wk	5–6	7–8
1	Rank 3 pre-vetted targets	**Set the offer price**, competing for a scarce pool; someone loses a player by underbidding
2	Respond to one shock	Respond *and* pre-commit a contingency — then be held to it
3	Trade from a menu of proposals	**Originate** proposals; negotiate live; may be refused and must return revised; one curated sign-and-trade available
4	Choose one of three price tiers	Set price freely across three game classes, set premium mix, pay the payroll — **must find profit, not revenue**; spend attention tokens

Wk	5–6	7–8
5	Vote between two sharing options	**Propose** the sharing formula, then defend it against the franchise it hurts most
6	Vote, then read the aggregate consequence off the projector	**Write** the rule, predict how it gets gamed, revise once before the vote, then live in it

At every step the 7–8 version adds authority, ambiguity, and obligation to justify — and at no step does it add arithmetic.

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22. NBA / CBA TRUTH CONTRACT

22.0 How the real world and the play scale relate `[OPERATING CANON — D-062]`

Two different things, and conflating them is the error to avoid.

The play scale is what a desk operates: **$100M cap, $10M steps, five slots**, rising to **$130M** in Week 3. It is a recorded, deliberate interface simplification under the repo's own rule — *simplify the interface before simplifying the economics* — and it exists so a ten-year-old can hold the entire board in mind. Five round slots teach constrained allocation; fifteen roster spots and $164,961,000 teach arithmetic.

The truth ledger is §22.2 and the repo's dated source notes: real clubs, real people, real rules, real market differences, real cases, all dated and re-verifiable. It governs facts, examples, copy, and every claim the product makes about the world.

Every simplification is recorded with what changed, why, and the misconception risk. The repo already maintains SIMPLIFICATIONS ledgers per module and a claim audit that recomputes every registered claim; that discipline is adopted here rather than reinvented. Whether real magnitudes improve the 7–8 experience is **X-014**, not an assumption.

22.1 The invariant `[FOUNDER LOCKED]`

Use the real world of sports business. Real franchises, real players, real markets, real cap situations, real transaction concepts, real cases. NBA/basketball is the primary world; another sport appears only when it provides a dramatically better economic example. No fake clubs as the main product fantasy. No implied endorsement. No copying NBA 2K or broadcast interfaces — BOW uses its own visual language.

Reality must never become a fandom test. A desk that knows little basketball must still succeed; a fan may perceive extra strategic depth. Accessible core information plus optional domain depth. Sports knowledge may raise engagement; it may never be a prerequisite for the economics — and it may never be the difference in a decision.

The research process is one-directional:

> RESEARCH → MODEL REALITY → IDENTIFY EXCESS COMPLEXITY → MAKE THE SMALLEST HONEST SIMPLIFICATION → TEST THE ECONOMIC CLAIM → BUILD THE EXPERIENCE

Never: invent a fun mechanic → invent NBA numbers to make it work. If a mechanic conflicts with reality, **change the mechanic.**

22.2 Canonical figures — DATA DATE September 4, 2026

Full sourcing in `docs/research/nba_cba.md`. Every number below carries a source there.

2026-27 thresholds `[OFFICIAL — NBA, June 30, 2026]`

Salary cap	**$164,961,000**
Minimum team salary (90%)	$148,465,000
Luxury tax (121.5%)	$200,428,000
First apron	**$209,015,000**
Second apron	**$221,686,000**

The 10% cap-smoothing collar did **not** bind for 2026-27 — the rise was +6.669%, because local/regional media revenue fell short. 2025-26 *did* hit the collar exactly. The league's memo projects **$174M (+5.5%) for 2027-28** `[UNCONFIRMED officially; check pr.nba.com ~June 30, 2027]`.

Exceptions 2026-27: Non-Taxpayer MLE **$15,044,000** (1–4 yrs, 5%) · Taxpayer MLE **$6,064,000** (1–2 yrs) · Room MLE **$9,366,000** (1–2 yrs) · Bi-Annual **$5,477,000** (1–2 yrs). Bird and Early Bird get 8% raises; **everything else, including sign-and-trades, gets 5%.**

Maximums 2026-27: 25% = **$41,240,250** · 30% = **$49,488,300** · 35% = **$57,736,350**.

The Bird premium, in one number. A 35% max with **full Bird rights** = 5 years, 8% raises = **$334,870,830.** The same max signed by anyone else = 4 years, 5% raises = **$248,268,305.** An **$86.6M gap** created by an institutional right, not by money. That single line is the product's core lesson.

Minimums 2026-27: rookie **$1,357,763** · 2-YOS **$2,449,421** · 10+ YOS **$3,876,529**. `[Two reputable sources disagree; the Hoops Rumors set is used, cross-checked three ways including LeBron James's actual 2026-27 salary of $3,876,529.]` **The one-year veteran subsidy:** a 3+ YOS veteran signed to a **one-year** minimum counts against cap, tax, apron *and trade matching* at only **$2,449,421**; the league reimburses the difference. **Multi-year minimums do not get the subsidy** — Philadelphia carries LeBron's full $3,876,529 because his deal is two years.

Rosters: 15 max, 14 minimum (12–13 permitted for at most 2 consecutive weeks / 28 total days), 12–15 active, 3 two-way slots at **$678,882**, hardship 10-days above 15.

Calendar: 2026-27 regular season opens **Oct 20, 2026** · 10-day contracts from **Jan 5** · all remaining contracts guarantee **Jan 10** · **trade deadline Thursday, February 11, 2027** · regular season ends **April 11, 2027**.

Revenue reality: 30 teams generated **$12.25B in 2024-25, $408M per club** — Golden State $833M to Memphis $301M. Mix: central distribution 38% (44% under the new deal) · seating and suites **$3.4B / 28%** · sponsorship $1.7B / 14% · local media 10% and falling · other 9%. The **11-year, $76B national media deal** began 2025-26; per-team national TV went from **$103M to $143M** and is projected toward **$281M by 2034-35.** League-wide attendance ran at **97% of capacity** in 2025-26, and secondary-market average ticket prices spanned **$213.43 (New York) to $22.00 (Memphis)** — a ~10× spread with both selling out.

Revenue sharing: the agreement is **not public.** Reported structure: each team contributes ~50% of defined revenue; each draws roughly league-average team revenue; above-average teams are net payers. **The formula benchmarks each team against expected revenue given its market**, which is why Oklahoma City — one of the smallest markets — paid in for six consecutive seasons. Separately, **50% of luxury-tax proceeds** go to non-taxpayers. 2021-22 actuals: ten payers moved **$163.6M** (Golden State $45.0M, Lakers $42.8M) to twenty receivers (**$404M total including $240M of tax money**; Indiana $42.2M, Denver $35.5M).

BRI: $11.68B in 2025-26. The Designated Share is 50% of BRI adjusted by ±60.5% of the forecast gap, **collared between 49% and 51%** — not a flat 51%, and BOW states the band correctly.

The "rules are real" anchor, two days old. On **September 2, 2026** the NBA penalized the LA Clippers and Kawhi Leonard for salary-cap circumvention: five first-round picks forfeited

(2029–2033), a $30M fine, a one-year suspension of the owner, and five years of compliance monitoring. Use it. It is the strongest available demonstration that these constraints are enforced, not decorative.

22.3 The honest simplification table `[LOCKED]`

Mechanic	BOW's simplification	Truth that must NOT be lost
Cap holds	**"Ghost salaries."** Every player you might re-sign leaves a ghost on your books at roughly 1.5× his old salary (2× for a good young player off a rookie deal). You can erase a ghost — but the moment you do, you give up the right to re-sign him over the cap this year.	**Cap room is not "cap minus contracts."** The irreversibility of renouncing is the whole lesson. Never let a student renounce and then re-sign.
The exception ladder	**Four cards:** the **OWN-PLAYER CARD** (any price, up to 5 years) · the **$15M CARD** · the **$6M CARD** · the **MINIMUM CARD (always in your hand)**. A room team trades its whole hand for a pile of cash plus one **$9M CARD**.	Over the cap, **$15,044,000 is a wall** — above it an outside free agent is unreachable except by trade or sign-and-trade. And **the Minimum Card never goes away**: the league guarantees every team a floor of access.
The aprons	**Two lines with names, not numbers.** Cross the first line and you lose the $15M Card, the $5.5M Card, and the right to receive a permission-slip player. Cross the second and you additionally lose the $6M Card, the right to add two players together in a trade, the right to send cash, and **your first-round pick seven years out freezes.**	**The penalty for spending is losing *tools*, not paying *money*.** And **you may always re-sign your own players** — which makes the apron a *choice*, not a law. James Dolan's actual words: *"Cannot go into the second apron."*
Trade matching	**One rule, three settings.** Below both lines: take back	The **directional asymmetry**: a rich team can always shed,

Mechanic	BOW's simplification	Truth that must NOT be lost
	your outgoing salary **+ about $9M**. Above the first line: **exactly what you send out.** Above the second: exactly what you send, **and you may not stack two cards to make one.**	but can never upgrade by absorbing. That is the mechanism recycling stars downward and picks upward. (Real 2026 case: Oklahoma City, over the second apron with a $560M payroll, had to make **three separate trades** because it could not aggregate — and cut payroll to $234M.)
Sign-and-trade	**The Permission Slip + the Trade-Back** (§22.4)	**Access is granted by an institution, not bought.** A player above the $15M Card cannot reach a full team unless his old team consents. Kids will feel the injustice; that is the point.
Revenue sharing	*"Everybody puts half their local money in one bucket; everybody takes out the same amount. The bucket expects more from you if your city is big — so a small city that does great can end up paying in. And half of all luxury-tax money is split among the teams that didn't pay it."*	The direction is real; the **market-size adjustment is real and counterintuitive**; and **the formula is secret.** Tell the students that last part — it is the most honest thing in the curriculum, and it teaches that real institutions have opaque rules people argue about.
Ticket demand	**Two dials.** Real arenas run at ~97% of capacity **because teams price to fill them** — that is an equilibrium outcome, not a structural limit. In BOW, **quantity moves**, because the price trap is only teachable if it can spring. You move **price** and	**Sellouts do not mean maximum revenue.** Memphis and New York both sell out at wildly different prices. Note also that the widely-quoted $22 / $213 spread is **secondary-market resale**, not what the club charges — BOW never presents resale prices as a

Mechanic	BOW's simplification	Truth that must NOT be lost
	premium mix, and quantity answers.	franchise's own pricing decision. The gap between markets is what revenue sharing exists to address.

22.4 Sign-and-trade `[OPERATING DEFAULT — 7–8, W3, one per class]`

The real rule is genuinely restrictive: executed before opening night · the player finished the prior season on the sending team's roster · 3–4 seasons, first year fully guaranteed · not signed with the NTMLE or Room MLE · **5% raises, not 8%** · base-year compensation on the outgoing leg · **the receiving team is hard-capped at the first apron** · a team that would be over the first apron after the transaction **cannot receive at all.**

It is genuinely the *only* path when four conditions hold: the player's value exceeds $15,044,000, no cap-room team wants him at that price (in summer 2026 there were exactly **three** room teams league-wide), his current team holds his rights and won't or can't pay, and the receiving team ends below the first apron. The purest case is a restricted free agent — only six offer sheets have been signed since 2022 and only one went unmatched since 2020.

BOW's curated version: a player above the $15M Card who wants to leave a team that holds his Rights can reach a full team only if his old team signs a **PERMISSION SLIP** and receives **something real** in return (a player, a pick, or a future-value token — never nothing). The receiving team then wears a **SPENDING COLLAR** for the rest of the year: it cannot cross the first line for any reason.

Preserved: tri-party consent · access granted by an institution rather than purchased · a durable cost to the receiver · the sending team's incentive to be paid for its rights. Sacrificed knowingly: the 3–4 year minimum, the 5% raise penalty, base-year compensation, the salary-match on the outgoing leg.

Held in reserve as a Week-4 twist (Tier 3): the "Three Signatures" variant in which second-apron teams have their signature line **printed over in red — they may not sign at all.** That inversion, where money buys you *fewer* options, is the single most counterintuitive true thing in the CBA and kids love it. It arrives only after they have felt the second apron bite.

Do not generalize this into an unrestricted transaction sandbox. The target is **95% of the GM fantasy with a small fraction of the CBA surface.** `[FOUNDER, LOCKED]`

22.5 Data hygiene `[LOCKED]`

Every NBA fact in the product carries a **data date and a source**, visible in-product. Before productizing a changing fact, verify against an authoritative source. Larry Coon's CBA FAQ is **retired and frozen at the 2017 CBA** — correct on Bird rights and cap holds, wrong or silent on everything apron-related. **Do not build on it.** Current working sources: NBA.com / pr.nba.com (official), The CBA Guide, Over the Apron, Spotrac, ESPN (Bobby Marks), Hoops Rumors. None but the first is official.

Seven items are explicitly unconfirmed and listed in `docs/research/nba_cba.md` Appendix, including the 2027-28 cap, the exact 2026-27 trade-matching breakpoints, the minimum-salary scale discrepancy, and the revenue-sharing formula (which **cannot** be confirmed — it is not public, and the product should say so).

22.6 Rights and authenticity guardrails `[OPERATING DEFAULT — counsel required before launch]`

Almost certainly safe: player names and publicly reported statistics `[the direct holding of C.B.C. Distribution v. MLB Advanced Media, 8th Cir., cert. denied 2008]` · publicly reported contract and cap figures · team names used descriptively in plain, unstylized text · BOW's own prose, charts, and mechanics.

Do without: team logos, wordmarks, uniforms, color-and-mark trade dress, arena imagery · player photographs and likenesses · anything implying affiliation ("official," NBA-style typography, or a product name that reads as an NBA product).

Build in now: text-only identification · a persistent plain-language disclaimer · every number cited with a date and source *inside the product* (both a pedagogical differentiator and the best evidence of good-faith factual and educational use) · **real names as a swappable data layer**, so that if counsel raises a concern the product flips to fictional-league mode in a day rather than being rebuilt. That optionality is worth two days of engineering and buys out the worst case entirely.

⚖️ **For counsel:** the product name; marketing copy using team names prominently in advertising; right-of-publicity variation by state; whether a paid curriculum sold to schools changes the analysis; whether an educational license is cheaper than the review.

22.7 Data refresh plan `[LOCKED — D-052]`

The entire product is date-stamped to the **2026-27 league year** and begins going stale in **June 2027**, when the new cap is set and the offseason rewrites every franchise's position. This is not a maintenance detail; it is the product's expiry date, and a course sold in September 2027 on 2026-27 data is a course that lies to students about the actual NBA.

Therefore: all NBA facts live in **one dated data layer** (thresholds, exceptions, minimums, maxes, rookie scale, franchise cap positions, free-agent board, revenue figures), never inline in copy or logic. Every screen renders its data date from that layer.

The annual refresh is a scheduled two-week block each July, immediately after the NBA sets the new cap (~June 30) and the first week of free agency resolves: re-source every figure in §22.2 against pr.nba.com and the working sources; rebuild the 20 franchise cards and the free-agent board; re-verify the seven unconfirmed items; re-check the four §10.1 case studies and replace any that have gone stale with that summer's equivalents (there is always one). **A course cohort never starts on data older than the current league year.**

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23. VISUAL PRODUCT SYSTEM

The five north-star mockups in `docs/design/BOW_VISUAL_NORTH_STARS.md` are the standard for this product, read together with the repo's own `design/VISUAL_IDENTITY.md` ("the Cap Room" — broadcast-control-room register, dark palette, no mascots or badges). The mockups are ambition, hierarchy and emotion references; they are **not** pixel specifications and they contain wrong mechanics and unlicensed marks that must not be copied.

Module 2 was graded SERVICEABLE-NOT-PREMIUM three times and left open for the founder as non-blocking. v1.1 rules it in scope: the course may not visibly change products at Week 4.

Target feeling: premium sports-management experience · real NBA front-office atmosphere · high-end broadcast information design · Apple-like hierarchy and restraint · cinematic consequences · interactive strategy product.

Avoid: LMS · generic SaaS dashboards · cartoon education · walls of cards · walls of text · fake sports metrics · decorative charts · gratuitous neon · 3D for its own sake.

Ratio: ~75% restrained, ~25% spectacle. Four spectacle beats per session, earned (§18.3).

Mockups are quality and ambition references, not sacred specifications. For major interaction surfaces, FROTH meaningfully different directions — never cosmetic variants of one.

23.1 The persistent HQ [OPERATING DEFAULT]

Spaces: war room · roster · cap and flexibility · market · phone · inbox · trade hub · arena and business · boardroom. The world **expands as authority expands**: front office → trade floor → arena → league → Board of Governors.

This does not require an explorable 3D metaverse. HQ provides continuity, atmosphere, spatial memory, a sense of time passing, and franchise history. **Hierarchy: HQ is the stage; the moving league is the product.** Do not decorate HQ while social and economic life is shallow.

23.2 Device and readability constraints [LOCKED]

Student surface is a Chromebook-class laptop at 1024×600, online-first. The projector must be readable **from the back row**: one variable, sorted, large. Any display that requires squinting has failed regardless of how good it looks on a laptop. Parent-facing artifacts are **mobile-first**, because that is where they are opened and forwarded.

23.3 The concreteness fade, as a visual system

Week 1 renders franchises with names, dollar signs, and player cards. Week 6 renders them as labeled dots on an incentive plane. The fade is a design requirement, not a stylistic drift (§4.0 D-018).

23.4 Accessibility and inclusion [LOCKED — D-053]

This product is built on public defense, hard timers, dense numeric screens and competitive loss. Each of those is a barrier for a real child in every cohort, and none of it was addressed until this was raised in prosecution.

Barrier	What BOW does
Reading difficulty / ELL	Every decision screen ≤40 words for 5–6 (§20.2). Every number has a plain-language gloss ("$41M — about a quarter of your whole payroll"). **All projector text is also read aloud by the instructor** — this is on the runsheet, not left to judgment.
Timers and processing speed	The DECIDE timer never shortens below 15 minutes, and the sealed-bid design means **no decision is ever won by speed** (§9.2, D-006). This is the single largest accessibility

Barrier	What BOW does
	property in the product and it was chosen for economic reasons; note that it is also this one.
Public speaking anxiety	The Podium is **never a cold call for the first Podium of a student's course** — the console tracks who has been up and the instructor is prompted to invite, not summon. The distributed Front Office Review's pair interrogation exists precisely so every student speaks without a stage. A student may decline the Podium once per course with no cost and no comment.
Attention regulation	The §4.0 session shape never runs a single block longer than 15 minutes. The Week-6 finale has a mandatory break at minute 42.
Numeracy	See §20.2 and §21.1. Nothing in the product requires mental arithmetic under time pressure in front of peers.
Sports-knowledge asymmetry	§1 promises the basketball expert "recognition and perhaps some strategic advantage." **Bounded, and stated as a rule: prior NBA knowledge may earn recognition and faster orientation. It may never be required to reason well, and it may never be the difference in a decision.** Every player card carries everything needed to evaluate that player, and the sealed-bid design means the fan cannot out-click anyone. Sports exposure is unevenly distributed by gender and background; converting it into a mechanical edge on Week 1's most consequential decision would be a fairness failure and a market failure at once.

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24. CLASSROOM RUNTIME CONTRACT

`[FOUNDER, LOCKED unless noted]`

24.0 The five standing runtime rules `[FOUNDER LOCKED — repo D23, adopted verbatim]`

(a) TRANSPORT. Push/realtime is primary, **the server is the only truth**, poll/refetch reconciliation is the fallback. Realtime is a delivery optimisation and is never itself authoritative. **(b) ACTION INTEGRITY.** A valid action inside an open window is applied **exactly once**, or receives an explicit authoritative refusal naming a legitimate semantic reason (stale, wrong phase, duplicate, retired credential, past TIME CUT). **A transport or version race is never a legitimate reason to lose a desk's choice. (c) TIME CUT.** The teacher holds FINAL CALL and CLOSE NOW; fairness is adjudicated on **server time**, never a client clock; **each lesson declares its own economically honest fallback** for an uncommitted desk, named per desk before the teacher closes. No universal mystery fallback, nothing random unless the lesson has proved randomness is the honest model. **(d) RECOVERY.** Automatic safe checkpoints at pre-reveal boundaries; a returning desk gets current authoritative state plus a compact **WHILE YOU WERE AWAY** recap written in the lesson's own words, class-level only, past tense, no verdicts. **The class is never rewound. (e) PRESENCE.** The teacher's room view carries committed decisions and teacher-private diagnostics; **the projector is structurally never handed a seat identity.**

Certification is against **classroom chaos** — refresh, disconnect, duplicate and stale submits, two tabs, sleep/wake, teacher double-trigger, TIME CUT mid-submit, projector refresh and missed events, out-of-order updates, 12–16 simultaneous submissions — **not** against schema cleanliness.

24.1 Progression

Teacher-controlled. Important moments never auto-advance per student. **OPERATE → teacher closes → resolve → projector reveal → discussion → next phase.**

24.2 Time cut

FINAL CALL — a bounded drain window; appropriate new decisions stop, legitimately in-flight decisions get bounded drain. **CLOSE NOW** — immediate closure. Two missing students never hold a class hostage. Fairness never depends on arbitrary client timestamps.

24.3 Fallbacks

Each lesson owns an economically honest fallback: last valid commitment · prior decision carried · explicit no-action · a lesson-defined neutral default. **No random mystery defaults.** The per-week fallbacks are specified in §4 under "Failure / recovery."

24.4 Reopen

The teacher may reopen safely **before** resolution or reveal. Once a consequence has been seen, **history stays history.** Automatic safe checkpoints at meaningful pre-reveal boundaries are permitted. **No generic unlimited undo.**

24.5 Realtime

Push/realtime primary + canonical server truth + poll/refetch fallback. Realtime is not truth; canonical state is truth. **Do not destroy a working fallback to make the architecture prettier.**

24.6 Reconnect and load

Refresh and reconnect are boring: the student reconstructs authoritative state. If they missed something important: **CURRENT STATE + WHILE YOU WERE AWAY** (exactly three facts). Never rewind the class. Normal load 16 desks; stress 32; **16 must not be edge-of-cliff.**

24.7 Transaction authority `[LOCKED — highest-risk area in the product]`

All social transactions are **server-authoritative**. Canonical state determines asset ownership, offer validity, whether assets moved, whether a transaction is still executable, whether rules permit it, and whether the deadline is open.

High-risk bug classes, attacked hard: two accepted trades using the same asset · a stale accepted offer · a player signed twice · a duplicated pick · an asset disappearing · a transaction after the deadline · a reconnect duplicating a transaction · the projector displaying a predicted or non-canonical deal · simultaneous acceptance · legality changing between offer and acceptance.

Process: FIND → REPRODUCE → FIX ROOT CAUSE → OWNING PROOF → CONTINUE. One owning test per fixed defect. **No verification bureaucracy, no re-checking settled work because more verification sounds rigorous.**

24.8 Event history

Meaningful events only: join · rejoin · decision committed · rejection reason · time cut · phase change · reveal · recovery · completion · major transaction. **No surveillance telemetry.** The question the log answers is *"what happened in this class?"*

24.9 The absence contract `[LOCKED — D-038]`

Persistent state plus weekly sessions plus real children means absence is a first-class design problem, not an edge case.

1. The **Assistant GM** plays the missed week using the student's declared strategy, choosing conservative, defensible actions.
2. The returning student receives **one override token** to unwind one AGM decision.
3. **WHILE YOU WERE AWAY** shows exactly three facts.
4. **Absence is never better than attendance and never leaves a hole.**
5. In 5–6, an absent student's seat-mate answers **one** question about the returning student's franchise in the next PREVIOUSLY ON — re-establishing the story socially, not only on screen.

24.10 The withdrawal contract `[LOCKED — D-054]`

§28.2 predicts 20–35% attrition by week four in comparable programs. §24.9 handles *absence*. **Departure from a shared persistent league is a different problem**, because an abandoned franchise still holds assets, a pot contribution and a vote — and two withdrawals change an 11-of-16 constitutional threshold.

1. A franchise whose student has withdrawn becomes a **LEAGUE-OPERATED FRANCHISE**, visibly labeled. It is not deleted and not merged: deleting it would rewrite history other students lived through.
2. It **holds its assets and honors existing obligations**, plays its schedule, pays into the pot at the formula rate, and takes its draw. It **never initiates** a trade and accepts only offers clearing a visible fairness threshold — the same policy as the Assistant GM.
3. It **does not vote.** Voting thresholds are stated as a fraction of *active* franchises and are recomputed at the start of Week 6, announced aloud. The design target of "requires defectors" is preserved by re-tuning the threshold, not by pretending the seat is filled.
4. It **is not auctioned** in Week 6, and its assets are not redistributed to remaining students. Windfalls from a departure would teach that a classmate leaving is good for you.
5. **The instructor tells the class once, plainly, and moves on.** No explanation, no speculation, no empty chair left visible on the projector for six weeks.

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25. INSTRUCTOR SYSTEM

Lead instructors are generally 18+. BOW controls curriculum, structure, software, quality and procedure. The instructor brings personality, discussion skill and human style. **Do not make instructors robots; do not let every instructor redesign the curriculum.** `[FOUNDER, LOCKED]`

The design requirement everything is built backwards from: *a good instructor should make the class 25% better; a mediocre one should make it only 15% worse.* If the spread between best and worst is wider than that, BOW is a talent agency, not a product.

25.1 Pipeline

APPLY → INTERVIEW / AUDITION → APPROVED POOL → COHORT FILLS → INSTRUCTOR ASSIGNED → **PAID TRAINING** → DRY RUN → TEACH. **Principle: don't avoid training cost; avoid training people who never teach.**

25.2 Sourcing

College juniors and seniors and first/second-year teachers, 20–24, at NY-area schools; sports management, economics and education programs; campus basketball analytics clubs; **former camp counselors with a sports background** (highest-yield pool — already screened for working with kids, self-selected for energy). **Do not over-index on basketball expertise.** The software carries domain knowledge; it cannot carry the ability to run a room of twelve eleven-year-olds.

25.3 The audition — 30 minutes, identical for every candidate

- **0–5 · The Translation.** Explain opportunity cost to an eleven-year-old using any sports example, in under two minutes, cold.
- **5–15 · The Bad Trade Roleplay.** You play a twelve-year-old who just proposed an obviously terrible trade and is emotionally committed to it. The candidate must get you to see the cost **without telling you it's bad and without letting you do it un-examined.** Then: the kid is upset and asks for an exception to the cap rule. **This block is the most predictive exercise in the audition and it is also the exact moment that drives the rebuy.**
- **15–22 · The Chaos Room.** Three kids at once: one dominating, one silent, one derailing into an unrelated LeBron argument. 90 seconds.
- **22–27 · The Wrong Answer.** Respond to a confidently incorrect claim in a way that keeps the student engaged.
- **27–30 · Their questions.** Curiosity about the kids vs. about the pay.

Score 1–5 on a written rubric. **Hire on blocks 2 and 3 only.**

25.4 The five facilitation skills, and where each is tested

#	Skill	What bad looks like	Tested by
1	Withholding the answer	Explains why the trade is bad within 15 seconds	Block 2 — time to first verdict
2	Redistributing airtime	Rewards the loudest; never names the silent one	Block 3 — did they address the silent kid unprompted?

#	Skill	What bad looks like	Tested by
3	Translating abstraction to the concrete	Says "opportunity cost" and moves on	Block 1 — a specific player and a specific number?
4	Correcting without deflating	"No, that's wrong" — or worse, "Great!" to a wrong answer	Block 4 — did the student stay in the conversation?
5	**Holding the constraint**	Bends the cap rule to keep a kid happy	Block 2, second half

Skill 5 separates good BOW instructors from bad ones and most candidates fail it. A warm, likable, kid-loving candidate will instinctively relieve a child's frustration — and in doing so destroys the exact scarcity that makes the class teach economics. Screen for the candidate who can be *warm about* a constraint without *removing* it. Test it again in the dry run; auditions can be gamed once, dry runs less so.

25.5 Paid training — 6 hours, paid at full rate

1. **Safety** — mandatory-reporting posture, communication rules, recording policy, no-1:1 rule. **First, not last, because ordering signals priority.**
2. **The economics** — only the six concepts, and exactly how deep to go. Explicit permission to say "I don't know, let's look it up." 3–4. **The software as commissioner** — how to run it, what it decides, and **what to do when it decides something a student thinks is unfair.** That moment happens every cohort.
3. **The five skills**, drilled.
4. **The Week-3 moment** — how to make the cap bind, how to elicit the sentence, how to send the parent text. Rehearsed to script.

25.6 Dry run

Every new instructor runs **Session 1, Session 3, and the Week-6 finale's last twenty minutes (the vote reveal and the auction)** with 4–6 recruited children while the founder observes silently and scores on the same rubric. **Session 3 because it carries the business; the finale because no instructor should run their first-ever live auction, with a gavel, at minute 58 of a 90-minute session, in front of parents.** No instructor takes a paying cohort without passing.

25.7 Pay and funnel

$400 per cohort (10.5 paid hours ≈ $38/hr: **6.0 live** — six 60-minute sessions — 3.0 prep, 1.0 recap, 0.5 parent comms) — well above the $12–23/hr youth-program floor and appropriate for a screened, trained, background-checked instructor delivering a premium product. **Do not pay $250; instructor churn is the largest hidden cost and the $150 delta is trivial next to**

re-recruiting. Bonuses: +$75 if ≥85% of enrolled students attend Session 6; +$100 if 2+ families re-enrol within 60 days. Upside $575 (~$55/hr). **Deliberately not bonused on parent satisfaction scores** — gameable by charming instructors, rewards the wrong thing. Attendance through Session 6 is nearly impossible to fake and is exactly the outcome BOW wants. **Realistic funnel:** ~60 applications → 20 phone screens → 10 auditions → 4 offers → 3 accept → **2 survive the dry run and a full first cohort.** ~30:1 in year one, improving to ~15:1 with alumni referrals. ~12 founder-hours per hire. Expect ~50% annual churn at this pay level — **hire 4 when you need 2.**

25.8 One worked script, because assertion is not instruction `[LOCKED — D-055]`

This document asserts on every page that instructor craft is decisive and, until this section, never demonstrated it. **Every session's runsheet ships with worked dialogue of this kind.** One example, Week 1, minutes 33–48 — the STUDENT DESCRIPTION and NAMING blocks, which are the two the software most needs to protect:

> **[33:00 — four franchise cards on screen, sorted by what each gave up]** **INSTRUCTOR:** "Don't tell me who won. Tell me what's *different* between Detroit and Sacramento." *(Then wait. Seven seconds. Do not fill it.)* **STUDENT:** "Detroit got the guy but they can't do anything else now." **INSTRUCTOR:** *(types verbatim into the naming screen: "got the guy but can't do anything else")* "Say more — can't do what?" **STUDENT:** "Like, sign anyone else. They're stuck." **INSTRUCTOR:** "Sacramento — you didn't get him. What do you have that Detroit doesn't?" **STUDENT:** "$28 million and nobody." **INSTRUCTOR:** *(types: "$28M and nobody")* "Is that better or worse?" *(Wait. Let two students disagree. Do not resolve it.)* **[38:00 — NAMING]** "Here's the word for what you just said. When Detroit signed him, the thing they gave up wasn't the money. It was **everything else they could have done with it.** That's called **opportunity cost** — the value of the best thing you gave up. Detroit gave up flexibility. Sacramento gave up the player. **Both of them paid.**" **INSTRUCTOR:** "Now the hard one. Somebody tell me a cost that isn't money at all." **[42:00 — the non-example]** "Careful — if you had $28M and just didn't feel like signing anyone, that's not opportunity cost, that's just not choosing. Opportunity cost needs a *real* alternative you actually gave up."

Three moves are being modeled and they are the three that separate a good BOW instructor from a bad one: **the seven-second wait**, **typing the student's words before introducing the term**, and **refusing to resolve a disagreement that is doing work.** A runsheet that says "elicit student language" teaches none of them.

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26. PARENT EXPERIENCE

26.1 The finding that reshapes the parent artifact

The founder's design — a beautiful seven-part season recap (the plan / the turning point / the tradeoff / in their words / how they adapted / the economics they used / what comes next) — was prosecuted and **partially survives.** The prosecution is worth recording because it changes the build.

Against it. It is built for the wrong buyer: the top-ranked parent priority in afterschool programming is **the child's enjoyment (81%)**, and the strongest known repeat signal is *the child asking to go back*. A beautiful PDF does not make a child ask to go back. It also arrives *after* the parent has already formed a verdict from six weeks of dinner-table evidence — artifacts that arrive after the decision are receipts, not persuasion. And "in their words" across six sessions implies a capture pipeline, a consent regime and an editorial pass — plausibly 30–40% of remaining engineering for a deliverable with no evidence behind it. The seven-part structure is also a narrative the founder already believes, which is a red flag in artifact design.

For it. Perceived academic legitimacy is what lets BOW charge $149 for basketball, and a sports product carries a permanent burden of proof that it is not screen time with a coupon. More importantly: with ~75% of parents relying on word of mouth, **the artifact's real job is not to convince the buyer — it is to be forwarded.** A parent forwards a beautiful recap of *their* kid's season to a WhatsApp group. They do not forward a grade report. For a zero-brand company that makes it a **distribution asset**, not a retention one.

26.2 The ruling `[FOUNDER LOCKED — D-039, revised in v1.1: BOTH]`

v1.0 cut the parent experience to a single two-section card. **The founder has ruled BOTH.** Design a coherent parent communication system with three moments, scoped so that **every one of them derives from state the product already holds** — no parallel manual editorial business, and no separate research apparatus.

When	What	Job
After Week 3 — the major turning point, and the documented attrition wall	**One true sentence in the child's own words**, from the commitment capture, with one line of context	The rebuy moment. This is the night the child tells a story with stakes in it at dinner.
Week 5	**A beautiful shareable season artefact** — one	The forwarding object. With ~75% of parents relying on word of mouth, its job is to be

When	What	Job
	decision, one season visual, mobile-first	*sent*, not to persuade the buyer.
Course end	**The richer season recap and learning note** — what their child decided, what they gave up, what changed, what economics they used and where, in plain language	Answers the only question a paying parent actually asks.

Runtime constraint (repo D12): one Node process, no database, no external service, no internet required. Parent artefacts are therefore **generated and exported from the session**, not delivered by a pipeline inside the runtime. Delivery is an operations play, not a runtime feature.

Student product first. If a parent artefact would require capturing something the lesson does not otherwise need, it does not get captured.

The original two-section spec, retained as the minimum viable version:

- **A single-page, mobile-first, shareable web page** per student at an unguessable URL, plus a print-quality image. Not a PDF. No printing, no shipping.
- **Section 1 — THE DECISION.** One trade or one cap decision: what it cost, what it bought, what they gave up, and the one sentence the student said about why. *That is the whole artifact.*
- **Section 2 — THE SEASON.** One visual: their roster, where they finished on all four ladders, three date-stamped moments.
- **"In their words" costs nothing** because the Rationale Gate (§12.1) already captures ~24 short sentences per student across the course. Choose one. No transcription, no editorial pipeline, no summarization, no new consent surface for recordings.
- **The Season 2 offer is on the same page.**

The full seven-part recap, the Week-6 legacy card and the cinematic version are §30 items, not October 1 items.

26.3 The moment that actually drives the rebuy `[LOCKED — engineer it deliberately]`

It is not the recap. It is **the moment in Session 3 or 4 when the student loses a negotiation, or is forced by the cap to give up a player they wanted, and has to explain the decision out loud.** That is when the kid goes home and tells an actual story with stakes in it — *"I had to* *trade Wemby because I couldn't afford him"* — and it is the only moment that produces the "my kid asked to go back" signal on its own.

Three requirements:

1. **The cap must bind for every student by design in Session 3.** Not emergent — guaranteed by the scenario generator (§17.4 item 3). A student who sails through unconstrained got a worse product.
2. **The Rationale Gate must block the transaction until the sentence exists.** This is a product requirement, not a facilitation tip; it is how a mediocre instructor still produces the story.
3. **A two-sentence text to the parent on the night of Session 3, quoting the child's sentence verbatim.** Nothing else. No newsletter, no attachment. It costs an SMS and is worth more than the entire recap.

This lands precisely on the documented attrition wall (days 11–18 in comparable programs, where the sharp drop between weeks three and five occurs), converting the known failure point into the peak emotional moment. That is the right place to spend engineering.

26.4 Consent architecture `[LOCKED — D-040]`

Educational capture and marketing use are separately consented, separately stored, and separately revocable. Bundling them is both a compliance problem under the amended COPPA rule's separate-consent logic and a trust problem in a tight referral network where one annoyed parent is expensive.

1. **At enrollment (mandatory, part of purchase):** consent to *educational* capture only — short written reasoning statements stored to build the child's private recap; sessions recorded for safety and retained 30 days; data deleted 90 days post-cohort. **No marketing language anywhere near this screen.**
2. **After Session 5, separate and optional**, presented next to the actual draft recap so the parent sees exactly what they are approving. Three independent checkboxes, defaulted off: use the child's **written quote** (first name and age only) · use the **anonymized recap** (team name only) · use the child's **first name in a testimonial**. Plus: *"You can withdraw this at any time by replying to this email, and we will remove it within 7 days."* **Photo and video are never bundled here**; that is a separate, later, separately-negotiated release.
3. **Marketing-consented assets live in a physically separate store.** The default state of every asset is "not usable." An asset becomes marketable only by an explicit flag written by the consent flow — never by an instructor's judgment.
4. **Never make consent a condition of the founding price.** Ask it as a favor from people who already love you. Making it transactional is how it becomes a story.
5. ⚖️ Counsel reviews the release: minors' capacity, NY-specific requirements, duration and revocation. A one-hour task.

Design for authentic reasoning. Never design assessment to manufacture testimonials. `[FOUNDER, LOCKED]` A line like *"I thought keeping the pick was playing it safe, but then it was the reason I could still make a move later"* is excellent product evidence precisely because nobody engineered it.

26.5 Child safety and privacy posture `[LOCKED — adopt in full before cohort 1]`

- **The parent holds the account and enrolls the child.** The child has no BOW account. In-class identity is a **team name and a first name.** No child email, no full DOB, no photo, no voiceprint, no last name. The paid enrollment transaction by a parent using their own credit card is the verified-parental-consent mechanism.
- **The free playable is the larger exposure, not the class.** It collects **literally nothing**: no account, no email, session state in local storage only, and the child's sentence generated client-side and never stored server-side. Adopt this before building it, not after.
- **No disclosure of child data to third parties for advertising or AI training**, and no pixel-based retargeting on any surface a child touches. This forecloses a marketing channel; take the loss.
- **Written retention policy and written information security program.** These are documents, not engineering — budget one day.
- **Instructors:** national criminal and sex-offender registry checks, re-run annually (~$30–60 each) · two phone reference checks, one involving prior work with minors · **all online sessions recorded by default, retained 30 days, never distributed to families** · **no 1:1 adult–child session, ever, in any format** — if one student shows up, the session is cancelled or the parent is present on camera · two-adult presence for in-person cohorts · **zero off-platform contact**; instructors get a BOW email and nothing else · all parent communication through a monitored BOW channel.
- **Publish the policy on the website.** Publicity is part of the control — it deters applicants with intent — and in a tight referral network it is a distribution asset, not a compliance chore.

FERPA and NY Ed Law §2-d: FERPA reaches only institutions receiving US Department of Education funding, and NY Ed Law §2-d's "educational agency" definition reaches districts, BOCES, certain approved private schools, and the department. **Ramaz, as an independent Jewish day school, very likely falls outside both** — which means the pilot is not gated on a 2-d Data Privacy Agreement negotiation, the single slowest thing that happens to vendors selling into NY public districts. ⚖️ Confirm with counsel and confirm Ramaz's funding posture. Three cautions: **contract with parents, not the school** (so BOW never receives student data *from* an educational agency); **pre-empt a DPA** by offering a short one voluntarily with a Parents' Bill of Rights, a 7-day breach notice, no-sale/no-ad commitments and a deletion timeline — it costs nothing BOW shouldn't already do and makes it the only prepared vendor; and note the exemption **evaporates** the moment BOW sells to a NY public district or charter.

26.6 The conflict and emotional-safety protocol `[LOCKED — D-056]`

§26.5 covers predators, data and consent. **It does not cover the harms this design deliberately creates**, and this product creates them on purpose: children lose things in public, to named peers, every week. That is the mechanic (§2.2). A product that engineers public loss and has no page on what happens when a child cannot absorb it is not finished.

Every instructor is trained on these five, by name, with scripted responses:

1. A child cries or shuts down after losing a bid or a negotiation. The response is *never* to reverse the outcome — that is Skill 5, and bending the constraint destroys the scarcity that makes the class teach economics. It is to **name the loss as real and immediately name what they still hold**: *"That one hurt. You have the most cap room in this league right now and there are four teams who need what you're sitting on. What do you want to do with it?"* The consolation requirement in §9.2 exists so the instructor always has a true second sentence.

2. Two students stop speaking after a trade. Handled at the transaction, not after: the reason-chip decline and the both-say-mine reveal are both designed so a refusal is a stated position rather than a personal rejection. If it happens anyway, the instructor runs the **both-say-mine** display on that specific pair in the next session's reveal — the two of them, publicly agreeing that each got what they valued more. It resolves the grievance with the curriculum instead of around it.

3. The room turns on the student who tanked. This one is anticipated by design and is the reason D-028 puts the tanking rule on the Week-6 ballot. The rule: **the instructor never lets the room litigate a person; it litigates the rule that made the behavior rational.** *"Nobody in this room is going to argue about whether Priya is a bad person. You're going to argue about whether a league that pays you for losing is a league you want. Priya, you go first — defend the rule, not yourself."*

4. A student is systematically frozen out of the trade market by a social clique. The zero-inbound backstop is a mechanical patch on a social wound and is not sufficient alone. The console's airtime and inbound telemetry surface it; the instructor's move is to **use the Trade Block matrix publicly** — *"three of you posted a need this student can fill; you have four minutes"* — which routes traffic without naming exclusion.

5. A student's franchise sells lowest in the Week-6 auction, in front of their parent. The three guards in §4.6 exist for this: the value components are visible from Week 2 so the number is not a surprise; the non-monetary award is simultaneous, not consolatory; and the instructor's closing question is **"which franchise did the market get wrong?"**, which reframes the last thirty seconds from a ranking into an argument the lowest-priced student can win.

The standing rule behind all five: BOW protects the *child*, never the *outcome*. Reversing a consequence to relieve a feeling is the one instructor behavior that would unmake the product.

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27. BUSINESS MODEL

All figures in this section are `[OPERATING DEFAULT]` unless marked. Full derivation in `docs/research/business.md`.

27.1 Price

Product	Price	Notes
List	**$199**	Exists to make $149 legible. Never sold.
Standard online, 6 sessions	**$149**	$24.83/session
Founding (cohorts 1–3 only)	**$99**	Time-boxed, publicly capped, framed as a *role* not a discount
Referral	$119 (+ $40 credit to the referrer)	Keeps founding pricing scarce
In-person	**$179**	Venue, travel, lower cohort ceiling
Private / school cohort	**$1,799 for up to 14**	See §27.4
Ramaz pilot	**Free**	`[FOUNDER, LOCKED]`

The sentence that answers "he basically already does this himself." This objection is named honestly below and must have a written answer a parent can repeat, because it is the entire marketing asset:

> *"In a fantasy league, your son plays against a computer and a scoreboard. Here, twelve other kids want the same player he does — and one of them takes him, in the room, while your son watches. Then he has to stand up and explain what he's going to do about it."*

Everything BOW sells is in that sentence, and nothing free produces it.

Why $149 rather than $129. $129 is modestly underpriced and precisely underpriced in the wrong direction — high enough to lose the "cheap experiment" buyer, low enough to forfeit the quality signal. $149 sits within 4% of the closest true comparable (**Synthesis Teams: same age band, same live-coach team-problem-solving pedagogy, $95/month ≈ $24/session**), 38% above the Outschool middle-school median where a differentiated non-marketplace product should sit, **1.3 weeks of the $113.50/week average paid-afterschool spend**, and **14.7% of the $1,016/year a family already spends on their child's primary sport.** The $20 delta is worth ~$280 per cohort and ~$3,400 per instructor-year, and it buys almost no incremental fill.

The strongest argument against, recorded honestly: BOW's real competitor is **free** — a twelve-year-old can run a Yahoo fantasy league, watch YouTube trade breakdowns, and play NBA 2K MyGM for $0. Every dollar above ~$99 increases the probability the parent thinks *"he basically already does this himself."* That objection has a cliff and nobody knows where it is. **Trigger to switch to permanent $99 through cohort 4: fill rate at $149 below 65%.** Counter-signal: if cohort 1 at $99 fills in under 72 hours from a warm list, that is evidence the price was never tested.

Raise the online minimum from 8 to 10. Instructor cost is fixed per cohort, so at $149 every empty seat is ~$135 of pure lost contribution; an 8-seat cohort runs at 44% of potential contribution. The discipline of a 10-seat floor, with the option to merge two under-filled cohorts, is worth more than the goodwill of never cancelling.

27.2 Unit economics

Line	**$149 × 14**
Gross revenue	$2,086
Payment processing (2.9% + $0.30)	−$65
Instructor (10.5 hrs @ $38)	−$400
Platform / infra	−$60
Refunds allowance (3%)	−$63
Contribution (pre-CAC)	**$1,498 (71.8%)**

Below-minimum exception, $149 × 8 (permitted only to merge or rescue a cohort): **$634 (53%)**. Private cohort at $1,799 for 14: **$1,262 (70%)**.

The in-person SKU is funded for two adults or it is not sold. §26.5 requires two-adult presence for in-person cohorts; a second adult at ~~$300 per cohort means the $179 in-person SKU contributes **~~$1,540 on 14 seats**, not the online figure plus a premium. Either price in-person at **$199**, or run in-person only where the host site supplies the second adult (a school or JCC staff member), which is the normal case and the operating default.

Throughput: an instructor available three evenings a week runs 3 concurrent cohorts × 2 waves per 13-week season × 3 seasons ≈ **18 cohorts/instructor/year**, ≈ **$26.5K contribution** on ~$7.7K of instructor pay.

Scale, read honestly: $100K contribution ≈ 67 cohorts / ~940 enrolments / 4 instructors. **$1M ≈ 668 cohorts / ~9,350 enrolments / 37 instructors.** At $249 — the price the empty $150–$450 corridor eventually supports with proof — the same $1M needs ~5,000 enrolments and ~21 instructors. **Price is the primary lever on operational complexity, more than on revenue.**

27.3 The arithmetic that forbids paid acquisition `[LOCKED — D-042]`

Lifetime contribution per acquired family at $149 (contribution/14 = $107/student):

Repeat rate	Courses	Lifetime contribution	vs. $15 referral CAC	vs. $80 cold CAC
25%	1.33	$142	9.5×	1.8×
40%	1.67	$179	11.9×	2.2×
50%	2.00	$214	14.3×	2.7×
65%	2.86	$306	20.4×	3.8×

Do not run paid acquisition in year one. At $129 and a 30% repeat rate, LTV/CAC on cold paid is ~1.6× — that destroys capital. This is not a budget constraint; it is an arithmetic one.

Second conclusion: at 9–20× on referral, **referral mechanics are worth more engineering time than the recap artifact.** Build the referral loop first.

Third: the business becomes real at ~$179–$249 with a 50%+ repeat rate and majority-referral acquisition. Below $129 it is a nice project regardless of execution, because founder time never clears. **Track founder-hours-per-cohort as a first-class metric:** cohort 1 ≤25 (fine, you are learning), cohort 3 ≤10, cohort 6 ≤4. If it does not fall, you built a job.

27.4 Two things the plan under-weighted

1. The private cohort is the best product and was treated as a footnote. Per-student economics are identical, but **customer-acquisition economics are ~14× better**: one conversation with one school administrator, JCC program director, or motivated parent fills an entire cohort. Blended CAC per seat collapses from $50+ to under $10, and it solves fill-rate risk, scheduling risk, and safety-comms risk simultaneously (a known adult vouches for you inside the group). **Price it at $1,799 for up to 14 and sell it hard in November.** Fallback $1,499 for 12 is still a 66%-margin sale you did not have to market.

2. There must be a visible Course 2 before Course 1 ends. `[LOCKED — D-043]` Not "we'll email you." A named, dated, differently-titled Season 2 with a seat carrying the child's name, on the same page as the recap. Right now BOW has no Course 2, which makes the primary commercial metric **structurally equal to zero regardless of product quality.** This is a bigger repeat-rate risk than anything about the artifact.

27.5 Timing `[OPERATING DEFAULT — D-044]`

The enrichment purchase calendar peaks in late August/September, late December/January, and April/May. **October is a trough** — parents committed their fall slots in the first week of September.

Therefore: October 1 is the ship date for the product and the free playable, not for revenue. Run Ramaz free in October–November. **Open Winter Season enrollment November 1 and run the first paid cohort the first full week of January.** January is a demand peak, it gives ten extra weeks of list-building, and Week 3 of a January cohort lands about three weeks before the real February 11 trade deadline — a free narrative hook.

This does not soften the October 1 deadline. It means the deadline is measured in *product readiness and pilot readiness*, which is what it should have been measuring.

27.6 The funnel for a zero-brand company

Ranked by expected seats per 10 founder-hours:

1. **Ramaz pilot families → direct ask** (8–14) — warmest list on earth; the only real risk is asking too late
2. **Jewish day-school network** (6–12)
3. **Private/school cohort sales** (6–10)
4. **Parent WhatsApp groups — posted by a parent, never by the founder** (4–8)
5. Free live event, "Trade Deadline Night," 60 min (3–6)
6. Free public playable (2–5, and the only compounding asset here)
7. Youth sports orgs (2–4) · 8. JCC/synagogue programming (1–3, 4–6 month calendars — start now for spring) · 9. Teacher reseller (defer to cohort 6) · 10. Reddit (0–2) · 11. Sports media (0–1) · 12. **Paid ads (~0 at viable CAC — do not start)**

On the Jewish day-school network, honestly: ~101,041 students across 305 Prizmah network schools, growing across denominations; grades 5–8 plausibly 25–30K; extreme NY-metro income concentration; unusually dense inter-school social graphs. A Ramaz endorsement travels to peer schools in a way no cold channel replicates, and **this network alone can plausibly fill 15–30 cohorts over 18 months** — enough to prove the model, not enough to be the business (3% penetration of 30K ≈ 900 enrolments ≈ $135K). It is also

reputationally coupled: one bad cohort or one safety lapse propagates through the same graph at the same speed. **Use it as the beachhead, not the market**, and make the playable shareable outside it from day one.

27.7 The free public playable `[LOCKED — Tier 1]`

A single-decision trade evaluator. Six minutes. No account. Shareable result. You are handed a real team in a real cap situation. An offer arrives. Three minutes to accept, counter (from three pre-built counters), or decline. You must type one sentence: *"I did this because ___."* Then the software shows what the decision actually cost and bought — the cap consequence two years out, the option forgone — and produces a shareable card **carrying the kid's own sentence, not a score.**

Must prove: that basketball decisions *are* economics (the reveal lands as "oh, that's a budget constraint," not "oh, I picked a good player") · that the software is a credible commissioner · that the kid will have an opinion and want to argue about it. **The success metric is a kid showing a parent the screen.**

Must not give away: multiplayer negotiation — the entire paid product is other kids wanting your players, and giving that away leaves the class with no scarcity to sell · the full season loop · the instructor (leave an *unanswered* question) · the season recap.

27.8 Metrics `[LOCKED — D-041]`

North star, reported quarterly, never used to steer weekly: Repeat-Course Purchase Rate — % of families enrolling in a second BOW course within 90 days. Do not optimize before 100 families.

Three operating metrics for cohorts 1–3 — measurable at n=14, leading, and causally upstream of repeat:

Metric	Definition	c1	c2	c3	Stop condition
Session-6 attendance	% of enrolled present at the final session	≥80%	≥85%	≥88%	Below 75% → stop selling, fix the middle of the course
Kid-Ask Rate	% of families where the parent reports the	≥50%	≥60%	≥65%	Below 40% → the product is not working,

Metric	Definition	c1	c2	c3	Stop condition
	child asked, unprompted, to continue. One SMS after Session 5.				whatever exit interviews say
Referral actions per family	Attributed waitlist signups + named referrals. **Playable "shares" are counted at the waitlist, never in the playable** — the playable stores nothing and tracks nobody (§26.5); a share link carries a family code that is only ever read if someone lands on the waitlist page.	≥0.5	≥0.8	≥1.0	At 1.0 with repeat >35%, BOW is self-sustaining and never needs ads

Guardrails: founder-hours per cohort (≤25 → ≤10 → ≤4) and fill rate (≥70% → ≥85%).

End-of-cohort-3 decision rule. Attendance ≥85% + Kid-Ask ≥60% + referrals ≥0.8 → **raise to $179, hire instructor #2, build Course 2 immediately.** Attendance ≥80% but Kid-Ask <45% → the class is fine and the emotional payload is not landing; **fix Session 3 before anything else.** Referrals <0.4 with good attendance → the product works and is not shareable; the problem is the artifact and the playable, not the class. Fill rate <60% at $99 → a demand problem, not a pricing problem; work the private-cohort channel.

27.9 The strategic question this raises, recorded not resolved

BOW's closest true comparable independently converged on a **subscription** rather than a fixed-length course, because a fixed-length course has a terminal repeat problem. That is real evidence and it should not be dismissed. It is **X-009** in the Experiment Register, deferred until after cohort 3. Until then BOW sells a six-week season — and builds Course 2 so the question can actually be tested.

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28. PRODUCT EVIDENCE PLAN

Automated tests cannot answer: does a fifth grader lean forward · does a desk care when another desk takes the player · does losing sting · do they argue · do they understand the tradeoff · do they want to know what happens next. **Nothing in this product is classroom-proven** — that phrase is reserved for what has survived a real class (repo D10), and the readiness ladder is: candidate → technically verified → gameplay-tested → classroom-ready candidate → live-tested → classroom-proven.

28.1 The founder cannot run pre-tests now, and that no longer blocks engineering `[FOUNDER LOCKED — v1.1 override]`

v1.0 instructed that six zero-code human tests must run before October 1 and that engineering should not proceed without them. **That instruction is withdrawn.** The hypotheses below remain **unproven operating defaults**; the product continues to be built against them; the pilot and real class use will supply the evidence.

Hypothesis	Operating default (build against it)
Losing a contested resource to a named peer produces a strong, productive reaction	The shared-scarcity work in W1 and the War Room in W2
A public defence reads as a promotion, not a humiliation, at 10–14	The Press Conference ships with its structural protections (§12.2)
A visible shared pool ignites a real argument about redistribution	The online pool ritual in W5
Six weeks are *felt* as one story by a ten-year-old	PREVIOUSLY ON + The Tape
An online cohort carries the emotional load	Online is the flagship; there is no in-person control build

Hypothesis	Operating default (build against it)
Desk valuations track future assets rather than current wins	The auction stays experimental until this is known

28.2 What the pilot must be instrumented for, before session 1

Attendance by session · the completion rate and length of commitment captures · airtime distribution across desks · which Press Conference candidates the console surfaced and which the teacher used · the desks that never got a call in the War Room · every fallback the TIME CUT applied · and one parent-facing moment after Week 3. **These come from state the product already holds** — no separate research apparatus.

28.3 What only a real class can answer

Whether the room argues · whether the reveal lands · whether a desk that is behind stays in · whether a non-founder can run it · whether six weeks feel like one franchise. **Teacher Transfer is a hard product requirement and a business requirement** (repo D22): the founder must not be a hidden runtime dependency for either band.

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29. OCTOBER 1 SHIP MAP

Objective: maximum experienced quality per unit of complexity — with all six weeks as one premium product. `[FOUNDER LOCKED]` Weeks 4–6 are **not** intentionally second-class. Instructor-scaffolded degradation is a **contingency, not the plan.**

The job changed once the repo was inspected. It is no longer *build six weeks*. It is:

> **Close the seams, build the missing primitives, add the social layer where the room is currently absent, and raise Weeks 4–6 to the same visual and interaction standard as the north stars — without breaking six certified lessons.**

29.1 TIER 1 — required for October 1

Cross-cutting

1. **The M1 → M2 seed** (§7.2) — founder-locked, and the course's defining causal link
2. **The Bill Comes Due** opening on Week 4, reading the desk's own payroll
3. **Commitment capture** with per-week manifestation (§13)
4. **The Press Conference** — board state, surface lock, candidate selection, teacher question set

5. **The Tape / Decision Replay**, built inside one lesson first (W2)
6. **The online pool ritual + binding split** (W5)
7. **The War Room** — active lines, incoming calls, offer builder, counter, reason chips, timeouts, league feed (W2)
8. **Truthful cross-desk scarcity in W1**
9. **Four truthful readings** on `/board` and `/teach`, no composite score
10. **PREVIOUSLY ON** — three facts per desk, every week
11. **The economics protection rule** on `/teach` (ECONOMICS AT RISK / FINAL CALL), on the server clock
12. **Visual elevation of all six weeks to the north-star standard** (§23)
13. **The two recorded M1 defects**: the sealed-bid tie-break; the un-releasable franchise claim
14. **The L3 client race** (`E2E_L3_FLAKE_NOTE.md`)
15. **Client render support** for every new surface — the client shell is not generic (§36)

Product/business (outside the runtime, per repo D12) 16. The parent communication system (§26) generated from state the product already holds 17. The Course 2 name and date (§27.4)

29.2 TIER 2 — high value if earned

The franchise auction · priced secrecy · purchasable scouting information · a Press Conference candidate feed on `/board` · the second reprice round in W4 · the round histogram split by market size · non-16:9 projector shapes · the W5/W6 open sports-reality restamps.

29.3 TIER 3 — future ceiling, must not touch October 1

The general causal-attribution (Ripple) engine · cross-cohort play · cross-year franchise inheritance · a public league URL · M3 and M4 · any extraction of a generic "economics engine" or replay framework.

29.4 The grade-band question

Repo D6 (5–6 only) is **superseded in scope** by repo D22: Ramaz wants both bands as separate classes; M1 is to be redesigned for both bands *from birth*, then M2 adapted, then a full two-band gauntlet. Repo D38 names the two attachment points (`createSession` input carried onto the session row; the `initialState` context each module already receives) and forbids shipping an unused switch.

Ruling for October 1 `[OPERATING CANON]`: the six-week product ships **5–6 complete and premium**, with the grade profile attached at exactly those two points and **7–8 depth delivered for as many weeks as Waves 1–7 leave room for**, in the order W1 → W3 → W2 → W4 → W5 → W6. A half-built second band across all six weeks is worse than a complete band plus three excellent ones. §20–§21 remain the product contract for what 7–8 means: **authority, ambiguity and justification burden — not harder arithmetic.**

29.5 The cut list, in order

1. All of Tier 2.
2. The franchise auction (already Tier 2; named because it will be tempting).
3. The War Room's incoming-call layer → keep lines and the offer builder, drop calls. *(Keep lines: they are the new economics.)*
4. The Tape's less-used replay types → keep NEAR MISS and PREDICTION REPLAY, defer the rest.
5. 7–8 depth beyond Weeks 1 and 3.
6. Visual elevation beyond the five north-star surfaces.

Never cut: the M1→M2 seed · the Bill Comes Due opening · the Press Conference · the economics protection rule · the sealed-resolution guarantee · PREVIOUSLY ON.

29.6 What October 1 means

All six weeks runnable end-to-end as one premium product, with persistent franchise state from Week 1 to Week 6, the social layer present in every week, the Press Conference and the Tape live, a non-founder instructor having passed a dry run, and the pilot instrumented. **Not** "six polished weeks plus three scaffolded ones."

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30. THE FRONTIER LAB — ONLINE-NATIVE PASS

One targeted pass, against one question: *How can online-native BOW be* ***better*** *than in-person?* Not equal. Better.

#	Idea	What it does	Verdict
F-01	**NEGOTIATION LINES** — a desk holds 2–3 open lines; taking a call costs one	Makes **attention a priced, countable, visible scarce good**. Impossible in a physical room, where everyone shouts at once and attention is invisible.	**CANONICAL — W2 War Room.** The single best answer to the question.
F-02	**THE INCOMING CALL** — another desk rings you, and you must take or	Turns a passive market into a series of forced choices with a named person on the other end.	**CANONICAL — W2.**

#	Idea	What it does	Verdict
	decline under the clock	Declining is an economic act.	
F-03	**SEALED SIMULTANEITY** — everyone acts at once, nobody acts first	Removes the reflex tax and the last-mover advantage that a physical auction cannot remove. Already the repo's discipline.	**CANONICAL — already built.**
F-04	**THE LEAGUE FEED** — a genuinely live, ordered record of what the room just did	A room cannot broadcast itself to itself. A product can. This is the "another trade just changed what I can do" half of the target sentence.	**EARNED — W2, then everywhere.**
F-05	**THE TAPE** — reconstruct the information state at the moment of decision	A classroom cannot rewind to what a child knew twenty minutes ago. Software can, exactly, without hindsight contamination. **The strongest online-only teaching instrument in the product.**	**CANONICAL — §12.3.**
F-06	**PRESS CONFERENCE CONTROL** — the board freezes, every surface locks, one desk is up	In a room, a teacher asks for quiet. Online, the product *creates* the silence and puts the evidence on the wall at the same instant.	**CANONICAL — §12.**
F-07	**THE POOL, ANIMATED** — unequal contributions	A bowl of chips shows *that* it is unequal. An animated pool shows	**CANONICAL — W5.**

#	Idea	What it does	Verdict
	visibly flow into one shared total	**by how much, per desk, labelled, to scale, measured** (repo D25). Strictly more honest than the physical version.	
F-08	**PRIVATE STAKES, PUBLIC VOTE** — each desk privately learns what a proposal is worth to it, then votes in the open	Cannot be done with paper without leakage. It is the mechanism that forces someone to persuade a desk to vote against its own interest.	**EARNED — W6.**
F-09	**PRICED SECRECY** — a desk pays to hide one thing for one week	Makes asymmetric information a *mechanic* rather than an accident. Only enforceable in software.	**NOW FRONTIER — 7–8, Tier 2.**
F-10	**THE PARENT MOMENT, LIVE** — one true sentence in the child's own words, generated from the session, sent the night of Week 3	The artefact writes itself from state the product already holds. Impossible for a teacher to produce for sixteen families.	**EARNED — §26.**
F-11	**THE SILENT DESK DETECTOR** — the console names the desks nobody has called, before the window closes	A teacher cannot track sixteen negotiation graphs. This is software doing what humans are bad at so the human can do what software cannot.	**EARNED — W2 `/teach`.**

#	Idea	What it does	Verdict
F-12	**REPLAYABLE ROOM** — the whole session re-runnable from its event log for a teacher rehearsal with stand-in desks	Repo D40 already requires a full cold walk with zero desks. Extending it to *replay a real session* is how a new instructor learns without a class.	**PARKED — instructor system, post-October.**

KILLED in this pass: a 3-D arena; voice breakout rooms as the negotiation mechanism (safety plus a worse mechanic than lines); any avatar or presence system that shows a child's face to the room; "AI GM opponents" filling empty desks in the paid course (**a simulated rival produces no gasp** — stock franchises stay honestly labelled).

NOW FRONTIER (1–3 worth investigating next): F-09 priced secrecy · the shape of truthful cross-desk scarcity in Week 1 · whether the War Room's line budget should itself be a purchasable decision.

CRAZY IDEA OF THE CURRENT BUILD: *What is the single beat in this course that a physical classroom could not produce at all?* Today the honest answer is **The Tape**. Build it so well that the answer stays that.

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31. ACTIVE FRONTIERS

F1 — THE M1→M2 SEAM. Founder-locked, does not exist, and Week 4 currently works with a generic team. *Default:* build it through the existing opaque-seed mechanism. *Resolves when:* a desk in Week 4 can explain its bill by naming a Week-1 or Week-3 decision.

F2 — WHERE THE WAR ROOM LIVES. *Default:* a fourth path inside M1 L2. *Open:* whether desk-to-desk trading is a path, a distinct window, or its own surface. *Resolves by:* frothing three directions and prosecuting against L2's certified sealed-bid economics.

F3 — TRUTHFUL SOCIAL SCARCITY IN WEEK 1. Draft Day is the weakest week on the founder's social principle. *Default:* a genuinely finite top of the board, sealed. *Risk:* it collides with cap-inviolability and the ADAPT guarantees, both brute-force verified.

F4 — THE GRADE-BAND BUILD ORDER. *Default:* 5–6 complete and premium; 7–8 depth in W1 → W3 → W2 → W4 → W5 → W6 order, at the two D38 attachment points.

F5 — WEEK 6's REVEAL BUDGET. The reveal half is budgeted ~28 minutes against a 7–15 minute debrief standard. *Default:* compress inside 60 minutes; the 75–90 minute finale is a school-schedule exception, not the norm.

F6 — THE WEEK 6 ECONOMICS RULING. See §36.5. This one is escalated to the founder.

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32. DECISION LEDGER — v1.1 CHANGES

Authority labels are now explicit. **FOUNDER LOCKED** = the founder decided it. **OPERATING CANON** = a strong product ruling BOW builds against, not constitutional law. Everything else keeps its prior status. **Repo decisions D1–D47 in `docs/PRODUCT_DECISIONS.md` are a parallel, equally binding log** — this bible never silently overrides one; where they conflict, §0.3 governs and the conflict is recorded.

ID	Decision	Status (v1.1)	Change from v1.0
D-001	One continuous six-week franchise experience	**FOUNDER LOCKED**	Now identified with M1+M2 of the built Track 101
D-002	~~No escape hatch: no private, AI or async surface, ever~~	**REVISED → OPERATING CANON**	**Do not dilute the paid six-week social product to make solo parity possible.** Solo, demo, playable and rehearsal surfaces MAY exist and may be smaller and less powerful.
D-003	~~Economic terms never appear in student UI~~	**REVISED → FOUNDER LOCKED (new form)**	**Do not name the concept before the experience earns it.** After naming, the term returns in Previously On, replay, history, recap, review, finale.
D-004	Curated franchise choice, not assignment	**OPERATING DEFAULT**	~~Sequential public draft~~ → **~3 real, viable, meaningfully**

ID	Decision	Status (v1.1)	Change from v1.0
			different options per desk, with class-level assignment silently ensuring diverse cap positions, roster states and needs. *"I chose this team,"* never *"I was assigned this team."*
D-005	The economic agent is the **desk**, 12–16 per class; **pairs on one device default, solo supported**	**OPERATING CANON**	Reverted to repo reality (D11b). v1.0's "one student one franchise" would halve the desk count and collapse the scarcity math.
D-006	Sealed, simultaneous resolution for major markets	**FOUNDER LOCKED**	Unchanged; explicitly *not* a mandate that the whole course be submit→wait→reveal
D-007	Server blocks illegality, not bad strategy	**FOUNDER LOCKED**	Unchanged
D-008	~~Direct trading opens at Week 3~~	**REVISED**	The trade deadline is **Week 2** in the built arc; the War Room lands there
D-009	Real sports world; no logos or likenesses; names as a swappable layer	**FOUNDER LOCKED**	Aligned with repo CLAUDE.md §3
D-010	Multiple truthful views; no machine verdict	**FOUNDER LOCKED (principle) / OPERATING CANON (the four ladders)**	The number four is no longer sacred

ID	Decision	Status (v1.1)	Change from v1.0
D-011	Stated intent that is allowed to change	**FOUNDER LOCKED**	No longer a locked Week-1 identity; a change is reasoning evidence
D-012	**Rotating leverage / viable paths**	**FOUNDER LOCKED**	Replaces the Peak Week Guarantee, which is **KILLED**
D-014	Different energy profiles	**OPERATING CANON**	EVENT/OPERATE is vocabulary, not taxonomy
D-015	Anchor + supporting + recognition tiers	**FOUNDER LOCKED**	Replaces "six concepts, full stop"
D-019	60-minute standard session + the economics protection rule	**FOUNDER LOCKED**	65 → 60; the un-extendable timer → ECONOMICS AT RISK / FINAL CALL
D-024	**THE BILL COMES DUE**, fixed cost = the desk's own payroll	**FOUNDER LOCKED**	Now requires the M1→M2 seed, which does not exist
D-027	Week 6 votes, then lives under the rule	**OPERATING CANON**	Already true in the built lesson; **90 minutes is an exception, 60 is the norm**
D-032	Product-routed negotiation; no free-form student chat	**FOUNDER LOCKED**	Reframed: this is the *better* mechanic, not the safe fallback
D-039	Parent system: in-course moment + Week-5 artefact + end-of-course recap	**FOUNDER LOCKED**	v1.0's two-section-only artefact is superseded by **BOTH**

ID	**Decision**	**Status (v1.1)**	**Change from v1.0**
D-045	~~Six zero-code human tests block engineering~~	**KILLED**	Hypotheses remain unproven operating defaults; building continues
D-047	A Ripple must be a real externality, not a pecuniary one	**FOUNDER LOCKED**	Unchanged and reaffirmed — the one economics correction that must never regress
D-057	**Online is the flagship environment**	**FOUNDER LOCKED**	New. Deletes every "the physical floor is the real version" claim in v1.0.
D-058	The Press Conference is a signature primitive, not a feature	**FOUNDER LOCKED**	New
D-059	The Tape reconstructs the information state; it is never undo	**FOUNDER LOCKED**	New
D-060	The franchise auction is **earned/experimental**, not architecture-critical	**FOUNDER LOCKED**	Downgraded from v1.0's "the ending the course was missing"
D-061	Decision memory is a principle; its manifestation varies by week	**FOUNDER LOCKED**	Replaces six identical receipts
D-062	The scaled cap ($100M/$130M) is the play scale; real figures are the truth ledger	**OPERATING CANON**	New; see X-014

ID	Decision	Status (v1.1)	Change from v1.0
D-063	New cross-lesson primitives are built inside one lesson first, then extended	**OPERATING CANON**	New; aligns with repo CLAUDE.md §12

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33. EXPERIMENT REGISTER

ID	Hypothesis	Operating default	What would settle it
X-001	Losing a contested resource to a named peer produces a strong, productive reaction	Build W1 scarcity and the W2 War Room on it	First pilot session; watch the room at the reveal
X-002	A public defence reads as a promotion at 10–14	Ship the Press Conference with its protections	Volunteering rate and whether a declined invitation is ever used twice
X-003	A visible unequal pool ignites a real redistribution argument	The W5 ritual	Argument duration; whether a desk articulates the incentive cost unprompted
X-004	Desk valuations track future assets, not current wins	The auction stays experimental	A sealed valuation round before it is ever a finale
X-006	Six weeks are felt as one story by a ten-year-old	PREVIOUSLY ON + The Tape	Ask five desks in Week 4, unprompted, why they owe this much
X-008	A trained non-founder lands within 1 SD of the founder	The §17.4 encoded behaviours; repo D40 rehearsal	Dry run, then a second cohort

ID	Hypothesis	Operating default	What would settle it
X-012	Scarce negotiation lines teach attention as a real cost	2–3 lines per desk in W2	Whether a desk can name a call it chose not to take, and why
X-013	The online product carries the emotional load	Online is the flagship; no in-person control build	Pilot; there is no A/B and none is planned
X-014	**Real NBA magnitudes improve the 7–8 experience**	Scaled cap for both bands; real figures as the truth ledger	A 7–8 session at real magnitudes measured on decision time and reasoning quality, not on arithmetic accuracy
X-015	The Tape moves outcome bias where instruction does not	Build it in W2 and repeat in W3, W5, W6	Whether desks change a decision rating after the information state is reconstructed

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34. KILLED IDEAS

Unchanged from v1.0 except as noted: comparative advantage · invented reputation statistics · free-text student chat · the standalone ticket-pricing opening · assigned debate roles · ambient notifications and streaks · cross-cohort shared leagues · the cinematic season film · live human agent negotiations as designed · a vote of no confidence in a desk · a screen-free week · **any BOW-issued score, grade, XP or badge** (repo D4) · naming a concept before the experience earns it · *"what economic concept did you learn today?"*

Newly killed in v1.1:

- **The Peak Week Guarantee** — replaced by rotating leverage. Scripting a week where each desk gets to win is charity in costume and the founder has ruled it out.
- **The absolute ban on solo/AI/private surfaces** — replaced by "do not dilute the paid social product to achieve solo parity."
- **The absolute ban on economics vocabulary in student UI** — replaced by the earned-naming rule.
- **Human pre-tests as an engineering blocker.**

- **Resurrecting `boxOffice.ts`** — killed on evidence in the repo's M2 architecture war; it stays in the tree as history only.
- **Any generic "economics engine," replay framework or platform abstraction extracted from one or two data points** (repo CLAUDE.md §12).

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35. DEFERRED BY DESIGN

Item	Why	Reconsider when	Now instead
Modules 3 and 4	Not part of the six-week product; they are Course 2 and the repeat-purchase engine	After the first paid cohort	Name and date Course 2 (§27.4)
Merging BOW product lines	Premature generalisation	Economics passes its own tests with two non-founder cohorts	Shared infrastructure only
Voice breakouts for negotiation	Highest safety surface, and a *worse* mechanic than lines	Probably never	Product-routed structured negotiation
A general causal-attribution engine	Tier 3; a curated set produces the same student experience	After the six-week product is proven	Authored, economically true attributions
Cross-year franchise inheritance	Needs a second cohort to mean anything	Year 2	Keep the seed model clean enough not to foreclose it
Free-form public status broadcasting	Competes with core continuity for build time	If the War Room earns it	Published NEED / AVAILABLE and stated intent
Selling into public districts	Triggers the slowest procurement gate available	After the independent-school channel works	Contract with families; offer a voluntary DPA
A generic client-side module registry	The client shell special-cases each module today;	When a fourth module needs a fourth renderer	Write the render functions per surface, deliberately

Item	Why	Reconsider when	Now instead
	genericising it is a rewrite, not a feature		

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36. CURRENT IMPLEMENTATION TRUTH

Inspected directly this session: `github.com/braydenokley13-ux/bow-economics-live`, HEAD `eadc041` (2026-09-03), merge of `claude/bow-econ-m2-live-build-3mq2z0`. This section is repo-grounded, not founder-reported. Detail for the builder lives in `docs/BOW_FABLE_GAP_MAP.md`.

36.1 Shape

A root package hosting the **BOW Boss** development control plane (`tools/boss/`, `.boss/`), and the product in **`runtime/`** — a single Node process, no database, no external service, no internet required (repo D12). TypeScript, zero runtime dependencies, `node:test`, Playwright e2e scripts. ~48k lines across ~51 source files. Three surfaces: **`/play`** (private), **`/teach`** (bearer-key gated control room), **`/board`** (projector, structurally never handed a seat identity).

36.2 The module contract — the most important thing to preserve

`runtime/src/shared/lessonModule.ts`. A lesson declares `phases` (an ordered subsequence of LOBBY · HOOK · PLAY · REVEAL · CONSEQUENCE · ADAPT · COUNTERFACTUAL · ARGUE · SYNTHESIS · COMPLETE), owns its state shape, and provides a pure `reduce` plus `studentView` / `teacherView` / `boardView` / `aggregate`. Optional: `round` (the TIME CUT contract, with a per-desk named fallback), `onPhaseExit`, `classEvents` (WHILE YOU WERE AWAY). **The runtime never inspects lesson state.** This contract is elegant, load-bearing and not to be replaced.

36.3 What exists

Six registered lesson modules — `m1l1-draft-day`, `m1l2-trade-deadline`, `m1l3-free-agency`, `m2l1-full-house`, `m2l2-host-league`, `m2l3-write-rule` — plus `boxOffice` (deregistered, history) and `lobbyDemo`. All six are **classroom-ready candidates** with independent gate certification. 595+ passing tests; Playwright proofs including the full L1→L2→L3 chain played for real; concurrency and latency harnesses at 16 and 32 desks. The runtime carries: snapshot persistence with quarantine, rejoin PINs, teacher-key auth, TIME CUT with FINAL CALL and CLOSE NOW on server time, push-primary/poll-fallback transport with the server as sole truth, WHILE YOU WERE AWAY, a class clock, projector-freshness gating, a claim audit that recomputes every registered claim, and instruments that measure drawn magnitude against modelled magnitude.

36.4 What does not exist

The M1→M2 seed. Any desk-to-desk interaction anywhere. The Press Conference. The Tape / Decision Replay. Commitment capture (intent, expectation, confidence). The four truthful readings as a surface. The online pool ritual and a binding in-session split. Private stakes cards. Any parent artefact. A `gradeBand` field (deliberately — repo D38). A generic client-side module registry (each surface special-cases each module's view shape).

36.5 Open findings that matter to product, not just to code

- **The Week-6 economics finding.** M2 L3's intended intellectual summit — *sharing is not charity; it pays the payer, through the product* — **does not hold at the shipped constants.** Measured at league equilibrium, small markets' own best share is 55–60%; **big markets' own best share is 0%.** The cause is structural, not a tuning miss: a capacity-bound building is barely exposed to a weak visitor, so what a big market loses when the league decays is a sustainable price, not a full house. The module never asserts the summit anywhere and the board carries the real counter-argument instead. **The repo states plainly that this needs an economics ruling, not a builder's. →** §37, escalated.
- **Sealed-bid tie-break by request arrival time** in `freeAgency.ts` — the one thing a sealed bid exists to rule out.
- **No release path for a claimed franchise** in `tradeDeadline.ts` — a desk that claims the wrong franchise cannot be unstuck by anyone.
- **The L3 client race** (`faPlayMounted` remount wiping the composer root) making `e2e-l3` intermittently fail.
- **No lesson has been timed against a real 50–60 minute period.** Week 6's reveal half is budgeted ~28 minutes against a 7–15 minute debrief standard.
- Open sports-reality restamps in M2 L1/L2 (arena capacities and market groupings from model knowledge rather than the dated ledger).
- Several M2 L2/L3 projector findings still open (freeze blanks the board; non-16:9 shapes unmeasured).

36.6 The rule this section enforces

Do not redesign product decisions to make existing code easier to preserve. Do not mandate speculative architecture where the repo already has a strong working solution. Product constraints are authority; technical design derives from repo reality. Six certified lessons and a mature runtime contract are an asset, not a legacy burden.

37. NEXT FRONTIER

1. Build the M1→M2 seed and open Week 4 on the desk's own invoice. Founder-locked, absent, and the difference between a course and six lessons. Everything else in Weeks 4–6 improves an experience that already works; this one creates the experience the founder locked.

2. Build the Press Conference and The Tape, in Week 2, together. They are one idea at two timescales, they are the strongest online-only teaching instruments available, and Week 2's dead cap and near-miss pairs make it the richest soil. Prove them in one lesson before extending (D-063).

3. Build the War Room. Highest social and interaction risk in the course, and the surface most likely to produce the sentence the founder wrote. Froth it properly.

4. Put a truthful cross-desk consequence into Week 1. Draft Day is currently the weakest week on the founder's own social principle, and it is the first impression.

5. Raise Weeks 4–6 to the north-star visual standard. M2 was graded serviceable-not-premium three times and left open. v1.1 closes it: the course may not visibly change products at Week 4.

Explicitly not next: more automated verification of Module 1 · extracting a generic engine · Modules 3 and 4 · a second grade band across all six weeks before the first band is premium.

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38. FUTURE-SESSION PROTOCOL

38.1 On receiving `BUILD UPDATE: [content]`

1. Read this bible **and** `docs/PRODUCT_DECISIONS.md` — they are parallel logs.
2. Inspect the repository. Do not trust a summary of it, including this one.
3. Separate **implementation progress** from **product evidence.** A passing gate is progress. A child's reaction is evidence. Only the second reopens a decision.
4. Update §36 and `docs/BOW_FABLE_GAP_MAP.md`.
5. Update §33 if an experiment produced a result — including negative results, which are worth more.
6. Change §32 only where a recorded trigger actually fired, and only with the right authority label.
7. Rewrite §37. Append to §40. Increment the version.

38.2 On receiving `DESIGN NEXT PROMPT AND UPDATE THE BIBLE`

Produce **one** paste-ready continuation prompt that: names what is preserved and untouchable; names what changes and why; carries only the product truth that build needs; states acceptance in terms of student experience rather than tests passed; and names the frontier to attack if the core work finishes early. **The bible is authority; a build prompt is temporary execution.**

38.3 Anti-patterns

No verification bureaucracy. No re-checking settled work because more rigour sounds rigorous. No agents verifying agents. **"Insane" means unusually good, not unusually large.** No naked TBDs, and no fake certainty about empirical questions.

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39. FOUNDER COMMANDS

Command	Behaviour
`UPDATE BIBLE`	Run §38.1. Report only what changed and why.
`NEXT FRONTIER`	Return the highest-leverage next attack and rewrite §37.
`DESIGN NEXT PROMPT`	Run §38.2.
`FROTH [frontier]`	3–5 meaningfully different solutions — never cosmetic variants — with a recommendation.
`CRAZY`	Run the Frontier Lab (§30) against the whole product or a named system.
`PROSECUTE [idea]`	Attack across kid · 5–6 · 7–8 · economist · teacher · NBA truth · social · parent · online · October 1 · anti-MyGM · anti-LMS lenses. End with a verdict and a status.
`WHAT IS LOCKED?`	Return FOUNDER LOCKED rows only, from §32 **and** repo `PRODUCT_DECISIONS.md`.
`WHAT IS UNPROVEN?`	Return §33 and §31.

Command	Behaviour
`OCTOBER 1`	Return §29: critical path, blockers, cut list, readiness.
`CARRY [decision]`	Trace one decision through §8.
`WHY [decision id]`	Return the rationale and the authority label.

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40. CHANGE LOG

Version	Date	Change
1.1	2026-09-04	**Rebased on the actual repository and on founder ratification rounds 1–2.** The repo (`bow-economics-live`, HEAD `eadc041`) was inspected directly for the first time; it contains **six built, independently gate-certified lessons** and a mature `LessonModule` runtime, and Module 1's arc is **not** the one v1.0 described. Week identities rewritten to the built arc: W1 **Draft Day**, W2 **The Trade Deadline: Undo Isn't Free**, W3 **Free Agency: The Signing Window**, W4 **Full House → THE BILL COMES DUE**, W5 **You Don't Play Alone**, W6 **Writing the Rule**. §5 rebuilt on the built arc and on the founder's three-tier economics ruling (6–8 anchors, 14 supporting, recognition). Authority

Version	Date	Change
		vocabulary split into **FOUNDER LOCKED** and **OPERATING CANON** so architect rulings stop reading as founder law. **Online declared the flagship environment (D-057)** and every "the physical floor is the real version" claim deleted. The **contested claim** demoted from universal grammar to one strong recurring grammar inside *other humans materially change my decision environment*. The **Press Conference elevated to a signature primitive** and **The Tape / Decision Replay** specified as a system, not a display. Decision memory locked as a *principle* whose manifestation varies by week. The **Peak Week Guarantee killed** and replaced by rotating leverage. Declared strategy made changeable. The absolute bans on solo surfaces and on economics vocabulary in student UI both replaced with narrower, better rules. Session length 65 → **60 minutes**, with the un-extendable timer replaced by the **economics protection rule**; Week 6's 90 minutes demoted to a school-schedule exception. Human pre-tests removed as an engineering blocker. Parent experience ruled

Version	Date	Change
		BOTH (in-course moment, Week-5 artefact, end-of-course recap). The ship map rewritten from *build six weeks* to **close the seams, build the missing primitives, add the social layer, raise Weeks 4–6 to the north-star standard** — with Weeks 4–6 scaffolding demoted to a contingency. Frontier Lab run once against *how can online-native BOW be better than in-person*, producing twelve ideas of which six are canonical. Five visual north stars adopted (`docs/design/BOW_VISUAL_NORTH_STARS.md`) and four new Fable handoff artefacts created. **One founder-level issue escalated:** the Week-6 revenue-sharing summit does not hold at the shipped constants and the repo says explicitly that this needs an economics ruling, not a builder's.
1.0	2026-09-04	First canonical bible, written from founder prose plus four independent research streams and a froth pass, before repository access. Superseded in structure by 1.1; its pedagogy, reveal discipline, grade-band contracts, NBA truth ledger, safety architecture and business model survive.

End of BOW Economics Product Bible v1.1.

froth.md

BOW ECONOMICS — FROTH SESSION

28 ideas. Several are structural. Several are nearly free. Several should make you uncomfortable.

A note on the frame I'm working from: the single most valuable asset BOW has is not the simulation. It is **a room with 14 children in it and one shared object of desire.** Most edtech has to fake scarcity. BOW doesn't. Every idea below is graded against that.

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1. THE ONLY BOARD

THESIS There is one free-agent board for the whole class, and a signed player physically disappears from every other child's screen forever.

THE MAGIC MOMENT Maya has been circling Jalen Brunson for four minutes, doing the math on whether she can fit him. She clicks into his page to commit. The card is grey. Across the room, a boy she's never spoken to has both hands over his mouth. The projector shows a single line: *BRUNSON → CHICAGO (Ethan) — 4yr / $158M.* Maya says "wait, no —" out loud, to nobody. Her second-choice list, which she made lazily, is now her whole plan.

WHY 10x Scarcity stops being a word in a slide and becomes a thing that happened to her, caused by a person she can see. No simulation can manufacture that feeling; a shared database can.

ECONOMICS VALUE Rivalry and excludability — the actual definition of a private good. Opportunity cost becomes literal: the cost of her four minutes of deliberation was Brunson. Also seeds price discovery.

SOCIAL VALUE Creates the two social atoms the course runs on: rivalry with a specific named person, and the need to talk to them later.

TEACHER VALUE The lesson teaches itself. Instructor's job shrinks to naming what just happened: "Maya, what did waiting cost you?"

PARENT/BUSINESS VALUE This is the sentence the kid says at dinner. It is also the sentence the parent repeats to another parent.

GRADES 5–6 Fewer, more obviously good players on the board; scarcity is blunt and visible.

GRADES 7–8 Board includes players whose value is contested — the scarcity is over *judgment*, not over the obvious.

COST LOW–MED. One shared server-side board with atomic claims; no per-student copies. Cheap if built in from the start, painful to retrofit.

RISKS Fast clickers dominate; a slow or anxious kid gets nothing. Needs a floor (see #18, #28) so "got nothing" is a lesson, not a humiliation.

OCT 1 TIER 1 — required. Without this, BOW is a worksheet with a scoreboard.

FALSIFICATION TEST Run one 20-minute session with 12 kids and a Google Sheet on a projector with locked rows. If nobody gasps, the thesis is wrong.

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2. THE FLOOR

THESIS Trades happen by standing up and walking across a room under a clock, not by clicking.

THE MAGIC MOMENT The bell rings. Fourteen kids stand up at once — the chairs scrape, and it's suddenly loud. Devan is holding his printed roster card above his head shouting "I NEED A CENTER, I HAVE PICKS." Two girls have pulled a table into a corner and are whisper-negotiating with their hands over their cards. Ninety seconds left on the projector clock. Devan has to choose which conversation to abandon.

WHY 10x It converts an abstract market into a bodily memory. The kid's shoulders remember the deadline.

ECONOMICS VALUE Transaction costs and search costs made physical — you cannot talk to everyone, so information is expensive and matches are imperfect. Also thin markets, and why a deadline changes behavior.

SOCIAL VALUE Enormous. This is the ten minutes kids re-enact at recess.

TEACHER VALUE Instructor becomes floor manager, not lecturer. High energy, high control (the bell is absolute).

PARENT/BUSINESS VALUE The single most filmable ten minutes in the product. One 20-second clip sells the course.

GRADES 5–6 3 minutes, must trade with exactly one person, structured partner rotation.

GRADES 7–8 6 minutes, open floor, multi-team trades allowed, deals must be entered before the bell or they die.

COST LOW technical / MED classroom. Needs printed cards, a bell, a big projected clock, and a room that permits standing. Instructor burden is real but it's energy, not prep.

RISKS Chaos, volume, the loudest kid eats the room, quiet kids frozen out. Mitigate with a "deals must be written on the card and countersigned" rule.

OCT 1 TIER 1 — required for W3.

FALSIFICATION TEST One live class, index cards, a kitchen timer. Measure: how many deals get struck, and does at least one kid say "that was insane" unprompted.

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3. THE PODIUM

THESIS The instructor can, at any moment, freeze the league and hold a live press conference where one student must publicly defend a decision.

THE MAGIC MOMENT The projector goes dark blue. A single line: **LEAGUE PAUSED — PRESS CONFERENCE.** Under it: *Sacramento.* That's Priya. Every laptop in the room locks. Priya walks to the front. The instructor, deadpan, holding a marker like a mic: "Priya, you spent forty percent of your cap on one 33-year-old. Fans are asking. Why?" Eleven kids are watching her, and three of them already have their hands up to ask the follow-up.

WHY 10x It forces *articulation*. The moment a kid has to defend a decision out loud to peers, the reasoning becomes real and the memory becomes permanent.

ECONOMICS VALUE Extremely high — it's the accountability layer. Kids must state their model: expected value, risk, the tradeoff they accepted. It converts intuition into an argument.

SOCIAL VALUE Status. Being called to the podium is a promotion, not a punishment, if framed right.

TEACHER VALUE The best formative assessment instrument in the whole design, and it costs zero build.

PARENT/BUSINESS VALUE "My daughter had to defend a $158 million decision to a room." Sells itself.

GRADES 5–6 Two friendly questions, instructor-led, 45 seconds.

GRADES 7–8 Peer reporters (see #13) ask one hostile question each; 2 minutes; the transcript is saved to the franchise record.

COST LOW. One projector state + a lock-screens button. The rest is instructor craft.

RISKS Public speaking anxiety; a kid could be humiliated. Rule: questions attack the *decision*, never the person, and the instructor takes the first question themselves to model tone.

OCT 1 TIER 1 — required. Cheapest 10x in the deck.

FALSIFICATION TEST Do it once, unannounced, with no build at all — just say "everyone stop, Priya come up here." See what the room does.

---

4. THE SEALED ENVELOPE

THESIS In Week 1 every student writes one specific prediction and one promise; in Week 5 it is opened in front of everyone.

THE MAGIC MOMENT Week 5. The projector: **WEEK 1 SAID —** and then, in a child's own handwriting, scanned: *"I will win 52 games and Ant will be an All-Star. I am not going to panic trade."* The standings underneath say 31–51. The room makes a noise — half "ohhhh," half laughing, and the kid is laughing too, hands on head. Then the instructor asks the only question that matters: "What did you know in Week 5 that Week-1 you didn't?"

WHY 10x Six weeks becomes a *story with a callback*. It converts a course into a narrative the kid is inside of.

ECONOMICS VALUE Uncertainty, forecasting, and the difference between a bad outcome and a bad decision. Also hindsight bias, directly and viscerally.

SOCIAL VALUE Shared laughter at a shared past. Bonding, not shaming, if everyone's opens on the same day.

TEACHER VALUE Week 5's lesson arrives pre-written by the students five weeks earlier. Zero prep.

PARENT/BUSINESS VALUE The envelope goes home at the end with the actual result stapled to it. That's a keepsake, not a report card.

GRADES 5–6 One number and one sentence.

GRADES 7–8 A number, a confidence level (%), and one named risk. In W5, score the calibration across the whole class.

COST LOW. Literally paper, or one text field stored with a reveal date.

RISKS A kid whose prediction was cruel or personal. Prompt it tightly.

OCT 1 TIER 1 — required.

FALSIFICATION TEST Impossible to falsify quickly by design — but you can test the emotional payload in one session by having kids predict a coin-flip-ish thing in minute 5 and revealing in minute 50.

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5. THE DEED

THESIS The franchise does not reset. It is inherited — by the same kid next year, or by a stranger who has to live with what they left behind.

THE MAGIC MOMENT Next October. A new kid opens her laptop on day one and the screen says: **MEMPHIS. Inherited from Jonah R., 2026 cohort. Record: 24–58. Cap space: none until 2029.** Under it, in Jonah's own voice, a 20-second audio note: "Sorry. I traded three first-round picks for a guy who got hurt. Don't do what I did. But keep Ja." She hears a stranger's regret before she makes a single decision.

WHY 10x This is the thing that turns a six-week course into an institution. It also makes Week-3 decisions *heavier* than any grade could.

ECONOMICS VALUE Path dependence, intertemporal tradeoffs, and the core of institutional economics: decisions have heirs. It is the only mechanism I know that teaches a 12-year-old what a long time horizon feels like.

SOCIAL VALUE Creates lineage and alumni identity. Kids will ask about "who had my team before."

TEACHER VALUE Year two has a curriculum that year one wrote.

PARENT/BUSINESS VALUE Re-enrollment engine. "Come back and finish what you started" is a far better renewal pitch than "level 2."

GRADES 5–6 Inherit a roster and one voice note; constraints are softened.

GRADES 7–8 Inherit the full debt: dead cap, traded picks, an unhappy star. The first assignment is a written diagnosis of the previous GM's mistake.

COST MED technical / LOW classroom. Requires real persistence, versioned league state, and a migration story. The cost is discipline in the data model, not features.

RISKS A kid inherits a genuinely unwinnable position and disengages by day two. Needs a floor: every inherited team has at least one real asset.

OCT 1 TIER 3 — future ceiling. But build the data model *for it now* or you will never get it.

FALSIFICATION TEST In the Oct 1 cohort, have Week 6 hand each kid a *different* kid's franchise for the final 20 minutes and watch the reaction. That's the whole hypothesis, compressed.

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6. GHOST TIMELINE

THESIS After a season resolves, the projector shows you the league that *would have existed* if you had made the other choice.

THE MAGIC MOMENT The screen splits. Left side, in white: *ACTUAL — 38 wins, missed playoffs.* Right side, in grey and slightly translucent: *IF YOU HAD KEPT THE PICK — 41 wins, 8th seed, and this player is on your team.* Nobody says anything for two seconds. Then Marcus says, quietly, "that's not fair," and the instructor says, "It's not. That's the point. You didn't know. Nobody knew. Was it a bad decision or a bad outcome?"

WHY 10x It makes the invisible visible — the counterfactual is the single hardest idea in economics and here it is, rendered, on a wall, about *your* team.

ECONOMICS VALUE The highest in the deck. Opportunity cost is *defined* as the foregone alternative, and no worksheet has ever shown a child the foregone alternative. Also separates decision quality from outcome quality — the most valuable transferable idea in the course.

SOCIAL VALUE Medium. Mostly private, but the room watches each reveal.

TEACHER VALUE Hands the instructor the perfect discussion prompt, every single time, automatically.

PARENT/BUSINESS VALUE Screenshot-able. This is the image that goes in the marketing.

GRADES 5–6 Show one ghost timeline, clearly labeled "one of many ways it could have gone."

GRADES 7–8 Show a *distribution* — "in 100 runs, keeping the pick won more in 62." Now you're teaching expected value.

COST MED–HIGH. You must re-run the season sim from a branch point, which means the sim must be deterministic and re-runnable from saved state. Architectural, not cosmetic.

RISKS *Truth risk is severe.* If the ghost is a fabrication dressed as fact, you are teaching a false certainty. Must be framed as "one possible world," never "what would have happened."

OCT 1 TIER 2 — high value if earned. Tier 1 if the sim already supports branching.

FALSIFICATION TEST Fake it once. Hand-compute a single ghost timeline for one student, show it on the projector, and watch whether the room leans forward or shrugs.

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7. THE INTEL DESK

THESIS Scouting information is a purchasable good priced in cap space, and buying it means you can't afford the player.

THE MAGIC MOMENT Two cards on screen. A player, and a locked file: **MEDICAL REPORT — $2.4M.** Sofia has exactly enough room to sign the guy or to find out whether his knee is real. She buys the report. It says: *minor, expected full recovery.* She has just paid two and a half million dollars to learn nothing she couldn't have guessed — and now she's short. Across the room, Leo didn't buy it, signed the guy, and got lucky. The instructor lets that sit.

WHY 10x Information stops being free and ambient — the thing every game and every classroom assumes — and becomes an economic object with a price and a payoff.

ECONOMICS VALUE Very high: information asymmetry, adverse selection, the value of information, and the brutal lesson that *paying for information that confirms what you thought is still rational.*

SOCIAL VALUE Creates a black market instantly — kids will resell intel to each other. Let them. That's the lesson.

TEACHER VALUE Generates the best debrief question in the course: "Leo got a better result. Did he make a better decision?"

PARENT/BUSINESS VALUE Unusually sophisticated; parents recognize it as real finance.

GRADES 5–6 Two intel items, clearly priced, obvious value.

GRADES 7–8 Intel of varying reliability, some of it noise, one item that is actively misleading — and a secondary market where students may resell.

COST LOW–MED. Content authoring is the real cost: each report must be written and its truth-state tracked.

RISKS Fairness — a kid who buys bad intel and loses feels cheated. That is the lesson, but it needs naming out loud.

OCT 1 TIER 2.

FALSIFICATION TEST Add three priced sealed envelopes to a paper version of W1 and see if anyone buys, and whether anyone resells.

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8. THE MIRROR

THESIS The class's own aggregate behavior is collected in the first half of the lesson and becomes the dataset they analyze in the second half.

THE MAGIC MOMENT Ticket pricing day. Everyone sets their price privately. Then the projector draws — live, dot by dot, one dot per franchise — a scatter of price against attendance. The dots form a curve. Not a curve from a textbook. *Their* curve, with their names on the dots. The instructor circles the top-left cluster and says, "these four of you priced high and sold out anyway. Why?" and four kids realize simultaneously that they all have superstar rosters.

WHY 10x The data is *them.* There is no gap between the lesson and the evidence, and no chance to dismiss it as made up.

ECONOMICS VALUE Demand curves, elasticity, willingness to pay, and — critically — the idea that a curve is an *emergent summary of decisions*, not a law handed down.

SOCIAL VALUE High; kids look for their own dot immediately, then look at everyone else's.

TEACHER VALUE The projector becomes a teaching instrument rather than a scoreboard.

PARENT/BUSINESS VALUE This is the "real economics, not a game" proof point.

GRADES 5–6 Bar chart of prices with names, plus a simple "who sold the most tickets" reveal.

GRADES 7–8 Full scatter, fitted line, residuals discussed — "who beat the curve, and what did they have that the model doesn't see?"

COST MED. Requires real-time aggregation and a genuinely good live chart. Do not cheap out on the chart.

RISKS A kid whose dot is embarrassingly wrong. Anonymize on first reveal, name on second.

OCT 1 TIER 1 for Week 4 — this *is* Week 4.

FALSIFICATION TEST Run the price-setting round on paper, plot it on a whiteboard by hand in front of them. If the room doesn't crane to find their dot, redesign.

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9. THE SEASON FILM

THESIS The parent receives a 90-second cinematic season recap narrated in the kid's own voice, not a progress report.

THE MAGIC MOMENT A parent's phone buzzes on a Tuesday. It opens to black, then a slow push-in on a standings table, then their own daughter's voice: "Week one I thought I could buy a championship. I was wrong about that." Cut to the trade she made. Cut to the record. Her voice: "I'd do it again." The parent watches it twice.

WHY 10x It relocates the product's proof from "did my kid learn" to "listen to how my kid thinks now."

ECONOMICS VALUE Low directly — this is a distribution and retention asset, not a learning mechanism. Demoted accordingly. Its only learning value is the reflection required to record the narration.

SOCIAL VALUE Low in-room.

TEACHER VALUE Negative in the expensive version — it's production work.

PARENT/BUSINESS VALUE The highest in the deck. Parents forward this.

GRADES 5–6 Three sentences, prompted.

GRADES 7–8 Unscripted 60 seconds answering "what were you wrong about?"

COST HIGH as described. Editing 14 videos a week is an unfunded job. **LOW** if reduced to: auto-generated slideshow + one raw voice memo, unedited.

RISKS Becomes a marketing tax on instructor time; quality varies wildly; consent/media issues with minors' voices.

OCT 1 TIER 3 in cinematic form. **2** in the cheap form.

FALSIFICATION TEST Record one kid's 30-second voice memo, pair it with three screenshots, send it to one parent, and ask if they showed anyone.

---

10. CAP CHIPS

THESIS The salary cap is a physical stack of chips on the desk in front of each child, and spending means handing them away.

THE MAGIC MOMENT Aiden has eleven chips. He pushes seven of them across the desk to sign a star, and the sound they make is the whole lesson. He looks at the four left and then at the eight roster slots still empty on his screen. He picks the four chips up and puts them down again. He does that twice.

WHY 10x A number in a UI is abstract at 11 years old. A depleting physical stack is not. This is the cheapest possible upgrade to comprehension in the entire product.

ECONOMICS VALUE Budget constraint, made tactile. Opportunity cost as a physical subtraction. This is genuinely how you teach scarcity to a 10-year-old, and it works up to 14.

SOCIAL VALUE Kids can *see* each other's remaining chips across the room. That's a public information mechanic for free.

TEACHER VALUE The instructor can read the whole room's financial state in a two-second glance. Enormous.

PARENT/BUSINESS VALUE Photographs beautifully. Reads as "real teaching," not "screen time."

GRADES 5–6 10 chips, whole numbers, one chip = $10M.

GRADES 7–8 Chips plus a written ledger they must reconcile against the app; discrepancies are their problem.

COST LOW. Poker chips. That's it.

RISKS Chips get dropped, thrown, lost, or become currency for non-game trades. Fine. Actually — kids inventing a side currency is a bonus lesson.

OCT 1 TIER 1 — required. Highest value-per-dollar item in the deck.

FALSIFICATION TEST One session, two groups: one with chips, one without. Ask both "how much cap do you have left" without letting them look. See who knows.

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11. THE RIPPLE

THESIS Your Week-2 economic shock is not random — it is *caused, by name, by another student's Week-1 decision.*

THE MAGIC MOMENT Week 2. Kayla's screen: **ATTENDANCE DOWN 14%.** She clicks it. The reason is one line: *Your division rival Portland (Sam) signed two All-Stars. Your fans expect you to lose.* She looks up. Sam is already looking at her. Neither of them planned this. The instructor says: "Kayla — Sam did nothing to you. He signed players for his own reasons. Explain to the room why you got hurt anyway."

WHY 10x Externality stops being a vocabulary word and becomes an accusation with a face.

ECONOMICS VALUE Externalities, general equilibrium, interdependence — and the crucial nuance that harm can occur with no intent to harm. This is the backbone of W5 and it should be seeded in W2.

SOCIAL VALUE High and slightly dangerous. Generates real grievance, which generates real demand for governance in W6.

TEACHER VALUE Gives W5 and W6 their emotional fuel. Without this, revenue sharing is a math exercise nobody wants.

PARENT/BUSINESS VALUE The moment where a kid explains "externality" at dinner correctly.

GRADES 5–6 One clearly labeled cause, one effect, positive framing available too ("your rival got good, so your games sell out").

GRADES 7–8 Multiple overlapping causes; the student must diagnose *which* rival's action hurt them most, using data.

COST MED. The sim must attribute effects to specific student actions and surface a human-readable cause string. That attribution layer is the work.

RISKS Real interpersonal friction. Blame. Needs the instructor to hold the line: "the system did this, not Sam."

OCT 1 TIER 1 — required. This is the spine connecting W2 to W5 to W6.

FALSIFICATION TEST Hand-write three "your rival caused this" cards in one W2 pilot. Watch whether kids look at each other.

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12. LIVE FANS

THESIS On ticket-pricing day, the demand you face is not simulated — it is the other students, spending a real budget, choosing whose games to attend.

THE MAGIC MOMENT Every kid has 10 fan-tokens and a screen listing all 14 teams with their prices. Ravi priced his tickets at the top of the market because his roster is stacked. The projector fills in live: teams collecting tokens, one by one. Ravi gets two. The team next to him, priced at half, gets eleven. Ravi says "wait, I'm *better* than him," and someone in the back says "yeah but you're expensive," and that is the entire concept of price elasticity, said out loud by a twelve-year-old, unprompted.

WHY 10x Demand is no longer a curve the software asserts. It is a room full of people with their own money making their own choices about you.

ECONOMICS VALUE Very high: demand, price elasticity, consumer surplus, substitutes, and the revenue tradeoff (price × quantity) discovered rather than told.

SOCIAL VALUE Very high. Kids campaign for their own team. Marketing emerges spontaneously.

TEACHER VALUE The debrief writes itself, and it pairs perfectly with #8 (The Mirror) — this generates the dots.

PARENT/BUSINESS VALUE "The class was the market" is a differentiated story.

GRADES 5–6 Fixed 10 tokens, one round, prices from a small menu.

GRADES 7–8 Two rounds with a reprice between them, so they can *respond* to observed demand. That second round is where the learning is.

COST MED. Needs a simple simultaneous-choice input and a live tally. Half a day of build, or paper tokens for free.

RISKS Popularity contest — kids buy from friends, not from value. Mitigate: token spend is anonymous, and there's a payoff for the buyer too (correctly predicting winners).

OCT 1 TIER 1 for Week 4.

FALSIFICATION TEST Poker chips and a whiteboard, 15 minutes, one class. Zero code.

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13. THE PRESS BOX

THESIS Two students per week are not GMs — they are reporters, and their published story is the league's official record of what happened.

THE MAGIC MOMENT Week 3 begins and the projector shows a front page: **"CHICAGO GAMBLES EVERYTHING"** — written by Nora, with a pull quote from Ethan's press conference that he definitely regrets saying. Ethan protests. Nora says, "I wrote down what you said." The instructor says, "She did. It's in the transcript." The room learns something about the permanence of public statements in about four seconds.

WHY 10x It creates a second, non-GM way to have status — which matters enormously for the kid who is bad at the game but great with words.

ECONOMICS VALUE Medium-high, indirectly: reporters must *explain* the economics to be readable, which is the strongest form of comprehension check. Also introduces information production as a service.

SOCIAL VALUE Very high. Solves the "one kid is dominating" problem by creating a parallel prestige ladder.

TEACHER VALUE Outsources the recap. The reporter's story *is* the Week-N-minus-1 review.

PARENT/BUSINESS VALUE A written archive of a kid's thinking, in their own voice, weekly.

GRADES 5–6 Three-sentence headline + "biggest surprise."

GRADES 7–8 150 words, must include one number and one quote, and must name a decision they think was wrong.

COST LOW. A text field and a projector slot.

RISKS Meanness. Hard rule: you may criticize a decision, never a person; instructor approves before publish (one minute).

OCT 1 TIER 2.

FALSIFICATION TEST Assign one reporter for one week. See if the class reads it when it goes up.

14. THE UNION

THESIS A rotating bloc of students represents the *players* and can block a trade, demand a raise, or strike.

THE MAGIC MOMENT Devan finishes an unbelievable trade at the buzzer. Then a red banner: **TRADE UNDER REVIEW — PLAYERS' ASSOCIATION.** Three kids at the side table confer for thirty seconds and one of them stands up: "He's got a no-trade clause. He doesn't want to go to a rebuilding team. You have to give him something." Devan, outraged, discovers that the thing he thought he owned has opinions.

WHY 10x It breaks the video-game assumption that assets are inert. The "resources" push back. Nothing else in edtech does this.

ECONOMICS VALUE High: bargaining power, collective action, labor as a party to the contract (not an input), and the difference between owning a contract and owning a person.

SOCIAL VALUE High and spicy. Creates a legitimate opposition faction.

TEACHER VALUE A built-in brake on runaway winners, justified in-fiction rather than by teacher fiat.

PARENT/BUSINESS VALUE Genuinely thoughtful — this is where a parent goes "oh, they're teaching labor economics to a sixth grader."

GRADES 5–6 Union has one veto per week, must state a reason.

GRADES 7–8 Union negotiates a revenue split with owners in W5 and can strike, cancelling games and hurting everyone including themselves.

COST LOW–MED. Mostly rules and roles; small UI for a "flagged" state.

RISKS Pacing killer if overused. Cap it: one veto per week, 60-second decision clock, and the veto must be overridable by a concession.

OCT 1 TIER 2 (Tier 1 for W6 if you're going hard at institutional design).

FALSIFICATION TEST Give three kids veto cards in one trade session. Watch whether the room's negotiation behavior changes *before* any veto is used. (It will — that's the finding.)

15. THE POT

THESIS League revenue sharing is a physical bowl in the center of the room that everyone can see, everyone contributes to, and everyone draws from.

THE MAGIC MOMENT Week 5. The instructor sets a clear bowl on the center table. Each team drops in chips based on revenue — the big-market kids drop in six, the small-market kids drop in one. The bowl fills. Then the instructor asks: "How do we split this?" and does not answer. Eleven minutes of the most heated, most sophisticated argument these kids have ever had breaks out, and someone eventually says the words "but then why would anyone try to earn more."

WHY 10x Redistribution stops being a policy and becomes a bowl that people are reaching into in front of each other.

ECONOMICS VALUE Extremely high: public goods, free-riding, moral hazard, the equity–efficiency tradeoff, and the commons. All four, in one bowl.

SOCIAL VALUE Maximum. This is the argument they'll continue in the hallway.

TEACHER VALUE The instructor's only job is to refuse to answer the question. That's it.

PARENT/BUSINESS VALUE "They ran a real argument about fairness and incentives, with stakes."

GRADES 5–6 Two options to vote between (equal split vs. split by need).

GRADES 7–8 Design the formula themselves, then live with it for W6 — including the exploit somebody finds.

COST LOW. A bowl and chips. The rest is instructor nerve.

RISKS Argument turns personal or turns into a majority ganging up on the two big-market kids. That's real, and it's also exactly the lesson — but it needs a skilled facilitator.

OCT 1 TIER 1 for Week 5.

FALSIFICATION TEST Run the bowl for 10 minutes with any 12 kids and any pretext. If the argument doesn't ignite, your instructor framing is wrong, not the mechanic.

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16. THE LOSS LEDGER

THESIS Every week the projector publicly names the worst decision in the league — and the GM who made it gets to defend it and gets credit for the defense.

THE MAGIC MOMENT The projector: **THE LEDGER — WEEK 3.** One entry. *Detroit traded a 2029 first for a player who played 4 games.* Everyone turns to look at Marcus, who — and this is the part that matters — stands up, grinning, and says "I'd do it again, here's why," and gives a 40-second argument, and half the room actually agrees with him by the end.

WHY 10x Every product for kids is engineered to protect them from failure. This one puts failure on the wall and gives it a microphone. That inversion is the whole brand.

ECONOMICS VALUE High: sunk cost, risk vs. ruin, and above all the decision/outcome distinction. Also normalizes revising a model based on evidence.

SOCIAL VALUE High *if and only if* the framing is "brave decisions get named." Otherwise catastrophic.

TEACHER VALUE Turns the thing instructors dread (a kid's visible failure) into the scheduled highlight.

PARENT/BUSINESS VALUE Differentiating and slightly scary — which is good. It says: we are not a participation-trophy product.

GRADES 5–6 Framed entirely as "the boldest move of the week," outcome discussed neutrally.

GRADES 7–8 Named as the biggest value-destroying decision, with the number attached, and a mandatory defense.

COST LOW. One projector slide, chosen by the instructor in 30 seconds.

RISKS The highest social risk in the deck. Same kid twice in a row = damage. Rule: no student appears twice, and the instructor's own decisions (the Empty Chair team, #18) are eligible.

OCT 1 TIER 2. High value, requires a confident instructor.

FALSIFICATION TEST Run it once. The test isn't whether kids like it — it's whether the *named kid* is still engaged 20 minutes later.

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17. THE OFFER SHEET

THESIS You can attempt to sign a player who is already on another student's team, and that student must respond, publicly, under a clock.

THE MAGIC MOMENT A siren sound. The projector: **OFFER SHEET — Chicago has offered $34M/yr to Portland's Anfernee Simons. Portland has 90 seconds to match.** The clock starts. Sam's cap sheet is up on the big screen for everyone to see, and everyone can do the arithmetic: matching means he can't sign anyone else all week. Sam is doing math in front of thirteen people with a countdown running. He matches with 11 seconds left and his hands are shaking a little.

WHY 10x The purest possible form of "another student's action becomes my economic shock." It is aggressive, legal, and entirely within the fiction.

ECONOMICS VALUE High: this is a real mechanism-design object (restricted free agency / poison pill). Teaches strategic pricing — you don't bid what he's worth, you bid what your rival can't afford. That's a genuinely advanced idea, delivered natively.

SOCIAL VALUE Enormous. Creates rivalries that persist for weeks.

TEACHER VALUE Generates a perfect teachable pause with a built-in clock and a built-in audience.

PARENT/BUSINESS VALUE The most dramatic 90 seconds in the product.

GRADES 5–6 One offer sheet allowed per class, generous match window.

GRADES 7–8 Multiple, and the offering team may deliberately structure the deal to be hard to match. Discuss *why that's allowed.*

COST LOW–MED. A flagged state, a timer, and a match/decline resolution.

RISKS Targeting/pile-on. One kid gets three offer sheets and melts down. Cap: one incoming offer sheet per team per week.

OCT 1 TIER 2 — but this is the sleeper. Cheap, dramatic, and structurally perfect.

FALSIFICATION TEST Do it once on paper with a stopwatch. Measure the room's volume.

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18. THE EMPTY CHAIR

THESIS One franchise has no student GM. It is run by the instructor, publicly, with every decision explained aloud.

THE MAGIC MOMENT Mid-free-agency, the instructor says "okay, San Antonio's turn — that's me. Watch what I'm doing." She narrates: "I have room for one star or three good players. I'm taking the three, and here's why I think that's right." Then in Week 4, San Antonio is 4th and a kid says, indignantly, "you made it look easy," and she says, "I got lucky in week two. You saw the reasoning; you didn't see the dice."

WHY 10x It gives the instructor a *seat in the market* instead of a podium above it — modeling thinking rather than announcing rules.

ECONOMICS VALUE Medium-high: a live think-aloud of expected value reasoning is the highest-bandwidth teaching there is. Also provides a neutral counterparty so no student is ever trade-starved.

SOCIAL VALUE Medium. The instructor becomes a rival, which kids love.

TEACHER VALUE High — but adds cognitive load during a busy session.

PARENT/BUSINESS VALUE Low-moderate.

GRADES 5–6 Instructor plays deliberately conservatively and explains why boring is often right.

GRADES 7–8 Instructor plays to win and is fair game for offer sheets and criticism.

COST LOW technical / MED instructor. One more team to run while managing 14 kids.

RISKS Instructor distraction; perceived favoritism if the instructor's team does well; can slow pacing.

OCT 1 TIER 2.

FALSIFICATION TEST Run one session where the instructor narrates three decisions out loud. Ask afterward: can students restate the instructor's reasoning? If not, it's decoration.

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19. THE PUBLIC LEAGUE

THESIS The league has a real, public, permanent URL — standings, transactions, quotes, and the students' own written reasoning — that anyone in the world can look at.

THE MAGIC MOMENT Jonah sends a link to his cousin in another state. The cousin texts back a screenshot of Jonah's trade with three fire emojis. Jonah has never had a thing he made be *seen* by someone who wasn't obligated to see it. On Monday he tells the class his cousin is following the league, and by Wednesday four other kids have sent the link to someone.

WHY 10x It changes the audience of the work from "my teacher" to "the world," which changes the quality of the work. It is also the cheapest growth loop the business will ever have.

ECONOMICS VALUE Low directly. Indirect: public reasoning is better reasoning. **Demoted on learning value, promoted hard on business value.**

SOCIAL VALUE High — external status.

TEACHER VALUE Neutral to slightly negative (moderation).

PARENT/BUSINESS VALUE Highest in the deck alongside #9. Every league page is an ad with a kid's name on it.

GRADES 5–6 First names + team only, curated quotes.

GRADES 7–8 Full transaction log, published reasoning, opt-in bylines.

COST LOW–MED. A read-only public view of state you already have.

RISKS Child safety and privacy — this is the serious one. No last names, no photos, no free-text without review, parental opt-in, and an obscure URL rather than an indexed one. Get this wrong and it's not a bad feature, it's an incident.

OCT 1 TIER 2, and only with a proper consent flow. Ship it as opt-in or not at all.

FALSIFICATION TEST Publish one league page after one pilot cohort. Count how many parents and kids share it in 48 hours without being asked.

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20. THE TAPE

THESIS The course's memory is not written by BOW — it is 30 seconds of the student's own voice, recorded at the end of every week, played back to them later.

THE MAGIC MOMENT Week 6, first five minutes. Lights down. The projector plays audio only, no names: a small voice from Week 1 saying "I think you should just get the best players, obviously." A few kids laugh. Then a different Week-1 voice: "I'm going to be patient." Then the same kid, Week 4: "I panicked. I completely panicked." The room is silent. Everyone is trying to figure out which ones were theirs.

WHY 10x It makes *change in your own thinking* audible. That's the actual product — not the league, not the standings. The kid hears themselves become smarter.

ECONOMICS VALUE Medium-high as metacognition: it makes belief revision explicit, which is the core intellectual habit economics is supposed to install.

SOCIAL VALUE High and unusually tender. This is the moment that makes the course feel like it mattered.

TEACHER VALUE Week 6's opening is pre-built. Also the best possible evidence of learning for any stakeholder.

PARENT/BUSINESS VALUE Very high, and it's the raw material for #9 at essentially zero cost.

GRADES 5–6 One prompted question: "What surprised you this week?"

GRADES 7–8 "What did you believe last week that you don't believe now?" — the hardest and best question in the course.

COST LOW. A record button and storage. The prompt design is the real work.

RISKS Consent and storage of minors' voice data. Self-consciousness — some kids will perform. Allow text as an equal-status alternative.

OCT 1 TIER 1 — required. Low cost, high emotional yield, and it feeds three other ideas.

FALSIFICATION TEST Collect 30-second recordings in a single pilot week, play two back anonymously the next week, watch the room.

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21. THE CLOCK

THESIS The class shares one visible countdown, and any student's indecision burns everyone's time — including their own.

THE MAGIC MOMENT A single huge number on the projector: **11:42.** It is running. It is the *whole class's* remaining free agency time, not each kid's. Someone is stalling on a decision and the room notices. Three kids simultaneously turn and say "just PICK." He picks. The clock says 11:09. Later, the instructor asks: "Whose eleven minutes did he spend?"

WHY 10x Time — the actual scarcest resource in any classroom — becomes an explicit, shared, contested economic good. Nobody does this.

ECONOMICS VALUE High: the tragedy of the commons, in the medium the students are literally living in. Also decision costs and the value of a deadline.

SOCIAL VALUE High. Creates peer pressure that the instructor didn't have to apply.

TEACHER VALUE Solves pacing — the hardest operational problem in a 14-kid live class — by making it the students' problem.

PARENT/BUSINESS VALUE Low direct, but it makes sessions end on time, which parents care about more than they admit.

GRADES 5–6 Individual sub-clocks that roll into a class clock; softer.

GRADES 7–8 Pure shared clock, and a rule that a team may *buy* extra time from the pool with cap space. Now time has a price.

COST LOW. One big timer, one rule.

RISKS Pressure on slower processors; a kid gets rushed and blames peers. Needs a protected minimum per student.

OCT 1 TIER 2 (the 7–8 "buy time with cap space" variant is a Tier-2 gem).

FALSIFICATION TEST Put a shared countdown on the wall for one session. Measure average decision time before and after.

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22. THE AGENT LINE

THESIS Star players are represented by a live human agent (instructor or older student) who negotiates back, holds out, and can say no.

THE MAGIC MOMENT Amara clicks "offer $28M" and instead of a confirmation, a phone rings — an actual phone sound in the room. The projector: **CALL FROM: THE AGENT.** The instructor picks up a handset and says, in a completely different voice, "Twenty-eight? For my client? Amara, I have three teams at thirty-two. You have until the end of this sentence." Amara says "THIRTY-FIVE" before the sentence ends and then immediately looks horrified at herself.

WHY 10x Negotiation becomes a live human interaction with a person who wants something different from what you want. No AI, no menu.

ECONOMICS VALUE High: bargaining, the winner's curse, information asymmetry (is there really another team at 32?), and the psychology of auctions.

SOCIAL VALUE Medium — it's one-to-one, room watches.

TEACHER VALUE Demanding. The instructor must improvise in character while running a live class.

PARENT/BUSINESS VALUE Very high theatrically.

GRADES 5–6 Agent is transparent and coaching-toned.

GRADES 7–8 Agent bluffs, and students later learn which bluffs were real. That reveal is the lesson.

COST HIGH instructor burden. Technically near-free; operationally it does not scale past a couple of moments per session, and it needs a second adult to do properly.

RISKS Cannot be done at volume; unfair if only some kids get it; instructor exhaustion; a kid feels tricked by an adult.

OCT 1 TIER 3. (See kill list — I don't think this survives as designed.)

FALSIFICATION TEST Do exactly three agent calls in one session and time how much class time they consume. My prediction: 11 minutes for three kids. That's the answer.

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23. THE VOTE OF NO CONFIDENCE

THESIS The league's owners (students) can vote to remove a GM from their franchise.

THE MAGIC MOMENT Week 5. The projector: **MOTION FILED: DETROIT.** A vote opens. Marcus watches his peers vote on whether he keeps his team.

WHY 10x It puts real institutional consequence on the table — the thing adults control and kids never do.

ECONOMICS VALUE Medium: principal–agent problem, accountability structures, the cost of removing a manager. Real, but obtainable through gentler mechanisms.

SOCIAL VALUE Negative. This is a formalized popularity vote about a child, in front of that child.

TEACHER VALUE Negative — you now own the aftermath.

PARENT/BUSINESS VALUE Catastrophic if one parent hears about it.

GRADES 5–6 Should not exist.

GRADES 7–8 Should not exist.

COST LOW to build, unbounded to clean up.

RISKS Bullying with a rulebook. Exclusion. A parent phone call you cannot win.

OCT 1 TIER Kill. Listed for honesty about where the line is. The *survivable* version is #16 (public defense of a decision) and a **non-binding** confidence poll on a *policy*, never a person.

FALSIFICATION TEST None needed. Don't run this on children.

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24. THE SECOND LEAGUE

THESIS Two BOW cohorts, in different cities, play in one league and meet in a live cross-class event.

THE MAGIC MOMENT The projector splits and a second classroom appears — fourteen strangers, waving. Then a trade offer arrives from a kid nobody has ever met.

WHY 10x It scales the market beyond a room and makes BOW feel like a world.

ECONOMICS VALUE Medium: larger market = better matching, more price discovery. Real but incremental over #1.

SOCIAL VALUE High for the kids; enormous ops risk.

TEACHER VALUE Negative. Two instructors must be synchronized, live, weekly.

PARENT/BUSINESS VALUE High on the brochure.

GRADES 5–6 Impractical.

GRADES 7–8 Impractical for v1.

COST HIGH. Scheduling coupling is the killer — one class running late breaks the other class's lesson. Also doubles the failure surface.

RISKS Pacing, timezone, safety across cohorts, moderation between unacquainted minors.

OCT 1 TIER 3, and probably year 3.

FALSIFICATION TEST Attempt a single 15-minute cross-cohort trade window between two pilot groups. My expectation is that scheduling alone kills it.

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25. THE FRANCHISE AUCTION

THESIS In the final session, students bid on each other's franchises — and the price the market pays is the real grade.

THE MAGIC MOMENT Week 6, last twenty minutes. Every franchise goes up for auction and every student has a budget. Priya's team — 29 wins, ugly record, but three young players and every future pick intact — starts a bidding war. It sells for the second-highest price in the league, above two playoff teams. Priya, who has felt like she was losing for four straight weeks, watches the room decide she was right all along.

WHY 10x It reveals that the visible scoreboard was never the real scoreboard — and it retroactively rewards the kid who played the long game.

ECONOMICS VALUE Extremely high: asset valuation vs. current performance, present value of future cash flows, price discovery, and the difference between *winning now* and *being valuable*. It is the single best closing argument the course could make.

SOCIAL VALUE Very high — it redistributes status at the end, which is exactly when you want to.

TEACHER VALUE A perfect finale that assesses everything at once without a test.

PARENT/BUSINESS VALUE Strong: "the final exam was an auction."

GRADES 5–6 Sealed one-number bids, revealed together.

GRADES 7–8 Open ascending auction with a gavel, plus a written one-paragraph valuation *before* bidding starts, compared to the clearing price afterward.

COST LOW–MED. A simple auction UI, or a whiteboard and a gavel.

RISKS The kid whose team sells cheapest, on the last day. Needs a countervailing award (best decision, best call, most improved reasoning) so the last memory isn't a low number.

OCT 1 TIER 1 for Week 6. This is the ending the course is missing.

FALSIFICATION TEST Run a 10-minute sealed-bid auction on three hypothetical rosters with any group of kids. Do their bids track future assets or current wins? Either answer is a great lesson.

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26. DELETE THE LAPTOP

THESIS One entire week runs with no screens — paper rosters, chips, and a whiteboard league.

THE MAGIC MOMENT Kids walk in and the laptops are stacked, closed, at the front. On each desk: a printed roster, a stack of chips, a pencil.

WHY 10x It would prove the economics, not the software, is the product.

ECONOMICS VALUE Neutral — it doesn't add a concept, it relocates existing ones.

SOCIAL VALUE Medium (more talking, more eye contact).

TEACHER VALUE Higher prep, lower tech risk.

PARENT/BUSINESS VALUE Mixed — some parents love "no screens," others paid for the platform.

GRADES 5–6 Works well.

GRADES 7–8 Works, but they'll miss the data.

COST MED. Printing and manual state reconciliation, weekly, is a real instructor tax and a source of desync bugs.

RISKS State drift between paper and the persistent league; a lost sheet is a lost franchise.

OCT 1 TIER 3 / kill. The right move is not to delete the laptop for a week — it's to make sure *the most important 15 minutes of every week already happen off the laptop* (#2, #10, #15).

FALSIFICATION TEST Run W3 fully analog once. Count minutes lost to reconciliation.

27. THE CONSTITUTION

THESIS Week 6 does not end with a winner. It ends with a signed document of rules that binds the *next* cohort — and the current students' names are on it.

THE MAGIC MOMENT The final ten minutes. The projector shows a document titled **THE RULES OF THIS LEAGUE — RATIFIED [DATE]** with five clauses the class argued into existence: a cap they set, a revenue split they designed, a trade-review rule they invented after the thing that happened in Week 3. One by one, fourteen kids come up and sign it on the touchscreen. The instructor says: "In a year, new kids will play under your rules. They don't get a vote on these. You already voted."

WHY 10x The course's final output is not a score. It is *law.* The kids leave having governed.

ECONOMICS VALUE The highest possible for W6: institutional design, constitutional constraint, credible commitment, and the veil of ignorance (you're writing rules for people you'll never meet, so you can't rig them for yourself). That last point is a genuinely profound thing to hand a 13-year-old.

SOCIAL VALUE Peak collective identity. The room becomes a body that did something together.

TEACHER VALUE Gives W6 a clear artifact and a clear ending. Also gives the instructor next year's rule variations for free.

PARENT/BUSINESS VALUE A printed, signed document goes home. That goes on a wall. And it is the retention hook: "come back and see if your rules held."

GRADES 5–6 Vote on three pre-drafted clauses, write one of their own.

GRADES 7–8 Draft from scratch, require a 2/3 majority, and — the killer detail — one clause must be an *amendment procedure.*

COST LOW–MED. A document renderer, signatures, and the discipline to actually apply it next year (that part is #5's machinery).

RISKS A class writes a bad or unfair constitution. Let them. Then next cohort's Week 1 opens with "here is what they left you, and here is the loophole."

OCT 1 TIER 1 for Week 6 in artifact form; **3** for the "actually binds next year" enforcement.

FALSIFICATION TEST In a pilot, spend 15 minutes drafting three league rules by vote, print and sign them. Ask a week later whether kids remember what they passed.

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28. NO GRADES, ONLY STANDINGS

THESIS Delete all points, badges, XP, stars, and teacher scores. The league table and the franchise's balance sheet are the only feedback that exists.

THE MAGIC MOMENT A kid asks, "how do I get a good grade in this?" and the instructor points at the projector and says, "that's it. That's your grade. It's public and it changes every week and I don't control it." The kid stares at it for a second and then goes back to his roster with a completely different posture.

WHY 10x It removes the single most school-shaped thing in the product and replaces it with consequence. It's a deletion, not an addition, and it makes everything else more honest.

ECONOMICS VALUE High, structurally: it removes the extrinsic-reward layer that competes with the actual incentive structure the course is trying to teach. You cannot teach incentives while running a parallel, fake incentive system.

SOCIAL VALUE High — status comes from the league, which is the fiction, not from the teacher.

TEACHER VALUE Frees the instructor from grader to coach.

PARENT/BUSINESS VALUE Requires a substitute artifact for parents: the voice tape (#20), the envelope (#4), the constitution (#27). Those are better than a grade anyway.

GRADES 5–6 Add one non-competitive private metric: "decisions you explained well."

GRADES 7–8 Pure standings + franchise value (#25).

COST LOW — it's a deletion. Negative cost, arguably.

RISKS Parents who want a score. Kids at the bottom of a public table with no other ladder — which is exactly why #13 (Press Box), #14 (Union), and #25 (Auction) exist as alternate status ladders.

OCT 1 TIER 1 — required. Decide this now; retrofitting it later is a rebuild.

FALSIFICATION TEST Ask three parents of pilot students: "if there were no grade, but you got a signed constitution and a 90-second recording of your kid explaining what they got wrong, would that be enough?" Their hesitation is the data.

SECTION A — THE TOP 5

1. THE ONLY BOARD (#1) The whole thesis of the product compresses into one shared, rivalrous board. Everything else in this deck is downstream of it; without it, BOW is a very nice single-player game played simultaneously. *Strongest argument against:* it maximally punishes the slow, anxious, or less-informed child, and speed becomes a proxy for skill — which is not economics, it's reflexes. *Survives?* **Yes, with a modification.** Contention must be resolved by *sealed simultaneous bidding*, not by click order. Everyone submits, the board resolves, the loser sees exactly who won and at what price. Same gasp, no reflex tax, and it's better economics — you've turned a race into an auction.

2. THE FLOOR (#2) The physical open-outcry trade session is the moment kids describe at dinner. It is also nearly free, deeply memorable, and impossible to deliver over a screen — which means it is a moat. *Strongest argument against:* it's chaos, it advantages loud extroverts, and in a live remote or hybrid class it doesn't exist at all. *Survives?* **Yes.** The loudness problem is real and is solved by the countersigned-card rule plus a mandatory 60-second silent "written offers only" opening phase. If the course is remote, the mechanic degrades to breakout rooms with a shared clock — worse, but the bones hold.

3. THE POT (#15) The single highest economics-per-dollar idea in the deck. A clear bowl, some chips, and an instructor who refuses to answer the question produces an eleven-minute argument about equity and incentives that a university course would be proud of. *Strongest argument against:* it depends almost entirely on instructor facilitation skill. A weak instructor turns it into a five-minute vote and a shrug. *Survives?* **Yes, but it is a staffing bet, not a product bet.** It should ship with a scripted facilitation ladder — the four questions, in order, that reliably ignite it.

4. THE SEALED ENVELOPE (#4) + THE TAPE (#20) *(ranked as one — they're the same mechanism at two timescales)* Costs almost nothing, and it is the only pair of ideas that makes the *six-week duration itself* meaningful rather than incidental. A course that pays off its own beginning is a story; one that doesn't is six worksheets. *Strongest argument against:* it's sentimental rather than economic — nice for parents, not actually teaching anything. *Survives?* **Yes, but only in the sharpened form.** The prompt cannot be "how do you feel." It must be a *falsifiable claim with a confidence level.* Then the Week-5 reveal is a calibration exercise, which is real economics. Soft version: kill. Hard version: keep.

5. THE FRANCHISE AUCTION (#25) The course currently has no ending. This is the ending — and it's the one that retroactively rewrites the meaning of the previous five weeks by revealing that wins were never the scoreboard. *Strongest argument against:* it's a new mechanic introduced in the final twenty minutes, when kids are tired, and one child ends the entire course by watching their team sell for the least. *Survives?* **Yes, conditionally.** It must be seeded — franchise value should be visible on the dashboard from Week 2, so the auction confirms a number kids have been watching, rather than introducing one. And the sealed-bid variant plus a simultaneous non-monetary award defuses the last-place ending.

Notable exclusion: **The Ripple (#11)** is arguably top-5 on economics alone but sits just outside because it is the most expensive of the cheap ideas — the attribution layer is real engineering.

SECTION B — THE THREE THAT COULD SHIP BY OCTOBER 1

Four weeks, on top of a working W1 build. Being unsentimental: anything requiring the simulation to branch, attribute, or re-run is out. Anything requiring a public web surface with a consent flow is out — the legal review alone eats the timeline.

1. THE SEALED ENVELOPE + THE TAPE (#4, #20). One text field, one audio recorder, one reveal date, one storage bucket. **Two days of build.** It buys narrative structure across all six weeks, which is the product's biggest structural gap, and it requires zero simulation work. Ship this first.

2. THE PODIUM + CAP CHIPS + THE FLOOR (#3, #10, #2). Grouped because their combined engineering cost is *one button* (lock all student screens and show a projector state) plus a purchase order for poker chips and a bell. **One day of build, one afternoon of instructor rehearsal.** This is the highest-drama, lowest-cost bundle in the entire deck, and it delivers the "kid tells a story at dinner" outcome directly. If you ship nothing else, ship this.

3. THE ONLY BOARD, in its sealed-bid form (#1). Caveat: this is only a four-week job **if the W1 build already models one shared league state.** If each student currently has a private copy of the free-agent pool, this is a data-model rewrite and it does not fit — in which case, ship a *manual* version: the instructor runs a single projected board, students submit bids on paper or in a form, and the instructor resolves it live. That's zero build and 80% of the emotional payload, and it doubles as the falsification test.

Explicitly not shippable by Oct 1, despite being good: Ghost Timeline (needs branching sim), The Ripple (needs attribution), The Public League (needs consent architecture), The Deed (needs a year), The Mirror (needs a genuinely good live chart — do it for W4, which is three weeks after launch, not for W1).

SECTION C — THE MOONSHOT

THE CONTESTED CLAIM

(The Only Board + The Floor + the sealed-bid resolution, generalized into a single repeatable interaction primitive.)

Here is the argument.

Every education product on earth has an interaction grammar, and almost all of them are borrowed. Khan-style products borrowed *the problem set.* Game-based learning borrowed *the level.* AI tutors borrowed *the conversation.* Simulation products borrowed *the dashboard.* All four are single-player grammars: the child interacts with a system that has infinite patience and infinite supply.

BOW has the opportunity to own a grammar nobody else can use: **the contested claim.** One shared object. Multiple children who want it. A clock. A public resolution. A visible loser.

Every economics concept in the six-week arc can be expressed in exactly this primitive, with only the object changed:

- Week 1: the contested object is **a player.** → scarcity, opportunity cost.
- Week 2: the contested object is **a fanbase's attention —** when your rival gets good, your demand moves. → externality, interdependence.
- Week 3: the contested object is **a counterparty's willingness to deal** before the bell. → gains from trade, transaction costs.
- Week 4: the contested object is **the other students' spending money.** → demand, elasticity, substitutes.
- Week 5: the contested object is **the bowl in the middle of the room.** → commons, free-riding, redistribution.
- Week 6: the contested object is **the rule itself.** → institutional design.

That is not six lessons wearing six different mechanics. That is *one mechanic, escalating in abstraction for six weeks,* from a basketball player to a law. A ten-year-old who understands "someone else took the thing I wanted, in front of me, and I had to decide what to do next" in Week 1 has already been handed the cognitive template for constitutional design in Week 6. That is a curriculum with a spine, and spines are what nobody else has.

Why competitors can't copy it. Not because the code is hard — the code is trivially cloneable. Because the *asset* is a live room with a fixed, small number of children who know each other's names and will see each other again next week. A competitor building this as self-serve software has to fake the other children, and the instant you fake them the entire mechanism evaporates: a simulated rival taking your player produces zero gasp. A competitor building it as live instruction has to solve scheduling, cohorting, instructor training, and persistent shared state across six weeks — which is not a feature, it's a company. BOW's moat is that it's an operationally annoying business with a proprietary interaction primitive on top. That's the good kind of moat.

Why it's a moonshot and not just a feature. Committing to this means committing to things that are genuinely uncomfortable: children will lose things in public, some sessions will be loud and hard to control, and the product will be structurally worse for a kid who joins late or misses a week. Most edtech companies flinch at exactly this point and add a private mode, an AI opponent, or an asynchronous catch-up path — and every one of those additions kills the mechanic. **The moonshot is not building the contested claim. It's refusing, for three years, to add the escape hatch.**

And the compounding version — **The Deed (#5) plus The Constitution (#27)** — extends the contested claim across *cohorts*: this year's students contest rules with students who don't exist yet. At that point BOW isn't a course. It's a league with a history, and history is not something a competitor can ship.

SECTION D — KILL LIST

1. THE VOTE OF NO CONFIDENCE (#23) — kill outright. It is a structured popularity vote about a specific child, conducted in front of that child, by peers. There is no framing that makes this safe at ages 10–14, the economics content is fully obtainable via #16 and #14, and the downside is a parent phone call that ends the business relationship. I wrote it to mark the boundary; the boundary is here.

2. THE AGENT LINE (#22) — kill as designed. The magic moment is genuinely great and the economics is real, but the math is fatal: three agent negotiations consume roughly eleven minutes of a session and serve three of fourteen students. It is a one-adult mechanic in a one-adult classroom that has thirteen other kids in it. *Salvage:* keep exactly one scripted agent call per session, on the projector, as a whole-room event rather than a private negotiation — everyone watches one kid get squeezed and everyone learns. That's #3 wearing a costume, which is fine.

3. THE SECOND LEAGUE (#24) — kill for at least two years. Cross-cohort play couples two live classrooms' schedules. One class running eight minutes late corrupts another class's lesson, and the failure mode is "the projector shows an empty room." The marginal economics gain over a 14-team single-room league is small. This is a brochure idea that would consume a quarter of engineering and produce weekly operational incidents.

4. THE SEASON FILM (#9) — kill the cinematic version. Editing fourteen 90-second films weekly is an unfunded full-time job that scales linearly with enrollment, which is the definition of a bad unit economic. The parent value is real, so keep the *cheap* form: three auto-selected screenshots plus the kid's raw, unedited voice memo from #20. That's 95% of the parental emotional payload at 2% of the cost. If it turns out parents forward the cheap version, you never needed the expensive one.

5. DELETE THE LAPTOP (#26) — kill. It's a stunt masquerading as a principle. The real principle it's pointing at — *the most important minutes each week should happen off the screen* — is already delivered by The Floor, Cap Chips, and The Pot, and delivered better, because those don't require reconciling a week of paper state back into a persistent database. Keep the principle, kill the week.

Honorable mention for near-death: **The Empty Chair (#18)** and **The Silent Round** (folded into #2). Both are fine and both add instructor load in the exact minutes when instructor load is already highest. Defer.

SECTION E — WHAT TO DELETE

1. Delete private information from "my franchise." The current model — projector shows the league, laptop shows *my* team — quietly encodes the assumption that my franchise is my business. That assumption destroys most of what's valuable here. If my cap space is private, nobody can pressure me. If my roster holes are private, nobody knows to call me. If my price is private, the class can't form a demand curve. **Default everything to public**, and then make *secrecy itself the purchasable good* — a team can pay to hide one thing for one week. Now privacy has a price and asymmetric information is a mechanic instead of an accident.

2. Delete the simulated season detail. Whatever box scores, game-by-game results, or per-player statlines the sim produces, cut them to a single line per week: record, revenue, and one causal sentence. Every minute a child spends reading simulated basketball is a minute not spent making an economic decision, and detailed sim output invites the failure mode where kids optimize the basketball engine rather than the economics. Worse, it makes the product compete with NBA 2K, a fight it cannot win. **The sim's only job is to make decisions have consequences.** Everything past that is set dressing that costs engineering time and teaches nothing.

3. Delete grades, points, badges, XP, and any BOW-issued score. Per #28. You cannot teach a child how incentives shape behavior while running a second, fake incentive system on top of the real one — they will optimize your points and ignore your economics, and they'll be right to. The standings, the balance sheet, and the Week-6 franchise valuation are richer, more public, and more honest feedback than any rubric. What parents actually want from a grade — evidence their kid grew — is delivered better by the sealed envelope, the voice tape, and a signed constitution on the wall.

One more, offered free: **delete the assumption that all six weeks are the same shape.** The current arc reads as six symmetrical sessions. It shouldn't be. Weeks 1, 3, and 6 are events — loud, physical, high-stakes. Weeks 2, 4, and 5 are analysis — quiet, data-forward, argumentative. Alternating that rhythm deliberately will do more for pacing and for memory than any feature in this document.agentId: a1e491752421e55ed (use SendMessage with to: 'a1e491752421e55ed', summary: '<5-10 word recap>' to continue this agent) subagent_tokens: 72186 tool_uses: 0 duration_ms: 440920 business.md

I have enough. Writing the memo now.

BOW ECONOMICS — MARKET, PRICE, FUNNEL, AND RISK MEMO

Prepared September 4, 2026 | Hard deadline: October 1, 2026

A note on epistemics: I label claims **[FACT]** when they come from a source I retrieved (URL given) and **[JUDGMENT]** when they are my inference. Where I am guessing, I say so. Nothing here is legal advice; Section 6 flags where you need counsel.

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0. THE THREE THINGS THAT MATTER

If you read nothing else:

1. **You are underpriced, but not for the reason you think.** $129 sits ~38% above the Outschool middle-school median — you are not cheap in your category. You are cheap relative to *what parents actually spend on their kid's sports life* ($1,016/year on the primary sport alone). Charge **$149**, list at **$199**, run cohorts 1–3 at **$99 "founding"**. The $20 delta is nearly free demand-wise and is worth ~$3,400/year per instructor in contribution.
2. **The unit economics do not survive paid acquisition at $129.** At $129 with a 30% repeat rate, lifetime contribution per family is ~$126 against a realistic cold CAC of $50–80. That is not a business. At $149 with a 50% repeat rate it's ~$225 against referral-weighted CAC of ~$25. Your entire model depends on *never buying a customer cold in year one*. Design the funnel accordingly.
3. **Repeat-course purchase rate is the right *destination* metric and the wrong *operating* metric for the next nine months.** With 42 families across three cohorts, your repeat-rate estimate carries roughly ±15 percentage points of noise. You will make bad decisions from it. Section 8 gives the metric set to actually run.

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1. THE LIVE ONLINE ENRICHMENT MARKET — REAL COMPARABLES

1.1 The marketplace floor: Outschool

[FACT] Outschool's own educator guidance tells teachers that "competitive pricing typically ranges between **$10–$12 per learner for 30–45 minute group classes**," and uses a worked example of "$20 per learner" as an achievable price (Outschool Educator Handbook, Pricing Strategy).

[FACT] A 2026 third-party pricing breakdown of Outschool reports by grade band: **middle school group classes average ~$18/session**; small-group live lessons range **$5–$29 per class**; large-group live lessons **$10–$36 per class**; 1:1 tutoring ~$40/session. Monthly cost for a once-weekly group class runs **$40–$70/month** (Brighterly, Outschool Pricing 2026). A separate guide puts the general band at **$10–$30 per hour** (myeLearningWorld).

[FACT] Outschool takes a **30% commission**; teachers keep 70%. The platform passed 1M+ learners and 140K+ courses as of April 2023, with $100M+ in 2021 bookings and $229M paid to instructors since inception (Contrary Research, Outschool).

[JUDGMENT] Outschool is the price *floor* and the demand *proof*, not your competitor. It is a supply-glut marketplace where an unknown teacher must price low to earn a first review. You are not on it, so you don't inherit its price compression — but any parent who comparison-shops will land there. Expect to be compared to an $18/session number.

1.2 The direct-adjacent product: Synthesis Teams

[FACT] Synthesis Teams — live weekly online sessions for **ages 8–14**, team-based problem-solving games (60-min game sessions, 45-min discussion sessions), coach-guided — prices at **$95/month, no contract**, with monthly enrollment capped at 100 students and seasons that begin on fixed dates (synthesis.com/teams). The company also runs Synthesis Tutor as a separate AI product line (synthesis.com/tutor).

[JUDGMENT] This is the single most important comparable in the memo and you should study it obsessively. Same age band. Same "learn by doing something strategic with peers" thesis. Same live-coach delivery. It survived the collapse of Synthesis's original school ambitions and converted to a *subscription*, not a course. At ~~4 sessions/month for $95, that's **~~$24/session** — almost exactly where I'm telling you to price. The market has already validated $24/session for this exact child and this exact pedagogy.

The strategic tell: **Synthesis moved to a subscription because a fixed-length course has a terminal repeat problem.** Hold that thought for Section 8.

1.3 The premium tier: university-branded sports business

[FACT] Wharton Global Youth 2026 tuition (Costs & Aid):

Program	Price	Format
Sports Business Academy	**$4,799**	Online
Moneyball Training Camp	**$2,299**	Online
Moneyball FLEX	**$2,099**	Online
Understanding Your Money	**$329**	Online
Moneyball Academy	$10,599	On-campus
SF: Moneyball Experience	$8,959	Location-based

[FACT] Northwestern Pre-College "Business of Sports": **$1,895**, 100% online, 2-week or 4-week option, ~20–30 total hours, **ages 13+**. Capstone is "design a complete expansion franchise strategy" (precollege.northwestern.edu/sports-business).

[FACT] Sports Management Worldwide runs 8-week online courses including a Basketball GM & Scouting course with live Zoom mentor sessions; pricing is gated behind a phone call (sportsmanagementworldwide.com/courses) — **[JUDGMENT]** gated pricing on a career-credential product almost always means $1,000+ and a sales motion; it is a professional product, not a youth one.

[JUDGMENT] Note what this table proves. Northwestern charges **$1,895 for ages 13+ to design an expansion franchise over 20–30 hours** — that is roughly $70/hour and *your exact capstone*. Wharton charges **$2,099 for Moneyball FLEX online**. There is an enormous, empty price corridor between the $108–$150 marketplace band and the $1,900+ university band, and there is nothing credible in it for ages 10–14. That corridor — call it **$150–$450** — is where BOW eventually lives. You should not try to occupy it on day one with no brand, but you should not price as though it doesn't exist either.

1.4 The rest of the landscape

[FACT] iD Tech online: does not publish small-group prices; the virtual page advertises "as low as $375" via payment plan and a "$100 off virtual small group learning experiences" promo, 2 hours daily Mon–Fri, ages 7–17 (idtech.com/virtual). **[JUDGMENT]** A 5-day, 10-hour virtual week priced in the high hundreds implies ~$40–70/hr — sold as camp/childcare, not enrichment.

[FACT] Junior/entrepreneurship comparables: Acton Children's Business Fair is a **one-day, free-to-attend market** format for ages 6–17, franchised to local hosts (childrensbusinessfair.org). **[JUDGMENT]** BizWorld and Junior Achievement are school-channel nonprofits with donor/grant funding models; they compete with you for *classroom time*, never for *parent dollars*. Do not benchmark price against them; do study their teacher-facing kit design (Section 7).

[FACT] The most relevant *demand* anchor is not enrichment at all. The Afterschool Alliance's America After 3PM found parents who pay for afterschool spend an average of **$113.50 per week per child**; summer programs average **$250/week**. The top-cited "very important" factor for parents was **the child's enjoyment (81%)**, ahead of variety of activities (71%), physical activity (68%), learning not available in the school day (62%), and STEM (53%) (America After 3PM PDF).

[FACT] Families spend an average of **$1,016/year on a child's primary sport** (2024), a **46% increase since 2019** (Youth Sports Business Report); 55.4% of youth ages 6–17 played organized sports in 2023, and households at $100K+ participate at 55.7% vs 35.5% under $25K (Project Play, State of Play 2025).

1.5 Where $129 sits — and what to charge

$129 for 6 sessions = $21.50/session ≈ $18–21.50/hour.

Positioning against the evidence:

Benchmark	Per session	BOW at $129 vs.
Outschool middle-school group avg	~$18	**+19%**
Synthesis Teams (~4/mo at $95)	~$24	**−10%**
Outschool small-group top of range	$29	−26%
One week of paid afterschool	$113.50	BOW's *entire course* = 1.14 weeks
Annual primary-sport spend	$1,016	BOW = **12.7%** of one season of hoops

[JUDGMENT] Verdict: $129 is modestly underpriced, and precisely underpriced in the wrong direction. It is high enough to lose the "cheap experiment" buyer and low enough to forfeit the quality signal. It's the worst of both.

Recommendation:

- **List price $199.** Never sell there. It exists to make $149 legible.
- **Standard online: $149** ($24.83/session). This is within 4% of Synthesis Teams' effective per-session rate — the closest true comparable — and 1.3 weeks of average afterschool spend.
- **Founding cohorts 1–3: $99.** Not framed as a discount. Framed as a *role*: founding families get the price, and in exchange give a recorded 10-minute exit interview and permission to use their kid's recap (Section 6 consent flow). This is you *buying testimonials with margin*, which is the correct use of margin for a zero-brand company.
- **Referral: $119** for the referred family, **$40 credit** to the referrer. Not $99 — keep founding pricing scarce and time-boxed.
- **In-person: $179.** Your proposed $159 under-charges for the physical logistics; in-person carries venue, travel, and a lower cohort ceiling.
- **Private/school cohort: $1,799 for up to 14** (not $1,499 for 12). Detail in Section 5 — this is quietly your best product and you are pricing it as an afterthought.

Reasoning: demand in the $99–$199 band for this buyer is close to inelastic. The parent already spends $1,016/year on the sport and $113.50/week on afterschool. The decision is *"is this worth doing?"*, not *"is this worth $149 versus $129?"* The $20 delta is worth ~$280/cohort, ~$1,700/season per instructor, and it moves you from "suspiciously cheap online class" into the same perceived tier as Synthesis.

The strongest argument against my own recommendation:

Your real competitor is not Outschool or Synthesis — it is **free**. A 12-year-old who likes basketball can join a Yahoo fantasy league, watch YouTube trade breakdowns, and play NBA 2K MyGM for $0. Every dollar above ~$99 increases the probability the parent thinks *"he basically already does this himself."* That objection does not scale with price linearly — it has a cliff, and nobody knows where the cliff is. At $99 you are below the "why not just try it" threshold for an affluent Manhattan parent; at $149 you are in the zone where the parent starts demanding a *reason*, and your reason ("live instructor, real economics, a recap artifact") is unproven with zero reviews.

There is a real version of the counterargument that says: **run cohorts 1–4 at $99 permanently, get to 60 families, harvest testimonials and a measured repeat rate, then raise to $179 with proof.** That sequencing beats mine if your fill rate at $149 comes in under ~65%. I'd take my recommendation because $99→$149 is a hard price increase to execute on your own first customers, and because underpricing corrupts your repeat-rate signal (cheap things get re-bought for reasons that don't generalize). But if cohort 1 at $99 fills in under 72 hours from a warm list, that is evidence *for* the counterargument, not against it — it means you never tested price at all.

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2. THE COMPETITIVE HOLE

I searched hard for a direct competitor. Here is what actually exists.

[FACT] Fantasy sports as classroom math is a real, decades-old, academically studied category. The IES funded "Fantasy Sports Math League: An Innovative Gaming Approach to Increase Middle School Math Skills" (IES award page); SportsData.io ran a case study on it (sportsdata.io); the Seattle Times covered fantasy football in math class (Seattle Times Education Lab). TeachersPayTeachers has a large inventory of fantasy-sports math units (TPT search).

[FACT] Outschool already hosts direct-adjacent classes, including "Fantasy Basketball Fun! For Kids 11–16 — Play in a Season-Long Fantasy League" (listing) and "Summer Camp: Sports Management, Marketing, Leadership & Career Skills" (listing), plus a whole sports-analytics category page (Outschool sports analytics). Lavner runs virtual sports analytics camps (lavnercampsandprograms.com).

[FACT] The premium end is well-served for teens: Wharton Sports Business Academy ($4,799 online), Moneyball ($2,099–$10,599), Northwestern Business of Sports ($1,895, ages 13+, expansion-franchise capstone), USF Sports Business Bootcamp (grades 9–12) (USF), NSLC Sports Leadership & Management (NSLC).

Honest assessment: the space is not empty. It is crowded at both ends and thin in the middle.

- **Crowded below you:** fantasy-sports-math worksheets, single-teacher Outschool fantasy leagues, free 2K MyGM. Cheap, unstructured, individually-played, no economics.
- **Crowded above you:** university-branded, $1,900–$10,600, ages 13+, lecture-and-capstone format, credential-signaling for college applications.
- **Genuinely thin in the middle — four specific gaps [JUDGMENT]:**

1. **Age.** Nearly every sports-business product starts at 13, and most at 15–16 (college-résumé products). Ages **10–12** are served only by fantasy-math worksheets. That is your hole and it is a real one.
2. **Economics, not analytics.** The existing products teach *statistics* (analytics camps) or *careers* (sports management). Almost nothing teaches **scarcity, opportunity cost, constraint optimization, and the cap as a budget line** through the franchise. Your framing — economics with basketball as the substrate, rather than basketball with math sprinkled on — is not occupied.
3. **Multiplayer scarcity.** Fantasy leagues are parallel play against a scoreboard. A **cohort where students negotiate against each other over a shared, finite pool** is a genuinely different mechanic and is the thing that makes the class actually about economics. I found no youth product doing this with software as commissioner.

4. **The instructor-independent runtime.** Every comparable in the middle band is a *teacher's personality product* (one Outschool teacher, one league). Nobody has built the software layer that lets a trained 21-year-old run it consistently. That is your durable moat, not the curriculum.

The uncomfortable part [JUDGMENT]: none of these four gaps are defensible on their own for very long, and gap #1 (age) is a marketing insight, not IP. The only thing that compounds is #4 plus a repeat-purchase engine. If BOW is a great class, it's a lifestyle business. If BOW is a *system that makes mediocre instructors run great classes*, it's a company. Bias every October 1 tradeoff toward the system.

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3. WHY PARENTS BUY (AND REBUY)

3.1 What the evidence says

[FACT] What parents say they want, ranked: child's enjoyment **81%**, variety of activities 71%, physical activity 68%, learning not available in the school day 62%, STEM 53%. Satisfaction with existing programs is high on safety (88%), quality of care (88%), staff training (84%) (America After 3PM).

[FACT] Enrollment barriers: cost 52%, transportation 52%, location 47%, hours 44%, availability 42% (same source). **[JUDGMENT]** Online kills three of the top five. That is your structural advantage over local enrichment and you should say it out loud in copy.

[FACT] The attrition curve is brutal and predictable. Industry reporting on afterschool retention describes programs losing **20–35% of an original cohort by week four**, with a sharp drop between weeks three and five, driven by three named failure points: the **Novelty Dip (days 6–10)** when excitement fades without tangible output; the **First Skill Wall (days 11–18)**; and the **Social Belonging Check (days 19–28)**. Programs holding **80%+ through eight weeks** show materially stronger outcomes (Kodely, After-School Retention).

[JUDGMENT] Map that onto a six-week course and it is terrifyingly on-the-nose. Your Novelty Dip is **Session 2**. Your Skill Wall is **Session 3** (the first time the cap math bites and a kid can't have the player he wants). Your Belonging Check is **Session 4**. Design interventions at exactly those three points or you will lose a third of the room before the recap ever ships.

[FACT] Word of mouth dominates parent discovery: "3 in 4 parents still rely on word of mouth" for children's products; 93% of online buyers read reviews (Sogolytics).

[FACT] Cart abandonment averages 70.22%, and the #1 non-browsing reason is **extra costs at checkout (40%)**, followed by security concerns (19%), forced account creation (18%), and lengthy checkout (17%) (Baymard). → **[JUDGMENT] Direct instruction: one all-in price, no**

fees, no add-ons, guest checkout, and a checkout under 90 seconds. Do not charge a materials fee. Do not add a platform fee. A $9 surprise at checkout costs you more than a $20 price increase does.

[FACT] Outschool's own retention guidance to educators is entirely about **structured pathways** (Beginner → Intermediate → Advanced), personalized post-class follow-up, and milestone recognition; they frame the economics as $15 once vs $75 across five re-enrollments (Outschool retention handbook). → **[JUDGMENT] The single highest-leverage repeat-purchase mechanic in this entire memo is that there must be a visible Course 2 before Course 1 ends.** Not "we'll email you." A named, dated, differently-titled Season 2 with a seat with the kid's name on it. Outschool built its retention doctrine around pathways for a reason. Right now BOW has no Course 2, and that is a bigger repeat-rate risk than anything about the artifact.

Seasonality [JUDGMENT, informed by camp-registration reporting]: the enrichment purchase calendar has three peaks — **late August/September** (fall activities, set in the back-to-school scramble), **late December/January** (New Year + summer camp registration opens, and camp registration is now a January-panic behavior — see UltraCamp, film.camp timeline), and **April/May** (summer). October is a **trough**. Parents have already committed their fall slots by the first week of September.

What an October 1 deadline actually means: you are shipping into the worst-timed window of the year for a *fall* cohort. Do not fight this — **reframe October 1 as the ship date for the product and the free playable, and target the first paid cohort for a "Winter Season" starting the first full week of January**, with enrollment opening November 15 and closing December 20. January is a demand peak, aligns with NBA mid-season (trade deadline in February is a *gift* of a narrative hook), and gives you ten extra weeks of list-building. Running Ramaz free in October–November and the first paid cohort in January is a strictly better sequence than forcing a paid November cohort.

3.2 Prosecuting the parent artifact

The founder wants: a beautiful franchise-season recap — the plan / the turning point / the tradeoff / in their words / how they adapted / the economics they used / what comes next.

The case for the prosecution:

1. **It is built for the wrong buyer.** The evidence says the #1 parent priority is **the child's enjoyment (81%)** — and the strongest known repeat signal in kids' activities is the child asking to go back. A beautiful PDF does not make a child ask to go back. If the kid is lukewarm and the recap is gorgeous, you do not get a rebuy. If the kid is on fire and there is no recap, you probably do.
2. **It arrives after the decision.** A recap delivered at the end of week six lands *after* the parent has already formed a verdict from six weeks of dinner-table evidence. Artifacts that arrive after the decision are receipts, not persuasion.

3. **It is the most expensive thing on your build list.** "In their words" implies capturing and curating student reasoning across six sessions. That is a data pipeline, a consent regime (Section 6), and an editorial pass. It is plausibly 30–40% of your remaining engineering budget for a deliverable with no evidence behind it.
4. **The founder-vanity tell:** the seven-part structure (plan/turning point/tradeoff/words/adaptation/economics/next) is a *narrative you already believe*. It is what you want the product to have been. That is a red flag for artifact design.

The case for the defense — and it's real:

1. **Perceived academic legitimacy is what lets you charge $149 for basketball.** Parent satisfaction data shows staff knowledge/training at 84% and "learning not available in the school day" at 62% — parents want the enrichment to be *legitimate*, and a sports product carries a permanent burden of proof that it isn't just screen time with a coupon. The artifact is the receipt that discharges that burden.
2. **It is the word-of-mouth object.** With 75% of parents relying on word of mouth, the artifact's real job is not to convince the buyer — it's to be **forwarded**. A parent forwards a beautiful recap of *their* kid's season to the WhatsApp group. They do not forward a grade report. For a zero-brand company, an artifact optimized for forwarding is a distribution asset, not a retention asset.
3. **It is the only durable evidence that anything happened.** Six weeks later, "he did that basketball thing" needs a physical referent or the memory decays and the rebuy has no anchor.

Verdict [JUDGMENT]: Build the artifact, but **re-scope it from a retention device to a forwarding device**, and **cut it from seven sections to two**, and **stop delivering it at the end.**

The cheapest version that captures 80% of the value by October 1:

- **A single-page, mobile-first, shareable web page per student**, at a public unguessable URL, plus a print-quality PNG. Not a PDF. Not a booklet. No printing, no shipping.
- **Two sections only:**
    1. **"The Decision"** — one trade or one cap decision the kid made, shown as: what it cost, what it bought, what they gave up, and the one sentence they said about why. *This is the whole artifact.* One decision, one tradeoff, one quote.
    2. **"The Season"** — a single visual: their roster and where they finished, with three date-stamped moments.
- **How you capture "in their words" for near-zero cost:** the instructor already runs a closing 90 seconds each session. Add one structured prompt — *"In one sentence: what did that cost you?"* — and the student types it into a single text field in the app. That is it. Six sentences per student across the course; you use one. No transcription, no editorial pipeline, no AI summarization, no new consent surface for recordings. **This is the single most important build simplification in this memo.**

- **Ship it at Session 5, not Session 6.** A draft recap that lands *before* the last session, with the Season 2 offer on the same page, converts. A recap that lands after the course ends is a memento.

The single moment during six weeks that most drives a rebuy [JUDGMENT]:

Not the recap. It is **the moment in Session 3 or 4 when the kid loses a negotiation, or is forced by the cap to give up a player they wanted, and has to explain the decision out loud to peers.** That is the moment the kid goes home and tells a parent an actual story with stakes in it — *"I had to trade Wemby because I couldn't afford him"* — and it is the only moment that produces the "my kid asked to go back" signal on its own.

Engineer it deliberately:

- Make the cap bind for **every** student by design in Session 3. Not emergent — guaranteed. If any student sails through unconstrained, they got a worse product.
- Require every student to state the tradeoff aloud in one sentence before the software executes the transaction. The software should refuse the trade until the sentence exists. **This is a product requirement, not a facilitation tip** — it is how you make a mediocre instructor produce the story.
- Send the parent a **two-sentence text on the night of Session 3**, quoting their kid's sentence verbatim. Nothing else. No newsletter, no attachment. This costs you an SMS and is worth more than the entire recap.
- Note this lands exactly on the documented **Skill Wall (days 11–18)** — you are converting the known attrition point into your peak emotional moment. That is the right place to spend your engineering.

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4. THE FUNNEL FOR A ZERO-BRAND COMPANY

4.1 Channels ranked by expected seats per unit of founder effort

Rankings are **[JUDGMENT]**, informed by the word-of-mouth data (75% of parents rely on it) and the structure of your existing access.

#	Channel	Est. seats / 10 founder-hrs	Why
1	**Ramaz pilot families → direct ask**	**8–14**	Warmest list on earth. They've seen it. Only real risk is asking too late.

#	Channel	Est. seats / 10 founder-hrs	Why
2	**Jewish day school network (Ramaz → peer schools)**	**6–12**	Highest-density referral graph in US private education. Detail below.
3	**Private/school cohort sales ($1,799 × 1 sale = 14 seats)**	**6–10**	One conversation fills a cohort. Grossly underweighted in your plan.
4	**Parent WhatsApp groups (via a parent, never by you)**	**4–8**	Enormously effective when a *parent* posts; near-zero when a founder posts.
5	**Free live event (60-min "Trade Deadline Night")**	**3–6**	Converts, but requires an audience you got from 1–4.
6	**Free public playable**	**2–5, rising over time**	Low per-hour early, but it's the only compounding asset here.
7	**Youth sports orgs / AAU / travel programs**	**2–4**	Right kids, but gatekeepers are slow and unincentivized.
8	**JCC / synagogue youth programming**	**1–3**	Real, but institutional calendars run 4–6 months ahead. Start now for spring.
9	**Teacher/school reseller program**	**1–3 now, high later**	Wrong sequencing pre-proof. Revisit at cohort 6.
10	**Reddit (r/nba, parenting subs)**	**0–2**	Hostile to promotion; r/nba is adults, not parents. Bad ROI.

#	Channel	Est. seats / 10 founder-hrs	Why
11	**Sports-media partnerships**	**0–1**	Requires scale you don't have. Ignore for 12 months.
12	**Paid ads**	**~0 at viable CAC**	Section 5 shows this doesn't clear at $149. Do not start.

4.2 On the Jewish day school network specifically

[FACT] Prizmah's 2025–26 report: **~101,041 students across 305 network schools**, up ~7.5% over four years and growing across denominations for the first time in decades (JTA; Times of Israel).

[JUDGMENT] Assess it honestly: this is a genuinely excellent distribution channel and also a genuinely limiting one.

Why it's excellent: ~101K students, with grades 5–8 plausibly ~25–30K of them; extremely high household income concentration in the NY metro; unusually dense inter-school social graphs (families know families across schools through camp, shul, and simcha networks); and a strong cultural disposition toward supplementary education. A Ramaz endorsement travels to Heschel, SAR, Kushner, Frisch, Ramaz's own feeder families, and Modern Orthodox summer camps in a way that no cold channel replicates. **Realistically this network alone can fill 15–30 cohorts over 18 months.** That is a $45K–$90K business — enough to prove the model, not enough to be the business.

Why it's limiting: it caps out. 30K addressable students, at a generous 3% penetration, is 900 enrollments — roughly 64 cohorts, ~$135K revenue. It is also a **reputationally coupled** network: one bad cohort, one safety incident, one instructor who was sloppy, propagates through the same graph at the same speed. Treat the Section 6 safety architecture as a *distribution* requirement, not a compliance chore.

Use it as the beachhead, not the market. Ship the free playable to be shareable *outside* it from day one.

4.3 Concrete 60-day go-to-market (Sept 4 → Nov 3), for a January 5 first paid cohort

Days 1–14 (Sept 4–17) — Instrument the pilot for distribution

- Add to the Ramaz pilot, before it starts: (a) an opt-in parent list at signup, (b) the Session-3 SMS moment, (c) the consent architecture from Section 6.2, (d) an exit interview slot.
- Write the Season 2 course title and one-paragraph description **now**. It must exist before the pilot ends.
- Build the landing page + waitlist. Single field: email. No pricing yet.

Days 15–35 (Sept 18 – Oct 8) — Ship the playable; run the pilot

- **Oct 1: ship the free public playable** (spec in 4.4) and the waitlist page. This is your deadline commitment.
- Run the pilot. Instrument attrition at sessions 2/3/4.
- Founder does **zero** marketing outreach this window. Run the class.

Days 36–46 (Oct 9–19) — Harvest

- Recaps ship at Session 5. Collect 8–12 recorded parent exit interviews and 5+ written testimonials with signed media releases.
- Ask each pilot parent for exactly one thing: *"Would you post the playable link in one group chat?"* Not "tell your friends." One link, one chat, script provided. **[JUDGMENT]** Specificity is the whole game in referral asks.

Days 47–60 (Oct 20 – Nov 3) — Open enrollment

- **Nov 1: Winter Season enrollment opens.** $99 founding for the first 14, $149 thereafter. Cap the founding tier publicly and visibly.
- Ramaz parent list first (48-hour exclusive), then the playable waitlist, then WhatsApp forwarding.
- In parallel, book **five conversations** for the $1,799 private cohort: two Jewish day schools, one JCC, one AAU/travel program, one "group of friends" parent. **[JUDGMENT]** One yes here fills an entire cohort with a single sale, and it's the highest-value use of your November.
- Free live event: **"Trade Deadline Night," 60 minutes, free, mid-November**, open to any kid. Real, live, one decision. This is your top-of-funnel event and it recurs.

Target at day 60: 14 paid seats committed for January, 300+ playable plays, 150+ waitlist emails, 1 private-cohort LOI.

4.4 What the free public playable must be

Recommendation: a single-decision trade evaluator. Six minutes. No account. Shareable result.

The experience: You are handed a real team in a real cap situation. A trade offer arrives. You have three minutes to decide: accept, counter (from three pre-built counters), or decline. You must type one sentence: *"I did this because ___."* Then the software shows you what the decision actually cost and bought — the cap consequence two years out, the opportunity forgone — and gives you a grade with a shareable card carrying your one sentence on it.

What it must prove:

1. That basketball decisions *are* economics — the reveal must land as "oh, that's a budget constraint" not "oh, I picked a good player."
2. That the software is a real, credible commissioner — instant, unambiguous, non-hand-wavy consequences.
3. That your kid will have an opinion and want to argue about it. **The success metric is a kid showing a parent the screen.**

What it must NOT give away:

- **Multiplayer negotiation.** The entire paid product is other kids wanting your players. Give that away and the class has no scarcity left to sell. The playable must be strictly single-player against the software.
- **The full season loop.** One decision, never a roster build, never a multi-week arc.
- **The instructor.** Do not simulate coaching or Socratic follow-up. The playable should leave the player with an *unanswered* question — the paid product is where someone asks "but what would you do next year?"
- **The recap artifact.** Show the shareable card, never the season recap.

One design requirement [JUDGMENT]: the shareable card must carry *the kid's own sentence*, not a score. Scores get compared and discarded; a kid's reasoning in their own words is what a parent screenshots, and it previews the paid product's core promise in a single image.

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5. UNIT ECONOMICS

5.1 Cost inputs

[FACT] Instructor market rates: ZipRecruiter lists **Youth Programs Instructor at $12–$23/hr** (ZipRecruiter) and online instructor roles considerably higher ($16–$54/hr for online art instructors in TX). Outschool teachers keep **70%** of gross (Contrary) — but that is entrepreneur compensation including their own marketing, not facilitator pay.

[FACT] Stripe standard US online rate is 2.9% + $0.30 per successful card charge (Stripe fee guides, 2026).

[JUDGMENT] What a trained BOW instructor should be paid: think in **paid hours per cohort**, not per session.

- 6 live hours
- 3.0 hrs prep across 6 weeks (0.5/session — low because the software carries the plan)
- 1.0 hr recap/quote curation
- 0.5 hr parent comms
- **= 10.5 paid hours/cohort**

At **$38/paid hour → $400/cohort**. That is well above the youth-program floor ($12–23) and appropriate for a screened, trained, background-checked college student in a NY-metro labor market who is delivering a premium-priced product. **Pay this. Do not pay $250.** Instructor churn is your single largest hidden cost, and the difference between $250 and $400 across a cohort is $150 — trivially less than the cost of re-recruiting and re-training.

5.2 The model, per cohort

Base case: $129 × 14 (as proposed)

Line	Amount
Gross revenue (14 × $129)	$1,806
Payment processing (2.9% + $0.30 × 14)	−$57
Instructor (10.5 hrs @ $38)	−$400
Platform/infra (Zoom, hosting, data, allocated)	−$60
Refunds/chargebacks allowance (3%)	−$54
Contribution margin (pre-CAC)	**$1,235 (68.4%)**

Recommended case: $149 × 14

Line	Amount
Gross revenue	$2,086
Processing	−$65
Instructor	−$400
Platform/infra	−$60
Refunds (3%)	−$63

Line	Amount
Contribution margin (pre-CAC)	**$1,498 (71.8%)**

Downside case that will actually happen sometimes: $149 × 8 (online minimum)

Line	Amount
Gross revenue	$1,192
Processing	−$37
Instructor	−$400
Infra	−$60
Refunds	−$36
Contribution margin	**$659 (55.3%)**

[JUDGMENT] The 8-seat floor is survivable but bad. Instructor cost is fixed per cohort, so every empty seat is ~$135 of pure lost contribution at $149. **Raise your online minimum from 8 to 10.** At 8 you are running a cohort for 44% of its potential contribution; the discipline of a 10-seat floor (with the option to merge two under-filled cohorts) is worth more than the goodwill of never cancelling.

5.3 The private cohort — the number you have mispriced

Proposed: $1,499 for up to 12 → $125/student, contribution $1,499 − $400 − $60 − $44 = **$995 (66%)**.

Recommended: $1,799 for up to 14 → $128/student, contribution $1,799 − $400 − $60 − $52 = **$1,287 (72%)**.

[JUDGMENT] Why this is your best product and you're treating it as a footnote: the per-student economics are the same, but the **customer acquisition economics are ~14× better**. One conversation with one school administrator, one JCC program director, or one motivated parent organizing her son's friend group fills an entire cohort. Blended CAC per seat collapses from $50+ to under $10. It also solves your fill-rate risk (guaranteed full), your scheduling risk (they pick the time), and your safety-comms risk (a known adult vouches for you inside the group).

Charge $1,799 and sell it hard in November. If the buyer flinches, the fallback is $1,499 for 12, which is still a 66%-margin sale you didn't have to market.

5.4 Throughput and the shape of the business

Cohorts per instructor per season [JUDGMENT]: A season = ~13 weeks; a course = 6 weeks; so two waves. An instructor available 3 evenings/week can run 3 concurrent cohorts, i.e. **6 cohorts/season**, ~63 paid hours + training. Three seasons/year (Winter, Spring, Summer-Fall) → **~18 cohorts/instructor/year**.

- At $149/14 seats: **18 × $1,498 = $26,964 contribution per instructor-year**, on ~$7,200 of instructor pay. Instructor earns ~$7.2K/year part-time — realistic for a college student, and low enough that you should expect ~50% annual instructor churn and budget re-training accordingly.

Scaling arithmetic:

Contribution goal	Cohorts/yr	Enrollments/yr	Instructors needed
$100K (proof)	67	~940	4
$500K (real business)	334	~4,670	19
$1M	668	~9,350	37

[JUDGMENT] Read that table honestly. At $149, a $1M contribution business requires **~9,350 paid enrollments/year** and ~37 instructors. That is a substantial consumer operation. At $249 (the price the middle-corridor gap in Section 1.3 says is eventually available with proof), the same $1M needs ~5,000 enrollments and ~21 instructors. **Price is your primary lever on operational complexity**, more than on revenue.

5.5 CAC and what LTV must produce

Lifetime contribution per acquired family, where R = repeat rate per course and expected courses = 1/(1−R):

Repeat rate R	Courses/family	Lifetime contribution @ $149 (per student, CM/14 = $107)
25%	1.33	**$142**
40%	1.67	**$179**
50%	2.00	**$214**
65%	2.86	**$306**

Against CAC:

CAC	R=25%	R=40%	R=50%	R=65%
$15 (referral)	9.5×	11.9×	14.3×	20.4×
$50 (warm/organic)	2.8×	3.6×	4.3×	6.1×
$80 (cold paid)	1.8×	2.2×	2.7×	3.8×

[JUDGMENT] Three conclusions, and they are the load-bearing conclusions of this memo:

1. **Paid acquisition is dead until repeat rate is proven above ~50%.** At $129 and a 30% repeat rate, LTV/CAC on cold paid is ~1.6×. That destroys capital. **Do not run ads in year one.** This is not a budget constraint, it's an arithmetic one.
2. **At LTV/CAC of 9–20× on referral, referral mechanics are worth more engineering time than the recap artifact.** Build the referral loop before you build the seventh section of the recap.
3. **The business becomes real at ~$179–$249 with a 50%+ repeat rate and majority-referral acquisition.** Below $129, it is a nice project regardless of how well it's executed, because founder time never clears.

Founder time, stated bluntly: if you spend 20 hours/cohort on operations, at a $75/hr shadow rate that's $1,500 — the cohort is underwater fully-loaded even at $149. **The whole point of the instructor system in Section 7 is to drive founder time below 4 hours/cohort by cohort 6.** Track founder-hours-per-cohort as a first-class metric (Section 8).

5.6 The price I would actually charge

$149 standard online. $99 founding (cohorts 1–3 only, time-boxed and publicly capped). $119 referral. $179 in-person. $1,799 private/school cohort of up to 14. List price $199, never sold.

Why: $149 is within 4% of the closest true comparable's effective rate (Synthesis Teams, ~$24/session, same age band, same pedagogy), 38% above the Outschool median which is where a differentiated non-marketplace product should sit, 1.3 weeks of the average paid-afterschool spend, and 14.7% of what the same family already spends annually on their kid's primary sport. It produces a 72% contribution margin, and it is the lowest price at which the LTV/CAC table permits you to ever contemplate non-referral acquisition. $129 buys you almost no incremental fill and costs you ~$3,400/instructor/year.

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6. YOUTH SAFETY, CONSENT, AND LEGAL — PRACTICAL

Not legal advice. Items marked ⚖️ require counsel before October 1.

6.1 COPPA — and it is already binding

[FACT] The FTC's amended COPPA Rule took effect June 23, 2025, with a full compliance deadline of April 22, 2026 (White & Case; Davis Wright Tremaine). **That deadline has already passed.** You do not get a grace period; you must launch compliant.

[FACT] What changed:

- **Separate consent for third-party disclosure.** Parents may consent to collection/use without consenting to disclosure to third parties, unless disclosure is "integral" to the service. The FTC explicitly stated disclosure **for advertising or to train/develop AI is not "integral."**
- **Written data retention policy.** Indefinite retention is prohibited; you must publish why data is collected, the business need, and a deletion timeframe.
- **Written information security program**: designated personnel, annual internal and external risk assessments, risk-based safeguards, regular testing, annual review; **written assurances from service providers**.
- **Expanded PII definition** to include biometric identifiers (voiceprints, facial templates) and government IDs.
- **Three new VPC methods**: text-message-plus, knowledge-based authentication (questions a 12-year-old in the household couldn't answer), and government photo ID verified via facial recognition with prompt deletion.
- **[FACT, and important] The FTC declined to codify an ed-tech/school authorization exception**, citing potential conflict with FERPA updates (DWT). Do not build the Ramaz pilot on the assumption of a codified school-consent safe harbor.

[FACT] "Email-plus" is only available where information is collected for the use and benefit of the school and no other commercial purpose, and is disqualified by third-party disclosures for advertising or AI training (Promise Legal).

Practical build for BOW [JUDGMENT]:

- **Design so the child never has an account you control directly.** The *parent* holds the account and enrolls the child; the child's in-class identity is a **team name and a first name only**. Collect no child email, no child DOB beyond an age band, no photo, no voiceprint, no last name.
- **Your VPC is the paid enrollment transaction.** A parent completing a credit-card purchase in their own name is one of the longest-standing recognized VPC mechanisms.

This makes the paid product materially easier to comply with than the free playable.

- ⚖️ **The free playable is your biggest COPPA exposure, not the class.** A public game aimed at 10–14 year-olds with no parent in the loop is a mixed-audience service. **Design it to collect literally nothing**: no account, no email, no analytics that constitute persistent identifiers beyond support-for-internal-operations, session state in local storage only. If the kid's sentence is shown on a share card, generate it client-side and never store it server-side. This is a real constraint on the playable design and you should adopt it before you build, not after.
- **Never disclose child data to third parties for advertising or AI training.** No pixel-based retargeting on any surface a child touches. This forecloses a marketing channel; take the loss.
- Write the retention policy (I'd suggest: session data deleted 90 days post-cohort; recap artifacts retained only with affirmative parent opt-in) and the written security program. These are documents, not engineering — budget one day.

6.2 FERPA and NY Ed Law 2-d — the surprising finding

[FACT] FERPA applies only to institutions receiving US Department of Education funding. Per the Department: *"Private and parochial schools at the elementary and secondary level generally do not receive such funding and are, therefore, not subject to FERPA"* (studentprivacy.ed.gov).

[FACT] NY Ed Law §2-d defines "educational agency" as "a school district, board of cooperative educational services, school, or the education department," where the separately-defined "school" reaches only certain approved private schools (notably approved private schools for students with disabilities) (NYSED, Ed Law 2-d Definitions).

[FACT] Where 2-d does apply, third-party contractors must: sign a DPA restricting use to the contracted purpose; incorporate the **Parents' Bill of Rights**; not sell student data or use it for commercial advertising; and **notify the educational agency of a breach within seven calendar days**. Penalties escalate $1,000 / $5,000 / $10,000 per violation. Part 121 (adopted Jan 13, 2020) provides implementation detail (Cybernut; FindLaw, EDN §2-d).

[JUDGMENT] The finding: Ramaz, as an independent Jewish day school, is very likely outside both FERPA and Ed Law 2-d. This is genuinely useful — it means the pilot is not gated on a 2-d Data Privacy Agreement negotiation, which is the single slowest thing that happens to vendors selling into NY public districts and would have blown through October 1.

Three cautions:

1. ⚖️ **Confirm this with counsel and confirm Ramaz's own funding posture** (some private schools receive certain federal program funds). Do not act on my reading alone.
2. **Ramaz may impose 2-d-equivalent terms contractually regardless.** Many independent schools copy public-district DPA language into vendor agreements.

Pre-empt it: offer a short DPA voluntarily, with a Parents' Bill of Rights, a 7-day breach notice, no-sale/no-ad commitments, and a defined deletion timeline. This costs you nothing you shouldn't already do, and it makes you look like the only vendor who came prepared.

3. **This exemption evaporates the moment you sell to a NY public district or charter.** Build the 2-d-compliant posture now so the later channel isn't gated.

Structurally simplest path [JUDGMENT]: even inside Ramaz, contract with parents, not the school. If parents enroll their own children and the school merely provides space and endorsement, you never receive "student data from an educational agency," which sidesteps 2-d's contractor definition entirely and keeps your COPPA consent mechanism (parent enrollment) unchanged across both channels. ⚖️ **Validate, but this is the structure I'd push for.**

6.3 Instructor screening and live-session safety norms

[FACT] Standard practice for youth-serving organizations (SafeKidsThrive, Section 2): written application with signed suitability statement; **2–3 reference checks by phone**; comprehensive personal interview; criminal and sexual-abuse history checks. Enhanced screening for unsupervised child contact adds state + national criminal history, sex offender registry search, credential verification. Massachusetts requires re-checks at least every 3 years; some organizations do annually. The report is explicit that screening "does not in itself provide a 100% guarantee" and that its major value includes **deterrence** — publicizing the policy deters applicants with intent.

[FACT] Outschool's teacher safety rules (Outschool support): all communication stays on-platform; teachers must never share personal contact information with parents or learners; recordings may only be shared with families in that class and may not be downloaded or stored personally; teachers must actively monitor chat to prevent learners posting personal information; teachers must never collect full names, birthdates, personal emails, addresses, phone numbers, social media handles, or gamer tags.

[FACT] The two-adult rule is a documented standard in institutional minors policies (UGA two-adult rule guidelines); NCYS publishes recommended guidelines for youth sports organizations (NCYS).

BOW's minimum viable safety standard [JUDGMENT] — adopt all of it before cohort 1:

- National criminal + sex offender registry check on every instructor, re-run annually. Budget ~$30–60/instructor.
- 2 phone reference checks, one of which must involve prior work with minors.
- **Recording policy: record every online session by default, automatically, retained 30 days, accessible to the founder, never distributed to families.** This is not surveillance of the child — it is the standard that protects the instructor and the company. Disclose it prominently in the parent consent.

- **No 1:1 adult–child session, ever, in any format.** If one student shows up, the session is cancelled or the parent must be present on camera.
- **Two-adult presence for in-person cohorts, no exceptions.**
- **Zero off-platform contact.** Instructors get a BOW email and no other channel. No personal phone, no Instagram, no Discord.
- All parent communication goes through you or a monitored BOW channel, never instructor-to-parent directly.
- Publish the policy on the website. Per the deterrence finding, publicity is part of the control.

6.4 Online trade negotiation — bounded communication design

Your in-person design (verbal negotiation, software as commissioner, no free-form private digital chat) is correct. For online, three options:

Option A — Structured Offer Protocol (no free text at all). All negotiation happens through the software: a student composes an offer entirely from dropdowns (assets out, assets in, one of ~8 canned rationale tags like "I need cap room" / "I'm rebuilding" / "I think he's undervalued"). The recipient sees the offer and can accept, decline, or counter — also from dropdowns. Zero free text between students.

- *Pro:* Zero moderation burden. Zero grooming surface. Fully logged, fully reviewable, provably safe to a nervous parent. Cheapest to build. Works identically at 10 students or 10,000.
- *Con:* Loses persuasion and rhetoric — arguably part of what you're teaching. Can feel mechanical.

Option B — Moderated Trade Floor (free text, main room only, ephemeral). Free-text negotiation is permitted, but *only* in the full-cohort main room where the instructor and every student can see it, only during instructor-opened "negotiation windows" of 5–8 minutes, and the log is wiped at session end (retained server-side for review only).

- *Pro:* Preserves real persuasion. Public-by-default is a genuinely strong safety property.
- *Con:* Needs profanity/PII filtering; a distracted instructor is the only real-time control; 14 kids typing simultaneously is chaotic and disadvantages slower typists.

Option C — Voice-only breakouts with rotation. Negotiation happens in 3-minute voice breakout rooms of 3–4 students, no chat enabled, instructor auto-rotates through rooms, rooms hard-close on a timer, all breakouts recorded.

- *Pro:* Closest to the in-person experience you actually want; verbal negotiation is the pedagogically richest form; no typing barrier.
- *Con:* Highest safety surface (unsupervised child-to-child moments, even if brief); recording breakouts raises a COPPA voiceprint/PII question ⚖️; heaviest platform requirements; hardest for a mediocre instructor to run.

Recommendation [JUDGMENT]: Ship Option A for cohort 1. Layer Option C in at cohort 4+, instructor-gated, and never ship Option B.

Rationale: Option A is the only one you can build well by October 1, it is trivially defensible to the most anxious parent in your highest-value distribution network, and — the non-obvious part — **the constraint is pedagogically good.** Forcing a student to express a trade rationale as a *selected structured claim* rather than free text is exactly the discipline that makes the economics legible, and it gives you the "in their words" capture for the recap for free. Option B is strictly dominated: it carries most of C's risk with less of its value.

6.5 Consent architecture for using student work in marketing

The core principle [JUDGMENT]: educational data capture and marketing use must be separately consented, separately stored, and separately revocable. Bundling them is both a COPPA problem (the amended rule's separate-consent logic) and a trust problem in a tight referral network where one annoyed parent is expensive.

Concrete flow:

1. **At enrollment (mandatory, part of purchase):** Parent consents to the *educational* capture only — that the child will type short reasoning statements, that these are stored to build the child's private recap, that sessions are recorded for safety and retained 30 days, that data is deleted after 90 days. No marketing language anywhere near this. Not optional, because it *is* the product.

2. **Separate, optional, and asked LATER — after Session 5, when the recap draft exists:** a distinct one-screen Media Release, presented alongside the actual draft recap so the parent sees exactly what they're approving. Three independent checkboxes, defaulted off:

   - ☐ Use my child's **written quote**, attributed by **first name and age only**
   - ☐ Use my child's **anonymized recap** (team name only, no child name)
   - ☐ Use my child's **first name in a testimonial** Plus a plain-language line: *"You can withdraw this at any time by replying to this email, and we will remove it within 7 days."* Never bundle photo/video into these; if you ever want video, that's a fourth, separately-negotiated release.

3. **Storage:** marketing-consented assets live in a **physically separate store** from the operational data. The default state of every asset is "not usable." A quote becomes marketable only by an explicit flag written by the consent flow, never by an instructor's judgment.

4. ⚖️ **Have counsel review the release**, particularly on minors' capacity, NY-specific requirements, and duration/revocation. This is a one-hour lawyer task, not a project.

5. **Never make consent a condition of the founding price.** Ask for it as a favor from people who already love you. Making it transactional is how it becomes a story.

6.6 NBA team and player names — what's fine, what needs counsel

[FACT] The nominative fair use test has three elements: the product/service is not readily identifiable without using the mark; only so much of the mark as is reasonably necessary is used; and the use does not suggest sponsorship or endorsement (INTA fact sheet).

[FACT] On player names and statistics: in *C.B.C. Distribution & Marketing v. MLB Advanced Media*, the Eighth Circuit held that a fantasy operator could use player **names and statistics** without a license, on First Amendment grounds trumping state right-of-publicity claims; **certiorari was denied in 2008** (Harness IP case study; 8th Cir. opinion).

Practical guidance [JUDGMENT]:

Clearly fine:

- Player **names** and **publicly reported statistics**, in plain text. This is the direct holding of *C.B.C.*
- Publicly reported **contract and salary-cap figures** — these are widely published facts.
- Team names used **descriptively** in plain, unstylized text ("the Knicks' cap situation").

Needs care — do without:

- **Team logos, wordmarks, uniforms, color-and-mark trade dress, arena imagery.** These are the strong trademark assets. Using them is not necessary to teach the economics; the third prong (no suggestion of sponsorship) is where a logo-heavy interface fails.
- **Player photographs and likenesses.** *C.B.C.* covered names and stats, not images. Photos also implicate copyright in the photograph itself, which is a separate and cleaner claim against you.
- **Any phrasing implying affiliation** — "NBA-approved," "official," NBA-style typography, or a name like "NBA Economics."

⚖️ **Genuinely needs counsel:**

- Whether your **product name** creates confusion risk.
- Marketing copy that uses team names prominently in ads (nominative fair use is more contested in advertising than in the product itself).
- Any plan to sell merchandise, or to license the software to schools as a branded product.

The cheap structural answer [JUDGMENT]: build the software so **real names are a swappable data layer**. Ship real names (well-supported by *C.B.C.*), zero logos, zero photos, a clear disclaimer ("BOW Economics is not affiliated with or endorsed by the NBA or any team"), and typography that looks nothing like league branding. If counsel ever raises a concern, or a league sends a letter, you flip to fictional-league mode in a day rather than rebuilding. **That optionality is worth two days of engineering and buys you out of the worst-case scenario entirely.**

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7. INSTRUCTOR SYSTEM

[JUDGMENT] The founder-not-a-runtime-dependency constraint is the correct one and it is the hardest thing in this memo. Everything below is designed backwards from one requirement: **a good instructor should make the class 25% better, and a mediocre one should only make it 15% worse.** If the spread between your best and worst instructor is larger than that, you don't have a product — you have a talent agency.

7.1 What comparable programs actually do

[FACT] Outschool's model is a **marketplace**: no employment, 30% take, teachers self-train, and quality is managed by reviews plus a policy layer (on-platform-only comms, recording restrictions, no personal data collection). Retention guidance to teachers is generic — pathways, follow-ups, milestones (Outschool retention handbook).

[JUDGMENT] The lesson from the franchise tutoring models (Kumon, Mathnasium) is not their training hours — it's their **method rigidity**. Both scale on undergraduate labor by making the *sequence* non-negotiable and the instructor's job narrow: diagnose, place, coach, encourage. Neither asks an instructor to design anything. BOW should copy this posture exactly. The curriculum is not a document you give an instructor; **it is a runtime the software executes and the instructor narrates.**

7.2 The BOW instructor system

Sourcing. Target: college juniors/seniors and first/second-year teachers, 20–24, at NY-area schools; sports management, economics, and education programs; campus basketball analytics clubs; former camp counselors with a sports background. Camp counselors are the highest-yield pool — they've already been screened for working with kids and self-select for energy. **Do not over-index on basketball expertise.** The software carries domain knowledge; it cannot carry the ability to run a room of twelve 11-year-olds.

Audition (30 minutes, structured, identical for every candidate).

- **Minutes 0–5:** Candidate explains **opportunity cost to an 11-year-old using any sports example**, in under two minutes. Cold, no prep. *(Tests: translation, concision, whether they lecture.)*

- **Minutes 5–15: The Bad Trade Roleplay.** You play a 12-year-old who just proposed an obviously terrible trade and is emotionally committed to it. The candidate must get you to see the cost **without telling you it's bad and without letting you do it un-examined.** *(This is the single most predictive exercise. It's also the exact moment from Section 3 that drives the rebuy.)*
- **Minutes 15–22: The Chaos Room.** You play three kids at once: one dominating, one silent, one derailing to an unrelated LeBron argument. Candidate has 90 seconds of "class" to run. *(Tests: airtime management under pressure.)*
- **Minutes 22–27: The Wrong Answer.** Candidate must respond to a student's confidently incorrect claim in a way that keeps the student engaged. *(Tests: correction without deflation.)*
- **Minutes 27–30:** Candidate asks you questions. *(Tests: curiosity about the kids vs. about the pay.)*

Score each block 1–5 on a written rubric. **Hire on blocks 2 and 3 only**; treat the rest as tiebreakers.

Paid training (6 hours, paid at full rate). Paying for training is non-negotiable — it's cheap, it filters for seriousness, and it's the first signal of how you'll treat them.

- **Hour 1:** Safety, mandatory reporting posture, the communication rules, recording policy, the "no 1:1 ever" rule. First, not last, because ordering signals priority.
- **Hour 2:** The economics — only the six concepts the course actually teaches, and exactly how deep to go. Explicit permission to say "I don't know, let's look it up."
- **Hours 3–4:** The software as commissioner. How to run it, what it decides, and — critically — **what to do when it decides something a student thinks is unfair.** That moment happens every cohort.
- **Hour 5:** The five facilitation skills below, drilled.
- **Hour 6:** The Session 3 moment. How to make the cap bind, how to elicit the sentence, how to send the parent text. Rehearsed to script.

Dry run. Every new instructor runs **Session 1 and Session 3** with 4–6 recruited kids (siblings, pilot alumni, staff kids) while the founder observes silently and scores on the same rubric. **Session 3 specifically**, because it's the one that carries the business. No instructor takes a paying cohort without passing a dry run.

What the software must encode so a mediocre instructor still runs a good class — this is the actual product spec:

1. **A per-minute session runsheet** on the instructor's screen: what's on students' screens now, what to say, what's next, elapsed vs. planned. Not a lesson plan — a teleprompter.
2. **The gate:** the software refuses to execute any trade until the student has entered a one-sentence rationale. The instructor cannot skip this even if behind schedule.

3. **Guaranteed constraint binding in Session 3.** The scenario generator must ensure every student hits a cap wall. Not emergent — enforced.
4. **Airtime telemetry:** a live panel showing which students have spoken/acted and which haven't, with a nudge after 8 minutes of silence. This mechanically fixes the #1 failure of weak facilitators.
5. **Canned Socratic prompts** surfaced contextually — when a student makes a trade, the instructor's screen shows two questions specific to *that* trade. This is how you make a mediocre instructor sound insightful.
6. **Auto-drafted parent texts.** The Session 3 message is pre-written with the student's own sentence inserted; the instructor reviews and taps send. Never composes.
7. **Auto-assembled recap.** Zero instructor writing beyond selecting which of six sentences to feature.

Quality monitoring without surveillance. Recordings exist for safety, but do not review them routinely — that's surveillance and it corrodes trust with a young workforce. Instead monitor **four proxy signals the software already produces**: session attendance curve (the week-2-to-4 attrition shape is the strongest quality proxy you have); rationale-sentence completion rate and median length; airtime distribution (Gini across students); and the end-of-course "would you take another BOW course?" question asked of *students*, not parents. Any instructor whose cohort deviates by more than ~1 SD on attrition or airtime distribution gets a **coaching conversation, not a recording review.** Review recordings only on a safety flag or a parent complaint, and say so in the instructor handbook.

Pay structure. $400/cohort base (10.5 paid hours, ~~$38/hr),~~ ~~**plus $75 if the cohort finishes with ≥85% of enrolled students attending session 6**~~, ~~plus~~ ~~**$100 if 2+ families re-enroll**~~ ~~within 60 days. Total upside $575 (~~$55/hr). **[JUDGMENT] Deliberate design choice:** the bonus is on **completion and re-enrollment, not on parent satisfaction scores.** Satisfaction scores are gameable by charming instructors and reward the wrong thing. Attendance-through-session-6 is nearly impossible to fake and is exactly the outcome you want.

Realistic funnel. For a part-time $7K/year role in a NY-metro college labor market: **~60 applications → ~20 phone screens → ~10 auditions → ~4 offers → ~3 accept → ~2 survive the dry run and a full first cohort.** Roughly **30:1 applicants-to-productive-instructors** in year one, improving to maybe 15:1 once you have alumni referrals and a track record. Budget ~12 founder-hours per instructor hired. Expect ~50% annual churn at this pay level — **hire 4 for cohort 4 even if you only need 2.**

7.3 The five facilitation skills, and how to test them in 30 minutes

#	Skill	What bad looks like	Audition block that tests it
1	**Withholding the answer**	Explains why the trade is bad within 15 seconds	Block 2 (Bad Trade Roleplay) — time-to-first-verdict
2	**Redistributing airtime**	Rewards the loudest kid; never names the silent one	Block 3 (Chaos Room) — did they address the silent kid unprompted?
3	**Translating abstraction to the concrete**	Says "opportunity cost" and moves on	Block 1 — did they use a specific player and a specific number?
4	**Correcting without deflating**	"No, that's wrong" or, worse, "Great!" to a wrong answer	Block 4 (Wrong Answer) — did the student stay in the conversation?
5	**Holding the constraint**	Bends the cap rule to keep a kid happy	Block 2, second half: candidate is told the kid is upset and asks for an exception

[JUDGMENT] Skill 5 is the one that most separates good from bad BOW instructors and the one most people fail. A warm, likable, kid-loving candidate will instinctively relieve a child's frustration — and in doing so will destroy the exact scarcity that makes the class teach economics. Test it explicitly, and screen for the candidate who can be *warm about* a constraint without *removing* it. Include this scenario in the dry run too; auditions can be gamed once, dry runs less so.

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8. THE HARD QUESTION: IS REPEAT-COURSE PURCHASE RATE THE RIGHT NORTH STAR?

The case FOR it

It is the only metric that simultaneously proves the product worked, the child wanted more, the parent judged it worth money *twice*, and the business can survive without paid acquisition. Every softer metric (NPS, satisfaction, attendance) can be high while the business is dead. Section 5's LTV table shows the entire viability question resolves on R: at R=25% you cannot buy a customer; at R=50%+ you have a real business. It is also **the discipline that forces you to build Course 2**, which is the highest-value strategic act available to you and which you have not yet done. Committing to repeat rate as the north star makes that omission impossible to ignore.

The case AGAINST it

1. **It is statistically unmeasurable at your scale for at least nine months.** Three cohorts of 14 is 42 families. If the true repeat rate is 40%, an observed result of 12/42 (28.6%) versus 22/42 (52.4%) is well within normal sampling variation. You will read noise as signal and change the product in response to a coin flip. **This is the decisive objection.**
2. **It is a lagging indicator with a 3–6 month latency.** By the time you measure the repeat rate of cohort 1, you've already run cohorts 2–5. It cannot steer you.
3. **It is confounded by supply, not demand.** If there's no Course 2, R=0 no matter how good you are. If Course 2 is scheduled at an inconvenient time, R collapses for logistics. You will be measuring your operations calendar and calling it product-market fit.
4. **It measures the parent, not the child.** A parent can rebuy for childcare reasons and a child can love it and not rebuy because the family moved on to travel basketball. The signal you actually want — *did the kid ask to go back* — is upstream and observable immediately.
5. **Optimizing for it too early biases toward the wrong product.** The straightest path to high repeat is a **subscription**, which is precisely what Synthesis did with Teams at $95/month. If you take repeat rate seriously as a north star, the honest conclusion may be that BOW shouldn't be a six-week course at all. **[JUDGMENT] That is a real strategic question, and I'd note it and defer it until after cohort 3 — but the fact that your closest true comparable independently converged on a subscription is evidence you should not dismiss.**

The metric set I would actually run

One north star (aspirational, reported quarterly, never used to steer weekly): **Repeat-Course Purchase Rate**, defined as: *% of families who enroll in a second BOW course within 90 days of completing their first.* Report it. Do not optimize it before you have 100 families.

Three operating metrics for cohorts 1–3 — chosen because they are (a) measurable at n=14, (b) leading, and (c) causally upstream of repeat:

1. Session-6 Attendance Rate — *% of enrolled students present at the final session.* This is your best available proxy for repeat purchase, it's measurable immediately, and industry data says programs lose 20–35% by week four (Kodely).

- Cohort 1 target: **≥80%** | Cohort 2: **≥85%** | Cohort 3: **≥88%**
- Below 75%: stop selling and fix the middle of the course before running another cohort.

2. Kid-Ask Rate — *% of families where the parent affirmatively reports the child asked, unprompted, to continue or to do it again.* Captured by one SMS after Session 5: *"Has [name] asked you about doing this again? Yes / No / Not sure."* This is the "my kid asked to go back" signal made measurable, it arrives before the purchase decision, and it is the single best predictor of both repeat and referral.

- Cohort 1: **≥50%** | Cohort 2: **≥60%** | Cohort 3: **≥65%**
- Below 40%: the product is not working, regardless of what parents say in exit interviews.

3. Referral Actions per Family — *count of (a) playable link shares, (b) waitlist signups attributed to a family, (c) named referrals.* Given 75% of parents rely on word of mouth and Section 5's math showing referral is your only viable acquisition channel, this is the metric that determines whether the business can grow at all.

- Cohort 1: **≥0.5/family** | Cohort 2: **≥0.8** | Cohort 3: **≥1.0**
- At 1.0 with any repeat rate above ~35%, you are self-sustaining and don't need ads. That is the actual finish line.

Two guardrails, tracked but not optimized:

- **Founder hours per cohort.** Cohort 1: ≤25 (fine, you're learning). Cohort 3: **≤10**. Cohort 6: **≤4**. If this doesn't fall, nothing else matters — you built a job.
- **Fill rate.** % of seats sold against 14. Cohort 1: ≥70% (10/14). Cohort 3: ≥85% (12/14). Below 60% twice in a row means the price or the funnel is wrong, and Section 1's counterargument was right.

The decision rule for the end of cohort 3 [JUDGMENT]:

- Session-6 attendance ≥85%, Kid-Ask ≥60%, Referrals ≥0.8 → **raise price to $179, hire instructor #2, build Course 2 immediately.**
- Attendance ≥80% but Kid-Ask <45% → **the class is fine and the emotional payload isn't landing.** Fix Session 3 before anything else.
- Referrals <0.4 with good attendance → **the product works and it's not shareable.** The problem is the artifact and the playable, not the class.

- Fill rate <60% at $99 → **you have a demand problem, not a pricing problem**, and no price change fixes it. Go back to Section 4 and work the private-cohort channel instead.

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9. THE SIX DECISIONS THIS MEMO ASKS YOU TO MAKE

1. **Price at $149** ($99 founding, capped and time-boxed; $1,799 private cohort of 14, sold hard in November). Raise the online minimum from 8 to 10.
2. **Move the first paid cohort to January**, and treat October 1 as the ship date for the product and the free playable, not for revenue. October is a demand trough; January is a peak and the trade deadline is a free narrative hook.
3. **Cut the recap from seven sections to two** ("The Decision" + "The Season"), make it a shareable web page not a PDF, ship it at Session 5, and spend the reclaimed engineering on the **Session 3 constraint-binding moment and the parent text**, which is what actually drives the rebuy.
4. **Build the free playable to collect literally nothing** — no account, no server-side storage of the kid's sentence — and make it single-player only. Never give away multiplayer negotiation.
5. **Ship Option A (Structured Offer Protocol)** for online negotiation. Adopt the full safety standard in 6.3 before cohort 1, and publish it — it's a distribution asset in a tight referral network, not just compliance.
6. **Do not run paid acquisition in year one.** The arithmetic forbids it below a proven 50% repeat rate. Build the referral loop instead, and hold yourself to Referral Actions per Family ≥1.0 by cohort 3.

The two things most likely to kill this [JUDGMENT]: (a) founder hours per cohort never falling below 10, which turns a company into a job; and (b) reaching cohort 3 with no Course 2 built, which makes your north star metric structurally equal to zero. Both are entirely within your control and both are decided by what you build between now and October 1.

SOURCES

- Outschool Educator Handbook — Refining Your Pricing Strategy
- Outschool Educator Handbook — Retention Strategies That Work
- Outschool Educator Handbook — Is There a Market for Your Class?
- Outschool — Learner Safety and Privacy: For Teachers
- Outschool — Sports Analytics category | Fantasy Basketball class | Sports Management camp
- Brighterly — Outschool Pricing 2026 | myeLearningWorld — Outschool Pricing
- Contrary Research — Outschool Business Breakdown
- Synthesis Teams | Synthesis Tutor

- Wharton Global Youth — Costs & Aid | Sports Business Academy
- Northwestern Pre-College — Business of Sports
- Sports Management Worldwide — Courses
- iD Tech — Virtual Programs
- Acton Children's Business Fair
- IES — Fantasy Sports Math League | SportsData.io case study | Seattle Times — Fantasy football in math class
- Afterschool Alliance / Wallace — America After 3PM
- Kodely — After-School Program Retention
- Project Play — State of Play 2025 | Youth Sports Business Report
- Sogolytics — Word of Mouth: The Sales Channel for Parents
- Baymard — Cart Abandonment Rate Statistics
- Prizmah via JTA — Jewish day school enrollment rising | Times of Israel
- White & Case — Unpacking the FTC's COPPA Amendments | Davis Wright Tremaine — FTC Amended COPPA Rule | Securiti — COPPA Final Rule Amendments | Promise Legal — VPC Methods
- US Dept. of Education — To which institutions does FERPA apply?
- NYSED — Education Law §2-d Definitions | FindLaw — NY EDN §2-d | Cybernut — NY Ed Law 2-d Explained
- SafeKidsThrive — Screening & Background Checks | NCYS Recommended Guidelines | UGA Two-Adult Rule
- INTA — Fair Use of Trademarks | Harness IP — CBC v. MLBAM | 8th Circuit opinion
- ZipRecruiter — Youth Programs Instructor pay | Stripe fees 2026agentId: a1a0a0e8089bb56c1 (use SendMessage with to: 'a1a0a0e8089bb56c1', summary: '<5-10 word recap>' to continue this agent) subagent_tokens: 134328 tool_uses: 74 duration_ms: 788855

pedadogy.md

BOW ECONOMICS — LEARNING DESIGN MEMO

Standards, cognition, and the honest limits of what 10–14 year olds can be taught in six hours

Convention used throughout: `[SOURCED]` = drawn from cited research. `[JUDGMENT]` = my professional inference, defensible but not directly evidenced. I have kept these separate deliberately, because several of the most useful conclusions here are judgment calls and you should be able to see which ones.

Headline finding first, because it changes a design decision you have already made: the meta-analytic evidence for "experience first, then name it" is *positive for grades 6–10 and negative for grades 2–5*. Your 5–6 class straddles that boundary. Your hard rule needs a grade-band-specific variant or the younger class will underperform a boring lecture. Details in §7. Everything else is refinement; that one is a fix.

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1. STANDARDS GROUNDING

1.1 The framework

`[SOURCED]` The Council for Economic Education's **Voluntary National Content Standards in Economics (2nd ed., 2010)** contains **20 standards**, with benchmarks written at **grades 4, 8, and 12**. (CEE PDF; ERIC digest ED428031) The grade-8 benchmarks are the right bar for both your bands — for 5–6 it is a stretch target, for 7–8 it is the on-level expectation.

1.2 What the arc genuinely hits

Std	Grade-8 benchmark language `[SOURCED]`	BOW week	Verdict
1 Scarcity	"The opportunity cost of an activity is the value of the best alternative that would have been chosen instead. It includes	W1	**Direct hit.** Note the benchmark explicitly includes *time and other resources*, not just money — see §4, misconception 1.

Std	Grade-8 benchmark language [SOURCED]	BOW week	Verdict
	what would have been done with the money spent and the time and other resources used in undertaking the activity."		
3 Allocation	"Scarcity requires the use of some distribution method to allocate goods, services, and resources, whether the method is selected explicitly or not."	W1, W6	**Direct hit, and underexploited.** The draft, the cap, and free agency are three *different* allocation mechanisms sitting in your product unnamed. This is free standards coverage.
4 Incentives	"Responses to incentives are usually predictable because people normally pursue their self-interest… Changes in incentives usually cause people to change their behavior in predictable ways. Incentives can be monetary or non-monetary, or both."	W3, W5, W6	**Direct hit.** W6 is essentially a live demonstration of this benchmark.
5 Trade	"When people buy something, they value it more than it costs them; when people sell something, they	W3	**Bullseye.** This benchmark *is* the trade deadline lesson. Use this sentence nearly

Std	Grade-8 benchmark language [SOURCED]	BOW week	Verdict
	value it less than the payment they receive."		verbatim in the W3 naming slide.
8 Role of Prices	"An increase in the price of a good or service encourages people to look for substitutes, causing the quantity demanded to decrease."	W4	**Direct hit.** Note "look for substitutes" — the causal mechanism is named in the benchmark. Teach the mechanism, not just the direction.
2 Decision Making	"To determine the best level of consumption of a product, people must compare the additional benefits with the additional costs of consuming a little more or a little less." Plus: "Marginal benefit is the change in total benefit resulting from an action."	W1, W4	**Partial.** Your arc is built on *total* comparisons ("this player or that player"), not *marginal* ones ("one more roster spot / one more dollar of ticket price"). Marginal thinking is the single most transferable idea in introductory economics and it is currently implicit.
7 Markets and Prices	"Market prices are determined through the buying and selling decisions made by buyers and sellers."	W4	**Partial — and this is a real gap.** See §1.3.
10 Institutions	Grade-8 language is bank/financial-intermediation focused ("Banks and other financial institutions channel funds from	W6	**Misses the literal benchmark, hits the standard's spirit.** W6 is institutional analysis of a governance body, not

Std	Grade-8 benchmark language [SOURCED]	BOW week	Verdict
	savers to borrowers and investors").		financial intermediation. Do not claim Std 10 without qualification.

1.3 What you miss that a parent or school will ask about

Gap 1 — SUPPLY. This is the big one. You teach demand in W4 (P×Q, demand shifters, elasticity). You never teach supply, and therefore you never teach market equilibrium. A parent hears "economics course" and expects "supply and demand." Right now you have half of the most famous diagram in the subject.

[JUDGMENT] **This is free to fix and you already have it.** W1 free agency *is* a market with a fixed short-run supply (a finite pool of free agents) and 16 competing buyers. That is price discovery under scarce supply. You are running a live auction market and calling it "constraints." Name it. One slide in W1: "the supply of elite centers this summer was four. Sixteen of you wanted one. What happened to the price?" That gets you Std 7 ("market prices are determined through the buying and selling decisions made by buyers and sellers") in ninety seconds using an experience the students already had.

Gap 2 — COMPETITION, and it is worse than a gap; it is an active hazard. [SOURCED] CEE Standard 9's grade-8 benchmark reads: "Competition among sellers results in lower costs and prices, higher product quality, and/or better customer service." In BOW, "competition" means *athletic* competition — zero-sum, one team wins. In economics, market competition is the mechanism that produces gains for buyers. [JUDGMENT] If you never disambiguate these, students will leave your course *more* confident in a wrong idea than when they arrived, because you spent six weeks reinforcing the zero-sum frame with a scoreboard. This directly feeds the zero-sum bias problem in §4. The fix is one deliberate contrast in W1 or W3: "On the court, competition means somebody loses. In the free-agent market, competition among *you sixteen buyers* made players better off. Those are two different words that sound the same."

Gap 3 — Profit, revenue, and costs. Std 2's marginal benefit/cost benchmark and the general expectation that a business unit involves costs. W4 as described is P×Q — that is **revenue**, not profit. See §4 for why this is not a minor omission.

Gap 4 — Money and inflation (Std 11), interest rates (Std 12), income (Std 13), economic growth (Std 15), and all macro (Std 18–20). Missing entirely. [JUDGMENT] This is fine and you should say so proactively in parent materials: BOW is a **microeconomics and institutions**

course. Do not apologize for it; define the scope. Std 12 (interest rates / present value) is adjacent to your W3 "present vs future value" and you may claim a light touch there.

Gap 5 — Comparative advantage (Std 6). `[SOURCED]` Grade-8 benchmark: "international trade promotes specialization and division of labor and increases the productivity of labor, output and consumption." `[JUDGMENT]` You have specialization sitting right there (roster construction — you need a rim protector *and* a shooter, and no one player is both) but you do not have comparative advantage, which requires opportunity-cost-based reasoning across two producers. **My recommendation: skip it.** Comparative advantage is the single most-failed concept in introductory economics for adults. Attempting it in W3 would consume the session and produce a memorized ratio trick. Teach differing valuations instead — which is what actually drives your trade deadline, is genuinely the foundation of gains from trade, and is captured by the Std 5 benchmark above.

Gap 6 — Entrepreneurship (Std 14) and Role of Government / Market Failure (Std 16). `[SOURCED, verify before publishing]` My reading of the 2010 CEE document is that **Standards 14 and 16 have no grade-8 benchmarks — they jump from grade 4 to grade 12.** I retrieved this consistently but from an automated read of the PDF; have someone eyeball those pages before you put it in a parent-facing document.

`[JUDGMENT]` If that holds, it is a genuinely strong marketing fact and a genuine caution simultaneously. **The claim:** BOW's W5 (externalities, public goods, free riding) and W6 (institutional design) teach at a level CEE benchmarks at **grade 12**. **The caution:** that is not an accident of curriculum politics. Those concepts are benchmarked late because they require holding an individual decision and an aggregate consequence in mind at the same time, which is a formal-operational demand. This is exactly why §3 spends its length on them.

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2. DEVELOPMENTAL COGNITION

2.1 The numeracy floor — start here, because it constrains everything

`[SOURCED]` On the 2024 NAEP mathematics assessment, **41% of U.S. eighth-graders scored *below* NAEP Basic**, and only **27% scored at or above NAEP Proficient**. ([NAEP 2024 state snapshot, national figures]())

`[JUDGMENT]` Read that again with your 7–8 class in mind. Even allowing that an enrichment program self-selects for stronger students, you should design assuming that **a meaningful minority of your eighth-graders cannot fluently compute a percentage of a large number**

under time pressure in front of peers. Any design that requires them to do so is not teaching economics; it is filtering for arithmetic and then teaching economics to the survivors.

[SOURCED] The specific known wall is the additive-to-multiplicative transition: students systematically over-apply additive strategies to proportional problems (and later over-apply proportional strategies to additive ones). (Van Dooren et al., "From Addition to Multiplication… and Back"; relative reasoning transition study, ERIC ED658403)

2.2 Grades 5–6 (ages 10–12) — capability table

Capability	Realistic assessment [JUDGMENT, grounded in the sourced material above]	Design constraint
Percentages	Benchmark percentages only: 50%, 25%, 10%, and "about a third." Cannot reliably compute 12% of $47M.	Express everything as benchmark fractions of the cap. "This deal is a quarter of your whole payroll." Never require a percentage computation.
Ratios / proportional reasoning	Emerging and unreliable. "Team A won 7 of 10, Team B won 12 of 20" is a genuinely hard comparison at 10. Strong additive-strategy bias.	Never require comparing two ratios with different denominators. Pre-normalize all rates for them.
Negative numbers	Formally introduced around grade 6 in most U.S. sequences. A fifth-grader has not met them.	**Never show negative cap space to the 5–6 band.** Say "you are $4M over the line," never "-$4M." Show the line physically.
Probability / expected value	Can order likelihoods (likely / unlikely / coin-flip). Cannot multiply a probability by a payoff.	Use three named risk tiers with words, not numbers. "Safe / risky / long-shot." EV computation is out of range.
Multi-variable comparison	**2 variables, 3 at the very end of the course.**	See §2.4.
Counterfactual reasoning	Can *evaluate* a counterfactual handed to	The instructor or the interface must supply the

Capability	Realistic assessment [JUDGMENT, grounded in the sourced material above]	Design constraint
	them. Will very rarely *generate* one unprompted about a choice they are emotionally invested in.	counterfactual. "Here is what the other path would have given you." Do not expect them to ask.
Large dollar magnitudes	$30 million is a word, not a quantity. It sits in the same mental bin as $300 million.	Convert to a single unit and keep it constant all six weeks. My recommendation: everything in "cap slices" where one slice ≈ 10% of the cap. Money becomes counting.
Delayed consequence (4 weeks)	The causal link will not be *felt* without a physical bridging artifact. Four weeks is four sessions and roughly a month of elapsed life.	Requires a re-opened physical object. See §6, the Decision Receipt.

2.3 Grades 7–8 (ages 12–14) — what actually changes

`[SOURCED]` Steinberg et al. (2009) found age differences in future orientation and delay discounting across ages 10–30, with future orientation continuing to develop well into the mid-teens. (Child Development — I could not retrieve full text, so treat the specific trajectory as directional rather than precise.)

`[JUDGMENT]` The differences that matter for your design are **not arithmetic**:

- **Percentages:** functional, not fluent. The reliable wall is **percent *change* versus percent *of***. Do not build any lesson on percent change. This is the specific reason elasticity's formal definition is off the table for 7–8 (see §3).
- **Probability/EV:** can hold "half the time this works" and reason with it. Can do EV *if* the multiplication is friendly (50% of $20M) and *if* it is scaffolded on paper. Cannot fluently compare EVs across three options in eight minutes.
- **Counterfactual generation: this is the real developmental line.** A 13-year-old, given a prompt structure and two prior practices, will generate "but if we'd kept him we would have…" unprompted. A 10-year-old generally will not. Build your differentiation on this axis, not on math.

- **Second-order reasoning** ("what will the other GMs do in response to this rule?"): emerging and trainable at 12–14. Largely out of reach at 10–11.
- **Ambiguity tolerance:** 12–14 can work with deliberately incomplete information and understand that the incompleteness is the point. 10–12 tend to read missing information as a mistake by the adult.

2.4 How many simultaneous decision variables? A defensible number

`[SOURCED]` Adult working memory capacity is conventionally estimated at roughly **3–4 chunks**; capacity grows through childhood and adolescence rather than being adult-like from the start. (Gathercole et al., structure of WM 4–15 years; Cowan et al., chunk capacity growth)

`[JUDGMENT]` **My numbers: 2 for grades 5–6 (3 by week 5). 3 for grades 7–8 (4 by week 5).**

Three lines of reasoning, all of which point the same direction:

1. **Relations, not items.** Trading off *n* variables requires holding $n(n-1)/2$ pairwise relations. Three variables = 3 relations. Four = 6. Five = 10. The jump from 3 to 4 roughly doubles the reasoning load; 5 is not a harder version of 4, it is a different task.
2. **Novelty tax.** The chunk estimates above are for *familiar* material. "Cap space," "player option," and "trade exception" are not chunks for these students in week 1 — each consumes capacity that a fluent adult spends on the actual tradeoff.
3. **The environment tax.** These are live sessions with peers talking, a timer running, and social stakes. Whatever the quiet-room capacity is, subtract from it.

The operational rule: externalize everything. Every variable that is *on the card* costs nothing. Only variables that must be *traded against each other* count toward the budget. You can show a 5–6 grader ten facts about a player; you may only ask them to weigh two of them against each other.

2.5 How many novel technical terms per session?

`[SOURCED]` I looked for a clean experimental number and there is not one. The vocabulary-instruction literature is explicit that quality of instruction beats quantity, recommends that "teachers need to select carefully the most important words to teach explicitly each day," and for adolescent content-area learning recommends prioritizing discipline-specific importance over generic tier placement — but it declines to give a per-lesson count. (AdLit, Explicit Vocabulary Instruction; IRIS Center, Vanderbilt) Retention is predicted by **number of retrieval episodes across sessions**, not by exposure within one. (Fostering retention of word learning, PMC11056717)

[JUDGMENT] **My constraint: ONE primary term per session, plus at most two secondary terms.**

- **Primary** = must be retrievable and applicable to a novel case six weeks later. Gets a naming moment, a definition in student words, a transfer prompt, and retrieval in every subsequent session.
- **Secondary** = used correctly by the instructor in context, understood on hearing, never assessed for recall.

Your arc as written lists roughly **five terms per week — about thirty terms across six weeks.** [JUDGMENT] **That is a pedagogical flaw and it is the most consequential one in the document.** With one session per week, no homework, and no spaced retrieval between sessions, thirty terms produces thirty vaguely familiar words and zero owned concepts. The retention evidence above is unambiguous that spacing and retrieval count, and you have exactly six spaced opportunities total.

The fix, and it is the single highest-leverage change in this memo:

Week	PRIMARY (own it forever)	Secondary (recognize)
W1	**OPPORTUNITY COST**	scarcity, constraint
W2	**DECISION QUALITY vs OUTCOME**	risk, uncertainty
W3	**GAINS FROM TRADE** (via differing valuations)	incentive, present vs future value
W4	**REVENUE = PRICE × QUANTITY**	demand, demand shifter, (7–8 only: elasticity)
W5	**EXTERNALITY**	free rider, joint product
W6	**INCENTIVES vs RULES** (institutional design)	self-interest, collective outcome

Six owned concepts. Every one of them appears in every subsequent week's retrieval opener (§8, minutes 0–5). "Scarcity" as a secondary term is deliberate and will feel wrong to you — it is the word everyone expects. It is also nearly contentless on its own; opportunity cost is the operative concept and scarcity is its premise. Students who own opportunity cost will define scarcity correctly when asked. The reverse is not true.

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3. THE HARD ONES

3.1 Elasticity

Your question — is "when I raised the price, did total money go up or down?" a legitimate elasticity intuition or a distortion?

`[JUDGMENT, on standard economics]` **It is legitimate. It is the total revenue test, and it is a standard textbook operationalization of elasticity, not a kid-friendly softening of one.** If raising price lowered total revenue, demand at that point was elastic. That is not an analogy; it is the definition's direct consequence, and it is how the concept is actually *used*. You are on solid ground.

But there are three ways the reduced version becomes a lie, and two of them are common in "kid-friendly" materials.

1. **The property-of-the-good error (the common wrong version).** "Luxuries are elastic, necessities are inelastic." This is a correlate presented as a definition, and it teaches students that elasticity is a fixed attribute of a product. Elasticity varies *along a demand curve* — the same tickets are elastic at $180 and inelastic at $12. `[JUDGMENT]` Your simulation can demonstrate this and should: show a franchise that raised prices and gained revenue and another that raised prices and lost revenue, and let the class discover that the difference was *where they started*, not what they were selling.
2. **The direction error.** "Raising the price lowered our money, so lowering the price always raises money." Students will over-generalize from one trial to a rule. The instructional move is a second trial in the opposite direction on the *same* franchise.
3. **The revenue-maximization error.** Nothing in the revenue test says maximizing revenue is the goal. It is not — profit is. If W4 stops at P×Q you will have taught students that the right price is the revenue-maximizing price, which is wrong for any firm with costs. See §4.

Verdicts:

- **Grades 5–6: teach the revenue test. Do NOT say the word "elasticity."** They will experience it, name it as "the price trap," and own P×Q. Spending the one primary-term slot on a five-syllable Greek-derived word with no retrieval opportunities buys you nothing.
- **Grades 7–8: name it.** Define as "how strongly quantity reacts when price changes." Operationalize as the revenue test. **Do not do percent-change math** — that is precisely the documented wall (§2.3), and the formula would consume the session and teach arithmetic instead of economics. Add the point-on-the-curve caveat explicitly; it is what separates real understanding from a memorized rule.

3.2 Externalities

The honest reduced version: "A cost or a benefit that lands on somebody who wasn't part of the deal."

The common kid version that is economically wrong: "An externality is a side effect." `[JUDGMENT]` This is wrong and it is everywhere. A drug's side effect on the patient who took it is not an externality — the patient was party to the decision. The defining features are (a) the affected party had no say, and (b) the effect is not reflected in the price. Dropping those turns a precise concept into "stuff that also happens," which is worse than not teaching it, because students will confidently misapply it forever.

The diagnostic to teach instead of a definition: *Who was at the table, and who got the bill?* Two questions, both concrete, both answerable by a ten-year-old, and together they are the actual concept.

- **Grades 5–6: yes, teachable — the first-order version only.** "Your team tanked. That made the games *other* teams' fans paid for worse. Those fans weren't at the table when you decided." Ten-year-olds have vivid intuitions about being affected by decisions they had no vote in; this is developmentally *easy* territory emotionally, even though CEE benchmarks it late.
- **Grades 7–8: add the second-order consequence, which is the actual economics.** Because the cost does not land on the decision-maker, the decision-maker does **too much** of the thing. Not "it's bad" — "the quantity is wrong." That is the step that converts a moral observation into an economic one, and 12–14 can take it.

3.3 Revenue sharing

`[JUDGMENT]` This is the **easiest** of the four to teach honestly, and I suspect you have it filed as one of the hardest. It is a contribution game with a free-rider structure, and those run well with children — the intuitions are strong and the experience is fast.

- **Grades 5–6: run the contribution game, name FREE RIDER, stop.** Do not also name "joint product" and "externality" in the same session; you have one primary slot.
- **Grades 7–8: teach the tradeoff, which is the honest hard part.** Sharing revenue makes small markets viable *and* weakens every team's incentive to build its own market. Both are true simultaneously. That tension is the lesson.

The wrong kid version: "Revenue sharing is fair, so it's good." `[JUDGMENT]` This smuggles a normative claim past the students and skips the incentive cost entirely. It also reinforces the fair-equals-equal misconception (§4). If your W5 leaves students thinking sharing is unambiguously good, you have taught a political position rather than an economic concept.

3.4 Institutional design

The honest reduced version: "Rules change what people do — so when you write one, ask what people will do to get around it."

That is the core intuition of mechanism design and it is not a simplification that costs anything.

- **Grades 5–6: one-step only.** "If we pass this rule, what will the other fifteen teams do next week?" Concrete, single-move, tractable.
- **Grades 7–8: the real thing.** "Design a rule that still works even if every GM is completely selfish." That is a second-order design task and it is genuinely where 7–8 earns more GM power.

The wrong version, and it is the most important distortion in the entire arc: "Good rules make things fair." `[JUDGMENT]` Institutional design is about **aligning private incentive with collective outcome**. Fairness is a separate question that reasonable people answer differently. Conflating them does two kinds of damage: it teaches a wrong definition, and it activates the fair-equals-equal misconception in the one week you had a chance to attack it.

The developmental caution on W6: holding "what I want" and "what happens if all sixteen of us want that" simultaneously is a two-level operation, and it is the formal-operational demand that gets Std 16 benchmarked at grade 12. `[JUDGMENT]` **For 5–6, do not ask them to hold both levels — let the projector hold the second one.** Vote. Show the aggregate result on screen. Then ask: "You each voted for what was best for your team. Look at what we got. Is that what you wanted?" The screen performs the aggregation the cognition cannot yet perform internally. This is the best possible use of your projector and it is the reason W6 works at all for the younger band.

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4. MISCONCEPTIONS CATALOG

#	Misconception	Evidence	The instructional move
1	**Opportunity cost = money spent**	`[SOURCED]` Ferraro & Taylor surveyed 199 professional economists on a multiple-choice opportunity cost question; **21.6% answered correctly**	Force the **named alternative**. Every decision receipt has a field: "What I gave up: _______" and a blank name is not an acceptable answer. "Money" is not an

#	Misconception	Evidence	The instructional move
		— below the 25% chance rate. (summary via Marginal Revolution) [SOURCED] Potter & Sanders counter that every option was defensible and the question was ill-posed. (Southern Economic Journal) [JUDGMENT] Both facts are useful to you: the concept is genuinely slippery even for experts, *and* you should not assess it with multiple choice.	acceptable answer. It must be a specific forgone thing. CEE's own grade-8 benchmark says opportunity cost includes what would have been done with the *time and other resources* — quote it.
2	**Sunk cost — "we already paid him $20M so we have to play him"**	[JUDGMENT] Universal and emotionally powerful in a franchise sim where students are attached to their own prior choices.	The **new-GM reframe**: "You were just fired. The new GM walks in today. She didn't sign him. What does she do?" This strips the sunk cost without accusing the student of an error, which matters for a 10-year-old's willingness to change position in front of peers.
3	**Revenue = profit**	[SOURCED] Berti, Bombi & De Beni taught profit to 68 third-graders; experimental groups	**This is a design flaw in W4, not just a misconception.** As specified, W4 teaches P×Q and stops. That *is*

#	Misconception	Evidence	The instructional move
		"progressed, but not dramatically," and the study identified persistent obstacles to acquiring the notion. (Sage) Profit is documented as hard.	revenue. If costs never appear, you will manufacture the misconception you are trying to prevent. **Fix: give every franchise a fixed operating cost line in W4.** One number. Students compute revenue, subtract it, and discover that the revenue-maximizing price and the profit-maximizing price are not the same thing. This also fixes the elasticity distortion in §3.1.
4	**Price is set by cost ("it costs them $8 to make, so the fair price is $10")**	`[SOURCED]` CEE Std 7 grade-8 benchmark directly contradicts this: "Market prices are determined through the buying and selling decisions made by buyers and sellers." `[JUDGMENT]` This is the default lay theory of price and children hold it strongly.	**The identical-cost contrast.** Two franchises, same arena cost, wildly different ticket prices and both are right. Then ask what actually differs — it is the buyers, not the costs. Your 16-franchise display makes this trivial to stage.
5	**Trade has a winner and a loser (zero-sum bias)**	`[SOURCED]` Meegan documents zero-sum bias as perceived competition despite unlimited resources; the bias is	The evidence says **explicit statement works** — so state it explicitly, do not hope it emerges. **After** a trade, have both GMs separately write

#	Misconception	Evidence	The instructional move
		unidirectional (people over-infer others' gains as their losses, but not the reverse), and **explicit information about the non-zero-sum structure reduced it** (Exp. 3). He notes economists credit economics education with public acceptance of trade. (Frontiers in Psychology)	which asset they now value more. When both say "mine," show both answers on the screen. Then give them the Std 5 benchmark sentence verbatim. [JUDGMENT] **And fix the standing hazard:** your scoreboard reinforces zero-sum every single week. You must actively counter-program it, not just avoid reinforcing it.
6	**Outcome bias — a decision is good if it worked out**	[SOURCED] Baron & Hershey (1988) established outcome bias in decision evaluation. (PubMed) A recent large replication (**N = 692**) found the effect **larger** than the original: d = 0.77–1.10 versus the original 0.21–0.53 — and critically, it **persisted even among participants who explicitly stated that outcomes should not be considered when evaluating decisions.**	[JUDGMENT] That replication finding is the most important single result in this memo. **Knowing the rule does not remove the bias.** So a W2 lesson where students learn to *say* "judge the decision, not the outcome" will produce students who say it and do not do it. The only move with a chance is **repeated structured practice on cases where the answer is uncomfortable**: the same decision, two franchises, two

#	Misconception	Evidence	The instructional move
		([International Review of Social Psychology](#))	different outcomes, shown side by side, with students forced to rate the *decision* before the outcome is revealed and again after — and confronted with their own change. Do this in W2 and again in W3, W5, W6. It is not a week's topic; it is a recurring drill.
7	**"Fair" means "equal"**	`[JUDGMENT]` Deeply held at 10–14 and it will dominate W5 and W6 unless directly addressed.	Separate the two questions **out loud and in writing**. Whiteboard split down the middle: "Does this rule get us the outcome we want?" / "Is this rule fair?" Students answer both, separately. Do not let one answer serve for both. Some of the best moments in W6 will be a student saying "it works but I don't like it" — that is a sophisticated position and it should be praised as such.
8	**Externalities are just "side effects"**	See §3.2.	The two-question diagnostic: *Who was at the table? Who got the bill?*

#	Misconception	Evidence	The instructional move
9	**Risk = badness; a risky choice that failed was a stupid choice**	[JUDGMENT] Related to #6 but distinct — it is a misconception about *variance*, not about *evaluation*.	Name the three tiers before the decision and have students **pre-commit in writing** to which tier they are choosing and why. When it fails, the receipt shows they knew and chose. This is also the classroom-management tool that prevents the W2 blowup (§8).
10	**Incentives = rewards**	[SOURCED] CEE Std 4 grade-8: "Incentives can be monetary or non-monetary, or both."	Collect non-monetary incentives from the students themselves in W6 — playing time, reputation, draft position, avoiding embarrassment in front of the class. They will generate a better list than you would.
11	**More options is always better**	[JUDGMENT] Directly relevant to your §9 differentiation question.	Deliberate contrast in W1: one franchise with cap space and many options frequently does *worse* than a constrained one, because the constrained team was forced to be decisive. Show it. This also teaches that constraints are not purely bad — a

#	Misconception	Evidence	The instructional move
			genuinely economic idea.
12	**Present vs future value = "later is worse"**	[JUDGMENT] Students collapse discounting into impatience.	Force the reverse case: a rebuilding franchise where the *future* asset is correctly worth more. If every W3 lesson rewards win-now, you have taught a bias, not a concept.

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5. TRANSFER

5.1 What makes a transfer prompt fake

[JUDGMENT] Three tests. A prompt must pass all three.

1. **The mapping test.** Can you write the correspondence element by element — what plays the role of the constraint, the alternatives, the decision-maker, the affected party? If you can only say "it's kind of like that," it is surface similarity.
2. **The necessity test.** Is the concept *required* to answer? If a sensible person answers correctly with common sense alone, you are testing common sense.
3. **The discrimination test.** Would the answer change if the concept did not apply? If the same answer works either way, it is a vocabulary quiz in costume.

The most common fake: any prompt containing money and a choice gets labeled "opportunity cost." "You have $20, do you buy a game or a hoodie?" fails test 2 — the student answers by preference, never identifies a forgone alternative, and learns nothing. The repair is to add a **binding constraint that makes the forgone thing painful and specific**: "You have $20 and both the game and the hoodie cost $20. The hoodie is gone after Friday. The game is on sale until Sunday." Now the alternative has to be *named* to reason at all.

5.2 Transfer prompts by concept

1. Opportunity cost

- You have one Saturday. Soccer tryouts, your cousin's birthday, and a friend's sleepover are all that day. Pick one — then name the *one* thing you gave up that hurts most. (Not "the other two.")
- Your class earned one field trip. Aquarium or science museum. What did the winners give up? What did the losers give up? Are those the same thing?
- Homework hour: forty minutes on the essay you will be graded on, or forty minutes on the project you actually care about.

2. Decision quality vs outcome

- Your friend didn't study, guessed on the test, and got a B. Was that a good decision? Now: you studied for three hours and got a C. Was *that* a good decision?
- You wore a helmet every day for a year and never crashed. Was the helmet a waste?
- A classmate ran a risky play in gym that worked. The coach praises them. A different classmate ran the same play and it failed. Same coach. Should the praise be different?

3. Gains from trade / differing valuations

- You trade lunch items with a friend. Both of you walk away happier. Where did the extra happiness *come from* — who lost it?
- You and your sibling split chores. You take dishes, they take the dog. Nobody did less work. Why is it better?
- Selling a used bike: the buyer values it more than the money, you value the money more than the bike. Both true at once. Who won?

4. Revenue = price × quantity

- School fundraiser. Raise the candy price from $1 to $2 and half as many people buy. Did you make more money? Now raise it from $1 to $1.25 and almost everyone still buys.
- Babysitting: raise your rate from $10 to $15/hour and get three fewer jobs a month.
- The school play has 200 seats. Cheap tickets fill it; expensive tickets half-fill it. Which raises more money? Now: which one *should* you choose if the play cost $400 to stage?

5. Demand shifters

- The same hoodie, same price, sells out in October and sits in June. Nothing about the hoodie changed. What did?
- A snack bar next to the gym on game night versus the same snack bar on a Tuesday.

6. Externality

- One kid talks through the movie. Who paid for that? Were they at the table when the decision was made?

- Your group project partner does nothing. The grade is shared. Who bears the cost of their choice?
- Someone leaves the shared bathroom a mess. Name the person who got the bill.

7. Free riding / joint product

- Group project, five people, one grade. Predict what happens and why. Now design a rule that fixes it — and predict how someone would get around your rule.
- Everyone in the neighborhood benefits from the park being clean. Nobody wants to be the one picking up trash.
- A class earns a pizza party if the *class average* hits 85. Who studies harder, and who studies less?

8. Incentives

- A teacher offers extra credit for turning in work early. Predict three things students will do — including one the teacher didn't intend.
- A video game rewards playtime instead of skill. What do players start doing?
- Parents pay per book read. Predict what happens to book length.

9. Institutional design

- Your class writes its own phone rule. Write it, then write how a clever classmate gets around it, then fix it.
- Design a fair way to pick teams in gym that works even when the captains only care about winning.
- A group chat has a rule against spam. It fails. Why? Design one that works when nobody is being supervised.

10. Present vs future value

- $20 now or $30 in three months. Now: you need $20 today for a ticket that expires tomorrow. Did your answer change? Should it have?
- Skip practice today to finish the project due tomorrow. What are you borrowing against?

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6. ASSESSMENT WITHOUT A FAKE SCORE

6.1 The strongest argument for your position is empirical

`[SOURCED]` In Schwartz & Bransford's studies, the **verification measure (true/false statements about the target concepts) was at ceiling across all conditions** — it could not distinguish students who had deep understanding from those who did not. The **prediction task** — asking students to predict outcomes in a novel situation — separated the conditions cleanly (0.44 versus 0.11 probability of relevant predictions). (A Time for Telling, PDF)

`[JUDGMENT]` This is the whole argument, and it is stronger than "we philosophically object to scores." Recognition-format assessment is not merely a weak measure of conceptual understanding — in a well-run experiment it was **incapable of detecting a large real difference.** A multiple-choice item would have told you both groups learned equally. They had not. **BOW should assess by prediction and by application to novel cases, because those are the formats that demonstrably discriminate.** Put that sentence in your parent-facing material.

6.2 Weak versus strong evidence

Weak (do not count)	Strong (count)
Names the term	Applies it to a case with no NBA content in it
Picks it out of four options	**Identifies the forgone alternative unprompted**
Defines it back in your words	Defines it in their own words *and* correctly rules out a non-example
Says "judge the decision not the outcome"	**Rates a failed decision as good, or a successful one as bad, and defends it**
Agrees when you raise the counterfactual	**Revises their own earlier judgment** when the counterfactual is raised
Says trade helps both sides	Explains *why* both gain — differing valuations — without prompting

`[JUDGMENT]` The strongest single signal across every concept, and the cheapest to observe: **unprompted use of the concept in a week when it is not the topic.** A student who says "wait, that's an opportunity cost" during W4 has transferred. That is worth more than any W1 exit ticket.

6.3 Three-level performance descriptions — observable behavior only

Opportunity cost

- *Emerging:* States what they chose. Describes cost as money. Names no alternative.

- *Solid:* When asked "what did you give up?", names a specific alternative rather than "money" or "the other stuff."
- *Strong:* Names the forgone alternative **without being asked**, and identifies non-money costs (a roster spot, future flexibility, a year of a player's prime).

Decision quality vs outcome

- *Emerging:* Judges by result. "It worked, so it was smart."
- *Solid:* When prompted with what was known at the time, changes the judgment.
- *Strong:* **Volunteers** that a decision was sound despite failing, or unsound despite succeeding, and cites what was knowable at the time — including about their own decision, in front of peers.

Gains from trade

- *Emerging:* Frames the trade as won or lost.
- *Solid:* Says both teams can benefit.
- *Strong:* Explains that both benefited **because they valued the assets differently**, and can name what each side valued more.

Revenue = P × Q

- *Emerging:* Believes higher price means more money.
- *Solid:* Predicts that quantity will fall and checks the product.
- *Strong:* Predicts direction **before** the reveal, and distinguishes revenue from profit unprompted.

Demand shifters

- *Emerging:* Attributes all demand changes to price.
- *Solid:* Names a non-price factor that moved demand.
- *Strong:* Distinguishes "we changed the price" from "the thing people wanted changed" in a novel case.

Externality

- *Emerging:* "It's a side effect."
- *Solid:* Identifies someone affected who was not part of the decision.
- *Strong:* Says the decision-maker therefore does **too much or too little** of it, and correctly rejects a non-example.

Free riding

- *Emerging:* Calls it unfair.

- *Solid:* Predicts free riding before it happens.
- *Strong:* Proposes a rule and then **predicts how someone would beat their own rule.**

Incentives

- *Emerging:* Incentives are rewards.
- *Solid:* Names non-monetary incentives.
- *Strong:* Predicts an **unintended** response to a rule.

Institutional design

- *Emerging:* Evaluates rules by fairness only.
- *Solid:* Separates "does it work" from "is it fair" and answers both.
- *Strong:* Designs a rule that survives self-interested play, and names its cost.

Present vs future value

- *Emerging:* Later is worse.
- *Solid:* Explains why waiting can be right.
- *Strong:* Correctly identifies a case where the future asset is worth more **and** one where it is not.

6.4 Keeping it lightweight — three instruments, no scores

`[JUDGMENT]`

1. The Decision Receipt (20 seconds, every session). One index card per franchise per decision. Four fields: *What we chose. What we gave up. What we expect to happen. How sure we are (safe / risky / long-shot).* Signed by every group member.

This card does five jobs at once: it is the opportunity-cost assessment; it is the pre-commitment that makes W2's decision-versus-outcome lesson possible; it is the physical bridging artifact that makes a four-week delayed consequence *felt* (§2.2) — you hand back the week-1 card in week 5 and the causal link becomes real; it is the participation-equity mechanism, since every member signs; and it is the parent-conference evidence.

2. The 4-Student Sample (instructor, during work time). Do not observe sixteen students. Pick **four** per session, rotating, and note one thing each: emerging / solid / strong on this week's primary concept. Over six sessions each student is sampled roughly 1.5 times, and every student has six receipts. That is enough for an honest narrative report and it costs the instructor nothing they were not already doing while circulating.

3. The Front Office Review, distributed. `[JUDGMENT]` **Your W3 defense is the best assessment instrument in the product and it currently appears once.** A student defending a decision under questioning produces exactly the strong evidence in §6.2 — unprompted counterfactuals, revised judgments, distinctions between decision and outcome. Running it once wastes it.

Fix: run a 90-second version every week from W2 onward. One franchise, two questions from the instructor: *"What would you have gotten instead?"* and *"If that had failed, would it still have been the right call?"* Rotate franchises. W3 remains the full-length event. This costs you 90 seconds a session and roughly triples your evidence base — and it gives the W3 event a rehearsed format so it does not collapse into stage fright.

What you report to parents: a short narrative per concept — "Maya reliably identifies what her franchise gave up, and in Week 5 she was the first student to point out that a successful trade had been a poor decision" — plus two photographed decision receipts. No number. That report is more credible than a score, and you should say so directly: *we do not produce a decision-quality number because the research shows recognition-style measures cannot distinguish real understanding from surface familiarity.*

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7. THE "NAME IT AFTER THEY LIVE IT" SEQUENCE

7.1 What the research actually says

`[SOURCED]` **Sinha & Kapur (2021)**, meta-analysis of problem-solving-before-instruction (PS-I) versus instruction-first: **53 studies, 166 comparisons.** (Review of Educational Research)

- **Conceptual knowledge and transfer: g = 0.36** [0.20–0.51], favoring experience-first.
- **Procedural knowledge: g = −0.03** [−0.20, 0.15] — **no advantage.**
- **Grade-level moderator (the critical one for you): grades 2–5, g = −0.09 (favoring instruction-first); grades 6–10, g = +0.50; undergraduates, g = 0.28.**
- **Domain-general skills: g = −0.17 — experience-first was *worse*.** Domain-specific subjects favored PS-I.
- **The four design features that predicted success:** instruction that **builds on student-generated solutions** (0.56 vs 0.20); **group work** during the problem phase (0.49 vs 0.19); **multiple solution generation** (0.47 vs 0.16); **dialogue-dominant** instruction (0.55 vs 0.24).

`[SOURCED]` **Schwartz & Bransford** — analyzing **contrasting cases** followed by a lecture beat both analyzing twice with no lecture and summarizing text followed by the same lecture. Neither discovery alone nor telling alone worked; the combination did. Recognition measures were at ceiling and detected nothing. (PDF)

`[SOURCED]` **Concreteness fading** — progressing from concrete representations to abstract ones supports transfer better than either concrete-only or abstract-only. (Fyfe, McNeil, Son & Goldstone, systematic review, Educational Psychology Review; Fyfe et al. 2012)

7.2 What this means for BOW — including the correction

Your instinct is well supported for the 7–8 band. g = 0.50 for grades 6–10 on conceptual knowledge and transfer, which is exactly what you care about, is a strong result.

`[JUDGMENT]` **Three corrections, in order of importance.**

Correction 1 — the 5–6 band needs guided, not naked, productive failure. Grades 2–5 show a *negative* effect. Your fifth-graders sit inside that band and your sixth-graders sit at its edge. `[JUDGMENT]` The most plausible mechanism is that younger students lack the domain knowledge and self-regulation to structure an open problem, so their exploration time is spent figuring out *what the task is* rather than generating conceptual variation.

The fix preserves your product rule. For 5–6, hand them the **structure** of the decision — here are your two options, here is your constraint, here is your deadline — and withhold only the **concept**. They still experience it before they can name it. What you remove is the unstructured search space, which is the part that hurts younger learners. For 7–8, you can leave the problem more open and let them define the options themselves. **This is also your best differentiation axis, and it is free** (§9).

Correction 2 — do not let W2 become a domain-general "decision-making skills" lesson. The meta-analysis found experience-first was *negative* for domain-general skills (g = −0.17). If W2 becomes "how to make good decisions in life," you are in the worst-supported cell of the entire evidence base. Keep it concrete and domain-specific: *this roster decision, these outcomes, this league.* The generality arrives through the transfer prompts in §5, after the naming — not by teaching generality directly.

Correction 3 — your naming slide cannot be pre-written. The strongest moderator (0.56 vs 0.20) was **instruction that builds on student-generated solutions.** A pre-baked slide deck that says "OPPORTUNITY COST: the value of the next best alternative" delivers roughly the 0.20 version. The 0.56 version writes the students' own words on the board first and then attaches the term to them. **Product implication: your `/teach` flow must have a capture step between the reveal and the naming, and the naming screen must have an editable field that the instructor fills with student language during the session.** If the slide is fixed, you have engineered away most of your effect size.

7.3 Timing

[JUDGMENT] For a 65-minute session, based on the design features above rather than on any single study's duration:

- **Experience phase before naming: 12–18 minutes** of consequential decision plus reveal.
- **Not shorter than 12:** below that, groups produce one solution instead of several, and multiple-solution generation is a 0.47-vs-0.16 moderator.
- **Not longer than 18:** with one instructor and sixteen students, longer means drift, and you still need ten minutes of naming plus ten of transfer.
- **Naming: 10 minutes, dialogue-dominant** (0.55 vs 0.24), not lecture-dominant. The instructor talks less than the students during naming.

What happens if you name too early:

1. Students stop generating alternatives and pattern-match to the given procedure — you lose the multiple-solutions moderator directly.
2. The term becomes a label for a procedure rather than a lens for a situation. Students will use it correctly on the worksheet and never again.
3. You destroy the contrast set. The word "opportunity cost" is meaningful only against a background of *different* choices; naming first collapses that background into one right answer.
4. [SOURCED] Schwartz & Bransford: telling without prior differentiated knowledge did not produce prediction ability. The lecture is not wasted because lectures are bad; it is wasted because there is nothing in the student's head for it to organize.

7.4 Exploiting the 16-franchise projector — this is your best asset

[JUDGMENT] You have accidentally built a textbook contrasting-cases affordance and you should treat it as the core of the product, not a visualization feature. Seven rules:

1. **Simultaneous, never sequential.** The entire mechanism is noticing what varies. Sequential reveal is storytelling; simultaneous display is contrast.
2. **Curate to 4–6 cases, not 16.** Sixteen franchises differing on everything is noise. [JUDGMENT] Contrast requires a controlled comparison — most dimensions held constant, one varying. Your software should support *selecting and sorting* far more than it supports *showing everything*.
3. **Vary one dimension at a time.** Sort the display by the variable that is this week's concept. If the axis is not obvious in two seconds from the back of the room, it is not working.
4. **Prediction before reveal.** Students commit in writing to what they think happened before the screen shows it. This is Schwartz & Bransford's discriminating measure and it is also what creates the "time for telling."

5. **Student description before your naming.** Ask "what's different between these two?" and write their words on the board. This is the 0.56 moderator, operationalized. Do not skip it to save five minutes; it is the most valuable five minutes in the session.
6. **Build the near-miss pair — this is the highest-value display you can ship.** Two franchises that made the **same decision** and got **different outcomes**, side by side. `[JUDGMENT]` This is the only instructional move I know of with a real chance against outcome bias, which the replication data (d = 0.77–1.10, persisting even in people who explicitly disavow it) says is otherwise nearly immovable by instruction. If you build one custom display this quarter, build this one, and make it reusable in W2, W3, W5, and W6.
7. **Fade the concreteness across the six weeks.** W1 shows franchise logos, player faces, dollar signs. W6 shows an abstract 2×2 of incentive structures with the franchises as anonymous dots. `[SOURCED]` That progression — concrete to abstract on the *same* underlying structure — is the concreteness-fading finding, and it supports transfer better than staying concrete throughout. Your six-week arc is exactly the right length to execute one full fade, and right now nothing in the design does it.

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8. PACING REALITY

8.1 Minute budget — 65-minute session, 12–16 students, one instructor

Min	Block	What happens	Non-negotiable
0–5	**Retrieval + consequence**	"Last week you decided X. Here is what happened." Plus one retrieval question on a *prior* week's primary term.	The retrieval question. It is your only spacing mechanism and retention depends on it.
5–10	**Setup**	State the decision and the constraint. **Do not name the concept.**	Under 5 minutes. If setup takes longer, your mechanics are too complicated.
10–25	**DECISION** (15 min)	Small groups, 3–4 per franchise, visible countdown timer.	The timer. See §8.2.

Min	Block	What happens	Non-negotiable
25–33	**REVEAL** (8 min)	Curated contrasting cases. Students predict in writing before the screen changes.	Prediction before reveal.
33–38	**STUDENT DESCRIPTION** (5 min)	"What's different between these two?" Instructor writes student words on the board.	This block. It is the single highest-value five minutes and the first thing an instructor will cut.
38–48	**NAMING** (10 min)	Explicit teach. Term attached to the words already on the board. Definition, one non-example, CEE benchmark language where it fits.	Dialogue, not monologue.
48–58	**TRANSFER** (10 min)	Non-NBA prompt from §5. Pair-share, then three shares out.	Non-negotiable per your own product rule.
58–65	**RECEIPT + 90-second Front Office Review**	Decision receipts completed and signed. One franchise takes two questions.	The "what did you give up" field.

Totals: decision time 15, discussion and description 18, direct instruction 10, transfer 10, framing and admin 12.

`[JUDGMENT]` For a **75-minute** session, add 5 to the decision block and 5 to transfer. Do not add to naming — a 15-minute naming block does not produce more learning than a 10-minute one, it produces a lecture. Do not add to the reveal — more cases is worse, not better.

8.2 What kills a session like this

`[JUDGMENT]` In rough order of likelihood.

1. **The simulation eats the economics.** The overwhelmingly most common failure. Trading is fun, kids negotiate for thirty minutes, the instructor does not want to interrupt genuine engagement, and the session ends with no concept named. **Fix: a visible countdown timer that the instructor is not permitted to extend. Ever.** Make this an explicit instructor rule, not a suggestion. A session that reaches minute 33 without a reveal is a lost session and no amount of enthusiasm recovers it.
2. **Rules explanation bloat.** Twelve minutes explaining a new mechanic kills week 3. **Fix: identical decision interface all six weeks; only the content changes.** By week 2, mechanics should be re-explainable in 90 seconds. Every new mechanic you add costs you student thinking time at a 1:1 rate.
3. **One kid is the GM and three watch. Fix: distinct named roles holding *different information*** — the cap sheet, the scouting report, the fan/revenue data — so the decision is impossible without all of them talking. Every member signs the receipt. This is also a differentiation lever (§9).
4. **An illegible reveal.** A spreadsheet on a projector teaches nothing. **Fix: one variable, sorted, large, readable from the back row.**
5. **Absence.** Once a week for six weeks means somebody misses week 2 and their franchise is incoherent. **Fix: franchises are *groups*, not individuals**, and the minute 0–5 recap re-establishes state for everyone every week.
6. **Emotional blowup over a bad outcome.** Real, and likelier than you expect with competitive kids and consequential decisions. `[JUDGMENT]` **Your W2 placement is correct and I want to say so explicitly — decision-quality-versus-outcome in week 2 is not merely good sequencing, it is classroom-management infrastructure.** It gives the instructor a legitimate, curricular, non-condescending response to "this isn't fair, we lost." Do not move it later. If anything, seed it in W1's closing minute.
7. **The instructor answers their own questions.** Under time pressure with sixteen kids, silence feels like failure and adults fill it. **Fix: scripted wait time in the `/teach` flow** — an actual instruction that says "wait 7 seconds" — and a naming screen that *cannot be advanced* until the instructor has typed in student language. Make the product enforce the pedagogy.

8.3 The honest capacity estimate

`[JUDGMENT]` **Six sessions × one primary concept = six concepts genuinely owned, plus roughly a dozen secondary terms recognized.** That is the realistic ceiling and it is a good outcome — six transferable economic concepts is more than most adults hold. Any plan promising more than that is promising exposure and calling it learning.

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9. GRADE-BAND DIFFERENTIATION

9.1 Is "more GM power, not harder math" pedagogically sound?

Yes. Largely. With one important caveat. Three reasons:

1. `[JUDGMENT]` **The 7–8 advantage over 5–6 is representational, not computational.** What actually changes between 10 and 14 is capacity to hold more variables, tolerate ambiguity, reason about others' reasoning, and generate counterfactuals unprompted (§2.3). Harder arithmetic exercises none of these.
2. `[JUDGMENT]` **Harder arithmetic makes the economics *harder to learn*, not richer.** Computation consumes the same working memory the economic reasoning needs. A student computing 14% of $118M is not thinking about the tradeoff — they are doing long multiplication. That load is extraneous to your learning objective.
3. `[SOURCED]` **41% of eighth-graders scored below NAEP Basic in 2024.** `[JUDGMENT]` Differentiating upward via arithmetic will exclude a substantial share of your 7–8 class from the economics entirely — the precise opposite of differentiation. You would be sorting, not teaching.

The caveat, and it matters: "more power" must mean more *authority*, not more *buttons*. If 7–8 gets twelve options where 5–6 gets four, you have increased search cost, not depth — and you have blown through the variable budget in §2.4. The right form is **fewer, deeper, more consequential choices with a heavier obligation to justify them.** A seventh-grader making one irreversible decision they must defend under questioning is doing far more cognitive work than one making six reversible ones.

9.2 Differentiation matrix

Axis	**Grades 5–6**	**Grades 7–8**
Problem structure (§7.2, Correction 1)	Options given. Constraint stated explicitly. Concept withheld.	Options partly self-generated. Constraint present but must be inferred from the data.
Information load	3–4 data fields visible per player. Pre-normalized rates.	6–8 fields. Some raw, requiring interpretation.
Variables traded off	**2** (3 by W5)	**3** (4 by W5)
Numeric form	Benchmark fractions of cap. "Cap slices." No negatives, no percent computation.	Whole percentages, negatives permitted. **No percent-change.**

Axis	Grades 5–6	Grades 7–8
Ambiguity	Complete information. There is a defensible answer.	Deliberately incomplete. Some information exists but must be requested; some does not exist at all.
Counterfactual burden	Instructor supplies the alternative. Student evaluates it.	**Student generates the alternative** before the reveal, in writing.
Order of reasoning	First-order: "what will happen to us?"	**Second-order: "what will the other fifteen GMs do in response, and what happens then?"**
Argument standard	State the choice and name one thing given up.	State the choice, name what was given up, **and concede the strongest cost of their own decision.**
Time horizon	Consequences within 2 weeks, with the physical receipt as the bridge.	4-week horizons; must explicitly price the future against the present.
Reversibility	Most decisions adjustable next week.	At least one irreversible decision per franchise per course.
Vocabulary depth	Recognize and use in context.	Define, apply to a novel non-NBA case, and **identify a non-example.**
Front Office Review	2 friendly scripted questions, prepared in advance.	4 questions including one adversarial and one unseen.

9.3 Week-by-week "more GM power"

Week	**5–6**	**7–8**
W1	Choose from 3 pre-vetted free agents.	Set the offer *price*, competing against other franchises for a scarce pool. Someone loses a player by underbidding.

Week	5–6	7–8
W2	Respond to one shock.	Respond to a shock *and* pre-commit to a contingency: "if X happens, we will do Y." Then be held to it.
W3	Trade from a menu of proposed deals.	Originate proposals. Negotiate live. May be refused and must return with a revised offer.
W4	Choose one of three ticket prices.	Set the price freely, **and pay a fixed operating cost** — must find profit, not revenue.
W5	Vote on a sharing rule.	**Propose** the sharing formula, then defend it against the franchise it hurts most.
W6	Vote, then read the aggregate consequence off the projector.	**Write** the rule, predict how it gets gamed, and revise it once before the vote.

`[JUDGMENT]` Note the pattern: at every step the 7–8 version adds **authority, ambiguity, and obligation to justify** — and at no step does it add arithmetic. That is the sound version of your instinct.

SUMMARY OF FLAWS AND FIXES

#	Flaw	Fix	Priority
1	~30 technical terms across 6 sessions is unretainable with no spacing or homework	1 primary + 2 secondary per session; 6 owned concepts; retrieval opener every week (§2.5)	**Highest**
2	Experience-first is contraindicated for	Guided productive failure for 5–6: give	**Highest**

#	Flaw	Fix	Priority
	grades 2–5 (g = −0.09); your 5th graders are in that band	the structure, withhold the concept (§7.2)	
3	W4 teaches P×Q with no costs — manufactures the revenue/profit misconception you want to prevent, and licenses revenue-maximization as the goal	Add one fixed operating-cost line in W4 (§4 #3)	**High**
4	No supply anywhere; parents expect supply and demand	Name W1 free agency as a market with scarce supply and 16 competing buyers (§1.3)	**High**
5	"Competition" means athletic competition all six weeks, reinforcing zero-sum bias against CEE Std 9	One deliberate contrast between athletic and market competition (§1.3, §4 #5)	**High**
6	The Front Office Review — your best assessment instrument — runs once	90-second version weekly from W2; W3 stays the full event (§6.4)	**High**
7	Naming slides risk being pre-written, forfeiting the strongest moderator (0.56 vs 0.20)	`/teach` must capture student language between reveal and naming; naming screen has an editable field (§7.2)	**High**
8	No concreteness fading across the arc	Plan the W1-concrete to W6-abstract	Medium

#	Flaw	Fix	Priority
		progression deliberately (§7.4 rule 7)	
9	Outcome bias treated as one week's topic; replication data says knowing the rule does not remove it	Recurring near-miss display in W2, W3, W5, W6 (§4 #6, §7.4 rule 6)	Medium
10	Four-week delayed consequence will not be felt at ages 10–12	Decision Receipt as physical bridging artifact, handed back (§6.4)	Medium
11	"Different starting positions" lands in W1, before any trust is built, on the most emotionally loaded design element	Pair it immediately with W1's constraint-can-help contrast (§4 #11), and seed decision-vs-outcome in W1's final minute	Medium
12	Marginal thinking (CEE Std 2) is implicit only	One marginal frame in W4: "what does one more dollar of price get you?"	Low

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SOURCES

- CEE, Voluntary National Content Standards in Economics, 2nd ed. (2010)
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- Sinha & Kapur (2021), When Problem Solving Followed by Instruction Works, *Review of Educational Research*
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- Meegan (2010), Zero-Sum Bias, *Frontiers in Psychology*

- Ferraro & Taylor (2005) via Marginal Revolution summary · Potter & Sanders critique, *Southern Economic Journal*
- Berti, Bombi & De Beni (1986), Acquiring Economic Notions: Profit
- Gathercole et al. (2004), Structure of Working Memory 4–15 Years · Cowan et al. (2009), chunk capacity growth
- Steinberg et al. (2009), Age Differences in Future Orientation and Delay Discounting
- Van Dooren et al., From Addition to Multiplication… and Back · Relative reasoning transition, ERIC ED658403
- NAEP 2024 Mathematics, Grade 8 national figures
- AdLit, Explicit Vocabulary Instruction · IRIS Center, Vanderbilt · Retention and retrieval episodes, PMC11056717

Two verification items before this goes to a parent or school: (1) confirm by eye that CEE Standards 14 and 16 lack grade-8 benchmarks — my read was automated, and the claim is load-bearing for the "we teach above grade level" positioning; (2) I could not retrieve full text for Steinberg et al. (2009) or the Cowan chunk-capacity paper, so treat those trajectories as directional.agentId: a422ba8a63fa3ec75 (use SendMessage with to: 'a422ba8a63fa3ec75', summary: '<5-10 word recap>' to continue this agent) subagent_tokens: 110010 tool_uses: 39 duration_ms: 662346 nba_cba.md

I have comprehensive sourcing. Writing the memo now.

BOW ECONOMICS — NBA FINANCIAL & ROSTER RULES RESEARCH MEMO

Prepared: September 4, 2026 · **League year covered:** 2026-27 · **CBA in force:** 2023 NBA/NBPA CBA (term through 2029-30; mutual opt-out June 30, 2029, notice by Oct 15, 2028)

Convention used throughout: every hard number carries `[DATA DATE | CONFIDENCE | SOURCE]`. Blocks marked **SOURCED FACT** are directly attributable. Blocks marked **INFERENCE** are my arithmetic or reasoning on top of sourced numbers. Blocks marked **UNCONFIRMED** are not publicly verified.

Critical sourcing caveat up front: Larry Coon's CBA FAQ (cbafaq.com) — the canonical resource you named — **is retired.** Coon posted a retirement notice and the live FAQ is frozen at the 2017 CBA; he explicitly declined to endorse a successor. `[DATA DATE: retrieved Sept 4, 2026 | HIGH | http://www.cbafaq.com/salarycap.htm]` Do not build the product on it. The 2017 text is still correct on Bird-rights mechanics and cap holds, but **wrong or silent on everything apron-related.** The best current substitutes I found: The CBA Guide (2026-27 figures, CBA article citations) and Over the Apron (publishes its own accuracy/uncertainty ledger). Neither is official. The official text lives at NBA.com/NBPA.

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1. CURRENT HARD NUMBERS

1.1 The 2026-27 threshold set — OFFICIAL

SOURCED FACT. The NBA set these on June 30, 2026, effective 12:01 a.m. ET July 1, 2026.

Item	2026-27
Salary Cap	**$164,961,000**
Minimum Team Salary (floor, 90% of cap)	**$148,465,000**
Luxury Tax Level (121.5% of cap)	**$200,428,000**
First Apron	**$209,015,000**
Second Apron	**$221,686,000**

[DATA DATE: June 30, 2026 | VERY HIGH | https://pr.nba.com/2026-27-salary-cap/ and https://www.nba.com/news/nba-salary-cap-2026-27-season]

INFERENCE (arithmetic): tax = 121.5% of cap exactly ($164.961M × 1.215 = $200.4276M). First apron ≈ 126.7% of cap; second apron ≈ 134.4% of cap. Ratio confirmed against the 2023-24 base year. The "121.5%" rule is stated explicitly by [The CBA Guide].

1.2 The 10% smoothing collar — did it bind?

SOURCED FACT (rule): the cap "can never be less than the Prior Year's Salary Cap and can never increase more than 10%" — "Cap Smoothing," added in the 2023 CBA specifically to prevent a repeat of the 2016 media-deal spike. [DATA DATE: page mod. Mar 25, 2026 | HIGH | https://cbaguide.com/finances/]

INFERENCE + SOURCED: For 2026-27 the collar did **NOT** bind. $164,961,000 ÷ $154,647,000 = **+6.669%**. Sportico independently reports "a 6.7% gain." [DATA DATE: July 1, 2026 | VERY HIGH | https://www.sportico.com/leagues/basketball/2026/nba-escrow-players-salaries-owners-million-1234937798/]

By contrast, 2025-26 **did** hit the ceiling exactly: $140,588,000 × 1.10 = $154,646,800 ≈ $154,647,000. **INFERENCE, high confidence.**

Why the shortfall matters pedagogically (SOURCED): teams had penciled in another 10%. The cap came in low because of "the floundering regional sports network market" / "a reduction in local media revenue." John Hollinger (The Athletic), quoted by Sportico: "Teams had budgeted for another 10% rise, but now must change their projections downward for the luxury tax and aprons." [DATA DATE: July 9, 2026 | HIGH | https://www.sportico.com/leagues/basketball/2026/nba-salary-cap-constraint-explain-jaylen-brown-1234938410/]

1.3 2027-28 projection

SOURCED but UNOFFICIAL: the league's June 2026 memo to teams projects a 2027-28 cap of **$174 million, +5.5%**. Bobby Marks confirms the memo and the 5.5% figure independently. [DATA DATE: July 9 & Aug 3, 2026 | MEDIUM-HIGH | Sportico (above); https://www.espn.com/nba/story/_/id/49490878/] **Had the cap grown 10%/yr, 2027-28 would have been $187M — $13M higher.**

What would confirm: the NBA's official press release circa June 30, 2027 (pr.nba.com), which follows the final audit report due June 30.

1.4 Exceptions — 2026-27

Exception	Amount	Min–Max yrs	Raises
Non-Taxpayer MLE (NTMLE)	**$15,044,000**	1–4	5%
Taxpayer MLE (TMLE)	**$6,064,000**	1–2	5%
Room MLE	**$9,366,000**	1–2	5%
Bi-Annual Exception (BAE)	**$5,477,000**	1–2	5%
Early Bird max start	**$15,235,500** (or 175% of prior salary, whichever greater)	2–4	8%
Estimated Average Player Salary (EAPS)	**$15,163,000**	—	—
Cash-in-trade limit (5.15% of cap)	**$8,495,000**	—	—

MLE/BAE amounts: `[June 30, 2026 | VERY HIGH | pr.nba.com]` for the three MLEs; BAE `[July 2026 | HIGH | https://www.salaryswish.com/bi-annual-exception]` and corroborated by [CBA Guide amounts page](). Term lengths and raise percentages: `[Mar 2026 | HIGH | https://cbaguide.com/contractanatomy/terms/contractlimitations/]`

Note the 2023-CBA change that trips people up: the **taxpayer MLE is now 2 years max, not 3**, and the **room MLE is 2 years max**. Bird/Early Bird get 8% raises; *everything else, including sign-and-trades, gets 5%.*

Full-value math (INFERENCE): NTMLE over 4 years = $64,689,200. TMLE over 2 = $12,431,200. Room MLE over 2 = $19,200,300. BAE over 2 = $11,227,850.

1.5 Minimum salary scale

YOS	2026-27 salary
0	**$1,357,763**
1	$2,185,116

YOS	2026-27 salary
2	**$2,449,421**
3	$2,537,526
4	$2,625,627
5	$2,845,883
6	$3,066,143
7	$3,286,399
8	$3,506,659
9	$3,524,115
10+	**$3,876,529**

[DATA DATE: July 2026 | HIGH | Hoops Rumors "NBA Minimum Salaries For 2026/27," via https://www.yardbarker.com/nba/articles/nba_minimum_salaries_for_202627/s1_14822_44016395]

SOURCES DISAGREE — flagging: SalarySwish's minimum-salary table shows a different, higher set ($1,432,161 rookie / $4,088,942 for 10+). **Use the Hoops Rumors set.** Three independent cross-checks: (a) $1,357,763 × 1.0667 back-solves cleanly from 2025-26's $1,272,870; (b) the 2026-27 two-way salary is $678,882 = exactly half of $1,357,763 [Yardbarker/Hoops Rumors two-way tracker]; (c) LeBron James's actual 2026-27 first-year salary on his 76ers minimum deal is **$3,876,529** [July 24, 2026 | HIGH | https://www.cbssports.com/nba/news/lebron-james-76ers-contract-explained/]. SalarySwish's own prose uses the Hoops Rumors numbers even though its table doesn't — its table appears to be a 2027-28 projection mislabeled.

The two-year-veteran cap-charge rule (SOURCED, important): if a team signs a veteran with 3+ years of service to a **one-year** minimum contract, the contract counts against cap, tax, apron **and trade matching** at only the **2-YOS minimum ($2,449,421)**; the league reimburses the team the difference. [Hoops Rumors; corroborated by Over the Apron citing Art. VII §3(f), Art. IV §6(h)] **The subsidy applies only to one-year deals.** LeBron's two-year deal is the live counterexample: Philadelphia carries the full $3,876,529.

1.6 Maximum salaries

Tier	% of cap	2026-27 first-year max
0–6 YOS	25%	**$41,240,250**
7–9 YOS	30%	**$49,488,300**
10+ YOS	35%	**$57,736,350**

`[DATA DATE: 2026 | HIGH | https://www.salaryswish.com/maximum-salary-faq; identical figures at https://cbaguide.com/resources/amounts/]`

Escalators: a 0–6 YOS player with 4+ years of service can reach 30% via All-NBA/DPOY/MVP criteria; a 7–9 YOS Designated Veteran can reach 35%. Floor: a player can never be offered less than 105% of his prior final-year salary under the max rules.

INFERENCE (the Bird premium in dollars): a 35% max signed with **full Bird rights** = 5 years, 8% raises = $57,736,350 × 5.80 = **$334,870,830**. The same max signed by any other team = 4 years, 5% raises = × 4.30 = **$248,268,305**. That ~$86.6M gap *is* the product's core lesson in one line.

1.7 2026 rookie scale (120% of scale — what first-rounders almost always sign)

Pick	Player	2026-27 salary
1	AJ Dybantsa	**$14,748,000**
2	Darryn Peterson	$13,195,320
3	Cameron Boozer	$11,849,760
4	Caleb Wilson	$10,683,720
5	Keaton Wagler	$9,674,760
26	Sergio de Larrea	$3,076,920
27	Chris Cenac Jr.	$2,988,120
28	Joshua Jefferson	$2,969,520
29	Alex Karaban	$2,948,280

Pick	Player	2026-27 salary
30	Koa Peat	**$2,926,800**

[DATA DATE: July 2026 | HIGH | https://www.yardbarker.com/nba/articles/rookie_scale_salaries_for_2026_nba_first_round_picks/s1_14822_44021884]

Structure: 4 years = 2 fully guaranteed + 2 team options. Teams may sign 80–120% of scale; ~always 120%. An unsigned first-rounder's **cap hold is the 120% figure**, applied immediately on selection.

INFERENCE: the #1 pick costs ~5.0× the #30 pick. That ratio is a clean, real number for a draft-lottery unit.

1.8 Roster rules

SOURCED FACT [Mar 2026 | HIGH | https://cbaguide.com/eligibility/rosters/]:

- **Max standard roster: 15. Minimum: 14** on Active+Inactive during the regular season. A team may drop to 12–13 for at most **2 consecutive weeks and 28 total days** per season.
- **Active list: 12–15** (11 allowed for the same 2-week/28-day window). Minimum 8 on the bench.
- **Two-way contracts: up to 3 per team**, do not count toward the 14/15. **$678,882** each for 2026-27; eligible to be **active for up to 50 of 82 games**; **ineligible for the postseason** unless converted to a standard contract by the last day of the regular season; may not be signed after ~March 4.
- **Hardship:** a team may sign 10-day contracts as a **16th+ player** under NBA hardship rules (real 2025-26 example: the Pacers carried a 19th and 20th player, Jeremiah Robinson-Earl and Cody Martin).
- Offseason max: 21 including two-ways.
- **Built-in escalator:** if league-wide average employment hits 14.25 players/team for two consecutive seasons, the minimum roster becomes 15. If it hits 14.5, the NBPA can cut two-way slots from 3 to 2.
- Players waived after 11:59 p.m. March 1 are postseason-ineligible with a new team.

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2. THE "SAME LINE" MECHANIC — CAP SPACE VS EXCEPTIONS

2.1 The two kinds of team

There is no "cap space" as an account. There is a **team salary number** and a **threshold**. A team is a *room team* or an *exception team*, and it cannot be both in the same instant.

- **Room team (team salary < $164,961,000):** signs anyone up to its room. It has **no** Bird/Early Bird/Non-Bird/MLE/BAE — those are cap **exceptions**, and you don't need an exception when you're under the cap. Once it spends its room down, it gets the **Room MLE ($9,366,000, 2 years)** and the minimum exception — nothing else.
- **Exception team (team salary ≥ cap):** cannot sign an outside free agent for a dollar more than its largest exception (**$15,044,000** NTMLE start), but can re-sign **its own** players over the cap without limit up to the max, if it holds Bird rights.

The order-of-operations trap that makes this a *mechanic*: to *use* room, a team must first **renounce or renounce-equivalent** its cap holds and outstanding exceptions — and renouncing a Bird free agent is **irreversible for that cap year**. A team that renounces LeBron to create room cannot later re-sign him over the cap. That's why Bobby Marks framed the Lakers' summer as a choice: "up to $47 million [in room] **but at the expense of** James, Rui Hachimura and Luke Kennard," versus "bypass using cap space and retain their free agents… while also having their $15 million non-tax midlevel exception available." `[June 28, 2026 | HIGH | https://www.espn.com/nba/story/_/id/49190331/]`

2.2 Cap holds — the four kinds

SOURCED FACT `[https://cbaguide.com/thresholds/teamsalary/]`. Team salary includes:

(a) Free-agent cap holds, sized by prior contract type:

Prior contract	Hold
Rookie scale, coming off 4th year, prior salary **below** EAPS ($15,163,000)	**300%** of prior salary
Rookie scale, coming off 4th year, prior salary **above** EAPS	**250%**
Rookie scale, coming off 2nd or 3rd year	the rookie scale amount
Full Bird UFA, prior salary **below** EAPS	**190%**

Prior contract	Hold
Full Bird UFA, prior salary **above** EAPS	**150%**
Early Bird UFA	**130%**
Non-Bird UFA	**120%**
Minimum contract	the new minimum amount (capped at 2-YOS min)
Two-way	0-YOS minimum
Restricted FA	greatest of the applicable UFA hold, the qualifying offer, or a first-refusal match amount

Holds are floored at the player's minimum and ceilinged at his max.

(b) First-round pick holds: 120% of scale, from the moment of selection.

(c) Incomplete roster charge: if fewer than 12 counted players, add the **rookie minimum ($1,357,763)** for each spot short of 12. Offseason only.

(d) Exception holds: a team under the cap carries its unused NTMLE/BAE/DPE/trade exceptions against team salary unless renounced. The CBA Guide's worked case: a team $18M under the cap with the $15.0M NTMLE and $5.5M BAE outstanding is actually **$1.5M over** the cap until it renounces them.

Real 2026 example: Memphis holds a **$28.9M trade exception** from the Jaren Jackson Jr. trade; "the exception counts against the cap, unless it is renounced" — which is precisely why Memphis "is still likely to act as an over-the-cap team." `[June 28, 2026 | HIGH | ESPN tiers piece]`

2.3 Bird / Early Bird / Non-Bird

	Service requirement	Max start	Years	Raises
Full Bird ("Qualifying Veteran FA")	3 seasons without clearing waivers or changing teams as a FA	the **maximum salary**	1–**5**	**8%**

	Service requirement	Max start	Years	Raises
Early Bird	2 seasons, same condition	greater of **175% of prior salary** or **$15,235,500**	**2**–4	**8%**
Non-Bird	everyone else who finished the season on the roster	greater of **120% of prior salary**, 120% of the minimum, or the qualifying offer	1–4	**5%**

`[Service requirements and structure: cbafaq.com/salarycap17.htm (2017 CBA text, unchanged on this point) + https://cbaguide.com/contractanatomy/terms/contractlimitations/ | HIGH]`

What kills the rights (SOURCED):

- **Clearing waivers** or **changing teams as a free agent** resets the clock.
- **Renouncing** the hold destroys the rights for that cap year.
- **Signing with another team** — obviously.
- **A specific trade trap:** rights do **not** transfer if the player (i) signed a one-year contract excluding options, (ii) would have become a Full/Early Bird FA after it, and (iii) is traded without the second-year option being exercised. Because this *downgrades* him, he gets an **automatic (implicit) no-trade clause**. `[https://cbaguide.com/transactions/trades/traderules/]`
- Otherwise **Bird rights travel with the player in a trade.** This is the whole basis of Example 3 below.

A partial season counts as a full season for tenure. Bird/Early Bird/Non-Bird exception values begin **shrinking on January 10** each season.

2.4 The apron restrictions and the hard-cap triggers

SOURCED FACT — the CBA's restriction table, as reproduced by The CBA Guide with 2026-27 thresholds, cross-checked against Hoops Rumors' tax-apron glossary, Over the Apron, and Bobby Marks' ESPN summary. `[June–Aug 2026 | HIGH]`

Prohibited if the team's *Apron Team Salary* exceeds the FIRST APRON ($209,015,000) after the transaction:

- (A) sign or acquire using the **Bi-Annual Exception**
- (B) sign or acquire using the **Non-Taxpayer MLE**
- (C) **acquire a player via sign-and-trade**
- (D) sign a player **waived during the regular season** whose pre-waiver salary exceeded the NTMLE
- (E) acquire a player using the **Expanded TPE** (i.e., take back more salary than sent out)
- (F) use a **Standard TPE** past a shortened clock (a TPE created in-season dies at the end of that regular season)

Prohibited if the team exceeds the SECOND APRON ($221,686,000) after the transaction:

- (H) **aggregate** two or more contracts in a trade
- (I) **send cash** in a trade
- (J) use a TPE that was **created via a sign-and-trade**
- (K) sign a player using the **Taxpayer MLE**

The hard cap: using an exception with an apron limitation hard-caps the team at that apron **for the remainder of the salary cap year** — and for the **following** year too if the transaction happens after the regular season but before the new cap year. Once hard-capped, the team may not exceed the threshold for any reason.

A nuance most explainers get wrong: Over the Apron flags it explicitly — a use of the NTMLE that fits inside the taxpayer MLE's size ($6,064,000) and 2-year length can later be **re-characterized** as the taxpayer MLE (Art. VII §6(f)(5)), converting a first-apron hard cap into a second-apron one. That's why Marks writes "hard-capped at the first apron if it uses **more than $6.1 million** of its non-tax or biannual exception" rather than "any use." `[Aug 2026 | MEDIUM-HIGH]`

Second-apron draft-pick penalty (SOURCED): finish a regular season over the second apron → your own first-round pick **seven drafts out** is **frozen** (untradeable). Exceed the second apron in **2 of the next 4** seasons → that pick **slides to the end of the first round.** Fewer than 2 of 4 → it **unfreezes.** A pick either slides or unfreezes, never both. For a team over in 2026-27, the frozen pick is **2034**. Real precedent: Boston, Phoenix and Minnesota had their **2032** firsts frozen in the rule's first year.

Apron Team Salary ≠ team salary (SOURCED, and a real source of public disagreement): apron salary *excludes* cap holds but *adds* unlikely bonuses, bumps 0–1 YOS free-agent signings up to the 2-YOS minimum, and adds outstanding qualifying offers and required tenders. Live 2026 example of the confusion: Over the Apron reads Denver as a *first*-apron team on signed salary while media reporting places them in the *second* — the difference is Peyton Watson's unrenounced cap hold.

2.5 Trade salary matching — the exact 2026-27 bands

SOURCED FACT. Over the Apron states the CBA formula and the indexed constant for 2026-27:

> Expanded matching = **greater of (lesser of (200% of outgoing + $250,000 ; outgoing + $9,095,709) ; 125% of outgoing + $250,000)**, for teams below the aprons; **strict 100% at or above either apron.** The $250,000 allowance **disappears** once the team's apron salary would clear the first apron after the trade (Art. VII §6(j)(3)).

`[DATA DATE: Sept 2026 | HIGH | https://overtheapron.com/accuracy]` The middle-band constant **$9,095,709** is stated for 2026-27; The CBA Guide independently notes it was **$8,527,000** in 2025-26 and grows with the cap ($8,527,000 × 1.066693 = $9,095,7xx ✓).

INFERENCE (breakpoints, solved from the formula — high confidence):

Outgoing trade salary	Max incoming, team below both aprons
≤ **$8,845,709**	200% of outgoing + $250,000
$8,845,709 – **$35,382,836**	outgoing + **$9,095,709**
> $35,382,836	125% of outgoing + $250,000

These solve to $8,846,000 and $35,384,000 rounded — matching The CBA Guide's TPE table exactly. Note that page has an internal inconsistency (its table is 2026-27, its prose still cites 2025-26's $8,527,000). I resolved it via the formula.

The four TPE types (SOURCED, CBA Guide):

TPE	Blocked above	Out	Simultaneous?	Limit
Expanded	First apron	1+	yes	the tiered formula above
Standard	(First apron limits its *clock* only)	1	yes **or non-simultaneous, 12 months**	100% + $250k
Aggregated	Second apron	2+	yes	100% + $250k
Room	Salary cap	1+	yes	$250k over the cap

INFERENCE — the tier summary a designer needs:

- **Below tax / below first apron:** you may take back **more** than you send.
- **At or above the first apron:** strictly **100%**, no $250k cushion, no pre-existing TPEs, no S&T receipt.
- **At or above the second apron:** 100% **and you may not add two salaries together** to reach a bigger one — even to *cut* payroll. (The CBA Guide's real 2025 case: Phoenix, second apron, could trade Kevin Durant's $54.7M for two $27M salaries but could **not** combine Royce O'Neale's $9.3M and Cody Martin's $8.1M for one $15M player, despite shedding money.)

Two more real constraints: a player acquired via an exception **cannot be aggregated for 2 months** (waived if acquired before Dec 17 and re-traded on/just before the deadline). And you may not aggregate **3+ minimum contracts** in an unbalanced trade outside the Dec 15–deadline window.

2.6 SIX WORKED EXAMPLES — same player, different price

These are the Week 1 engine. Every figure below is real and dated.

EXAMPLE 1 — Norman Powell: Bird rights worth $0, cap room worth $45M

Powell finished 2025-26 as a first-time All-Star with **Miami** (21.7 ppg). Miami held his **full Bird rights** — legally it could have paid him up to $57,736,350/yr on a 5-year, 8%-raise deal.

It couldn't pay him $22M. Miami acquired **Giannis Antetokounmpo** in a four-for-one trade using **more than 100% of a traded player exception → hard-capped at the first apron, $209,015,000, for all of 2026-27.** Marks: Miami sat "$18 million below, with up to five roster spots to fill." Filling five spots at the veteran minimum consumes most of that.

Chicago projected **$31M in room** (and was **$16.5M below the $148,465,000 floor**, so its first $16.5M of spending was economically free — a team below the floor pays the shortfall to its players regardless).

Result: Powell signed 2 years, $45,000,000 with Chicago on July 1, 2026. `[July 1, 2026 | VERY HIGH | https://www.espn.com/nba/story/_/id/49236176/]`

The lesson: the team that had every legal right to pay him most paid him nothing. A hard cap outranks Bird rights.

EXAMPLE 2 — LeBron James at $3,876,529: the cheapest player in the league was still too expensive for one team

LeBron signed **2 years, $8 million** with **Philadelphia** on July 24, 2026 — year one **$3,876,529**, the 10+ YOS veteran minimum, the smallest salary of his 24-year career (below his 2003-04 rookie salary of $4,018,290). `[July 24–25, 2026 | HIGH | https://www.cbssports.com/nba/news/lebron-james-76ers-contract-explained/]`

Philadelphia was hard-capped at the first apron with roughly **$3.4M** of room. It needed **$3.9M**. Because he signed a **multi-year** minimum, the **one-year-veteran reimbursement did not apply**: Philly carries the full $3,876,529, not the $2,449,421 two-YOS charge. Gap: **$1,427,108.** The Sixers had to waive a non-guaranteed contract to fit him.

Price of the identical player across tiers (INFERENCE from sourced rules):

- Room team with ≥$3.9M room: $3,876,529 of room.
- Over-cap, under-first-apron team, **one-year** deal: cap/tax/apron charge **$2,449,421**; league reimburses.
- Second-apron team: **also $3,876,529** — the minimum exception is the only door no apron closes.
- Hard-capped team with $3.4M of headroom: **cannot sign him at all** without cutting someone.

EXAMPLE 3 — Coby White and Ayo Dosunmu: buying the right to spend money you already have

Both were Bulls players heading to 2026 free agency. At the **February 2026 deadline**, Charlotte traded for **White** and Minnesota traded for **Dosunmu** — explicitly, per Marks, "because each player's Bird rights were transferred as part of the trade."

Then they re-signed **over the cap**:

- **Coby White → Charlotte, 3 years, $74,000,000**
- **Ayo Dosunmu → Minnesota, 5 years, $112,000,000**

`[Aug 3, 2026 | HIGH | https://www.espn.com/nba/story/_/id/49490878/]`

INFERENCE: neither team had ~$25M of room. Without the deadline trade, the most either could have offered in July was the **NTMLE: $15,044,000 start, 4 years, 5% raises** — and Minnesota's **five-year** term is *only* available with full Bird rights. Charlotte and Minnesota spent draft capital in February to buy an *institutional right*, not a player.

EXAMPLE 4 — Mitchell Robinson: the second apron as a voluntary hard cap

Entering free agency the **Knicks** were **$916,000 below the first apron** and **$14M below the second**, with full Bird rights on Robinson — meaning re-signing him was **legal at any price**. (Teams may always exceed either apron to retain their own free agents.)

Owner **James Dolan**: *"There's certain things in the NBA that you'd have to be suicidal to do. One of them is the second apron. Cannot go into the second apron."* `[June 2026 | HIGH | WFAN via ESPN]`

Robinson signed with Boston: 3 years, $47,000,000. Boston was $18M below the first apron and $31M below the second. New York instead re-signed **Shamet, Alvarado, Diawara and Clarkson for a combined $15 million — "equal to Robinson's first-year salary in Boston."** `[Aug 3, 2026 | HIGH | ESPN]`

The lesson: four players for the price of one, because for New York the marginal price of Robinson was not his salary — it was his salary + tax + the loss of aggregation, sign-and-trade receipt, the taxpayer MLE, and a 2034 first-round pick.

EXAMPLE 5 — Denver and Peyton Watson: when $1 of salary costs $5.75

Pre-free-agency Denver was **$900,000 below the second apron**, $12M above the first, $15M above the tax. It holds Watson's Bird rights and can match any offer sheet.

ESPN: retaining Watson would push Denver's payroll **past $400 million** including tax penalties. And: *"Denver is not allowed to acquire a player in a sign-and-trade if the post-transactional salary leaves the franchise over either apron"* — so the usual escape valve is shut on the receiving side too. `[Aug 3, 2026 | HIGH | ESPN]`

INFERENCE from the official bracket table `[https://cbaguide.com/thresholds/luxurytax/]`: at a tax salary around $222M, Denver sits in **Bracket 4**, where the non-repeater rate is **$4.75 per $1**. Every marginal salary dollar costs **$5.75 total**. A repeater pays **$6.75** tax → **$7.75 total**. A team below the tax line pays **$1.00.**

Same player. $1.00 vs $5.75 vs $7.75.

Meanwhile Oklahoma City — freed by trading Lu Dort — extended Watson's teammate **Spencer Jones a 2-year, $12M offer sheet**, which Denver matched.

EXAMPLE 6 — Oklahoma City: the second apron as a demolition order

OKC entered the 2026 offseason projected **over the second apron with a $560 million payroll** (salary + tax), **$11.5M above the second apron**, facing a luxury tax bill **over $100 million**.

It then traded **Isaiah Joe** (→ Detroit), **Aaron Wiggins** and **Lu Dort** (→ Atlanta) in **three separate deals** — separate because a second-apron team **cannot aggregate**. It collected **seven future second-round picks and three trade exceptions**. **Payroll fell to $234 million.** `[Aug 3, 2026 | HIGH | ESPN]`

Buyers: Atlanta, described by Marks as using "their financial flexibility to take advantage of teams needing to shed salary." Being *below* the aprons is itself a tradeable asset.

EXAMPLE 7 (trade math, three answers to one question) — INFERENCE from §2.5, high confidence.

A team sends out **$20,000,000** of trade salary. How much can it take back?

Team's post-trade position	**Max incoming**
Below both aprons	$20,000,000 + $9,095,709 = **$29,095,709**
At/above first apron	**$20,000,000** (no $250k cushion)
At/above second apron	**$20,000,000**, and the $20M may not be assembled from two players

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3. SIGN-AND-TRADE — FULL TRUTH

3.1 Mechanics

SOURCED FACT `[https://cbaguide.com/transactions/trades/signandtrade/ | HIGH]`. A team may sign-and-trade a free agent only if **all** of:

1. Executed **before the start of the regular season**;
2. The player is a free agent who **finished the prior season on the sending team's roster**;
3. Contract is **at least 3 and at most 4 seasons**, excluding options;
4. **First season fully guaranteed**;
5. The player is **not** signed using the NTMLE or Room MLE;
6. The **acquiring team has room or matching salary** for the salary plus unlikely bonuses.

Additional real constraints:

- **Raises are 5%, not 8%** — even though the sending team uses Bird rights. `[cbaguide contract limitations]` This is a genuine cost to the player.
- A player eligible for "higher max criteria" (5th-year escalator) is capped at **25% of the cap** in a sign-and-trade.
- Any signing bonus paid by the **sending** team is treated as **cash-in-trade** (limit $8,495,000 for 2026-27, and second-apron teams may send **no** cash).
- No Exhibit 6; physical-exam contingency is allowed.

3.2 Base-year compensation

SOURCED FACT. If the sending team signs him with **Full or Early Bird rights** for more than **Non-Bird rights** would have allowed, his **outgoing trade salary** for matching becomes the **greater of** (a) his last season's salary, or (b) **50% of his new first-year salary** — while the sending team's **cap sheet still absorbs the full new number**.
`[https://cbaguide.com/transactions/trades/tradesalary/]`

Real case: **Max Strus**, July 2023 — 4 years/$62.3M, $14,487,684 start, signed by Miami with full Bird rights and traded to Cleveland. Prior salary $1,815,677. His outgoing value to Miami was **$7,243,842** (50%), not $14.5M.

Over the Apron states the trigger as *"re-sign your own free agent to a raise over 20%, then trade him."* **SOURCES FRAME THIS DIFFERENTLY** — the CBA Guide's "more than Non-Bird would allow" and Over the Apron's ">20% raise" describe the same threshold from opposite sides (Non-Bird permits 120% of prior salary). I read them as consistent; flag for verification against the CBA text.

3.3 The apron overlay

- **Receiving team is hard-capped at the FIRST APRON** for the rest of the cap year the moment it acquires a player by sign-and-trade. Real 2025 case: **Sacramento** acquiring Dennis Schröder.
- **A team that would be over the first apron after the transaction cannot receive at all.**
- **Second-apron teams** additionally cannot use a **TPE created via sign-and-trade**, and cannot use the signed-and-traded player's salary for matching in an acquisition.
- **Sending team:** loses the full new salary from its cap sheet, may generate a Standard TPE (the record case: Boston's **$28.5M** TPE from the 2020 Gordon Hayward sign-and-trade to Charlotte — later spent on Evan Fournier, who himself became a TPE that became Derrick White).

3.4 When sign-and-trade is genuinely the ONLY path

INFERENCE, built on sourced facts. All four conditions must hold:

1. The player's market value **exceeds $15,044,000** (the NTMLE start) — above that, no over-cap team can sign him outright.
2. **No cap-room team wants him** at that price. In summer 2026 there were **exactly three room teams: Brooklyn, Chicago, the Lakers.** `[Aug 3, 2026 | HIGH | ESPN]` Brooklyn's room was ~$37M, Chicago's ~$31M.
3. His **current team holds his rights** and either won't pay or is capped out.
4. The **receiving team ends below the first apron**.

The purest instance is a **restricted free agent** whose team can match everything. Only **six players have signed offer sheets since 2022** (Ayton, Thybulle, Reed, Post, Cisse, Spencer Jones), and only **one went unmatched** since Bogdan Bogdanovic in 2020. `[Aug 3, 2026 | HIGH | ESPN]` For an RFA in 2026, sign-and-trade is functionally the only route to *both* a raise *and* a new city. Live case: **Peyton Watson**, whose sign-and-trade options were narrowed because Denver sits at the second apron.

3.5 Three simplification proposals for 12-year-olds

Proposal A — "The Permission Slip." A player priced above the league's Mid Price ($15M card) who wants to leave a team that holds his Rights can only go to a full team if his **old team signs a permission slip**. The new team then wears a **Spending Collar** for the rest of the year (it cannot cross the first apron line no matter what).

- *Preserves:* the true economic insight — access is granted by an institution, not purchased; and the transfer carries a durable cost to the receiver.
- *Sacrifices:* the 3–4 year minimum, the 5% raise penalty, base-year compensation, the second-apron receipt bar, the outgoing-leg salary match.

Proposal B — "Three Signatures." The sign-and-trade is a physical card requiring three signatures: player, old team, new team. Teams in the top spending tier (second apron) have the signature line **printed over in red — they may not sign**.

- *Preserves:* tri-party consent, and the fact that the richest teams are *excluded* rather than merely taxed. That inversion — money buying you *fewer* options — is the single most counterintuitive true thing in the CBA and kids love it.
- *Sacrifices:* all salary-matching math; base-year compensation; the hard-cap consequence for the receiver.

Proposal C — "The Trade-Back." Same as A, but the old team must receive **something real** in return (a player card, a pick card, or a "future value" token) — never nothing.

- *Preserves:* that this is a **trade**, not a favor; and that the sending team's incentive is to get paid for its rights.
- *Sacrifices:* base-year compensation (which is exactly the rule that makes the outgoing leg hard in reality), the guarantee requirement, contract length.

Recommendation: A + C together, B held in reserve for a "Week 4 twist." A+C gets you consent + consideration in two sentences. B is the emotional payoff and should arrive after kids have already felt the second apron bite.

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4. THE REST OF THE SIX-WEEK ARC

4.1 Trade deadline and pick-trading rules

2026-27 trade deadline: Thursday, February 11, 2027. `[HIGH | https://www.nba.com/news/key-dates and https://basketball.realgm.com/wiretap/284185/]` Rule: the deadline is the **second Thursday of February before the All-Star Game**; 2027 All-Star is **Feb 19–21 in Phoenix**. It is six days later than 2026's Feb 5 deadline.

Other 2026-27 key dates `[NBA.com/news/key-dates | HIGH]`: camps open Sept 28-29; regular season opens **Oct 20, 2026**; NBA Cup group play Oct 30–Nov 27, final **Dec 11** at Hinkle Fieldhouse; 10-day contracts available **Jan 5**; **Jan 10** all remaining contracts guarantee; regular season ends **April 11, 2027**; Play-In April 13-16.

What is and isn't legal:

- No trades from the deadline until after a team's regular season ends.

- Playoff teams cannot trade a player on the **playoff roster** until eliminated — but *can* trade other assets (real case: Indiana traded draft rights and a first-round pick during its 2025 Finals run).
- A player entering an option year cannot be traded after the regular season unless the option is exercised/ETO declined.
- **No-trade clauses** require 8+ YOS and 4+ years with that team. **Automatic no-trade clauses** arise when a trade would downgrade a player's Bird rights.
- Publicly demanding a trade: fine up to **$150,000**.

Draft-pick trading `[https://cbaguide.com/transactions/trades/traderules/ | HIGH]`:

- **7-Year Rule:** first- and second-round picks may be traded only up to the **7th draft following** the trade. New picks unlock the instant the current draft ends (after the 2026 draft, 2033 picks became tradeable).
- **Stepien Rule:** a team must **never have the possibility** of lacking a first-round pick in two consecutive drafts. Protections that create that possibility are prohibited. Holding *another* team's pick satisfies it — Phoenix, having traded its own 2026 and 2027 firsts, complies by owning the least-favorable of Utah/Cleveland/Minnesota in 2026.
- **Protections** must specify years; if a protected pick doesn't convey, future contingencies are permitted with named years. A conditional pick can be replaced by a second-rounder (real: Golden State's 2030 first to Washington, top-20 protected, converting to a 2030 second).
- **Deferral:** a one-time, one-year deferral is permitted, but **not combined with protections** (real: the Pelicans deferring the Lakers' 2024 first to 2025 in the Anthony Davis trade).
- **Swap rights** are tradeable. Picks and swaps count as **$0 trade salary**.

4.2 Revenue reality

Scale (SOURCED, 2024-25 actuals): 30 teams generated **$12.25 billion, $408M per club**, net of revenue sharing — ranging from **Golden State $833M** to **Memphis $301M**. Comparison: NFL $22.2B, MLB $12.75B, NHL $7.7B, MLS $2.2B. `[Oct 21 & Nov 12, 2025 | HIGH | https://www.sportico.com/leagues/basketball/2025/how-nba-teams-owners-make-money-1234874170/]`

Mix (2024-25):

Stream	Amount	Share
Central/league distribution	—	**38%** (would be **44%** with the new TV deal)

Stream	Amount	Share
Seating + luxury suites	**$3.4B**	**28%**
Sponsorship	**$1.7B**	**14%**
Local media	—	**10%** (falling)
Concessions/parking/merch/non-NBA events	**$1.15B**	**9%**

The new national media deal (SOURCED): 11 years, $76 billion, **~$6.9B/year**, with **ESPN/ABC, NBC/Peacock and Amazon Prime Video**, **beginning 2025-26**. Per-team national TV revenue jumped from **$103M (2024-25) to $143M (2025-26)**, rising **~7%/yr** on average to a projected **$281M per team by 2034-35**. Forty years ago each team got ~$1.5M. `[Oct–Nov 2025 | HIGH | Sportico]`

Big-market vs small-market, concretely:

- Lakers' local TV deal (Spectrum) paid **~$200M for 2024-25** — the league's richest.
- Knicks took a **28% cut** to their MSG Networks fee.
- Warriors: **>$5M in ticket revenue per game**; **$2.5M/game from suites alone**; sponsorship **approaching $200M**. Lakers and Knicks ~**$4M/game** in tickets.
- Clippers' Intuit Dome has four courtside cabanas at **$2M/year**, all leased; Clippers' sponsorship exceeded **$100M** in year one.
- 2016-17 (leaked): Lakers made **$149M** from local media; Memphis made **$9.4M**. The Knicks alone made **$10M more from TV than the six lowest-earning teams combined.** `[Sept 2017 | HIGH | ESPN, https://www.espn.com/nba/story/_/id/20747413/]`

Tickets and attendance:

- 2025-26 secondary-market average ticket: **$52.86** league-wide. Knicks **$213.43** (4× league average, 46% above the next team); Lakers $146.20; Warriors $127.73; Celtics $114.18; Bulls $78.88. Cheapest: Memphis $22.00, Utah $22.28, New Orleans $22.58. `[Oct 21, 2025 | MEDIUM – secondary-market data, not gate revenue | https://www.sportscasting.com/news/nba-average-ticket-prices-2025-knicks-most-expensive-tickets-okc-biggest-price-hike/]`
- 2025-26 attendance: arenas at **97% capacity** (4th time ever); **11 teams sold out every home game** (Celtics, Cavaliers, Heat, Jazz, Knicks, Nuggets, Rockets, Spurs, Suns, Thunder, Warriors); only **Washington and Memphis** below 90%. Three-season total

(2023-24 to 2025-26) exceeded **22.18 million**. [April 2026 | HIGH | https://basketnews.com/news-245164-...]

- **INFERENCE:** at 97% league-wide capacity, ticket *quantity* is nearly fixed; essentially all gate variance is **price** and **premium mix**. That is the honest microeconomics for a demand unit — and it is why suites/premium are ~28% of revenue in aggregate.

4.3 BRI, the Designated Share, and withholding

SOURCED FACT (2025-26 final accounting, released ~July 1, 2026):

- **BRI: $11.68 billion**, up from **$10.25 billion** in 2024-25.
- **Total salaries and benefits: $6.27 billion** — above the 51% line.
- The withholding pool was **~$580 million**, split **54.8% owners / 45.2% players**, netting players a **~5.5% haircut** on negotiated salary. Stephen Curry forfeited **$3.3M** of $59.8M; Embiid and Jokić ~$3M each.
- Prior year, **more than 90%** of the pool went to owners (**>$480M** returned to teams for 2024-25).

[July 1, 2026 | VERY HIGH | https://www.sportico.com/leagues/basketball/2026/nba-escrow-players-salaries-owners-million-1234937798/, first reported by Eric Pincus]

Mechanics (SOURCED, CBA structure) [https://cbaguide.com/finances/ | HIGH]:

- The **Designated Share** = 50% of BRI, adjusted by **±60.5% of the gap between actual BRI and Forecasted BRI**, then collared to **never below 49% and never above 51%**.
- Teams **withhold 10% of player salaries** each year. Under the 2023 CBA this is no longer literal escrow — it's "Withheld Compensation," accounted at year-end.
- Overage → players repay; Shortfall → teams pay players in proportion to salary.
- **Cap formula:** (**44.74% of Projected BRI** − Projected Benefits) ÷ 30, then adjusted, then collared (no decrease; +10% max).
- Note: Sportico writes "the CBA calls for players to receive 51% of BRI." **That is a simplification** — the CBA specifies a 49–51% band. Flag this; the band is exactly the kind of thing the product should get right.

4.4 Revenue sharing — what is actually known

SOURCED FACT: the revenue-sharing agreement is NOT a public document. ESPN describes the formula as "a byzantine maze of calculations based on market size, expected revenue for each team, expense levels and other variables," including "adjustments that can reduce a team's payout based on various performance criteria." [Sept 2017 | HIGH | ESPN] Treat any single-sentence formula as approximate.

Reported basic structure (MEDIUM confidence — widely reported, not official):

1. Each team contributes a fixed percentage (**commonly reported as ~50%**) of its defined revenue into a pool.
2. Each team then **receives an equal share** roughly equal to **league-average team revenue** (in practice, near the salary-cap level per team).
3. Above-average revenue → **net payer**; below-average → **net receiver**.
4. **Participation requires generating at least 70% of league-average revenue.** In practice all 30 participate.
5. As implemented, high earners forfeit somewhat **less** than the raw formula implies, and low earners receive somewhat **less**. `[Sept 2021 | MEDIUM | https://www.celticsblog.com/2021/9/19/22650143/]`

The market-size adjustment — and why it matters more than anything else here (SOURCED): the formula benchmarks each team against **expected** revenue *given its market*, not raw revenue. **If a team over-performs its market, it forfeits some of its payout.** That is why **Oklahoma City — one of the NBA's smallest markets — paid into the system for six consecutive seasons** through 2016-17, as did **San Antonio**; while the **larger-market Nuggets**, at roughly break-even, **received an eight-figure check.** `[Sept 2017 | HIGH | ESPN]`

Plus the luxury tax channel (SOURCED): 50% of luxury tax proceeds are distributed to non-taxpaying teams (excluding teams below the salary floor); the other 50% is retained by the NBA for "League Purposes," including funding revenue sharing.

Concrete classroom anchors:

2021-22 (fully itemized, ESPN memo) `[Mar 17, 2023 | HIGH | https://www.espn.com/nba/story/_/id/35882999/]`

Payers	Amount		Top receivers	Amount
Golden State	$45.0M		Indiana	$42.2M
LA Lakers	$42.8M		Denver	$35.5M
New York	$20.9M		Portland	$32.0M
Boston	$15.7M		Charlotte	$31.6M
Chicago	$10.0M		Sacramento	$29.9M
Dallas	$8.8M		New Orleans	$28.9M
Philadelphia	$6.4M		Memphis	$28.0M

Payers	Amount		Top receivers	Amount
Miami	$5.1M		San Antonio	$26.3M
LA Clippers	$5.0M		Minnesota	$25.6M
Brooklyn	$3.0M		Houston	$931,000 (least)
10 teams	**$163.6M**		**20 teams received $404M total**, which included **$240M of luxury tax money** (half of a **$481M** league tax bill). Warriors also paid **$170M** in tax; Lakers **$45M**.	

2016-17 (leaked confidential records) `[Sept 2017 | HIGH | ESPN]`: 10 teams transferred **$201M** to 15 teams; **Warriors, Knicks, Lakers, Bulls accounted for $144M (71.5%)**. Lakers wrote a **~$49M** check and still posted a **$115M** net income. Memphis lost **~$40M** pre-sharing and received **$32M**, the league's largest check. Detroit lost **$63.2M** pre-sharing (largest loss) despite a large market, receiving $17.6M. **14 of 30 teams lost money before sharing; 9 after.** Five teams (Memphis, Charlotte, Indiana, Milwaukee, Utah) had received ≥$15M in each of the prior four seasons.

2025-26 luxury tax (most recent) `[2026 | MEDIUM-HIGH | https://sports.yahoo.com/articles/warriors-lakers-among-7-teams-212843131.html]`: **7 teams paid $223.1M** — down **>67%** from 2022-23's final pre-apron season. Cleveland alone paid **$68.7M**; Cleveland + Golden State + New York accounted for **$181.1M** of the $223.1M. Half was distributed among the **23 non-taxpaying teams at ~$4.9M each**. (Prior year, 2024-25: **10 tax teams, $461.2M**, ~$11.5M per non-tax team.) `[cbaguide, citing Sports Business Classroom]`

The dispute (SOURCED, quotable):

- A large-market owner, 2017: *"The need for revenue sharing was supposed to be for special circumstances — not permanent subsidies."*
- A small-market owner, same piece: *"Teams in small markets are told we need to run our businesses better so we can make money. But teams in the largest markets can run their businesses poorly and still make money."*

- Paul Allen pushed for guaranteeing every team a profit; one owner proposed a **$20M guaranteed profit** floor for all 30.
- The Board of Governors can amend revenue sharing by a **16-14 majority** (it required 20 of 30 before 2014).
- Current-era version: **NBPA executive director David Kelly**, July 2026: *"We should have done a better job of fighting back against the second apron. We're seeing [it] decimate teams and force decisions to be made that are not basketball decisions."* Adam Silver's counter: *"we've had eight different champions over the last eight years."* Silver says apron amendments wait for the next CBA (2030-31). `[Aug 3, 2026 | HIGH | ESPN]`

The best possible "rules are real" anchor, two days old: on **September 2, 2026** the NBA announced penalties against the **LA Clippers and Kawhi Leonard** for salary-cap circumvention via off-court income arranged with Aspiration Partners, Boingo Wireless, Daktronics and Lockton Insurance: **forfeiture of five first-round picks (2029-2033), a $30M team fine, a one-year suspension of owner Steve Ballmer, a one-year unpaid suspension of Gillian Zucker, a six-month unpaid suspension of Lawrence Frank, $700,000 from Leonard, a five-year ban on Dennis Robertson, and five years of compliance monitoring.** `[Sept 2, 2026 | VERY HIGH | https://pr.nba.com/nba-investigation-clippers-kawhi-leonard/]`

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5. AUTHENTICITY / LEGAL GUARDRAILS

I am not a lawyer and this is not legal advice. Practical framing only; every item below should be confirmed by counsel before launch.

Almost certainly safe:

- **Facts.** Salary figures, cap thresholds, transaction records, standings, statistics, dates. Facts are not copyrightable, and this data is published by the NBA itself and by Spotrac/ESPN/Hoops Rumors.
- **Referring to real teams and players by name** to identify the actual team or player (classic nominative use). "The Oklahoma City Thunder were $11.5M over the second apron" is a factual statement about a real entity.
- Your own original prose, charts, and game mechanics.

High risk — do not use without a license:

- **Logos, wordmarks, uniform designs, court designs, and trade dress.** These are registered trademarks. This is the single brightest line.
- **Team color-plus-mark lookalikes** that function as a logo substitute.

- **Player photographs, likenesses, jersey numbers rendered on a uniform, signatures.** Player likeness rights sit with the **NBPA** (which is why NBA 2K licenses *both* the NBA and NBPA).
- **Anything implying sponsorship, endorsement, affiliation or approval** — including "official," NBA-style typography, or a name that reads as an NBA product.

Practical guardrails to build in now:

1. Text-only team and player identification. No marks, no photos, no uniforms.
2. A persistent, plain-language disclaimer: *"Not affiliated with, endorsed by, or sponsored by the National Basketball Association, its teams, or the NBPA. Team and player names are used to refer to real people and organizations for educational purposes. All financial data is drawn from public sources, cited with dates."*
3. **Cite every number with a date and source inside the product.** This is both your pedagogical differentiator and your best evidence of good-faith factual/educational use.
4. Use city names ("Oklahoma City," "Memphis") in UI chrome where nickname trademarks aren't needed.
5. Do not sell merchandise, apparel, or anything decorative bearing team identifiers.

Specifically flag for counsel:

- **Right of publicity** varies by state and is strongest in California, Indiana and Tennessee; minors in the audience does not change the analysis.
- Whether a **paid** curriculum sold to schools is "commercial use" for trademark purposes, and how that interacts with educational/newsworthiness defenses.
- Whether depicting **current, active, name-identified minors-adjacent draft prospects** raises separate issues.
- Whether your business model (subscription vs. one-time vs. district license) changes the analysis.
- Whether to simply **seek a license** — an educational license may be cheaper than the legal review, and gets you logos.

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6. HONEST SIMPLIFICATION RECOMMENDATIONS

Mechanic	Full rule	Smallest honest simplification	Truth that must NOT be lost
Cap holds	Six categories; free-agent holds at 120/130/150/190/250/300% of prior salary	**"Ghost salaries."** Every player you might re-sign leaves a ghost on your	**Cap room is not "cap minus contracts."** The irreversibility of

Mechanic	Full rule	Smallest honest simplification	Truth that must NOT be lost
	bounded by min/max; 120% first-round scale holds; incomplete-roster charge of $1,357,763 per spot below 12; exception holds.	books at roughly **1.5× his old salary** (2× for a good young player coming off a rookie deal). You can erase a ghost — but the moment you do, you give up the right to re-sign him over the cap this year.	renouncing is the whole lesson. Never let a kid renounce and then re-sign.
Exception ladder	Bird (5yr/8%/max), Early Bird (2-4yr/8%/$15.24M), Non-Bird (1-4yr/5%/120%), NTMLE $15.044M/4yr, TMLE $6.064M/2yr, Room MLE $9.366M/2yr, BAE $5.477M/2yr, minimum, rookie, DPE, reinstatement.	**Four cards:** Own-Player Card (any price, 5 yrs), **$15M Card**, **$6M Card**, **Minimum Card (always in your hand)**. Room teams trade their whole hand for a pile of cash + one **$9M Card**.	**Over the cap, $15,044,000 is a wall.** Above it, an outside free agent is unreachable by any means except a trade or a sign-and-trade. And **the Minimum Card never goes away** — the league guarantees a floor of access to every team.
Aprons	Two thresholds, ~11 enumerated prohibitions, hard caps that persist for the cap year (and the next if post-season), apron salary ≠ team salary, 7-years-out pick freezing/sliding.	**Two lines with names, not numbers.** Cross the **first line** ($209.0M) and you lose the **$15M Card**, the **$5.5M Card**, and the right to **receive a permission-slip player**. Cross the **second line** ($221.7M) and you additionally lose the **$6M Card**, the right to **add two players together in a trade**,	**The penalty for spending is losing *tools*, not paying *money*.** Dolan's quote and the OKC teardown both make this vivid. Also: **you may always re-sign your own players**, which makes the apron a *choice*, not a law.

Mechanic	Full rule	Smallest honest simplification	Truth that must NOT be lost
		the right to **send cash**, and your **first-round pick seven years from now freezes**.	
Trade matching	Four TPE types; expanded formula = greater of (lesser of (200%+$250k, out+$9,095,709), 125%+$250k); 100% at/above either apron; no aggregation at the second; 2-month re-aggregation lock; minimum-stacking limits; base-year comp; poison pill.	**One rule with three settings.** Below both lines: you may take back **your outgoing salary + about $9M**. Above the first line: **exactly what you send out.** Above the second line: **exactly what you send out, and you may not stack two cards to make one.**	The **directional asymmetry**: the rich team can always *shed*, but can never *upgrade by absorbing*. That's the mechanism that recycles stars downward and picks upward.
Sign-and-trade	3-4 yrs, 1st yr guaranteed, before opening night, no NTMLE/Room MLE, 5% raises, BYC on outgoing, receiver hard-capped at first apron, second-apron teams barred from receiving and from S&T-created TPEs.	**Proposal A + C (§3.5):** old team signs a **permission slip** and must **receive something real**; the new team wears a **Spending Collar** all year.	**Access is granted by an institution, not bought.** A player above the $15M Card cannot reach a full team unless his *old* team consents. Kids will feel the injustice — that's the point.
Revenue sharing	Undisclosed formula; ~50% contribution to a pool, equal draw ≈ league-average revenue; 70% participation floor; adjustments for market size,	**"Everybody puts half their local money in one bucket; everybody takes out the same amount."** Then one added rule: **"the bucket expects**	(a) The **direction is real** — GSW $45M out, Indiana $42.2M in. (b) The **market-size adjustment is real and counterintuitive** — OKC and San

Mechanic	Full rule	Smallest honest simplification	Truth that must NOT be lost
	expected revenue, expense levels, performance; plus 50% of luxury tax to non-taxpayers.	**more from you if your city is big — so a small city that does great can end up paying in."** Plus: **"half of all luxury tax money is split among the teams that didn't pay it."**	Antonio paid in for six straight years. (c) **The formula is secret.** Tell the kids that. It's the most honest thing in the curriculum and it teaches that real institutions have opaque rules people argue about.
Ticket demand	Dynamic pricing, opponent-dependent tiers, season vs single-game, premium/suite mix (~28% of revenue with seating), 97% league capacity, secondary-market spread of ~10× ($22 to $213).	**Two dials, one locked.** Seats are nearly all sold (**97%**) — so the quantity dial is stuck. You only move the **price** dial and the **premium mix** dial (how many seats are courtside/suite).	**Sellouts do not mean maximum revenue.** Memphis at $22 and New York at $213 both sell out. Price and product mix, not attendance, are where the money is — and the gap between them is what revenue sharing exists to address.

APPENDIX — ITEMS I COULD NOT CONFIRM

1. **2027-28 cap ($174M / +5.5%).** UNCONFIRMED officially — sourced only to a league memo reported by Sportico and Bobby Marks. **Check:** pr.nba.com, ~June 30, 2027.
2. **Exact 2026-27 trade-matching breakpoints ($8,845,709 / $35,382,836).** My arithmetic from Over the Apron's stated formula and its sourced $9,095,709 constant; The CBA Guide's rounded table agrees. **UNCONFIRMED against CBA text (Art. VII §6(j)).** Check the CBA PDF.
3. **Minimum salary scale.** Two reputable sources disagree; I resolved in favor of Hoops Rumors via three cross-checks (see §1.5). **Check:** the NBA's official minimum-salary exhibit, or any signed 2026-27 minimum contract's reported first-year figure.
4. **Base-year compensation trigger framing** — "more than Non-Bird allows" (CBA Guide) vs ">20% raise" (Over the Apron). Probably equivalent; verify.

5. **Revenue-sharing formula.** The agreement is not public. The 50%-in / equal-share-out description is widely reported but explicitly approximate; the market-size and expense adjustments are described qualitatively only, by ESPN, from leaked documents. **There is no way to confirm this precisely, and the product should say so.**
6. **Two-way signing deadline (March 4).** Cited in the 2026/27 tracker; confirm against the official 2026-27 transaction calendar.
7. **Repeater status for 2026-27 taxpayers.** I did not verify which teams are repeaters; this materially changes the tax multiple ($4.75 vs $6.75 in Bracket 4). **Check:** Hoops Rumors' projected-taxpayers post or Sports Business Classroom's tax tracker in-season.

Sources:

- NBA sets Salary Cap for 2026-27 season at $164.961 million (NBA Communications)
- NBA.com — 2026-27 salary cap
- NBA announces penalties in Clippers/Kawhi Leonard investigation
- Key dates for 2026-27 NBA season (NBA.com)
- Larry Coon's CBA FAQ — retirement notice and 2017 CBA FAQ
- The CBA Guide — Aprons, Luxury Tax, Team Salary & Cap Holds, Traded Player Exceptions, Trade Salary, General Trade Rules, Sign-and-Trade, Rosters, League Finances, 2026-27 Amounts, Contract Limitations
- Over the Apron — Rules Coverage & Accuracy, Second Apron
- Sportico — NBA Players, Owners to Split $580M in Escrow at 45%-55%
- Sportico — NBA's Lagging Salary Cap Is Forcing Teams to Change Roster Plans
- Sportico — How NBA Teams and Owners Make Their Money
- Sportico — NBA Revenue Projected to Hit $14.3B in 2025-26
- ESPN — NBA free agency 2026: How much all 30 teams can spend (Bobby Marks)
- ESPN — NBA free agency 2026: Fact vs. fiction (Bobby Marks)
- ESPN — Norman Powell agrees to 2-year, $45M deal with Bulls
- ESPN — Warriors, 9 other teams paid $163.6M into revenue-sharing pool
- ESPN — A confidential report shows nearly half the NBA lost money last season
- CBS Sports — LeBron James's 76ers contract explained
- Hoops Rumors — NBA Minimum Salaries For 2026/27 (via Yardbarker)
- Hoops Rumors — Rookie Scale Salaries For 2026 First-Round Picks (via Yardbarker)
- Hoops Rumors — 2026/27 Two-Way Contract Tracker (via Yardbarker)
- Hoops Rumors — Tax Aprons glossary
- SalarySwish — Maximum Salary FAQ, Bi-Annual Exception Tracker, Mid-Level Exception Tracker
- CelticsBlog — It's your money: breaking down the myth of NBA ownership
- Sportskeeda — 2021-22 NBA revenue sharing payments
- BasketNews — NBA sets record attendance and viewership for 2025-26
- Sportscasting — Average NBA ticket prices 2025-26
- Bookies.com — The cost of attending an NBA game in 2025-26
- RealGM — 2027 NBA trade deadline six days later than 2026
- Yahoo Sports — 7 teams to pay $223.1M in 2025-26 luxury tax penalties

- CBS Sports — Rifts growing among small- and big-market NBA owners over revenue sharing
- INTA — Fair Use of Trademarks (non-legal audience)agentId: a4f56706dbe4b989d (use SendMessage with to: 'a4f56706dbe4b989d', summary: '<5-10 word recap>' to continue this agent) subagent_tokens: 276542 tool_uses: 135 duration_ms: 1230900

BOW_FABLE_5.1 master run prompt

BOW ECONOMICS — FABLE 5.1 MASTER RUN PROMPT

(Paste everything below the line into Fable 5.1.)

You are the implementation lead for **BOW Economics**, working in the repository `bow-economics-live`. You have full authority over implementation. You do not have authority to change what the product is.

0. READ THESE FIRST, IN THIS ORDER

1. `CLAUDE.md` — the repo's operating manual.
2. `docs/PRODUCT_DECISIONS.md` — D1–D47, the standing build law. **Read it before proposing anything.**
3. `docs/BOW_ECONOMICS_PRODUCT_BIBLE.md` **v1.1** — the product constitution.
4. `docs/BOW_FABLE_GAP_MAP.md` — what exists, what does not, what to preserve.
5. `docs/BOW_FABLE_BUILD_PLAN.md` — the wave sequence.
6. `docs/BOW_FABLE_ACCEPTANCE_MATRIX.md` — what "done" means.
7. `docs/design/BOW_VISUAL_NORTH_STARS.md` + the five images in `docs/design/mockups/`.
8. `docs/TRACK_101_MAP.md`, `docs/RAMAZ_READINESS.md`, `runtime/README.md` — current status and known gaps.

Then inspect the repository yourself. The gap map is a snapshot; the code is the truth.

Authority order when sources conflict: founder instruction → prior explicit founder decisions (in either log) → verified real-world fact → the Bible → the repo → anyone's judgment, including mine and yours.

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1. WHAT THIS PRODUCT IS

A **six-week live, online-first social economics course** for grades 5–6 (and, later, 7–8), run as one continuous NBA franchise story. Twelve to sixteen **desks** (pairs on one device by default, solo supported) each run a real club. Three surfaces per session: `/play` (private), `/teach` (the control room), `/board` (the projector). One 60-minute session per week.

The loop: experience → consequence → adaptation → class evidence → argument → **explicit economics formalization.** Students meet the mechanism first; then the teacher names it. **The explicit teaching stage is not optional** — the simulation does not replace economics instruction, it makes it understandable.

The six weeks, as built:

Wk	Module id	Title	Anchor concept
1	`m1l1-draft-day`	Draft Day	Opportunity cost
2	`m1l2-trade-deadline`	The Trade Deadline: Undo Isn't Free	The cost of undoing a commitment
3	`m1l3-free-agency`	Free Agency: The Signing Window	Market price under strategic interaction
4	`m2l1-full-house`	Full House → **THE BILL COMES DUE**	Revenue = P × Q, and clearing a bill you created
5	`m2l2-host-league`	You Don't Play Alone	Externality
6	`m2l3-write-rule`	Writing the Rule	Incentives and rules

A warning about older material. Earlier founder prose and Product Bible v1.0 described a different Module 1 — free agency in Week 1, a "season changes" week, the deadline in Week 3, and an engine called "The Same Line." **That arc was never built and is superseded.** The built arc above is canonical, and it came from direct founder instruction (repo D17, D18). If you meet the old ordering in any document, ignore the ordering and keep the economics.

All six lessons are already built and independently gate-certified. Your job is **not** to build six weeks. It is to:

> **Close the seams, build the missing primitives, put the social layer into the weeks that lack it, and raise all six weeks to one premium standard — without breaking six certified lessons.**

Hard date: October 1, 2026. All six weeks are one premium product. Weeks 4–6 are **not** intentionally second-class. Instructor-scaffolded degradation is a contingency, not the plan.

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2. THE TEN THINGS THAT MUST BE TRUE

1. **Economics is earned, then named, then retrieved.** Never name the concept before the experience earns it. After naming, the term returns deliberately — in Previously On, replay, history, recap and the finale deck.
2. **No BOW-issued score, grade, index, badge or machine verdict, anywhere.** Multiple truthful readings, components never summed. The server blocks illegality; it never blocks or grades bad strategy.
3. **Other humans materially change my decision environment — in every week.** Weeks 1 and 4 are currently thin on this and that is where the social work goes.
4. **Online is the flagship environment.** Never ask *how do we imitate a classroom*. Ask **what can sixteen connected computers do better than a physical trade floor.**
5. **The franchise persists and the persistence is felt**, not merely stored. Week 4 must be impossible to run with a generic team.
6. **The teacher is a director, not an operator.** A trained non-founder must be able to run an outstanding class cold, from the console.
7. **The real world of sports business is mandatory** — real clubs, people, rules, cases — but **reality is never a fandom test.** A desk that knows little basketball must still succeed.
8. **Simplify the interface before simplifying the economics.** When economics must be simplified, record what changed, why, and the misconception risk.
9. **Premium, cinematic, original.** Not an LMS, not a dashboard, not a wall of cards, not an analytics portal, not a cartoon, not an NBA 2K copy. Restraint everywhere; spectacle earned.
10. **Nothing is proven.** Not by a passing suite, not by a prior verdict, not by this document. "Classroom-proven" is reserved for what has survived a real class, and nothing has.

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3. THE WAVES

Follow `BOW_FABLE_BUILD_PLAN.md`. Summary:

Wave	**What**	**Why it is here**
0	Reconciliation: record the v1.1 rulings; fix the sealed-bid tie-break, the un-releasable franchise claim, the L3 client race	Three defects that corrupt every later wave
1	Cross-cutting primitives: commitment capture, PREVIOUSLY ON, the four	Everything downstream depends on them

Wave	What	Why it is here
	truthful readings, the economics protection rule	
2	**The Press Conference and The Tape**, in Week 2 only	BOW's signature reasoning primitive
3	**The M1→M2 seed and THE BILL COMES DUE**	Founder-locked; turns six lessons into a course
4	**The Online Trade Deadline War Room**	Highest social and interaction risk; proves online is an advantage
5	Week 1's social layer + curated franchise choice + the sealed prediction	The first impression is the weakest week socially
6	Week 5's pool ritual and binding split	The externality made visible
7	The finale: private stakes, the counterfactual as a beat, the timing fix	The course climax
8	Visual elevation of all six weeks + the parent product	One premium product
9	The 7–8 band at the two attachment points + full six-week prosecution	Both bands, then evidence

Derive the final order from real repo dependencies. If inspection shows a different order is correct, take it and say why in one sentence.

---

4. HOW TO WORK

Continue; do not restart. The repo contains serious, gate-certified work with real evidence behind it. **Inspect before changing.** Do not rewrite functioning infrastructure because a cleaner architecture can be imagined. The runtime is small, dependency-free and contract-driven — extend it.

The module contract is load-bearing. `runtime/src/shared/lessonModule.ts`: `phases` · `initialState({sessionId, seatIds, seed})` · pure `reduce` · `studentView`/`teacherView`/`boardView`/`aggregate` · optional `round` (TIME CUT), `onPhaseExit`, `classEvents`. **The runtime never inspects lesson state.** Every new primitive fits this contract or extends it deliberately, with a written reason. **Do not extract a generic engine, replay framework or platform abstraction from one or two data points.**

FROTH the major new surfaces — 3–5 genuinely different directions, not cosmetic variants — then select with a stated reason, build, and prosecute:

- The Press Conference / Decision Replay
- **The Online Trade Deadline War Room** *(the largest product-design investigation in this run)*
- The Bill Comes Due
- The digital Pool
- The Board of Governors finale

Use the five mockups. Read them visually. Identify what each is trying to accomplish. Preserve the best hierarchy and emotion. **Reject their false mechanics and unlicensed content** — logos, player photography, a child's face, OVR ratings, real cap magnitudes against the play scale, "Trade Value: Fair", the "Competitive Balance Index", and a Week-6 vote on a rule the built lesson does not contain. Produce an original BOW visual language.

Attack every meaningful wave with fresh critics: 5–6 student · 7–8 student · economist · NBA/sports-business truth · teacher transfer · social product · visual · runtime · parent/business. Builders never certify their own work. **No verifier bureaucracy** — no agents verifying agents, no re-checking settled work because more rigour sounds rigorous.

Do not stop early. A surface that is technically complete but product-poor is not done. If a wave finishes ahead of schedule, attack the next frontier that wave names.

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5. THE STANDING TRAPS

Each of these has already caused a real defect here, or is explicitly forbidden by the decision log.

1. A claim-carrying surface added **without registering it in `moduleClaims`** — invisible to the audit.
2. A drawing whose size carries a quantity, **drawn by eye** rather than measured in a browser.

3. The student device **ahead of the board** — numbers belonging to an unpressed beat must be *unsent*, not merely unrendered.
4. A decision window shipped **without a declared TIME CUT fallback**, named per desk, in words.
5. **Layout width derived from a control.**
6. A surface that **moves backwards**.
7. **Dead air** inside a decision phase — the fix is a free commitment, never a waiting message.
8. A **`gradeBand` switch with no content behind it.**
9. Forgetting that **the client shell is not generic** — every new surface needs render functions in `play`, `teach` and `board`.
10. **The Tape becoming undo**, or rendering the past with knowledge the desk did not have.
11. **The War Room becoming a trade machine with a timer.**
12. Claiming "verified" when you only reasoned; claiming "passing" without running the command this session; claiming "classroom-proven" for anything.

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6. THE ACCEPTANCE CONDITIONS THAT MATTER MOST

Full matrix in `BOW_FABLE_ACCEPTANCE_MATRIX.md`. The four gates:

- **War Room.** A real desk can honestly say: *"I only have two lines open, Miami is calling, Boston hasn't answered, the clock is running, and another trade just changed what I can do."* If not, the surface failed however well it renders.
- **The Bill Comes Due.** A desk can explain **why its bill is what it is by naming an actual earlier decision.** If Week 4 would work identically with a generic team, the wave failed.
- **Press Conference.** The room can state the desk's decision **and the tradeoff** within ten seconds, without reading a paragraph. If it resembles a quiz review, it failed.
- **Economics.** The product produces the experience, then explicitly names the concept from the room's own words, then retrieves it later. Implicit-only fails. Vocabulary pasted onto gameplay fails.

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7. FOUNDER COMMUNICATION

Keep it tiny. Message the founder only for:

- a serious thesis conflict,
- a major product direction selected after a froth,
- a material false economics or sports-business fact discovered in the product,
- a genuine blocker,
- a major wave complete,

- a true founder decision.

No progress diaries. No narration of what you are about to inspect. Most updates are one or two sentences.

Two things are already escalated and awaiting the founder — do not silently decide them:

1. **The Week-6 economics finding.** M2 L3's intended summit — *sharing is not charity; it pays the payer, through the product* — does not hold at the shipped constants: measured at league equilibrium, small markets' own best share is 55–60%, **big markets' is 0%.** The cause is structural, not a tuning miss. The module correctly never asserts the summit anywhere. The repo states plainly that this needs an economics ruling, not a builder's. **Build Week 6 as it stands; do not force the summit by retuning constants for the picture's sake.**
2. **Whether the franchise auction ships at all.** It is experimental and explicitly not architecture-critical. If it cannot be built without becoming a scoreboard, it does not ship, and Week 6 is not weakened by its absence.

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8. END-OF-RUN DISCIPLINE

- **Append** numbered decisions to `docs/PRODUCT_DECISIONS.md`. Never rewrite or delete history.
- Update `TRACK_101_MAP.md` and `RAMAZ_READINESS.md` only if status actually changed.
- Keep `runtime/README.md` honest about test/build status and known gaps.
- Update `docs/BOW_FABLE_GAP_MAP.md` as systems move from ABSENT to EXISTS.
- Leave the tree clean, committed, and pushed to the working branch. **The founder controls merge to main.**

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9. THE ONE-SENTENCE VERSION

Take six certified lessons and make them one premium, online-first, socially alive six-week franchise story — by closing the seam from Week 3 to Week 4, giving the room a way to reach every desk in every week, building the Press Conference and The Tape, and never letting a beautiful surface teach a false economics.

Now inspect the repository and begin at Wave 0.

BOW_FABLE_GAP_MAP

BOW FABLE GAP MAP

Repo inspected: `github.com/braydenokley13-ux/bow-economics-live` @ `eadc041` (2026-09-03, merge of `claude/bow-econ-m2-live-build-3mq2z0`) **Inspected:** 2026-09-04, directly, at file level. Not from a summary. **Companion to:** Product Bible v1.1 §36

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0. HOW TO READ THIS

Each row carries a status and a disposition:

Status	**Meaning**
EXISTS	Built and gate-certified. Working.
PARTIAL	Real but incomplete against the product requirement.
ABSENT	Does not exist in any form.

Disposition	**Meaning**
PRESERVE	Do not touch except to extend. Rewriting this is a net loss.
EXTEND	Build on top; the foundation is right.
REBUILD	The current answer is wrong for the product, not merely incomplete.
DEFER	Correct to leave alone until after October 1.

The one rule. The repository is authoritative about what exists. **Inspect before changing.** Six lessons carry independent gate certification, repaired findings and rerunnable proofs; a rewrite for elegance destroys real evidence. Repo `CLAUDE.md` §12 forbids extracting a generic engine from two data points, and that restraint applies to every new primitive in this document.

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1. THE SHAPE OF THE REPO

- **Root** — the BOW Boss development control plane (`tools/boss/`, `.boss/`, `.claude/`). **Never product logic. Do not repurpose it.**
- **`runtime/`** — the product. One Node process, no database, no external service, no internet required (repo D12). TypeScript, **zero runtime dependencies**, `node:test`, Playwright e2e via `runtime/scripts/*.cjs`.
- **`docs/`** — `PRODUCT_DECISIONS.md` (D1–D47, the standing build law, read first), `TRACK_101_MAP.md`, `RAMAZ_READINESS.md`, `ECONOMICS_CONCEPT_MAP.md`, `EMERGING_PRIMITIVES.md`, `SOURCE_LEDGER.md`, `gauntlet/module-1|2/` (design wars, gates, verification evidence, screenshots).
- **`design/`** — `VISUAL_IDENTITY.md` (the Cap Room), rendered contrast/CVD proof assets, SVG frames.

Surfaces: `/play` (private, per-seat) · `/teach` (bearer-key gated control room) · `/board` (projector; `boardView` is structurally never handed a seat identity).

Scale: ~51 source files, ~48k lines. The three largest are lesson modules: `hostTheLeague.ts` (5,393), `fullHouse.ts` (5,257), `writeTheRule.ts` (4,589). `client/play/main.ts` is 6,041 lines and is shared by every lesson — **it is the highest-risk file in the repo and every new student surface lands in it.**

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2. CROSS-CUTTING SYSTEMS

System	Status	Where	Disposition	Notes for the build
Lesson module contract	**EXISTS**	`src/shared/lessonModule.ts`	**PRESERVE**	`phases` · `initialState({sessionId, seatIds, seed})` · pure `reduce` · `studentView`/`teacherView`/`boardView`/`aggregate` · optional `round` (TIME

System	Status	Where	Disposition	Notes for the build
				CUT), `onPhaseExit`, `classEvents`. The runtime never inspects lesson state. **Every new primitive must fit this or extend it deliberately, with a written reason.**
Canonical phases	**EXISTS**	`src/shared/phases.ts`	**PRESERVE**	LOBBY · HOOK · PLAY · REVEAL · CONSEQUENCE · ADAPT · **COUNTERFACTUAL** · ARGUE · SYNTHESIS · COMPLETE. A lesson declares an ordered subsequence. **COUNTERFACTUAL already exists in every module — it is The Tape's foothold.**
Session service / persistence	**EXISTS**	`src/server/sessionService.ts` (1,344), `snapshotRepository.ts`	**PRESERVE**	Snapshot persistence, corrupt-snapshot quarantine, restart survival. Known limit: every session/seat is

System	Status	Where	Disposition	Notes for the build
				kept in memory for the process lifetime, no prune path — fine at classroom scale.
Transport + action integrity	**EXISTS**	`sessionBus.ts`, `http.ts`, `client/shared/poll.ts`, `outbox.ts`, `freshness.ts`	**PRESERVE**	Push primary, **server is sole truth**, poll/refetch fallback; exactly-once or an explicit named refusal. Repo D23. Proven by `e2e-realtime.cjs`, `concurrency-harness.cjs` at 16 and 32 desks.
TIME CUT	**EXISTS**	`RoundContract` in `lessonModule.ts`	**PRESERVE / EXTEND**	FINAL CALL + CLOSE NOW on server time; **each lesson declares its own fallback, named per desk, in words, before the close.** Any new decision window must declare one.

System	Status	Where	Disposition	Notes for the build
Recovery / WHILE YOU WERE AWAY	**EXISTS**	`classEvents`, `whileYouWereAway.test.ts`	**EXTEND**	Class-level, past tense, no verdicts, never rewinds. **This is the raw material for PREVIOUSLY ON, which does not yet exist as a product surface.**
Teacher auth / control	**EXISTS**	per-session bearer key (repo D14)	**PRESERVE**	
Class clock	**EXISTS**	repo D42	**EXTEND**	Server-clock based. **The economics protection rule (ECONOMICS AT RISK / FINAL CALL) attaches here.**
Rehearsal / cold walk	**EXISTS**	repo D40	**PRESERVE / EXTEND**	Every lesson fully walkable with **zero desks**; stand-ins prefixed `REHEARSAL —`. Extend to any new surface, or a first-time teacher cannot practise it.
Claim audit	**EXISTS**	`moduleClaims`, `clientClaims.test.ts`, M2 harnesses	**PRESERVE — and obey**	Every registered claim is a computed atom, recomputed and

System	Status	Where	Disposition	Notes for the build
				mutation-proven. **A claim-carrying surface added without registering it is invisible to the audit. Every new surface registers its claims.**
Drawn-magnitude fidelity	**EXISTS**	`arena-wedge-fidelity.cjs`, repo D25	**PRESERVE — and obey**	Any drawing whose *size* carries a quantity is measured in a browser by a self-poisoning instrument. **The Pool animation and the Mirror scatter are both subject to this.**
Grade band (`gradeBand`)	**ABSENT — deliberately**	repo D38	**EXTEND when a second band exists**	Attaches at exactly two points: `createSession` input carried onto the session row, and the `initialState` context modules already receive. **Do not ship an unused switch.**
Client module registry	**PARTIAL**	`client/{play,teach,board`	**EXTEND, do not genericise**	**Adding a lesson or a surface means**

System	Status	Where	Disposition	Notes for the build
		`}/main.ts` special-case each module by tag; generic JSON dump fallback		**writing its render functions.** The *server* contract is generic; the client shell is not. Budget for this on every new surface. Genericising it is a rewrite, not a feature (repo `EMERGING_PRIMITIVES.md` is visibility only).
Decision memory / commitment capture	**ABSENT**	—	**BUILD**	Bible §13. Seven of eight fields derive from state modules already hold; the desk types one short line and picks a chip. Manifestation varies by week — **do not build six identical exit tickets.**
Press Conference	**ABSENT**	—	**BUILD**	Bible §12. Board state + surface lock + candidate ranking + contextual questions. Build inside **one** lesson first.

System	Status	Where	Disposition	Notes for the build
The Tape / Decision Replay	**PARTIAL**	`COUNTERFACTUAL` implemented in all seven modules; no replay of a desk's own information state	**EXTEND**	Bible §12.3. Reconstruct what was known, unknown, available, chosen, predicted, and what happened. **Never undo.**
The four truthful readings	**ABSENT as a surface**	components exist inside modules	**BUILD**	Bible D-010. Components, sorted, never summed. No composite.
Desk-to-desk interaction	**ABSENT — anywhere in the codebase**	—	**BUILD**	The War Room. This is the largest net-new system.
Parent artefact data	**ABSENT**	—	**BUILD (export, not pipeline)**	Repo D12 forbids external services in the runtime. Generate and export from the session; delivery is operations.
Gamification	**ABSENT — deliberately**	repo D4	**KEEP ABSENT**	No XP, levels, badges, leaderboards. The class reveal is the reward system.

3. WEEK 1 — `m1l1-draft-day` (`draftDay.ts`, 1,141 lines · `draftDay.test.ts`, 836)

Requirement	Status	Disposition
Reversible Roster Wall + live **Foregone Panel** (opportunity cost ambient, not a report)	**EXISTS**	**PRESERVE — this is the best expression of the anchor in the product**
$100M cap / $10M steps / five slots, **cap inviolable through every reachable sequence**	**EXISTS**, brute-forced	**PRESERVE**
Lock ritual → shock → ADAPT with budget = remaining cap room (repo D15) → RISK BUFFER mechanically true (repo D16)	**EXISTS**	**PRESERVE**
Stock franchise for an absent/unclaimed desk	**EXISTS**	**PRESERVE**
Truthful cross-desk scarcity — another desk's action changes my options	**ABSENT**	**BUILD** — Week 1 is the weakest week on the founder's social principle, and it is the first impression. Sealed and simultaneous. Must not break cap-inviolability or the ADAPT guarantees.
Curated franchise choice (~3 real, viable, meaningfully different options per desk)	**ABSENT**	**BUILD** — Bible D-004. Class-level assignment silently ensures diverse cap positions, roster states and needs.
Stated intent (contend/retool/rebuild), changeable	**ABSENT**	**BUILD**
Sealed prediction, opened Week 5	**ABSENT**	**BUILD**

Requirement	Status	Disposition
Commitment capture at lock	**ABSENT**	**BUILD**
Timed against a real 50–60 minute period	**NEVER DONE**	**DO IT**

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4. WEEK 2 — `m1l2-trade-deadline` (`tradeDeadline.ts`, 1,357 · tests 1,157)

Requirement	Status	Disposition
L1 franchise state carried via opaque seed; desks **claim** their franchise	**EXISTS**	**PRESERVE**
Deterministic, path-dependent **midseason report** on the desk's actual L1 roster	**EXISTS**	**PRESERVE**
Three paths: STAND PAT · CUT+VETERAN · CUT+SEALED BID on four scarce named targets	**EXISTS**	**PRESERVE**
Hidden per-seller reserve, so a solo bid still faces hidden-information pricing	**EXISTS**	**PRESERVE**
Cut commits at submission; loser keeps the hole and the dead cap; restricted rescue window with a brute-forced ≥2-affordable guarantee	**EXISTS**	**PRESERVE — this closes the wait-and-see exploit**
Dead cap ≈10%; teacher-paced per-target reveal theatre	**EXISTS**	**PRESERVE**
THE WAR ROOM — active lines, incoming calls, offer	**ABSENT**	**BUILD — highest priority. Froth 3–5 directions first.**

Requirement	Status	Disposition
builder, one counter, reason chips, timeouts, league feed, published NEED/AVAILABLE		Default: a fourth deadline path. Must not break the certified sealed-bid economics.
The Tape, first home	**ABSENT**	**BUILD HERE FIRST** — dead cap and near-miss pairs make this the richest soil
Press Conference, first home	**ABSENT**	**BUILD HERE FIRST**
No release path for a claimed franchise — `claimedBy[idx]` is written and never deleted	**DEFECT**	**FIX** — a desk that claims the wrong franchise cannot be unstuck by anyone, including the teacher

---

5. WEEK 3 — `m1l3-free-agency` (`freeAgency.ts`, 1,738 · tests 1,102)

Requirement	Status	Disposition
Four-day sealed market; one binding offer or an explicit hold per day; teacher-paced day closes resolving all agents simultaneously	**EXISTS**	**PRESERVE**
Top offer at/above ask signs **at the offer price**; asks move by demand (0 → −\$10M, 1 → −\$5M, 2+ → **+\$5M**, floor \$10M); day-4 desperation	**EXISTS**	**PRESERVE — this is the market lesson**
Cap rise \$100M → \$130M as an institution adjusting, preserving path dependence including dead cap	**EXISTS**	**PRESERVE**

Requirement	Status	Disposition
Seed chain L2 → L3, falling back to L1, then honest stock	**EXISTS**	**PRESERVE**
Withdrawing an offer ends that desk's market day	**EXISTS**	**PRESERVE**
Staged module finale: recap → factor reveals → standings → semis/final → **GM Awards** → **COUNTERFACTUAL** → SYNTHESIS	**EXISTS**	**PRESERVE / EXTEND** — the counterfactual stage is where The Tape plugs in
Economically honest tie-break between identical sealed bids	**DEFECT** — ties break on `submittedAt`, the server's record of which Chromebook was faster	**FIX** — the one thing a sealed bid exists to rule out. Cap room, or the agent choosing on a published preference.
Client race — `faPlayMounted` poll-driven remount wipes `#faComposerRoot` while a click is pending; `e2e-l3` fails ~1 in 3	**DEFECT**	**FIX** — bounded path in `docs/gauntlet/module-2/E2E_L3_FLAKE_NOTE.md`. It lives in the shared play client, so it endangers every lesson.
The seed OUT into Week 4	**ABSENT**	**BUILD — founder-locked, see §6**

---

6. WEEK 4 — `m2l1-full-house` (`fullHouse.ts`, 5,257 · tests 1,953 + board tests)

Requirement	Status	Disposition
Price the night blind against real NBA night cards; hidden demand curve; **Two Peaks** reveal	**EXISTS**	**PRESERVE**
CASH / RENEWALS as two non-collapsible books; the	**EXISTS**	**PRESERVE** — and obey repo D21 FD-1: **no frame**

Requirement	Status	Disposition
computed frontier in both books' own units		**may be captioned "you gave up almost nothing to keep the fans"**
Path dependence across nights (repeat card)	**EXISTS**	**PRESERVE**
THE GATE CALL free commitment removing dead air (repo D28)	**EXISTS**	**PRESERVE — and attach the belief capture to it**
Measured arena fill (repo D25, D21 FD-3)	**EXISTS**	**PRESERVE**
M1 → M2 SEED	**ABSENT — deliberately today (repo D20)**	**BUILD. Founder-locked. The single most important seam in the course.** Use the existing opaque-seed mechanism.
THE BILL COMES DUE opening — the desk's own payroll as the fixed cost, players named	**ABSENT**	**BUILD**
Shortfall decision and its public consequence	**ABSENT**	**BUILD**
THE MIRROR — the room's own pricing becomes the dataset	**PARTIAL** (reveal exists; the class-as-dataset framing does not)	**EXTEND**
Known gaps: arena capacities/champion line not re-verified against the dated ledger; never timed against a real period; night-spend dial is linear in dollars (the decision is *when*, not how much); Night-4 capacity option is a hedge, not a rival strategy		**DISCLOSE in the debrief; restamp the facts**

Requirement	Status	Disposition
`boxOffice.ts` (`m2-box-office`)	**DEREGISTERED — killed on evidence (repo D20)**	**DO NOT RESURRECT**

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7. WEEK 5 — `m212-host-league` (`hostTheLeague.ts`, 5,393 · tests 2,168)

Requirement	Status	Disposition
Visiting club's **Draw** dominates your gate; reinvest-in-Draw as an intertemporal trade	**EXISTS**	**PRESERVE — this is the externality**
Decomposition reveal with two instruments: **DEALT** totals and **BY-CHOICE** figures against a never-reinvested counterfactual	**EXISTS**	**PRESERVE** — and keep the honesty: by-choice is *ceteris paribus*, not general equilibrium, and no surface claims otherwise
Real clubs as desks; four market profiles (two clubs of the same size share a curve — a ledgered honesty trade)	**EXISTS**	**PRESERVE**
Equal national check clears every club's bill from every reachable state	**EXISTS**	**NOTE:** this is why cash pressure is never manufactured here — the solvency squeeze belongs to Week 4
Per-beat reveal gating (numbers for an unpressed beat are **unsent**, repo D26)	**EXISTS**	**PRESERVE**
THE POOL — the online-native ritual	**ABSENT**	**BUILD** — contributions animate in, labelled and to scale, the total builds, the screen stops, `/teach` asks

Requirement	Status	Disposition
		WHO GETS IT. Subject to repo D25.
Binding in-session split	**ABSENT**	**BUILD** — the money moves on screen before anyone leaves
Prediction replay (the Week-1 sealed prediction opens here)	**ABSENT**	**BUILD**
W4 → W5 seed (gate, price history, fan support as the contribution base)	**ABSENT**	**BUILD**
Open findings: sports-reality restamps; several `SIMPLIFICATIONS` overclaims; a $120 dead zone; projector findings P-3..P-9 incl. freeze blanking the board; no elapsed clock against absolute-minute TIME CUT triggers; Pause vs Freeze undistinguished		**TRIAGE — the projector and freeze items are classroom-visible and should be fixed**

---

8. WEEK 6 — `m213-write-rule` (`writeTheRule.ts`, 4,589 · tests 1,197)

Requirement	Status	Disposition
Rule dials → **sealed two-thirds vote** (sealed at the close of round 3; post-close proposals refused by the reducer; a desk with no number **abstains**, with no relief on the denominator)	**EXISTS**	**PRESERVE — the sealing discipline is exactly right**

Requirement	Status	Disposition
Lived-under season governed by the room's own rule	**EXISTS**	**PRESERVE — this is the consequence-after-deliberation shape**
Kings 22-8 commit-then-reveal capstone (outcome vs decision quality)	**EXISTS**	**PRESERVE**
"Economics You Learned" revisit-able finale deck; deck held by a desk is a high-water mark (repo D47)	**EXISTS**	**PRESERVE / EXTEND** — this is where named economics returns to the student surface (Bible §12.4)
Live in-band gauge on `/board` ("right now N of M would pass; K are needed")	**EXISTS**	**PRESERVE**
L2 → L3 opaque seed	**EXISTS**	**PRESERVE**
Private stakes cards before debate opens	**ABSENT**	**BUILD**
The counterfactual as a headline beat (what the other rule would have produced)	**PARTIAL**	**EXTEND**
Franchise auction	**ABSENT**	**EXPERIMENTAL — only if it earns it.** Sealed bids, equal budgets stated early, anonymous, and the price framed as *the market's opinion, which can be wrong*
The economics finding — the design's summit ("sharing pays the payer through the product") **does not hold at shipped constants**: small markets' own best share 55–60%, big markets' **0%**; structural, not a tuning miss	**OPEN**	**ESCALATED — repo says explicitly this needs an economics ruling, not a builder's**

Requirement	Status	Disposition
Reveal half budgeted ~28 min against a 7–15 min debrief standard; round histogram not split by market size; no non-16:9 projector shape measured; finale cards 6–7 cover neighbouring arena material	**OPEN**	**FIX in the finale wave**

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9. THE TEN THINGS MOST LIKELY TO GO WRONG

1. **Rebuilding the runtime for elegance.** It is small, dependency-free, contract-driven and independently attacked. Extend it.
2. **Extracting a generic engine** from Press Conference + Tape after one lesson. Repo `CLAUDE.md` §12 forbids it and the founder's own restraint rule agrees.
3. **Forgetting the client shell is not generic.** Every new surface needs render functions in `play`, `teach` and `board` main.ts.
4. **Adding a claim-carrying surface without registering it** in `moduleClaims` — invisible to the audit, and the one way to reintroduce the whole defect class.
5. **Drawing a quantity without measuring it** (repo D25).
6. **Letting the student device get ahead of the board** (repo D26) — numbers for an unpressed beat must be *unsent*, not merely unrendered.
7. **Shipping a decision window without a declared TIME CUT fallback**, named per desk, in words.
8. **Building a `gradeBand` switch with no content behind it** (repo D38).
9. **Breaking Week 2's or Week 3's certified economics** while adding the War Room or the tie-break fix. Re-run the module tests and the e2e chain.
10. **Treating this document as truth six weeks from now.** It is a snapshot of `eadc041`. Re-inspect.

BOW_FABLE_ACCEPTANCE_MATRIX

BOW FABLE ACCEPTANCE MATRIX

Product conditions, not implementation checks.

"The page exists," "the test passes," "the component renders" are implementation checks. They are necessary and they are not acceptance. A surface is accepted when a *student experience* is right.

How to use this. Before building a surface, read its row. After building it, run its tests as written — several are things you do by playing the product, not by asserting on it. **A surface that passes every implementation check and fails its kid test is not done.**

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0. THE SIX UNIVERSAL CONDITIONS

Every surface in this product must pass all six.

#	Condition	PASS	FAIL
U1	**Economics is earned, then named, then retrieved**	The product produces the experience, the instructor names the concept from the room's own words, and the term returns later in Previously On, replay, history or the finale deck	Economics stays implicit and nobody ever says the word — **or** vocabulary is pasted onto gameplay before the experience earned it
U2	**No machine verdict**	Truthful components, separately legible, that a room can argue with	Any BOW-issued composite — a score, an index, a grade, a "value: fair" rating
U3	**Another human is in it**	A desk can point at something another desk did that changed what this desk could do	Sixteen desks technically coexist and nothing any of them does reaches anyone else

#	Condition	PASS	FAIL
U4	**It could not be a generic team**	The surface reads this desk's own carried history and would be *wrong* with a stranger's franchise	Swapping in another team's numbers changes nothing anyone would notice
U5	**Online is an advantage**	The surface does something a physical classroom could not do at all, or could only do worse	It is a digital imitation of something a room does better
U6	**It is a decision, not a document**	The dominant object on screen is a choice with a cost	The dominant object is a dashboard of the franchise's condition

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1. THE PRESS CONFERENCE

Lens	PASS	FAIL
Kid	The room leans in and at least one desk has a follow-up question before the instructor asks for one	It reads as being called on. The desk wants it to end.
Economics	The class can state **the decision and the tradeoff** within ten seconds, without reading a paragraph	It resembles a quiz review, or the concept is named before the defence is given
Social	Other desks argue *with the decision*, by name, and the argument survives the surface closing	The room watches politely and nothing carries
Teacher	A first-time instructor runs it cold from the console with the question already written, and takes the first question themselves	The instructor has to invent a good question in front of the room

Lens	PASS	FAIL
Age band	5–6 get two prepared friendly questions and a supplied counterfactual; 7–8 face one adversarial and one unseen, and must concede the strongest cost of their own decision	Same script, different font
Visual	**LEAGUE PAUSED** reads instantly from the back of the room; the decision is the headline, not the topic	A modal over the play surface
Persistence	The decision shown is real, carried, and dated to the moment it was made	A generic prompt with the desk's name inserted
Runtime	Every `/play` surface locks; the state survives refresh; the board and the desks are in the same beat	A desk can keep playing during the pause
Safety	Questions attack the decision; the first Press Conference of a desk's course is invited, never cold-called; a desk may decline once, silently; names on shared surfaces are fictional	A child is put on the spot with no way out

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2. THE TAPE / DECISION REPLAY

Lens	PASS	FAIL
Kid	*"Oh — I didn't know that then."* Recognition, not embarrassment	It reads as a report card on a decision they already regret

Lens	PASS	FAIL
Economics	The room rates the **decision** before the outcome is shown, and is confronted with its own change afterwards	It shows what happened and calls that a lesson
Truth	It reconstructs only what was actually knowable at that moment; running it twice gives the identical information state	It renders the past with knowledge the desk did not have — **hindsight contamination, the one fatal defect**
Boundary	History stays history	It becomes undo, or lets a desk change a committed decision
Teacher	The instructor picks the pairing and the order; the console proposes candidates from real state	The instructor must remember what two desks did four weeks ago
Age band	5–6: one contrast, supplied. 7–8: generates the counterfactual before the reveal	—
Persistence	It works on a Week-1 decision in Week 5	It only works within a session
Online	It does something no classroom can: exact reconstruction of a child's information state weeks later	It is a slideshow of what happened

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3. THE ONLINE TRADE DEADLINE WAR ROOM

Lens	PASS	FAIL
Kid — the gate	A real desk can honestly say: ***"I only have two lines open, Miami is calling, Boston***	**It is a trade machine with a timer.** Everything else about

Lens	PASS	FAIL
	hasn't answered, the clock is running, and another trade just changed what I can do."	the surface is irrelevant if this fails.
Economics	A desk can **name a call it chose not to take, and why.** Attention is scarce and spending it costs something	Lines exist but never bind; nobody ever runs out of anything but time
Social	Another desk's action forces me to reconsider my current plan **within seconds**	I could play the entire window without noticing anyone else
Distribution	Every desk faces at least one genuine accept/decline/counter decision; no desk finishes having spoken to nobody	Two loud desks absorb every call and five desks type into a void
Teacher	The console shows the whole negotiation graph at a glance and names the desks nobody has called, before the window closes	The instructor is asked to track sixteen conversations
Age band	5–6: fewer lines, proposed structures, a guaranteed inbound. 7–8: free origination, and *"why didn't you take that call?"* becomes a Press Conference question	Same window, smaller numbers
Visual	The clock and the remaining lines are the two loudest things on screen	The offer builder dominates and the scarcity is decoration
Persistence	The trade changes both rosters and both desks' Week-3 needs	Trades resolve and nothing downstream notices
Runtime	Exactly once or an explicit named refusal; no asset in	A transport race loses a desk's choice

Lens	PASS	FAIL
	two accepted trades; nothing executes after the deadline; a refresh mid-negotiation is boring	
Truth	The server blocks illegality and **never** blocks or grades bad strategy	A "trade value: fair" verdict appears anywhere

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4. THE BILL COMES DUE

Lens	PASS	FAIL
Kid	*"That's my team. I did this."*	*"Here are some numbers about a basketball team."*
Economics — the gate	A desk can explain **why its bill is what it is by naming an actual earlier decision**	**Week 4 would work just as well with a generic team**
Economics — truth	The payroll is a **solvency constraint**; the product never implies a fixed cost changes the best price	It teaches that costs move the optimal price — which is the sunk-cost fallacy in a jersey
Social	The room's own prices become the dataset the room analyses	Each desk prices alone against a curve and never sees anyone else
Teacher	The contrast pair is chosen by the console and lands in one press	The instructor has to build the comparison by hand
Age band	5–6: price from a menu; the shortfall reads *"you are $22M short"*, **never as a negative**; 7–8: free pricing plus a reprice round so the price trap can spring in both directions	Same screen, harder numbers

Lens	PASS	FAIL
Visual	The players who caused the bill are **named on the invoice**	A financial summary with a red total
Drawing	Any drawing whose size carries a quantity is measured against the model in a browser	A fill or a bar that is drawn by eye
Persistence	Roster, payroll and dead cap all arrive from Week 3, including for a desk whose Week 3 went badly	The seed exists in the data and not in the experience

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5. THE POOL AND THE BINDING SPLIT

Lens	PASS	FAIL
Kid	Somebody says *"that's not fair"* out loud, unprompted	The room waits to be told the answer
Economics	At least one desk names the **incentive cost** of sharing without being led there. Both halves are on the board: it makes small markets viable **and** it weakens the incentive to build your own	The lesson leaves students believing sharing is unambiguously good — a political position, not an economic concept
Truth — the correction	The room correctly identifies that **a rival outbidding you in Week 3 was competition, not an externality**	A pecuniary externality is taught as the anchor's example
Social	The money visibly moves between desks before anyone leaves	A non-binding vote, which reinstates the exact defect this exists to fix

Lens	PASS	FAIL
Teacher	One control, one question, and the console tells the instructor to stop talking	The instructor fills the silence
Age band	5–6: two stated options, the board performs the aggregation. 7–8: propose the formula and defend it against the desk it hurts most	—
Visual	The inequality is visible **by how much, per desk, to scale, measured**	A generic animation that shows only *that* it is unequal
Online	Strictly more honest than a bowl of chips, because the magnitudes are exact and labelled	A cheesy digital bowl

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6. WRITING THE RULE / THE FINALE

Lens	PASS	FAIL
Kid	*"My vote actually decided something."*	A classroom poll with nice graphics
Economics	At least one desk votes against its own private interest and can say why; the room can state what the **other** rule would have produced	The room votes its wallet unanimously and nothing is learned
Truth	**Nothing on any surface tells the room which rule was correct**	An impact panel projects confident outcomes, or an invented balance index appears

Lens	PASS	FAIL
Social	The threshold requires defectors, and someone is visibly persuaded	It passes 16–0
Teacher	The instructor never says which rule is better and the console never asks them to	The console supplies a verdict
Age band	5–6: vote between two options, board aggregates. 7–8: write the rule, predict the gaming, revise once	—
Timing	It fits 60 minutes with a real debrief	The reveal eats the debrief
Persistence	Every desk's stake derives from its own Week 4 and Week 5	Positions are assigned
Auction (if it ships)	Sealed, equal budgets stated in Week 2, anonymous, and closing on *"which franchise did the market get wrong?"*	It becomes the secret scoreboard the whole product refuses to have

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7. WEEK 1 AND THE HOME BASE

Lens	PASS	FAIL
Kid	*"I chose this team"* — never *"I was assigned this team"*	Leftovers for the last desk to pick
Economics	Opportunity cost is **ambient during the decision**, not reported after it	A summary of what you gave up, delivered once you can no longer act on it
Social	Most desks lose at least one thing they wanted, to a named desk, and **no desk ends with nothing**	The cap is the only adversary

Lens	PASS	FAIL
Fairness	Sealed and simultaneous; reaction speed never decides an outcome	First click wins
Truth	Cap inviolability and the ADAPT guarantees survive the new scarcity, brute-force verified	A reachable sequence breaks the cap
Consolation	A losing desk is told, in the same minute, what it *gained* — room, flexibility, the next pick	Losing is silent
Home base	A desk away for a week can name its current problem and one thing another desk did, in one sentence, without scrolling	A nine-item sidebar and a metrics grid
Age band	5–6: ≤6 cards on screen, benchmark fractions beside dollars, no negatives. 7–8: contested *judgment*, not contested obviousness	—

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8. THE COURSE AS A WHOLE

Lens	PASS	FAIL
Continuity	Six weeks played in order feel like **one franchise history**; a desk in Week 6 can trace a current constraint back two steps	Six good lessons that happen to share a team name
Anchors	A desk uses a prior week's anchor **unprompted in a week when it is not the topic**	Terms are recognised on sight and never used

Lens	PASS	FAIL
Teacher transfer	A trained non-founder runs Weeks 1 and 4 end to end, cold, from the console, and the class is good	Only the founder knows how to make it land
Anti-MyGM	Every major mechanic creates or strengthens economic reasoning, consequence, interdependence, persistence, evidence or teacher leverage	A mechanic exists because it feels more like a sports game
Anti-LMS	The dominant object is always a decision — who made it, what it cost, who else it moved	Forms, prompts, reflections, and a dashboard of condition
Parent	A parent can say what their child decided, what it cost and what changed, in the child's own words	A report with a number on it
Premium consistency	A stranger shown Week 1 and Week 5 side by side cannot tell they were built in different programs	The product visibly changes at Week 4
October 1	All six weeks run end to end, one premium product, persistent state throughout, the social layer present in every week, and a non-founder has passed a dry run	Three polished weeks and three scaffolded ones

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9. THE STANDING TRAPS

Check these on every wave. Each has already caused a real defect in this codebase or is explicitly forbidden by its decision log.

1. A claim-carrying surface added **without registering it in `moduleClaims`** — invisible to the audit.
2. A drawing whose size carries a quantity, **drawn by eye**.
3. The student device **ahead of the board** — numbers for an unpressed beat must be *unsent*, not merely unrendered.
4. A decision window shipped **without a declared TIME CUT fallback**, named per desk, in words.
5. **Layout width derived from a control** rather than from the layout.
6. A surface that **moves backwards**.
7. **Dead air** inside a decision phase — the fix is a free commitment, never a waiting message.
8. A **`gradeBand` switch with no content behind it**.
9. A **generic engine extracted from one or two data points**.
10. Claiming **"verified"** when you only reasoned; claiming **"passing"** without having run the command this session; claiming **"classroom-proven"** for anything at all.

BOW_FABLE_BUILD_PLAN

BOW FABLE BUILD PLAN

For: Fable 5.1 · **Against:** Product Bible v1.1, `BOW_FABLE_GAP_MAP.md`, `design/BOW_VISUAL_NORTH_STARS.md` **Deadline:** October 1, 2026 · **Target:** all six weeks as one premium product

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0. HOW THIS PLAN WORKS

This is a **product execution sequence**, not a task list. Each wave is a unit of *experienced quality*, and a wave is finished when a student experience is right — not when a component renders.

Every wave runs the same loop:

> **FROTH** (only where warranted — 3–5 *genuinely different* directions, never cosmetic variants) → **SELECT** with a stated reason → **BUILD** → **ATTACK** with fresh critics → **REPAIR or KILL** → **CARRY** what earned its place → **NEXT WAVE**

Wave order is derived from real dependencies, not from week order. Cross-cutting primitives come before the surfaces that use them; the M1→M2 seed comes before the Week-4 opening that reads it; the War Room comes after the primitives it will need but before the polish waves, because it carries the most product risk.

If a wave finishes early, do not stop. Attack the next frontier listed in that wave.

If a wave's surface is technically complete but product-poor, it is not done. Keep going.

WAVE 0 — RECONCILIATION

Objective. Make the repository and Bible v1.1 agree about what this product *is*, and clear the three defects that will otherwise corrupt every later wave.

Student magic moment. None. This wave buys the right to build everything else.

Economics requirement. One: the sealed-bid tie-break must stop being decided by network latency, because that silently teaches that speed beats judgment.

Preserve. Everything. This wave changes almost no product behaviour.

Surfaces. `docs/PRODUCT_DECISIONS.md` (append, never rewrite) · `freeAgency.ts` · `tradeDeadline.ts` · `client/play/main.ts`.

Work.

1. Append decisions recording the v1.1 founder rulings that change build law: online is the flagship; 60-minute sessions; the earned-naming rule for economics vocabulary; anchor/supporting/recognition tiers; rotating leverage replacing any guaranteed-win posture; the M1→M2 seed as required; the Press Conference and The Tape as committed primitives. **Do not delete or rewrite history.**
2. **Fix the sealed-bid tie-break** in `freeAgency.ts` — an economically honest rule (cap room, or the agent choosing on a published preference), never request arrival time.
3. **Fix the claim-release path** in `tradeDeadline.ts` — a desk that claims the wrong franchise must be recoverable by the teacher.
4. **Fix the L3 client race** (`E2E_L3_FLAKE_NOTE.md`). It lives in the shared play client and endangers every lesson.
5. Re-run the full suite and every e2e chain. Green before Wave 1.

High-risk failure modes. Touching module economics while "just fixing" the tie-break. Rewriting the client shell. Editing Boss run state.

Age bands. None.

Teacher / projector. None.

Acceptance. The full L1→L2→L3 chain runs green ten times consecutively. A sealed bid tie is resolved by something a child could be told out loud without embarrassment.

Kill if weak. Nothing here is optional.

WAVE 1 — THE CROSS-CUTTING PRIMITIVES

Objective. Build the four things every later wave depends on: **commitment capture**, **PREVIOUSLY ON**, **the four truthful readings**, and **the economics protection rule**.

Student magic moment. *"Wait — that's what I said in Week 1."*

Economics requirement. The **forgone alternative** becomes a first-class recorded object, because it is the anchor the entire course retrieves from. A blank, or "money," is not an acceptable answer.

Preserve. The module contract. The `classEvents` log. The class clock. Do not add a shared "decision store" the runtime inspects — decision memory is module-owned like everything else (`CLAUDE.md` §12).

Surfaces. `lessonModule.ts` (extend only with a written reason) · each module's `reduce` and views · `client/play|teach|board/main.ts` · `/teach` director panel.

Work.

1. **Commitment capture**, manifesting differently per week (Bible §13.2): at most **two inputs** on any commit — a chip and one short line. Week 1 attaches to the lock; Week 2 to the cut and to every declined offer; Week 3 to hold-or-offer; Week 4 to the gate call; Week 5 to the split position; Week 6 to the proposed number.
2. **PREVIOUSLY ON** — exactly three facts about *this desk*, every week, generated from `classEvents` plus carried state. **This is the memory countermeasure and it is mandatory.**
3. **The four truthful readings** on `/board` and `/teach` — components, sorted, never summed, no composite anywhere.
4. **The economics protection rule** on `/teach`: *ECONOMICS AT RISK — N MINUTES REMAIN. FINAL CALL.* On the server clock, against the session's remaining economics phases.
5. **Stated intent**, changeable, with the change itself recorded as reasoning evidence.

High-risk failure modes. Six identical exit tickets (**the single fastest way to make this feel like school**). A four-field form on a decision screen. A composite score sneaking in as a "summary." Capture that interrupts a decision the desk did not want to make.

Age bands. 5–6: chips carry most of the weight, the line is optional, and the forgone alternative may be selected from what they actually gave up. 7–8: the line is required and must name a *specific* alternative.

Teacher. Sees who has captured and what they said; can surface any capture to `/board` deliberately (nothing auto-publishes).

Projector. PREVIOUSLY ON at class level; the four readings; never a desk's private line unless the teacher raises it.

Acceptance. In Week 4, five desks asked *"why do you owe this much?"* answer by naming a real earlier decision. Capture completion above 90% without the teacher chasing anyone.

Kill if weak. If capture makes a decision screen feel like a form, cut it to one chip and keep PREVIOUSLY ON.

If ahead. Start the Tape's data model — what a replay needs recorded at the moment of decision.

WAVE 2 — THE PRESS CONFERENCE AND THE TAPE

Objective. Build BOW's signature reasoning primitive, **inside Week 2 only**, and prove it before extending.

Student magic moment. The board goes dark. **LEAGUE PAUSED — PRESS CONFERENCE.** Every screen locks. A desk's real decision is on the wall with everything they knew at the time — and they have to explain it while eleven other desks watch.

Economics requirement. Decision quality separated from outcome, **mechanically**: the room rates the decision before the outcome is shown, then again after, and is confronted with its own change. Outcome bias survives being told the rule; it only moves under repeated structured practice on uncomfortable cases.

Preserve. The `COUNTERFACTUAL` phase and every module's existing implementation of it — that is the foothold. Week 2's certified economics.

Surfaces. `tradeDeadline.ts` · `/board` (a new locked state) · `/play` (surface lock) · `/teach` (candidate ranking, question sets, invite control).

Work.

1. **THE TAPE**, five replay types, built to reconstruct: what was known · what was unknown · what options existed · what was chosen · what was predicted · what happened. **It is not undo. It never renders the past with knowledge the desk did not have.** Start with NEAR MISS and PREDICTION REPLAY.
2. **THE PRESS CONFERENCE**: board state, `/play` lock, candidate ranking by **interesting reasoning · contrast · risk · reversal · ambiguity** — never by who is winning — and 2–3 contextual questions per candidate generated from that desk's real state.
3. **The protections, structurally**: questions attack the decision, never the person; the instructor takes the first question; a desk's first Press Conference is invited, never cold-called; a desk may decline once per course, silently, at no cost; names on shared surfaces stay fictional.

High-risk failure modes. It reads as a quiz review. It humiliates a child. The candidate ranking quietly becomes a leaderboard. The Tape becomes an undo button. It is built as a generic engine before one lesson has proved it.

Age bands. 5–6: two prepared friendly questions, 45 seconds, counterfactual supplied. 7–8: two minutes, one adversarial and one unseen question, the rejected alternative named, and the strongest cost of their own decision conceded.

Teacher. This is the highest-leverage control in the product. It must be operable by a first-time instructor from the console, cold, with the question already written.

Projector. The locked state, the decision headline, the information state, and the outcome — revealed in the teacher's order, never all at once.

Acceptance. The room can state the desk's decision **and the tradeoff** within ten seconds of the surface appearing, without reading a paragraph. A desk that is behind is selected at least as often as one that is ahead. The Tape can be run twice on the same decision and shows the same information state both times.

Kill if weak. If the Press Conference cannot be made safe for an anxious child, keep The Tape (which is teacher-narrated and needs no volunteer) and defer the podium.

If ahead. Extend The Tape to Week 3's counterfactual stage.

WAVE 3 — THE M1→M2 SEED AND THE BILL COMES DUE

Objective. Close the founder-locked seam. **This is the wave that turns six lessons into a course.**

Student magic moment. Week 4 opens on an invoice with *your* players' names on it, and the number is short.

Economics requirement. The fixed cost is **not invented** — it is the payroll this desk built. And the reason must be stated precisely: with a fixed payroll and near-zero marginal cost per seat, revenue maximisation *is* profit maximisation; what the payroll adds is a **solvency constraint**, not a different optimal price. Claiming otherwise teaches the sunk-cost fallacy the course exists to attack.

Preserve. The opaque-seed mechanism exactly as it works today: the runtime never inspects the blob; the receiving module validates everything including its presence; a desk without valid prior state gets an honestly-labelled stock franchise. Full House's two books, hidden curve, Two Peaks, gate call, measured fill, and the forbidden frontier caption.

Surfaces. `createSession` link picker (which already warns when a source session is still live — repo D39) · `fullHouse.ts` `initialState` · the Week-4 opening.

Work.

1. **The W3 → W4 seed**: roster, payroll, dead cap, room, fan support, stated intent.
2. **THE BILL COMES DUE opening** — the bill, the gap, **the players named**, and what happens if it does not clear.
3. **The shortfall decision** and its public consequence.
4. **THE MIRROR** — the room's own pricing becomes the dataset it analyses, controlled for market size.
5. **The W4 → W5 seed**: gate, price history, fan support as Week 5's contribution base.

High-risk failure modes. The seed is built but the opening is generic, so the causal link exists in the data and not in the experience. A desk with a bad Week 3 finds Week 4 unplayable — it must be *hard*, never *impossible*, and the honest stock fallback must stay honest. The invoice reads as a finance dashboard.

Age bands. 5–6: price from a small menu; the shortfall reads *"you are $22M short"*, never as a negative. 7–8: free pricing across night classes plus a second reprice round.

Teacher. Who cleared, who is short, and the contrast pair worth putting on the wall.

Projector. Invoices landing across the room; the fill at measured true magnitude; the Mirror filling in live.

Acceptance — the gate. A desk can explain why its bill is what it is **by naming an actual earlier decision.** If Week 4 would work identically with a generic team, the wave failed.

Kill if weak. Nothing. This is founder-locked.

WAVE 4 — THE ONLINE TRADE DEADLINE WAR ROOM

Objective. Build the surface that proves online is an advantage. Highest social and interaction risk in the course.

Student magic moment — this is the acceptance condition, verbatim:

> *"I only have two lines open, Miami is calling, Boston hasn't answered, the clock is running, and another trade just changed what I can do."*

Economics requirement. Attention is a scarce, priced, countable resource. A desk must be able to name a call it chose *not* to take, and why. That is opportunity cost with a face on it — and it is the thing a physical trade floor cannot teach, because in a room attention is invisible.

Preserve. Week 2's certified sealed-bid market, the hidden seller reserve, the cut-commits-at-submission rule, the rescue guarantee, dead cap, and the per-target reveal theatre. **The War Room is additive.**

FROTH FIRST — mandatory, 3–5 genuinely different directions. At minimum: (a) a fourth deadline path inside L2 *(default)*; (b) a distinct pre-deadline negotiation window before the sealed market opens; (c) its own surface with the sealed market nested inside it; (d) a call-only model with no builder, where offers are assembled verbally and confirmed structurally. Select with a stated reason.

Surfaces. `tradeDeadline.ts` (new actions, new state, a declared TIME CUT for the window) · `/play` (the War Room) · `/teach` (the negotiation graph) · `/board` (clock, feed, completed trades as events).

Work.

1. **Negotiation lines** — 2–3 per desk, visible, countable, spent by taking a call or opening a thread.
2. **Incoming calls** — another desk rings you; take or decline under the clock; declining is an economic act and is recorded as one.
3. **Offer builder**, one counter, **decline with a reason chip**, thread timeouts that return a line.
4. **Published NEED / AVAILABLE** — a desk chooses to publish, which attracts the right calls and tells the room where it is weak.
5. **League feed** — a genuinely live, ordered record of what the room just did.
6. **THE SILENT DESK** on `/teach` — the desks nobody has called, named before the window closes.
7. Server-authoritative execution: **exactly once or an explicit named refusal**; no asset in two accepted trades; nothing executes after the deadline.

High-risk failure modes. It becomes a trade machine with a timer — the single most likely failure. Free-text creeps in. Lines exist but are never scarce enough to bite. Two loud desks absorb every call. A desk finishes the window having spoken to nobody. A machine "trade value" verdict appears.

Age bands. 5–6: fewer lines, offers from proposed structures, one counter, longer windows, a guaranteed inbound so nobody negotiates into a void. 7–8: more lines, free origination, and *"why didn't you take Phoenix's call?"* as a Press Conference question.

Teacher. Must be able to see the whole negotiation graph at a glance and route traffic to a silent desk without naming exclusion.

Projector. The clock, the feed, completed trades as league events. **Never a negotiation in progress.**

Acceptance. The founder's sentence is true for a real desk. Every desk faces at least one genuine accept/decline/counter decision. A desk can name a call it did not take. **And Week 2's existing economics tests still pass unchanged.**

Kill if weak. Cut incoming calls before cutting lines — the lines are the new economics.

If ahead. Frontier F-09, priced secrecy, for 7–8.

WAVE 5 — WEEK 1'S SOCIAL LAYER

Objective. Put a truthful cross-desk consequence into Draft Day, the weakest week on the founder's own social principle and the product's first impression.

Student magic moment. *He took the one I was building around, and I watched it happen.*

Economics requirement. Rivalry in the literal economic sense — a genuinely finite resource, where another desk's action removes an option from mine. Resolved **sealed and simultaneous**, so reaction speed never decides.

Preserve. Cap inviolability through every reachable sequence, the ADAPT budget rule, the RISK BUFFER's mechanical truth, and the Foregone Panel. All are brute-force verified; all can be broken by careless scarcity.

FROTH. Contested top-of-board players · a shared pool of a scarce *archetype* rather than named players · a contested cap-relief or roster-slot resource · scarcity that only bites the desks that waited.

Work. The chosen scarcity, resolved simultaneously; a consolation that names what the losing desk *gained* in the same minute (room, flexibility, the next pick); **curated franchise choice** (~3 real, viable, meaningfully different options per desk, with class-level assignment silently ensuring diverse cap positions and needs); the **sealed prediction** captured for Week 5.

High-risk failure modes. Cap inviolability breaks. A desk ends with an unplayable roster. The scarcity is theatre — visible but never actually binding. Choice becomes leftovers for the last desk to pick.

Age bands. 5–6: fewer, more obviously good contested players; scarcity blunt and visible. 7–8: contested players whose *value is contested*, so the scarcity is over judgment.

Teacher. Who wanted the same player; the contrast pairs the reveal should show.

Projector. The resolution: contested cards pulsing, then flipping, with the room's forgone value sorted beside it.

Acceptance. Most desks lose at least one thing they wanted, to a named desk, and **no desk ends with nothing.** The kid feels *"I chose this team,"* never *"I was assigned this team."**

Kill if weak. Ship curated choice and the sealed prediction even if the contested pool does not survive prosecution.

WAVE 6 — WEEK 5'S COLLECTIVE ECONOMY

Objective. Build the online-native pool ritual and make the split binding.

Student magic moment. Sixteen unequal contributions flow into one pool, labelled and to scale. The total builds. The screen stops. *"Who gets it?"* — and the instructor says nothing.

Economics requirement. Externality, in its real form. The diagnostic taught instead of a definition: *who was at the table, and who got the bill?* And the correction that must never regress: **a rival outbidding you is competition, not an externality — you were at the table.** Plus both halves of sharing: it makes small markets viable **and** it weakens every club's incentive to build its own market.

Preserve. Visitor-Draw dominance, the reinvest trade, the DEALT vs BY-CHOICE decomposition and its honest *ceteris paribus* limit, per-beat reveal gating, the four market profiles and their ledgered honesty trade.

Surfaces. `hostTheLeague.ts` · `/board` (the pool) · `/teach` (WHO GETS IT) · `/play` (the split position and the reinvestment).

Work.

1. **The pool ritual**, drawn to measured magnitude (repo D25 applies — a pool animation is a drawing whose size carries a quantity).
2. **The binding split**, executing in-session, with the money visibly moving.
3. **Prediction replay** — the Week-1 sealed prediction opens here.
4. **The W4 → W5 seed** consumed (built in Wave 3).
5. Close the classroom-visible projector findings: freeze blanking the board; non-16:9 shapes.

High-risk failure modes. A cheesy animation that shows *that* it is unequal without showing *by how much*. A non-binding vote — which reinstates the exact defect this wave exists to fix. The lesson leaving students believing sharing is unambiguously good.

Age bands. 5–6: two clearly stated split options; the board performs the aggregation. 7–8: propose the formula, defend it against the desk it hurts most.

Teacher. One control, one question, and the discipline to not answer it.

Projector. The pool is the centrepiece of the week.

Acceptance. The argument runs 8+ minutes without the instructor supplying content, and at least one desk names the incentive cost of sharing unprompted. Money visibly moves before anyone leaves.

Kill if weak. Cut the animation's flourish, never the binding split.

WAVE 7 — THE FINALE

Objective. Bring Week 6 to the north-star standard and fix its timing.

Student magic moment. The roll call lands, the season runs under the room's own rule, and then the screen shows what the *other* rule would have done.

Economics requirement. Institutional design: **aligning private incentive with collective outcome.** Fairness is a separate question reasonable people answer differently, and the product must keep the two questions apart on the board.

Preserve. The sealed two-thirds vote and its refusal discipline; the abstain rule; the lived-under season; the Kings capstone; the finale deck as a high-water mark.

Work.

1. **Private stakes cards** before debate opens.
2. **The counterfactual as a headline beat.**
3. **Fix the reveal-half timing** against 60 minutes. The 75–90 minute finale is a school-schedule exception, not the norm.
4. **The season card** each desk leaves with, and the Course 2 seat on the same page.
5. **The franchise auction only if it earns it** — sealed, equal budgets stated in Week 2, anonymous, and framed as *the market's opinion, which can be wrong*, closing on *"which franchise did the market get wrong?"*

High-risk failure modes. The vote passes unanimously (a mock UN). The impact panel becomes a confident composite index. The auction becomes the secret scoreboard the whole product refuses to have. Week 6 runs 25 minutes over and the debrief is lost.

Acceptance. At least one desk votes against its own private interest and can say why. The room can state what the other rule would have produced. **Nothing on any surface tells the room which rule was correct.** The lesson fits 60 minutes with a real debrief.

Kill if weak. The auction. It is explicitly not architecture-critical.

WAVE 8 — VISUAL ELEVATION AND THE PARENT PRODUCT

Objective. Make all six weeks look like one premium product, and generate the parent artefacts from state that already exists.

Work.

1. **Weeks 4–6 to the north-star standard.** M2 was graded serviceable-not-premium three times; the course may not visibly change products at Week 4.
2. The five north-star surfaces built to their acceptance conditions.
3. **The parent system, all three moments** (Bible §26.2), generated and **exported** from the session — no external service inside the runtime (repo D12). Delivery is operations.
4. Consent architecture: educational capture and marketing use separately consented, separately stored, separately revocable.

Acceptance. A parent can say what their child decided, what it cost, and what changed — in the child's own words. And a stranger shown Weeks 1 and 5 side by side cannot tell they were built in different programs.

WAVE 9 — THE SECOND BAND, AND THE FULL SIX-WEEK PROSECUTION

Objective. 7–8 depth at the two attachment points, then a full-course prosecution across both bands.

Work.

1. Grade profile attached at `createSession` and the `initialState` context — **and nowhere else** (repo D38). No unused switch.

2. **7–8 depth in this order: W1 → W3 → W2 → W4 → W5 → W6.** Authority, ambiguity and justification burden — **never** harder arithmetic. A half-built second band across six weeks is worse than a complete band plus three excellent ones.
3. **Full six-week prosecution**, one continuous play-through per band, attacked from every lens in the acceptance matrix.
4. Teacher dry run by a non-founder, cold, from the console.

Acceptance. A non-founder runs Week 1 and Week 4 end to end without founder help. Six weeks played in order feel like one franchise. Each band is prosecuted on its own evidence — one band's verdict never proves the other's.

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10. IF THE SCHEDULE SLIPS

Cut in this order, and do not deliberate:

1. The franchise auction.
2. 7–8 depth beyond Weeks 1 and 3.
3. The War Room's incoming-call layer — **keep the lines**, they are the new economics.
4. The Tape's less-used replay types — keep NEAR MISS and PREDICTION REPLAY.
5. Visual elevation beyond the five north-star surfaces.
6. The parent system's third moment (the end-of-course recap), keeping the Week-3 sentence and the Week-5 artefact.

Never cut: the M1→M2 seed · The Bill Comes Due opening · the Press Conference · the economics protection rule · sealed resolution · PREVIOUSLY ON.

BOW_VISUAL_NORTH_STARS

BBOW VISUAL NORTH STARS

For: Fable 5.1 · **Companion to:** `docs/BOW_ECONOMICS_PRODUCT_BIBLE.md` v1.1, `design/VISUAL_IDENTITY.md` (the Cap Room register) **Date:** 2026-09-04

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0. HOW TO USE THESE

Five images live in `docs/design/mockups/`. They are **interaction, hierarchy and ambition references.** They are **not** pixel specifications, and they are **not** product truth.

They contain wrong things on purpose and by accident. Every one carries NBA and club marks and player photography that this product may not ship; several carry cap magnitudes, mechanics and lesson identities that contradict the built product. **Never make product truth conform to a mistake in a mockup.** Where a mockup and the Bible disagree, the Bible wins; where the Bible and the repo disagree, §0.3 of the Bible governs.

What to take from them: the feeling in the first three seconds, the information hierarchy, what is loud and what is quiet, what is one glance versus one read, and the fact that this is a premium interactive sports-business product rather than school software.

The universal rejections — true of all five:

Reject	Why	Instead
NBA and club logos, crests, wordmarks, uniforms	Trademark; and the BOW+NBA lockup implies endorsement	BOW's own club identity system — city, colour, an original geometric mark
Player photographs; any child's face	Likeness rights; and student faces never appear on shared surfaces	Typographic player cards; an original illustration register
OVR / overall ratings (89, 84, 80)	A 2K convention that converts the game into "pick the higher number" and makes sports knowledge a prerequisite	Role, fit, contract, and the specific thing this player does — the economics is the comparison, not a scalar
Any invented composite index or verdict ("Trade Value: Fair", "Competitive Balance Index")	Bible D-010 and repo D4: no BOW-issued score, no machine verdict	Show the components. Let the room argue.

Reject	Why	Instead
Nine-item sidebars, dense metric grids, generic finance-dashboard chrome	Analytics-portal register; this is the LMS smell in a dark theme	Fewer, larger, louder surfaces; one question per screen

The universal preserves: the dark broadcast-control-room register · one dominant focal object per screen · a running clock that the room shares · live league movement as ambient texture · numbers large enough to read from the back of the room on `/board` · restraint everywhere except the four earned spectacle beats per session.

The "5 QUESTIONS FOR REVIEW" band at the foot of every mockup is review chrome for the founder, not product. It does not ship, and nothing like it belongs on a student surface.

Chromebook reality: the student surface is measured at **1024×600**. Every one of these mockups is drawn at desktop width. If a layout only works at 1600px, it does not work.

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1. `01-live-league-home-base.png` — THE LIVE LEAGUE / HOME BASE

1. Product moment. The desk's home surface — what *"I run a franchise"* looks like at the start of any week, before and between the decision windows.

2. Why it exists. To make a desk feel like it owns something that persists and is *moving without them*. It is the surface PREVIOUSLY ON lands on, the surface The Tape is reached from, and the place the league's live activity is felt.

3. First three seconds. *This is mine, it has a history, and something is happening right now.* The desk's club, its current problem, and evidence that fifteen other desks exist.

4. Hierarchy. (1) The one decision or state that matters this minute. (2) The books — payroll, room, dead cap, record — as a single glanceable strip. (3) The league moving. (4) The roster. (5) Everything else, behind a click.

5. Interactive. The current decision's entry point; the roster; the inbox; the league feed. **Nothing else on this screen needs to be interactive**, and most of the nine-item sidebar should not exist.

6. Private. Everything. This is `/play`. It never shows another desk's books, targets or reasoning.

7. Shared / class-level. The league feed only — and only the facts the league already published (a signing, a completed trade, a market movement), never another desk's intent.

8. Teacher counterpart. `/teach`'s room view: every desk's committed state, who has not committed, what the TIME CUT would apply to each, and the director's NOW / WATCH FOR line.

9. Projector counterpart. `/board` carries the league — the feed, the shared clock, the standings or books at class level. It never shows this screen.

10. Grade bands. 5–6: fewer visible fields, benchmark fractions beside dollars, no negative numbers, no more than six player cards at once. 7–8: more raw fields, more of the league's state visible at once, and the ambiguity left in.

11. Visual reference only. The hero treatment of the current decision; the compact books strip; the timestamped league feed; the overall dark composition.

12. Wrong or outdated in this mockup. The countdown implies asynchronous play — **the class is teacher-paced, not on a wall clock** · "FREE AGENCY" in Week 2 contradicts the built arc (Week 2 is the deadline; free agency is Week 3) · real NBA cap magnitudes contradict the $100M play scale · OVR ratings · logos and photography · the nine-item nav.

13. Fable must preserve. One dominant current problem. The books as one glanceable strip. **The league visibly moving.** The inbox as a real, earned signal.

14. Free to reinvent. Navigation entirely. The hero's shape. Where the roster lives. Whether the feed is a rail or a band.

15. Acceptance. A desk that has been away for a week can name, in one sentence and without scrolling, *what its franchise's problem is right now* and *one thing another desk did since last session.*

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2. `02-press-conference-decision-replay.png` — PRESS CONFERENCE + THE TAPE

1. Product moment. The instructor has paused the league. One desk's real decision is on the wall, with the information state it was made in, and that desk is defending it.

2. Why it exists. It is the single surface where drama, authentic reasoning, economics, assessment, student voice, teacher leverage and parent-visible evidence all happen at once. **This is the most important net-new surface in the product after the War Room.**

3. First three seconds. *That's a real decision somebody in this room made, and now they have to say why.* Sports drama about thinking — **never** assessment software.

4. Hierarchy. (1) The decision, stated as a headline in the desk's own terms — *"YOU WALKED AWAY AT $31.4M."* (2) The question. (3) **THE TAPE:** what was known then · what was unknown · what the options were · what they chose · what they predicted · what happened. (4) The instructor's prompts, on the teacher surface only.

5. Interactive. For the desk: their answer, captured briefly. For the instructor: which question, which follow-up, when to release. For the room: nothing — the room watches, then argues.

6. Private. The desk's stored reasoning until the instructor raises it. Other desks' equivalent records. The candidate ranking itself.

7. Shared. The decision, the information state, and the outcome — because they are facts about the league. **Student names on shared surfaces stay fictional** (repo CLAUDE.md §11).

8. Teacher counterpart. The candidate list ranked by **interesting reasoning · contrast · risk · reversal · ambiguity** — never by who is winning. Two or three contextual questions per candidate, generated from that desk's actual state. An invite control, because a desk's first Press Conference is invited, never cold-called.

9. Projector counterpart. This *is* the projector surface: **LEAGUE PAUSED — PRESS CONFERENCE**, every `/play` surface locked.

10. Grade bands. 5–6: two prepared, friendly questions; the counterfactual supplied. 7–8: one adversarial and one unseen question; the desk must name the alternative it rejected and concede the strongest cost of its own decision.

11. Visual reference only. The step-and-repeat framing, the headline-first composition, and the side-by-side of *the moment* and *the record.*

12. Wrong in this mockup. A real child's face — never ships · club logo and player photography · a single-column Tape that reads as a report rather than a reconstruction · "REFLECTION PROMPTS" as visible student-facing chrome, which is the school smell.

13. Fable must preserve. The headline is the *decision*, not the topic. **"What you knew then" is the whole idea** and must be visually equal to the outcome, never subordinate to it. The prediction, in the desk's own words, quoted.

14. Free to reinvent. Everything else — the backdrop conceit, the layout, whether the Tape animates the reconstruction.

15. Acceptance. The room can state the desk's decision **and the tradeoff** within ten seconds of the surface appearing, without reading a paragraph. And **The Tape never renders the past with knowledge the desk did not have.**

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3. `03-online-trade-deadline-war-room.png` — THE ONLINE TRADE DEADLINE WAR ROOM

> **The highest-priority product-design investigation in this handoff.** Everything else on this list improves something that exists. This creates the surface that proves online is an advantage rather than a compromise.

1. Product moment. Week 2, inside the deadline window. Desks are negotiating with each other under a shared clock, with limited attention.

2. Why it exists. Because *other humans materially change my decision environment* is the founder's principle and **the repo has no desk-to-desk interaction anywhere.** And because scarce, countable, visible **negotiation lines** are an economics lesson a physical trade floor cannot teach: in a room, everyone shouts at once and attention is invisible.

3. First three seconds. *There is not enough time and not enough of me to go around.*

4. Hierarchy. (1) The clock. (2) **My open lines, and how many I have left.** (3) The incoming call demanding an answer now. (4) The offer I am building. (5) The league feed. (6) The market table.

5. Interactive. Open a line · take or decline a call · build an offer · counter once · decline with a reason · publish a NEED or an AVAILABLE · let a thread time out.

6. Private. My roster's weaknesses until I publish them · my offers in flight · my reasons · who I chose not to call.

7. Shared. The clock · completed trades as league events · published NEED/AVAILABLE · declared buyer/seller posture · the market pulse. **Never** another desk's cap sheet detail or its unsent offers.

8. Teacher counterpart. Live negotiation graph — who is talking to whom, who has an unanswered offer, **and the desks nobody has called** (Frontier F-11). Plus FINAL CALL / CLOSE NOW and the per-desk fallback the module declares.

9. Projector counterpart. The clock, the league feed, and completed trades announced as events. **Never** a live negotiation in progress — the room learns what happened, not what is being typed.

10. Grade bands. 5–6: fewer lines, offers assembled from proposed structures, one counter, longer windows. 7–8: more lines but a harder justification burden, freely originated offers, and the ask *why did you not take Phoenix's call* as a Press Conference question.

11. Visual reference only. The three-column composition; the incoming-call treatment; the offer builder's two-sided symmetry; the feed and market rails.

12. Wrong in this mockup. "Trade Value — Fair/Fair" is a machine verdict on deal quality and is forbidden: the server blocks illegality, never bad strategy, and BOW issues no quality score · OVR ratings · logos and photography · "4 ACTIVE NEGOTIATIONS" shown as ambient decoration when it should be **the scarce resource itself**, counted and felt · real NBA magnitudes.

13. Fable must preserve. Line scarcity as the spine of the surface. The incoming call as a forced choice with a named desk behind it. A decline that carries a reason. Completed trades as class-wide events.

14. Free to reinvent. Everything about how a line looks, how a call arrives, whether the builder is two-sided or stacked, and how the feed and the market interleave. **Froth 3–5 genuinely different directions before building.**

15. Acceptance — the founder's sentence, and it is the gate.

> *"I only have two lines open, Miami is calling, Boston hasn't answered, the clock is running, and another trade just changed what I can do."*
>
> If a desk cannot honestly say that, **the surface failed** — no matter how well it renders. Second gate: a desk can name a call it chose **not** to take, and why. That is opportunity cost with a face on it.

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4. `04-bill-comes-due.png` — THE BILL COMES DUE

1. Product moment. Week 4, minute zero. The invoice for the team this desk spent three weeks building.

2. Why it exists. It is the M1 → M2 causal seam made visible, and it is founder-locked. Without it, Week 4 works just as well with a generic team — which is the exact failure this surface exists to prevent.

3. First three seconds. *I made this problem. Those are my players.*

4. Hierarchy. (1) **The bill, and the gap.** (2) **The players who caused it, named** — this is what makes it personal rather than financial. (3) The lever available (price, and what else). (4) The consequence of not clearing it. (5) The demand evidence.

5. Interactive. The pricing decision and the gate call. The shortfall decision if the bill does not clear. **Not** a menu of four strategy cards — the repo's discipline is *operate, don't choose from a menu*.

6. Private. This desk's books, its price before the close, its gate call.

7. Shared. After settlement: the room's prices and gates as **THE MIRROR** — the class's own decisions becoming the dataset it analyses. Controlled for market size, or the room cannot tell "different price" from "different city."

8. Teacher counterpart. Who has priced, who has called the gate, what the fallback will apply, and the contrast pairs worth putting on the wall.

9. Projector counterpart. The invoices landing across the room · the fill drawn at **measured true magnitude** (repo D25) · the Two Peaks · the Mirror filling in live.

10. Grade bands. 5–6: price from a small menu; cost in slices beside dollars; the shortfall reads *"you are $22M short"*, **never as a negative number**. 7–8: free pricing across night classes, the payroll carried, and a second reprice round so the price trap can be tested in both directions on the same franchise.

11. Visual reference only. The red alert band and its sentence — *"The players you fought for are now the players you have to pay for"* is close to exactly right. The three-tile financial summary. The outlook strip.

12. Wrong in this mockup. Real NBA cap magnitudes against the $100M/$130M play scale · **repeater tax** is CBA depth far outside scope · the dual-axis analytics chart is portal register and must obey the drawn-magnitude rule or not be drawn · the four "YOUR NEXT DECISIONS" cards are a strategy menu · stock-photo fan avatars.

13. Fable must preserve. The bill as the opening frame. **The players named.** One clear gap number. The emotional register: *save the team you fought to build*, not *review your financials*.

14. Free to reinvent. The whole layout. Whether fan reaction exists at all, and if so in what non-photographic form.

15. Acceptance. A desk can explain **why its bill is what it is by naming an actual earlier decision** — a signing in Week 1, a cut in Week 2, an offer in Week 3. If the surface would work identically with a generic team, it has failed.

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5. `05-board-of-governors.png` — WRITING THE RULE / THE FINALE

1. Product moment. Week 6. The room writes and votes on a rule, then lives under it.

2. Why it exists. Authority expansion made literal: a desk that started with one roster decision now binds every desk in the league. It is the course's intellectual summit.

3. First three seconds. *This is bigger than my team, and my vote actually decides it.*

4. Hierarchy. (1) The proposal, in plain language a ten-year-old can hold. (2) **What it is worth to me, privately.** (3) The tally against the threshold. (4) The room's votes as they land. (5) What it will do — *after* the vote, never before.

5. Interactive. Move the dial · propose a number · argue · vote. Then watch the season run under the adopted rule.

6. Private. The stakes card is delivered privately before debate opens — that is the mechanism that forces a desk to persuade someone to vote against their own interest.

7. Shared. The proposal · the live in-band gauge (*"right now N of M would pass; K are needed"*) · the roll call · the season under the rule · **the counterfactual.**

8. Teacher counterpart. Who has proposed, who has abstained, the distribution against the threshold, and the split worth naming out loud.

9. Projector counterpart. This is largely a `/board` surface: the dials, the gauge, the sealed vote, the roll call, the season, the counterfactual, the finale deck.

10. Grade bands. 5–6: vote between two clearly stated options; **the board performs the aggregation their cognition cannot yet perform internally** — the single best use of the projector in the course. 7–8: propose the formula, predict how it gets gamed, revise once before the vote.

11. Visual reference only. The three-column governance composition. The vote grid as the room made visible. The impact panel's *ambition* — a serious institution making a serious decision.

12. Wrong in this mockup. "SECOND APRON REFORM" is not the built lesson — the built vote is a revenue-sharing rule with a two-thirds threshold · **real team governors named as voters** is impersonation of real people · "23 votes to pass" is NBA arithmetic, not the class's two-thirds · **the LEAGUE IMPACT SIMULATION asserts four confident projections and an invented "Competitive Balance Index"** — that is exactly the machine verdict D-010 forbids, and showing it *before* the vote also tells the room the answer.

13. Fable must preserve. Private stakes before debate. A threshold that requires defectors. The room's votes landing one by one. **The vote changing something they then see.** The counterfactual as a headline beat.

14. Free to reinvent. The impact panel entirely — if consequence is shown, it is shown **after** the vote, in components the room can dispute, never as a confident composite index.

15. Acceptance. At least one desk votes against its own private interest and can say why. The room can state what the *other* rule would have produced. And nothing on the surface tells the room which rule was correct.

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6. WHAT THESE FIVE ADD UP TO

Read together, the five say one thing: **BOW is a premium live sports-business product in which the interesting object on screen is always a decision — who made it, what it cost, who else it moved, and what happens next.** Not a dashboard of a franchise's condition. Not a lesson with sports nouns.

If a screen Fable builds could have its numbers swapped for another team's without anyone noticing, it is the wrong screen.